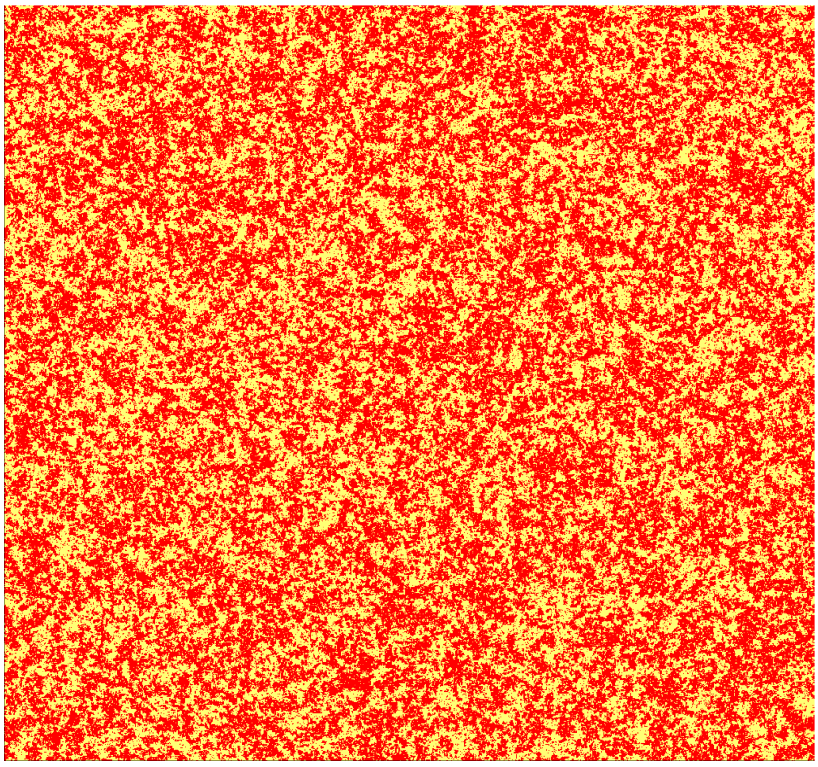
Computational Probability and Statistics (APMA 1690 @ Brown)

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These lecture notes are written for undergraduates who may not yet have encountered graduate-level probability or the advanced mathematical machinery it requires. Consequently, many discussions throughout these notes are intentionally not mathematically rigorous; they prioritize intuition over technical completeness. This choice is deliberate: setting aside certain technical details keeps the central ideas and the intuition behind them in clear view. The following quotation captures that philosophy:

> "*Computer compilers don't tolerate any mistakes or gaps, regardless of how minor they are. Human brains, on the other hand, are more flexible—minor glitches don't impede our progress. Moreover, humans typically don't process information in the same manner as computer compilers, except perhaps for some first-year graduate students. When these students initially delve into rigorous mathematics, they cannot tolerate any deviations from absolute rigor, sometimes causing them to lose sight of the big picture. Therefore, when dealing with mathematical concepts, it is advisable to start at a heuristic level initially. Once the big picture is established, we can then proceed to construct every detail in a mathematically rigorous manner.*"
>
> — believed to be rephrased sentences of Terence Tao

These sentences are from a post by a student who took Dr. Tao's MATH 247B course (Classical Fourier Analysis) at UCLA. The original version online is in Chinese; Kun Meng translated it into the English version herein.

Preface to the Fall 2022 Version

"*When you teach, you do something useful. When you do research, most days you don't.*"

— June Huh (2022 Fields medalist)

The lecture notes have been prepared for the course APMA 1690 at Brown University. The content is based on the notes from Professors Matthew Harrison, Hui Wang, and Stuart Geman, whose assistance I sincerely appreciate. Additionally, I would like to thank my teaching assistants, students, and friends who helped me refine these notes.

Background

I audited APMA 1690 in the fall of 2018 when I was a first-year Ph.D. student at Brown, which greatly helped me prepare these lecture notes. My educational background blends pure mathematics and biostatistics, positioning me somewhere between applied mathematics and statistics. Consequently, I believe I am well suited to teach APMA 1690, but time will tell.

The first draft of these lecture notes was written in the summer of 2022, shortly after I graduated with my Ph.D. from Brown. I poured my whole heart into crafting every sentence, equation, theorem, section, and chapter. To echo a sentiment from Taylor Swift's 2022 NYU commencement speech: "*Everything is connected by my love of the craft, the thrill of working through ideas and narrowing them down and polishing it all up in the end, editing, waking up in the middle of the night, throwing out the old ideas because you just thought of a newer, better one or a plot device that ties the whole thing together.*" This quote perfectly captures my experience while writing these lecture notes—the very notes you are now reading. Fall 2022 was the first time I taught APMA 1690. I am coming to realize that having brilliant ideas is one thing; conveying them clearly to others is quite another challenge.

Overview of the Lecture Notes

The lecture notes contain six substantive chapters, organized into the following three parts:

- **Part I—iid simulation and Monte Carlo**: Chapters 1 (*Simulating Randomness*) and 2 (*Stochastic Approximation*).
  - ♢ Chapter 1 reviews the probability and asymptotic theory needed for simulation, distinguishes true-random samples from pseudorandom output, constructs pseudorandom number generators, and develops transformations for sampling from univariate and multivariate distributions. It also introduces statistical tests for evaluating pseudorandom output.
  - ♢ Chapter 2 develops Monte Carlo integration from the law of large numbers and the central limit theorem. It studies estimation error and confidence intervals, compares Monte Carlo integration with deterministic Riemann sums, and introduces importance sampling as a response to the curse of dimensionality.

- **Part II—Markov chains and MCMC**: Chapters 3 (*Markov Chains*) and 4 (*Markov Chain Monte Carlo*).
  - ♢ Chapter 3 introduces transition probabilities, the Markov property, constructions of Markov chains, and random walks. It then develops recurrence and transience, stationary distributions, asymptotic behavior, and the spectral theory used to understand convergence and its speed.
  - ♢ Chapter 4 uses the two-dimensional Ising model to motivate sampling from complicated target distributions. It establishes the relevant convergence theory and develops the Metropolis–Hastings algorithm and the Gibbs sampler for approximating distributions and expectations.
- **Part III—structured high-dimensional models and data**: Chapters 5 (*Gibbs Random Fields*) and 6 (*Dimension Reduction*).
  - ♢ Chapter 5 revisits the Ising model to show how local graphical structure makes Gibbs updates computationally efficient. It then introduces Gibbs and Markov random fields, clique factorizations, the local Markov property, and the Hammersley–Clifford theorem.
  - ♢ Chapter 6 introduces the manifold hypothesis and the distinction between linear and nonlinear dimension reduction. Its main focus is principal component analysis, developed through both reconstruction error and explained variance, together with intrinsic-dimension selection and practical computation.

Since time is limited, we will not cover everything in the lecture notes. For instance, all materials marked with an asterisk "$*$" will be skipped in class and are not required.

In addition to purely applied topics, applied mathematics and statistics comprise two main components: methodology and its mathematical foundations. Since APMA 1690 is designed for undergraduate students, providing a complete mathematical foundation is nearly impossible unless all students have prior knowledge of real analysis (APMA 2110) and measure theory-based probability theory (APMA 2630 and 2640). Therefore, our primary focus will be on methodology, which includes various methods in computational probability and statistics. While we do cover the mathematical foundations of these methods, the presentation may not be rigorous or general. This approach is beneficial—it allows us to focus directly on the core of the course without getting bogged down by mathematical technicalities. From my own experience, and possibly shared by others, when a student from a pure mathematics background approaches statistics, a major hurdle can be an overemphasis on mathematical rigor and detail. Indeed, mathematical rigor solidifies methodology but can obscure the broader perspective. Just as tears bring realism to life yet blur the stars reflected in your eyes, rigorous mathematics solidifies methodology but may blur your view of the bigger picture.

Anyone familiar with multivariable calculus, linear algebra, and undergraduate-level statistics (APMA 1650/1655 or equivalent) should have enough prerequisite knowledge to read the lecture notes and take APMA 1690. Some basic coding skills will be helpful. In this course, we mainly focus on R. Supplementary materials will be provided

to help beginners in R and MATLAB. Programming languages are essentially nothing but calculators; it does not matter which one you use. Feel free to use any programming language you prefer.

Monologue

When I was in high school, I was extremely poor at both mathematics and English, although I excelled in Chinese literature and aspired to become an antique dealer. Ironically, I now earn my living by teaching mathematics in English.

Epilogue

To students who have been at Brown for one, two, or several semesters, I say, welcome back! To those who are new to Brown, I would say: welcome to Brown, it's been waitin' for you!

Meng, Kun (Michael †)

Providence, R.I.

Sept 1st, 2022.

† The traditional way of saying my full name is Meng Kun, which sounds like Michael. Both my first name **Kun** (`/Kwu:n/`) and my English name **Michael** work for me.

Preface to the 2026 Version

"*The future generations will look upon us, just like we look upon our past.*"

— Wang Xizhi

The lecture notes are intended for undergraduate students who are not familiar with graduate-level probability theory. I wrote the lecture notes while teaching *APMA 1690—Computational Probability and Statistics* at Brown University.

While writing the lecture notes, I referenced my very first paper on statistics (Meng and Eloyan, 2021). Now, this paper seems absolutely childish and clumsy, although I thought it was excellent when I was writing it. Similarly, I regard these lecture notes as childish and clumsy despite having devoted my heart to them. This sentiment applies not just to my first paper and these lecture notes but also to nearly everything I have done in my life. My future self will look back on me just as I now look back on my past. This process may be what we call growth, but growth is painful.

Kun Meng
(Mike)

I wrote a follow-up paper a few years later (Meng and Perez, 2026).

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Chapter 1

Simulating Randomness

"*Everything is connected by my love of the craft, the thrill of working through ideas and narrowing them down and polishing it all up in the end, editing, waking up in the middle of the night, throwing out the old ideas because you just thought of a newer, better one or a plot device that ties the whole thing together.*"

— Taylor Swift (NYU 2022 commencement speech)

1.1 Introduction

Simulation studies are an important tool in statistics. They are "computer experiments that involve creating data by pseudo-random sampling from known probability distributions" and are especially useful for evaluating and comparing statistical methods (Morris et al., 2019). Their foundation is the generation of random numbers, which also underlies the stochastic-approximation methods developed in Chapter 2.

This chapter develops that foundation. We first review the probability and statistics concepts needed to describe randomness. We then introduce pseudorandom-number generators and methods for transforming their output into draws from specified distributions (Sections 1.5 and 1.8). Section 1.9 uses hypothesis testing to assess the resulting sequences, and the final sections extend the discussion from individual random variables to random vectors. The ideas and results introduced here will be used throughout the later chapters.

1.2 Review of Probability and Statistics

"*There are only two kinds of math books: Those you cannot read beyond the first sentence, and those you cannot read beyond the first page.*"

— Chen-Ning Yang (1957 Nobel Laureate in Physics)

To prepare for the chapter's main topic, we briefly review the mathematical language used to describe randomness. The review is also useful in later chapters. To preserve intuition and avoid subtleties treated in *real analysis* (APMA 2110) and *probability theory* (APMA 2630 and 2640), **we will not be mathematically rigorous**. Optional remarks marked with an asterisk indicate where a more careful treatment is needed.

Two ideas organize the review: (i) recognizing the common features of random phenomena and (ii) representing those features through mathematical abstraction. These representations form the basic building blocks of probability theory.

Random phenomena arise whenever an outcome is uncertain before an experiment is performed or observed. Probability theory provides a framework for modeling and characterizing this uncertainty. The following experiments, interpreted broadly, illustrate such phenomena:

- Before flipping a coin, you do not know which side will show when the coin lands. There are two alternatives—heads or tails.
- Schrödinger's cat. A cat is in a closed box, and it can be either alive or dead. Before opening the box, you do not know whether the cat is alive or dead. The same story—two alternatives.
- Rolling two 6-sided dice has 36 possible outcomes. One does not know the outcome before rolling the dice.
- The amount of winnings from a lottery ticket.
- Your score on the APMA 1655 final.

These examples share the following features:

i) There is more than one possible outcome.

ii) Before doing or observing the experiment of interest, you do not know which outcome you will see. That is, there is some uncertainty.

iii) Some possible outcomes are likely, and some other outcomes are quite unlikely, e.g., winning one billion dollars by buying a lottery ticket.

Probability theory formalizes these features, helps us quantify uncertainty, and supports decisions made in its presence, such as whether to buy a lottery ticket tomorrow.

1.2.1 The Three Building Blocks of Probability Theory—$\Omega, X, \mathbb{P}$

"*Maybe if musicians got off their blogs and picked up their guitars, the music business would be in better shape.*"

— Gossip Girl (Episode 2, Season 1)

Probability theory rests on three building blocks: a sample space Ω, a random variable X, and a probability $\mathbb{P}$. We begin with motivating examples and then give formal definitions of these objects.

Building Block I: Sample Spaces

Each motivating example has more than one possible outcome. We seek a single framework that includes all these random phenomena as special cases. The following definition is the first step toward that framework.

Definition 1.2.1. *In probability theory, the* ***sample space*** *of an experiment or random trial is the set of all possible outcomes or results of that experiment. A sample space is usually denoted by* Ω.

Examples include:

- Flipping a coin. $\Omega = \{\texttt{H}, \texttt{T}\}$.
- Schrödinger's cat. $\Omega = \{\texttt{alive}, \texttt{dead}\}$.
- Rolling two 6-sided dice. $\Omega = \{(1,1), (1,2), \ldots, (1,6), (2,1), (2,2), \ldots, (2,6), \ldots, (6,5), (6,6)\}$ (there are 36 elements in Ω).
- Buying a lottery ticket. $\Omega = \{$"`to win`", "`not to win`"$\}$.
- The score you will get in your final exam of APMA 1655. $\Omega = \{0, 1, 2, \ldots, 99, 100\}$.

Before defining probability, we need one preliminary concept.

Definition 1.2.2. *Suppose* Ω *is a sample space.*

i) Each subset E *of* Ω *is called an* ***event****.*

ii) Additionally, the empty set $\emptyset$ *is called the* ***impossible event****.*

iii) The sample space Ω*, viewed as an event, is called the* ***certain event*** *(or* ***inevitable event****).*

Remark: Definition 1.2.2 is not adequate for graduate-level probability theory (e.g., APMA 2630). There, the definition of an event involves the real-analysis concept of a σ-algebra. In this undergraduate course, we use Definition 1.2.2, which is sufficient whenever Ω is finite.

We use the following example to explain Definition 1.2.2.

- Experiment: You take the final exam of APMA 1655 and get a score (out of 100).

- The sample space of this experiment is $\Omega = \{0, 1, 2, \ldots, 100\}$. Since you have not taken the final exam, the outcome—your score—is uncertain. Perhaps your score is available to the Lord but not available to us.
- The event "your score is higher than 50" is represented by the subset $E = \{51, 52, \ldots, 100\} \subset \Omega$. Additionally, the events "your score is a negative number", "your score is greater than 80 but less than 70", "your score became sentient and blew up the Sun", and all other impossible events are represented by $\emptyset$.

Building Block II: Probabilities

Set-Theoretic Preparation. Before defining probability, we need the following terminology.

Definition 1.2.3. *i) Let A and B be events. We say A and B are **disjoint** if $A \cap B = \emptyset$.*

*ii) Suppose we have a sequence of events $A_1, A_2, A_3, \ldots, A_n, A_{n+1}, \ldots$. We say that these events are **(mutually) disjoint** if all the pairwise intersections are empty, i.e., $A_n \cap A_m = \emptyset$ if $n \neq m$. For simplicity, we sometimes omit the word "mutually."*

Definition of Probability. In 1933, Andrey Kolmogorov introduced the axiomatic foundations of probability theory. The resulting definition is powerful enough to model randomness and serves as the foundation from which probability theory can be developed.

Definition 1.2.4. *Let Ω be a sample space. Suppose $\mathbb{P}$ is a real-valued function of subsets (events) of Ω, i.e.,*

$$\mathbb{P}: \quad \{\textit{subsets of } \Omega\} \to \mathbb{R}, \quad A \mapsto \mathbb{P}\{A\},$$

*where A is an input and $\mathbb{P}(A)$ is the corresponding output. If $\mathbb{P}$ satisfies the following three **axioms**,† the pair $(\Omega, \mathbb{P})$ is called a **probability space***

i) $\mathbb{P}\{A\} \geq 0$ for any subset $A \subset \Omega$; (interpretation: the probability of an event has to be non-negative;)

ii) $\mathbb{P}\{\Omega\} = 1$; (interpretation: the probability of "the inevitable event", which is just the event that literally something happens, is 100%;)

iii) For any sequence of disjoint subsets $\{A_i\}_{i=1}^{\infty}$, i.e., $A_i \cap A_j = \emptyset$ if $i \neq j$, we have $\mathbb{P}\{\bigcup_{i=1}^{\infty} A_i\} = \sum_{i=1}^{\infty} \mathbb{P}\{A_i\}$.

*Additionally, the map $\mathbb{P}$ is called a **probability**.*

Remark 1.2.1. *i) In the definition above, we used the phrase "a probability" instead of "the probability." On the same sample space, one may construct different probabilities.*

ii) $\mathbb{P}$ is the "blackboard bold font" of the capital letter P, which is a conventional notation in probability theory.

†: An axiom is a statement or proposition which is regarded as being established, accepted, or self-evidently true.

iii) One may show that $\mathbb{P}(A) \le 1$ *for all events* A *(see Theorem 1.2.1). Interpretation: the probability of an event is at most 1, i.e., 100%.*

*iv) (**σ-Algebras and Measurability*, Optional**) Definition 1.2.4 is not fully rigorous. It is not always possible to define* $\mathbb{P}$ *for all subsets of* Ω*. In graduate-level probability theory courses (e.g., APMA 2630), you will learn that* $\mathbb{P}$ *is well-defined only for certain special collections of subsets known as "σ-algebras." Therefore, in graduate-level probability theory courses, a probability space is defined as a triplet* $(\Omega, \mathscr{F}, \mathbb{P})$*, where* $\mathscr{F}$ *is a σ-algebra—a specific collection of subsets of* Ω*—and* $\mathbb{P} : \mathscr{F} \to \mathbb{R}$ *is a real-valued map defined on* $\mathscr{F}$ *satisfying the three conditions in Definition 1.2.4.*

In short, $\mathbb{P}\{A\}$ *makes sense only when* $A \in \mathscr{F}$*. If* $A \in \mathscr{F}$*, then* A *is called measurable. The discussion of σ-algebras and measurability is beyond the scope of APMA 1655. More details on the rigorous definition of probability spaces can be found in Chapter 1.2 of Klenke (2020) (Definition 1.38 therein).*

To avoid the constraint "$A \in \mathscr{F}$*," one needs a concept called "outer probability" (see Chapter 1.2 of van der Vaart and Wellner (1996)). Outer probability is a fundamental concept in the theory of "empirical processes," and empirical process theory is fundamental for modern nonparametric statistics. However, empirical process theory is beyond even the scope of APMA 2640, the Probability Theory II course for Ph.D. students.*

Example 1.2.1. $\Omega = \{1, 2, \cdots, n\}$ *with a large* n*. We can define a function* $\mathbb{P}$ *of subsets of* Ω *as follows*

$$\mathbb{P}(A) \stackrel{\text{def}}{=} \frac{\#A}{n}, \qquad \text{for any } A \subset \Omega, \tag{1.2.1}$$

where $\#A$ *denotes the number of elements in* A*. One may verify that the* $(\Omega, \mathbb{P})$ *defined herein is a probability space (left for homework). The probability* $\mathbb{P}$ *is just one probability, which is why we use the phrase "a probability" in Definition 1.2.4. We may define other probabilities, e.g., the probability* $\tilde{\mathbb{P}}$ *defined by the following*

$$\tilde{\mathbb{P}}(A) = 1 \quad \text{if } 1 \in A; \quad \tilde{\mathbb{P}}(A) = 0 \quad \text{if } 1 \notin A.$$

Properties of Probability. As noted above, Definition 1.2.4 is the foundation of probability theory. The following theorem collects several immediate consequences of the definition.

Theorem 1.2.1. *Let* $(\Omega, \mathbb{P})$ *be a probability space. Then the following statements hold:*

i) $\mathbb{P}(\emptyset) = 0$*, i.e., the probability of any impossible event is zero;*†

ii) if two events E_1 *and* E_2 *satisfy* $E_1 \cap E_2 = \emptyset$*, we have* $\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2)$*;*

iii) suppose $A, B \subset \Omega$*. If* $A \subset B$*, then* $\mathbb{P}(A) \le \mathbb{P}(B)$*;*

†: An event with probability zero is called a zero-probability event. Impossible events have probability zero, but zero-probability events need not be impossible. We discuss this distinction in later sections.

iv) $0 \le \mathbb{P}\{A\} \le 1$ *for any subset* $A \subset \Omega$*;*

v) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

vi) for any $A, B \subset \Omega$*, we have* $\mathbb{P}\{A \cup B\} = \mathbb{P}\{A\} + \mathbb{P}\{B\} - \mathbb{P}\{A \cap B\}$*;*

vii) for any countable collection $\{A_n\}_{n=1}^{\infty}$ *of subsets of* Ω*, we have* $\mathbb{P}\{\bigcup_{n=1}^{\infty} A_n\} \le \sum_{n=1}^{\infty} \mathbb{P}\{A_n\}$.‡

Proof. i) Let $A_1 = \Omega$ and $A_n = \emptyset$ for all $n \ge 2$. If $m \ne n$, at least one of A_m and A_n is empty, so $A_n \cap A_m = \emptyset$. Hence, $\{A_n\}_{n=1}^{\infty}$ is a sequence of disjoint sets, and

$$A_1 = \Omega = \bigcup_{n=1}^{\infty} A_n,$$

Therefore, $\mathbb{P}(A_1) = \mathbb{P}(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathbb{P}(A_n) = \mathbb{P}(A_1) + \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots$. Canceling $\mathbb{P}(A_1)$ gives

$$0 = \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots + \mathbb{P}(\emptyset) + \cdots \ge \mathbb{P}(\emptyset) \ge 0,$$

By axiom (i) of Definition 1.2.4, $\mathbb{P}(\emptyset) \ge 0$. Thus, $0 \ge \mathbb{P}(\emptyset) \ge 0$, so $\mathbb{P}(\emptyset) = 0$.

ii) Let $A_1 = E_1$, $A_2 = E_2$, and $A_n = \emptyset$ for $n \ge 3$. Then, $\{A_n\}_{n=1}^{\infty}$ is a sequence of disjoint sets (left for homework). We have the following identity (left for homework)

$$E_1 \cup E_2 = \bigcup_{n=1}^{\infty} A_n,$$

which implies

$$\begin{aligned}
\mathbb{P}(E_1 \cup E_2) &= \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(A_n) \\
&= \mathbb{P}(E_1) + \mathbb{P}(E_2) + \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots + \mathbb{P}(\emptyset) + \cdots \\
&= \mathbb{P}(E_1) + \mathbb{P}(E_2).
\end{aligned}$$

The proofs of other results are left for homework. □

‡: Separately, the finite inclusion–exclusion formula states that

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} (-1)^{k-1} \left\{ \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \mathbb{P}\left(\bigcap_{l=1}^{k} A_{i_l}\right) \right\};$$

This identity is not usually used in applications.

Definition 1.2.5 (Indicator functions)**.** *Let A be a subset of $\mathbb{R}^d$. The* ***indicator function*** *of A, denoted by $\mathbb{1}_A$, records whether a point belongs to A:*

$$\mathbb{1}_A(x) := \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

We also write $\mathbb{1}\{x \in A\}$ for $\mathbb{1}_A(x)$. Indicator functions translate statements about events into numerical expressions and will be used throughout these notes.

Building Block III: Random Variables

Suppose we are interested in the total SAT scores, reported on the 400–1600 scale, of all Brown undergraduate students. Associate each student ω with a total SAT score $X(\omega)$. For privacy reasons, the identity of ω is unavailable; we know only that ω is a Brown undergraduate. Let Ω denote the collection of all Brown undergraduates. Then X is the following function on Ω:

$$\begin{aligned} X: \ \Omega \to \mathbb{R} = \ & \text{the collection of all real numbers,} \\ & \omega \mapsto X(\omega). \end{aligned}$$

Furthermore, we assume that a probability $\mathbb{P}$ is defined on Ω, e.g.,

$$\mathbb{P}(A) \overset{\text{def}}{=} \frac{\#A}{\#\Omega}, \quad \text{for all } A \subset \Omega.$$

Suppose we are interested in the probability of the event

$$\begin{aligned} A = \ & \text{“a student's total SAT score is no higher than 1000”} \\ = \ & \{\omega \in \Omega \text{ such that } X(\omega) \le 1000\}. \end{aligned}$$

Then, the probability of interest is $\mathbb{P}(A)$, which may be expressed as $\mathbb{P}(X \le 1000)$ (just a notational convention).

After going through the motivating example, we do some abstraction and introduce the following definition.

Definition 1.2.6. *Let Ω be a sample space and $\mathbb{R}^d$ denote d-dimensional space.*

- *Any map $X : \Omega \to \mathbb{R}^d$, $\omega \mapsto X(\omega)$ is called a ($\mathbb{R}^d$-valued)* ***random variable****; when $d \ge 2$, the $\mathbb{R}^d$-valued random variable is also referred to as a* ***random vector****.*
- *If there exists a fixed $x \in \mathbb{R}^d$ such that $X(\omega) = x$ for all $\omega \in \Omega$, i.e., $X(\omega)$ is a constant function of ω, we call X* ***deterministic*** *or* ***degenerate****.*

- *If random variable X is not deterministic, we call X* ***truly random****.*

The term *pseudorandom* has a different meaning: it describes the deterministic output of an algorithm designed to mimic random samples. Pseudorandom sequences and their generators are introduced in Section 1.5.

Remark 1.2.2. *The concept of a random variable is very general: any function on a sample space Ω is a random variable. A potentially confusing point is that probability theory also treats deterministic quantities as random variables, a convention adopted for mathematical convenience. Thus, random variables include both genuinely random and deterministic cases.*

One fundamental and obvious question about Definition 1.2.6 is, *what is the underlying probability space $(\Omega, \mathbb{P})$?* We have provided several explicit examples of probability spaces. Generally, constructing probability spaces is a deep and complicated question in mathematics, which involves some machinery in *real analysis* (APMA 2110) and is partially the reason why the rigorous probability theory was not available until 1933. We avoid specifying probability spaces in this course. When we talk about the three building blocks (i.e., $\Omega, X, \mathbb{P}$), we only need to know the following in this course

- Whenever we talk about a random variable, we implicitly assume that there is an underlying probability space $(\Omega, \mathbb{P})$, and this random variable X is a function defined on the underlying space $(\Omega, \mathbb{P})$. In most scenarios, we will only emphasize the existence of such an underlying probability space without specifying what this space is. We sometimes briefly denote a random variable $X(\omega)$ as X, suppressing ω, for convenience.
- For a quantity X, to know whether it is truly random or deterministic, one just needs to answer the following question, *is it a nonconstant function of some "ghost" variable ω, i.e., $\omega \mapsto X(\omega)$?*

1.2.2 Probability Theory versus Statistics

There is an essential difference between Probability Theory and Statistics, although the two fields seem to discuss the same objects:

i) Data—objects observed by us, e.g., a collection of numbers.

ii) Data-generating mechanisms—the blueprints designed by God, e.g., a normal distribution that generates numbers.

The contrasting worldviews of probability theory and statistics may not be clear at first, and many people mistakenly treat the two fields as the same. This section helps distinguish the two fields. Probability theory and statistics apply to different scenarios and have different goals (that is, they answer different questions), which are summarized as follows:

- Probability theory: Suppose we understand the data-generating mechanism of interest, as if the blueprint had been given to us, but we have not observed any data generated from this mechanism. The question that probability theory helps answer is, *What will the data look like?*
- Statistics: Suppose we have observed a considerable amount of data. However, we are in a realistic scenario: we do not know the underlying data-generating mechanism. The question that statistics helps answer is, *What mechanism generated the data?*

The difference between the two fields is loosely summarized in the following table, i.e., Table 1.1.

Table 1.1: Probability Theory versus Statistics

	Probability Theory	Statistics
Data	Before observation—we do NOT have data.	After observation—we have data.
Data-generating mechanism	Known	Unknown—we want the mechanism.

1.2.3 Distributions of Random Variables

One may have noticed that the definition of random variables $X(\omega)$ has nothing to do with the probability $\mathbb{P}$. We also have claimed that we avoid specifying Ω. So, what do we really care about? We care about the combination of $\{X(\omega)\}_{\omega\in\Omega}$ and $\mathbb{P}$, instead of each single one of them. Specifically, for each $\mathbb{R}^d$-valued random variable $X(\omega)$, we care about all the following values

$$\mathbb{P}\{\omega \in \Omega : X(\omega) \in A\}, \quad \text{for any subset } A \text{ of } \mathbb{R}^d. \tag{1.2.2}$$

For each subset A, the value $\mathbb{P}\{\omega \in \Omega : X(\omega) \in A\}$ measures the rareness/commonness (according to $\mathbb{P}$) of the elements ω such that $X(\omega) \in A$. The so-called "distribution" of random variable X is defined as follows.

Definition 1.2.7. *Suppose $(\Omega, \mathbb{P})$ is the underlying probability space. Let $X(\omega)$ be a $\mathbb{R}^d$-valued random variable defined on Ω. We call the collection of all the values $\mathbb{P}\{\omega \in \Omega : X(\omega) \in A\}$, for all subsets A of $\mathbb{R}^d$,* ***the distribution of random variable X (according to $\mathbb{P}$).***

Cumulative Distribution Functions (CDFs)

Going through all the subsets A in Eq. (1.2.2) is a tough job. Fortunately, when $d = 1$, the real analysis machinery ("measure extension theorem," which is far beyond the scope of APMA 1690) shows that we only need to focus on the following specific subsets—intervals

$$(-\infty, t], \quad \text{for all } t \in \mathbb{R}.$$

These intervals result in the following concept.

Definition 1.2.8. *Let X be an $\mathbb{R}$-valued random variable defined on an underlying probability space $(\Omega, \mathbb{P})$. The function F_X defined as follows*

$$F_X(t) := \mathbb{P}\left(\{\omega \in \Omega : X(\omega) \leq t\}\right) = \mathbb{P}\left(\{\omega \in \Omega : X(\omega) \in (-\infty, t]\}\right), \quad \text{for all } t \in \mathbb{R}, \tag{1.2.3}$$

*is called the **cumulative distribution function** (CDF) of X. Thus, F_X is the CDF of X. (F_X is sometimes briefly denoted by F.)*

Remark: $F_X(t)$ is defined for all real numbers $t \in \mathbb{R}$.

The discussion above implies that **the CDF F of X contains all the information of the distribution of X**. Suppose we have another random variable Y, which is defined on a different probability space $(\Omega', \mathbb{P}')$. At first glance, X and Y are defined on different probability spaces and have nothing to do with each other. But we mentioned that we care neither just a single probability space nor just a single random variable. We care about the combination of them—the distribution of the random variable of interest. A potential relationship between random variables X and Y defined on different probability spaces is the following

Definition 1.2.9. *Let X be a random variable defined on a probability space $(\Omega, \mathbb{P})$, and let Y be a random variable defined on a possibly different probability space $(\Omega', \mathbb{P}')$. If X and Y have the same distribution, equivalently if $F_X(t) = F_Y(t)$ for every $t \in \mathbb{R}$, we say that X and Y are equal **in distribution** and write $X \overset{d}{=} Y$.*

Probability Density Functions (PDFs)

The concept of CDF will play an important role in simulating randomness (see the inverse CDF method in Section 1.8.1). Another concept in the following definition helps us simulate random numbers as well.

Definition 1.2.10. *Suppose $(\Omega, \mathbb{P})$ is the underlying probability space, and X is a random variable defined on this space. If there exists a non-negative function $p_X(\xi)$ on $\mathbb{R}$ such that*

$$F_X(x) = \mathbb{P}\{\omega \in \Omega : X(\omega) \leq x\} = \int_{-\infty}^{x} p_X(\xi) d\xi,$$

*then the function $p_X(\xi)$ is called the **probability density function** (PDF) of X.*

As reviewed in APMA 1650/1660, when the CDF F_X is **continuous and piecewise differentiable**, the CDF and PDF of X are related by

$$F_X'(x) = \frac{d}{dx} F_X(x) = p_X(x). \tag{1.2.4}$$

(I have to emphasize that Eq. (1.2.4) is just a sloppy formula, not rigorous.)

Remark 1.2.3 (*). *You will learn a general version of Eq. (1.2.4) in APMA 2110/2630—the Radon–Nikodym theorem. In the general version, F_X is only assumed to be absolutely continuous instead of piecewise differentiable (i.e., "continuous and piecewise differentiable" can be replaced with "absolutely continuous."). In this course, we stick to the differentiability assumption of F_X when applying Eq.(1.2.4).*

1.2.4 Continuous and Discrete Random Variables

Definition 1.2.11. *Let $(\Omega, \mathbb{P})$ be a probability space. Suppose X is a random variable defined on Ω, and F_X is the CDF of X. We say X is a* ***continuous random variable*** *if its CDF F_X is* ***continuous and piecewise differentiable****.*

Let us go back to the concept of CDFs. We need to know the following

- **All CDFs are nondecreasing** (a CDF is not necessarily strictly increasing). (The proof of this fact is left for homework.)
- **All CDFs F are right-continuous, and $\lim_{t\to-\infty} F(t) = 0$, $\lim_{t\to\infty} F(t) = 1$.** The rigorous proof of this fact involves some real analysis ideas, and we skip this proof. This right continuity comes from the "$\leq$" in the definition $\mathbb{P}\{\omega \in \Omega : X(\omega) \leq t\}$; if we replace the "$\leq$" herein with "$<$," the right continuity will be replaced with left continuity. Throughout this course, we stick to "$\leq$" and the right continuity, which is a widely adopted convention.

Continuous and Discrete Distributions

We now give an example of a CDF. Suppose X is a random variable taking values in the discrete set $\{x_1, x_2, x_3, \ldots\}$. Write

$$p_j = \mathbb{P}\{\omega \in \Omega : X(\omega) = x_j\}, \quad j = 1, 2, \ldots.$$

Because of the condition i) of Definition 1.2.4, we have $p_j \geq 0$ for all j; the condition ii) of Definition 1.2.4 indicates that $\sum_{j=1}^{\infty} p_j = 1$. One can verify that the following function is the CDF of X (left for homework)

$$\sum_{j=1}^{\infty} p_j \cdot \mathbb{1}(x_j \leq t).$$

Definition 1.2.12. *Let $(\Omega, \mathbb{P})$ be a probability space. Suppose X is a random variable defined on Ω, and F_X is the CDF of X.*

*i) We say X is a **discrete random variable** if its CDF F_X is of the following form*

$$\boxed{F_X(x) = \sum_{k=0}^{K} p_k \cdot \mathbb{1}_{[x_k, +\infty)}(x),} \tag{1.2.5}$$

where $\{x_k\}_{k=0}^K$ are distinct real numbers, $p_k \geq 0$ for all $k = 0, 1, \ldots, K$, and $\sum_{k=0}^K p_k = 1$; the K is allowed to be ∞.

*ii) If X is a discrete random variable whose CDF is of the form in Eq. (1.2.5), we call the ordered sequence $\{p_k\}_{k=0}^K$ as the **probability mass function (PMF)**† of X.*

The CDF of the distribution Bernoulli(p) can be represented as follows

$$F_X(x) = (1-p) \cdot \mathbb{1}_{[0, +\infty)}(x) + p \cdot \mathbb{1}_{[1, +\infty)}(x).$$

Hence, if $X \sim$ Bernoulli(p), then X is a discrete random variable; in addition, the PMF of X is the ordered sequence $\{1-p, p\}$.

Remark 1.2.4. *i) If the CDF of random variable X is the one in Eq. (1.2.5), we have $p_k = \mathbb{P}\{\omega \in \Omega : X(\omega) = x_k\}$ for all $k = 0, 1, \ldots, K$.*

ii) You may view the "ordered sequence" $\{p_k\}_{k=0}^K$ as the vector $(p_0, p_1, p_2, \ldots, p_K)$ if you are comfortable with the concept of vectors. When K is large, the vector may not have intuitive geometric meaning.

iii) (Not required) Some textbooks defined discrete random variables as follows: "A random variable X is discrete if its CDF is a step function" (e.g., Definition 1.5.7 of Casella and Berger (2002)). However, based on the definition of step functions on Wikipedia, the definition of discrete random variables in these textbooks is not fully accurate.

The following sentence on the Wikipedia page of step functions is accurate: "A discrete random variable is sometimes defined as a random variable whose cumulative distribution function is piecewise constant. In this case, it is locally a step function (globally, it may have an infinite number of steps). Usually however, any random variable with only countably many possible values is called a discrete random variable, in this case their cumulative distribution function is not necessarily locally a step function, as infinitely many intervals can accumulate in a finite region."

†: The ordered sequence $\{p_k\}_{k=0}^K$ is conventionally called as a function. You may view the map $k \mapsto p_k$ as a function. I think the reason $\{p_k\}_{k=0}^K$ is called a function is to make the names "PMF" and "PDF" look similar. In addition, if you are comfortable with the concept of vectors, you may view the ordered sequence $\{p_k\}_{k=0}^K$ as a vector $(p_0, p_1, \ldots, p_K)$; if $K = \infty$, this vector is infinitely long.

iv) (Not required) Suppose K is a finite positive integer. The following set is called a probability simplex

$$\Delta = \left\{ (q_0, q_1, \ldots, q_K) \,:\, \sum_{k=0}^{K} q_k = 1 \text{ and } q_k \geq 0 \text{ for all } k = 0, 1, 2, \ldots, K \right\}. \tag{1.2.6}$$

If the ordered sequence (you may refer to it as a vector) $\{p_k\}_{k=0}^K = (p_0, p_1, \ldots, p_K)$ is a PMF, then $(p_0, p_1, \ldots, p_K) \in \Delta$. The probability simplex defined in Eq. (1.2.6) is the fundamental concept in the homology theory (see MATH 2410—Algebraic Topology). The Δ defined in Eq. (1.2.6) can be used to prove the existence of stationary distributions of Markov chains (see Section 3.3).

You may try to visualize the Δ for $K = 1$ and 2.

v) (Not required) If $F_X(x)$ is the function defined in Eq. (1.2.5), we have the following "derivative"

$$F_X'(x) = \sum_{k=0}^{K} p_k \cdot \delta(x - x_k),$$

where $\delta(x)$ is the Dirac function in quantum mechanics, and the "derivative" is understood in an advanced sense. We may formally have the following formula

$$F_X(x) = \int_{-\infty}^{x} F_X'(t)dt = \int_{-\infty}^{x} \left[\sum_{k=0}^{K} p_k \cdot \delta(t - x_k) \right] dt.$$

The following are examples of continuous and discrete random variables.

Example 1.2.2. *i) If the PMF $\boldsymbol{p} = (p_0, p_1, p_2, \ldots)$ of a distribution is defined as follows*

$$\mathbb{P}\{\omega \in \Omega : X(\omega) = k\} = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \ldots,$$

for some fixed $\lambda > 0$, then this distribution is called the Poisson distribution with rate parameter λ and is denoted by $\text{Pois}(\lambda)$. The quantity $\frac{\lambda^k e^{-\lambda}}{k!}$ is interpreted as the probability that k events occur in a fixed interval of time, e.g., that k patients arrive in an emergency room between 10 and 11 pm.

*ii) Suppose X is a discrete random variable with PMF $\boldsymbol{p} = (p_1, p_2, \ldots)$. If there exists a positive integer k such that $p_j = 0$ for all $j > k$, then the distribution of X is called a **categorical distribution**, and the corresponding PMF is simply denoted by $\boldsymbol{p} = (p_1, p_2, \ldots, p_k)$. This distribution is denoted by $\text{Categorical}(\boldsymbol{p})$. When $k = 2$ and $x_1 = 0, x_2 = 1$, this special categorical distribution is called the Bernoulli distribution and is denoted by $\text{Bernoulli}(p)$, where $p = p_2$. The CDF of $\text{Bernoulli}(p)$ is*

$$(1 - p) \cdot \mathbb{1}(0 \leq t) + p \cdot \mathbb{1}(1 \leq t).$$

iii) The real-valued normal distribution with mean μ and variance σ^2, denoted by $N(\mu, \sigma^2)$, has the PDF $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$. The CDF of $N(\mu, \sigma^2)$ is

$$\Phi(t \,|\, \mu, \sigma) := \int_{-\infty}^{t} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx. \tag{1.2.7}$$

iv) If the PDF of a distribution is the following

$$\frac{1}{b-a} \cdot \mathbb{1}(a \le x \le b) \tag{1.2.8}$$

with $a < b$, then the distribution is called the uniform distribution on $[a, b]$ and is denoted by $\mathrm{Unif}(a, b)$. An important special case is $a = 0, b = 1$, which gives the distribution $\mathrm{Unif}(0, 1)$ with PDF $\mathbb{1}(0 \le x \le 1)$. We refer to $\mathrm{Unif}(0, 1)$ as "basic randomness" in Section 1.8.

v) We introduce a 2-dimensional generalization of $\mathrm{Unif}(a, b)$. Let B be a region in $\mathbb{R}^2$ with positive area, i.e., $\mathrm{Area}(B) > 0$. If the distribution of a $\mathbb{R}^2$-valued random variable (i.e., 2-dimensional random vector) X is defined as follows

$$\begin{aligned} \mathbb{P}\{\omega \in \Omega : X(\omega) \in A\} &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbb{1}\Big((x, y) \in A\Big) \cdot \left[\frac{1}{\mathrm{Area}(B)} \cdot \mathbb{1}\Big((x, y) \in B\Big)\right] dxdy \\ &= \frac{\mathrm{Area}(A \cap B)}{\mathrm{Area}(B)} \end{aligned} \tag{1.2.9}$$

for all subsets A of $\mathbb{R}^2$, then X has the uniform distribution on B, denoted by $\mathrm{Unif}(B)$. We use this distribution in the rejection sampling method for generating random numbers (see Section 1.8.3).

vi) We introduce a high-dimensional PDF through the two-dimensional case. Let $f(x, y)$ be a nonnegative bivariate function on $\mathbb{R}^2$ satisfying $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)dxdy = 1$. Let (X, Y) be a random vector such that

$$\mathbb{P}\Big\{\omega \in \Omega : \Big(X(\omega), Y(\omega)\Big) \in A\Big\} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbb{1}\Big((x, y) \in A\Big) \cdot f(x, y)dxdy \tag{1.2.10}$$

for all subsets A of $\mathbb{R}^2$, then $f(x, y)$ is called the PDF of (X, Y). Combining Eq. (1.2.9) and Eq. (1.2.10) implies that the PDF of $\mathrm{Unif}(B)$ is

$$\frac{1}{\mathrm{Area}(B)} \cdot \mathbb{1}_B(x, y). \tag{1.2.11}$$

More examples of distributions/CDFs/PDFs are available on the list of probability distributions in Wikipedia.

1.2.5 Random Variables That Are Neither Continuous Nor Discrete*

There are infinitely many distributions that are neither continuous nor discrete. For example, the distribution of the random variable X defined as follows

$$\begin{aligned} &X = Y + (1-Y)\cdot Z, \quad \text{where} \\ &Y \sim \text{Bernoulli}\left(\frac{1}{2}\right), \\ &Z \sim N(0,1), \\ &Y \text{ and } Z \text{ are independent.} \end{aligned} \tag{1.2.12}$$

Figure 1.1 displays the CDF F_X and illustrates that it is neither continuous nor of the form $\sum_{j=1}^{\infty} p_j \cdot \mathbb{1}(x_j \le t)$. Deriving an analytic expression for F_X is left for homework.

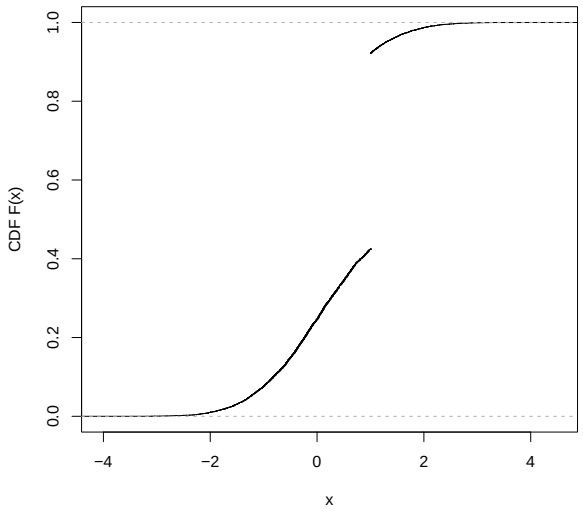


Figure 1.1: The CDF of the distribution of X defined in Eq. (1.2.12).

```r
n=10000
Y=rbinom(n, size = 1, prob = 0.5)
Z=rnorm(n)
X=Y*1+(1-Y)*Z
plot(ecdf(X), main="", xlab="x", ylab = "CDF F(x)")
```

Remark 1.2.5. *Lebesgue decomposition theorem implies that every CDF F can be decomposed as follows*

$$F(t) = p\cdot F_c(t) + q\cdot F_d(t) + r\cdot F_{sc}(t),$$

The "*" in the title of this subsection indicates that this subsection is not required in APMA 1690.

where p, q, r are non-negative and satisfy $p+q+r=1$, F_c is the CDF of a continuous random variable, F_d is the CDF of a discrete random variable, and F_{sc} is the CDF of a singular continuous random variable.

Singular continuous random variables are less common in elementary statistical modeling, but they are mathematically and scientifically relevant. Because this course focuses on common statistical simulation settings, we restrict attention to distributions without an F_{sc} component and use the following decomposition of F

$$F(t) = p \cdot F_c(t) + (1-p) \cdot F_d(t), \tag{1.2.13}$$

for some $p \in [0,1]$. When $p=1$, CDF F corresponds to a continuous random variable; when $p=0$, CDF F corresponds to a discrete random variable.

A random variable with the CDF F in Eq. (1.2.13) can be constructed as follows:

$$\begin{aligned} &X = Y \cdot W + (1-Y) \cdot Z \sim F = p \cdot F_c + (1-p) \cdot F_d, \quad \textit{where} \\ &Y, W, \textit{ and } Z \textit{ are mutually independent}, \qquad Y \sim \text{Bernoulli}(p), \quad W \sim F_c, \quad Z \sim F_d. \end{aligned} \tag{1.2.14}$$

The proof of Eq. (1.2.14) is left for homework. An example of Eq. (1.2.14) is provided in Figure 1.2.

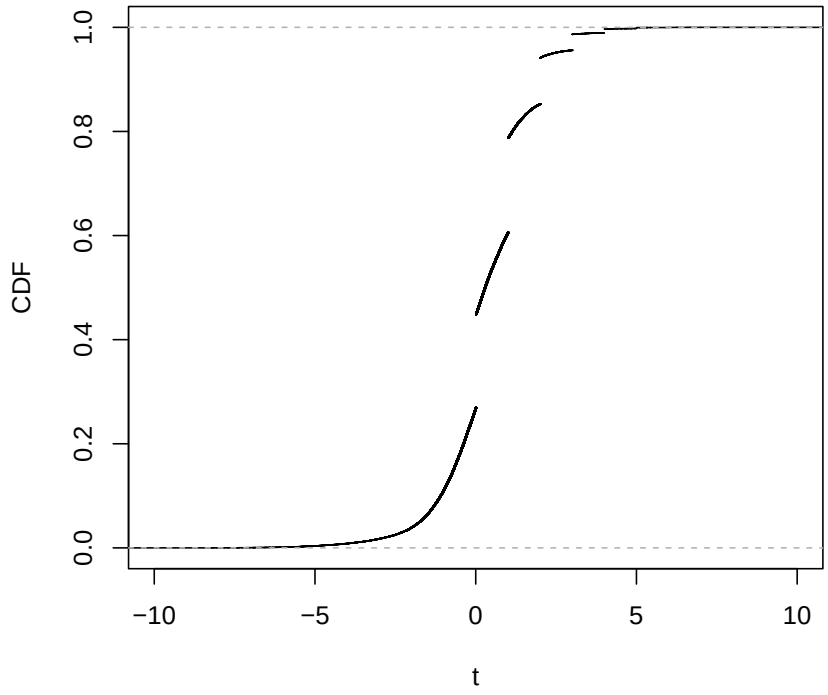


Figure 1.2: The CDF of $X = Y \cdot W + (1-Y) \cdot Z \sim F = p \cdot F_c + (1-p) \cdot F_d$, where Y, W, and Z are mutually independent, $Y \sim$ Bernoulli(0.3), $W \sim F_c = N(0,1)$, and $Z \sim F_d = \text{Pois}(1)$ = the Poisson Distribution with parameter $\lambda = 1$.

Because of Eq. (1.2.14), generating a random variable from the CDF F in Eq. (1.2.13) suffices to generate the continuous and discrete random variables considered in this course. For simulation methods based on densities, we assume that the continuous distributions of interest are absolutely continuous and therefore admit PDFs. Not every continuous distribution has a PDF; singular continuous distributions are the standard counterexamples.

1.2.6 Expectation and Variance

The expectation, also called expected value, of a random variable is a generalization of weighted average. Let X be a random variable defined on the probability space $(\Omega, \mathbb{P})$.

Suppose g is a real-valued function defined on $\mathbb{R}$, i.e.,

$$\begin{aligned} g: \ & \mathbb{R} \to \mathbb{R}, \\ & x \mapsto g(x). \end{aligned}$$

Composing the random variable X with the function g gives

$$\begin{aligned} \Omega \ & \xrightarrow{X} \ \mathbb{R} \xrightarrow{g} \ \mathbb{R}, \\ \omega \mapsto & X(\omega) \mapsto g\left(X(\omega)\right). \end{aligned}$$

According to the definition of random variables, the following map is a random variable

$$\omega \mapsto g\left(X(\omega)\right). \tag{1.2.15}$$

For simplicity, the map in Eq. (1.2.15) is denoted by $g(X)$.

For example, let Ω be the collection of all students taking APMA 1655 this semester, and let $X(\omega)$ denote the final-exam score of student ω. Suppose the professor curves the scores according to

$$Y(\omega) \stackrel{\text{def}}{=} \min\{X(\omega)^2, 100\},$$

where $Y(\omega)$ is the curved score of the student ω. If we define the function $g(x) \stackrel{\text{def}}{=} \min\{x^2, 100\}$, then $Y(\omega) = g(X(\omega))$.

A natural question is the following: If we know the distribution of X, can we compute or define the expected value of $g(X)$? The question is answered by the following definition.

Definition 1.2.13. *Suppose g is a real-valued function defined on $\mathbb{R}$.*

i) Let X be a discrete random variable whose CDF is $\sum_{k=0}^{K} p_k \cdot \mathbb{1}_{[x_k,\infty)}(x)$. If $\sum_{k=0}^{K} |g(x_k)| \cdot p_k < \infty$, then the

following sum is called the mean or expected value of $g(X)$ and is denoted by $\mathbb{E}[g(X)]$:

$$\boxed{\mathbb{E}\left[g(X)\right] \stackrel{\text{def}}{=} \sum_{k=0}^{K} g(x_k) \cdot p_k.} \tag{1.2.16}$$

If $\sum_{k=0}^{K} |g(x_k)| \cdot p_k = \infty$, we say the expected value of $g(X)$ does not exist.

ii) Let X be a continuous random variable with PDF $p_X(x)$. If $\int_{-\infty}^{+\infty} |g(x)| \cdot p_X(x)dx < \infty$, we call the following integral the mean or expected value of $g(X)$ and denote it by $\mathbb{E}[g(X)]$:

$$\boxed{\mathbb{E}[g(X)] \stackrel{\text{def}}{=} \int_{-\infty}^{+\infty} g(x) \cdot p_X(x)dx.} \tag{1.2.17}$$

If $\int_{-\infty}^{+\infty} |g(x)| \cdot p_X(x)dx = \infty$, we say the expected value of $g(X)$ does not exist.

Remark: In graduate-level probability theory courses, you will learn that Definition 1.2.13 is actually a theorem called the law of the unconscious statistician (LOTUS).†

Remark 1.2.6 (*). *For simplicity, assume $g(x) = x$ in this remark. The conditions $\int_{-\infty}^{\infty} |x| \cdot f(x)dx < \infty$ and $\sum_{j=1}^{\infty} |x_j| \cdot p_j < \infty$ are needed for three reasons.*

i) The absolute convergence of $\int_{-\infty}^{\infty} |x| \cdot f(x)dx < \infty$ and $\sum_{j=1}^{\infty} |x_j| \cdot p_j < \infty$ implies that the integral and infinite series in Eq. (1.2.17) and Eq. (1.2.16) are well defined. Recall that absolute convergence implies convergence.

ii) If the infinite series is conditionally convergent, i.e., the limit $\lim_{n\to\infty}(\sum_{j=1}^{n} x_j \cdot p_j)$ exists but $\sum_{j=1}^{\infty} |x_j| \cdot p_j = \infty$, then the Riemann series theorem implies the following "unreasonable" result: for any fixed real number M, we may permute the terms $x_j \cdot p_j$ (i.e., $1, 2, 3, 4, 5, \ldots$ are permuted to $7 = \sigma(1), 294 = \sigma(2), 27 = \sigma(3), 43 = \sigma(4), 1690 = \sigma(5), \ldots$, where $\sigma(j)$ is the j-th number in the permuted sequence) such that

$$\sum_{j=1}^{\infty} x_{\sigma(j)} \cdot p_{\sigma(j)} = M.$$

This result indicates that, if we did not assume the absolute convergence $\sum_{j=1}^{\infty} |x_j| \cdot p_j$, we could permute the terms $x_j \cdot p_j$ so that the value of $\mathbb{E}X$ in Eq. (1.2.16) would be any prescribed real number. In that case, the concept of expectation would be meaningless.

iii) In modern probability theory, $\int_{-\infty}^{\infty} x \cdot f(x)dx$ is understood in the Lebesgue integral sense. In this sense, $\int_{-\infty}^{\infty} x \cdot f(x)dx$ is well-defined only if $\int_{-\infty}^{\infty} |x| \cdot f(x)dx < \infty$.

As discussed at the beginning of this course, the value of a random variable X can be uncertain. Variance quantifies this uncertainty.

†: "The naming is sometimes attributed to Sheldon Ross' textbook Introduction to Probability Models, although he removed the reference in later editions."

Definition 1.2.14. *If $\mathbb{E}(|X|^2) < \infty$, we define the following value as the **variance** of random variable X*

$$\mathbb{V}X := \mathbb{E}\left[\Big(X - \mathbb{E}X\Big)^2\right] = \mathbb{E}(X^2) - \Big(\mathbb{E}X\Big)^2. \tag{1.2.18}$$

If $\mathbb{E}(|X|^2) = \infty$, then X does not have a finite variance; when $\mathbb{E}X$ exists, we often regard $\mathbb{V}X$ as infinite.

1.3 Asymptotic Theory

"*Computer compilers don't tolerate any mistakes or gaps, regardless of how minor they are. Human brains, on the other hand, are more flexible—minor glitches don't impede our progress. Moreover, humans typically don't process information in the same manner as computer compilers, except perhaps for some first-year graduate students. When these students initially delve into rigorous mathematics, they cannot tolerate any deviations from absolute rigor, sometimes causing them to lose sight of the big picture. Therefore, when dealing with mathematical concepts, it is advisable to start at a heuristic level initially. Once the big picture is established, we can then proceed to construct every detail in a mathematically rigorous manner.*"

— believed to be rephrased sentences* of Terence Tao

Having reviewed individual random variables and their distributions, we now turn to the behavior of statistics built from increasingly large samples. In statistics, the word "asymptotic" refers to the limiting regime in which the sample size n tends to infinity. We begin with the two most widely used asymptotic results, the **law of large numbers** (LLN) and the **central limit theorem** (CLT), and then discuss the law of the iterated logarithm (LIL) and the Glivenko-Cantelli theorem.

1.3.1 Law of Large Numbers

The LLN explains why sample averages stabilize. Several versions are available; see, for example, Chapter 5 of Klenke (2020). We use Etemadi's strong LLN because it gives almost-sure convergence under the relatively weak assumption of pairwise independence.

Theorem 1.3.1 (Etemadi's strong LLN, 1981, see Etemadi (1981) or Theorem 5.17 of Klenke (2020))**.** *Let $Z_1, Z_2, \ldots$ be real-valued random variables defined on probability space $(\Omega, \mathbb{P})$. Suppose they are identically distributed and pairwise independent (i.e., Z_i and Z_j are independent for all i, j with $i \neq j$). If $\mathbb{E}Z_1$ exists, then $\lim_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^n Z_i\right) = \mathbb{E}Z_1$ with probability one, i.e.,*

$$\mathbb{P}\left\{\omega \in \Omega : \lim_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^n Z_i(\omega)\right) = \mathbb{E}Z_1\right\} = 1. \tag{1.3.1}$$

* These sentences are from a post by a student who took Prof. Tao's MATH 247B course (Classical Fourier Analysis) at UCLA. The original version online is in Chinese; Kun Meng translated it into the English version herein.

A proof of Theorem 1.3.1 can be found in Etemadi's original 1981 paper. The proof is short—a one-page proof—if some advanced mathematical tools taught in APMA 2630/2640 are available.

In Theorem 1.3.1, $Z_1, Z_2, \ldots$ are assumed to be pairwise independent rather than mutually independent. The distinction between these concepts is left for homework; see also the Wikipedia article "Independence (probability theory)." Every collection of mutually independent random variables is pairwise independent, but the converse need not hold. If $X_1, X_2, \ldots$ are independent and identically distributed (iid) with finite mean, then they satisfy the conditions of Theorem 1.3.1.

Remark 1.3.1. *i) The statement that $X_1, X_2, \ldots$ are independent and identically distributed according to a distribution with CDF F is usually written as $X_1, X_2, \ldots \overset{iid}{\sim} F$.*

ii) The concept "with probability one" is usually referred to as "almost surely."

1.3.2 A Theoretical Example of the LLN

The LLN states that the following event has probability one, but an event of probability one need not equal the entire sample space. The next example makes this distinction explicit:

$$\left\{\omega \in \Omega : \lim_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^{n} Z_i(\omega)\right) = \mathbb{E}Z_1\right\}$$

- The experiment: We flip a fair coin infinitely many times.
- Since there are infinitely many flips, one possible outcome ω is an infinitely long sequence of H and T, e.g.,

  $$\omega = (H, H, T, H, H, \ldots).$$

  Then, the sample space Ω of this experiment is a collection of all such infinitely long sequences of H and T, i.e.,

  $$\Omega \overset{\text{def}}{=} \left\{\omega = (\omega^{(1)}, \omega^{(2)}, \omega^{(3)}, \ldots) \,\middle|\, \omega^{(i)} \text{ is either } H \text{ or } T, \text{ for all } i = 1, 2, 3, \ldots\right\}.$$

- For each fixed positive integer i, we define the following

  $$X_i(\omega) = \begin{cases} 1 & \text{if } \omega^{(i)} = H, \text{ i.e., the i-th flip gives } H, \\ 0 & \text{if } \omega^{(i)} = T, \text{ i.e., the i-th flip gives } T. \end{cases}$$

  That is, all the $X_1, X_2, X_3, \ldots$ are defined on the sample space Ω. Hence, they are random variables.

Suppose there exists a probability $\mathbb{P}$ defined on Ω satisfying the following conditions[†]

i) Since we flip a fair coin for each flip, we may assume that each $X_i \sim \text{Bernoulli}(1/2)$, i.e.,

$$\mathbb{P}(\{\omega \in \Omega \,:\, X_i(\omega) = 0\}) = \mathbb{P}(\{\omega \in \Omega \,:\, X_i(\omega) = 1\}) = 1/2.$$

That is, all the random variables $X_1, X_2, \ldots$ are identically distributed and follow Bernoulli(1/2).

ii) Since any one single flip does not influence other flips, we may assume that the random variables $X_1, X_2, X_3, \ldots$ are independent.

The two conditions above indicate that the random variables $X_1, X_2, X_3, \ldots$ are iid. Then, the LLN (Theorem 1.3.1) implies the following

$$\begin{aligned} &\mathbb{P}(A) = 1, \quad \text{where} \\ &A \overset{\text{def}}{=} \left\{\omega \in \Omega \,:\, \lim_{n\to\infty} \frac{X_1(\omega) + X_2(\omega) + \cdots + X_n(\omega)}{n} = \frac{1}{2}\right\}. \end{aligned}$$

That is, with probability one, the limiting proportion of flips that give heads is 1/2.

However, $A \neq \Omega$, which can be shown by considering the following possible outcome

$$\begin{aligned} \omega^* &= (T, T, T, \ldots, T, \ldots) \\ &= \text{“You are so unlucky that you get tails for all the infinitely many flips.”} \end{aligned}$$

For this outcome, $X_i(\omega^*) = 0$ for every i, so

$$\frac{X_1(\omega^*) + X_2(\omega^*) + \cdots + X_n(\omega^*)}{n} = 0, \quad \text{for all } n \implies \lim_{n\to\infty} \frac{X_1(\omega^*) + X_2(\omega^*) + \cdots + X_n(\omega^*)}{n} = 0 \implies \omega^* \notin A,$$

which implies $\omega^* \in A^c \implies A^c \neq \emptyset \implies A \neq \Omega$.

1.3.3 A Numerical Example for the LLN

To illustrate the LLN, we generate $Z_1(\omega), Z_2(\omega), \ldots \overset{iid}{\sim} N(0,1)$ and compute $X_n(\omega) := \frac{1}{n}\sum_{i=1}^n Z_i(\omega)$ for $n \in \{11, 12, \ldots, 5000\}$ and 50 different ω. Figure 1.3 displays the resulting 50 sample paths $\{X_n(\omega)\}_{n=11}^{5000}$. Over the displayed range, the sample averages approach the true mean 0, behavior that is consistent with the LLN but does not by itself establish convergence.

†: Such a probability $\mathbb{P}$ exists. Proving its existence requires the Kolmogorov extension theorem, which is far beyond the scope of APMA 1690. For our purposes, it is enough to know that a probability satisfying the stated conditions exists.

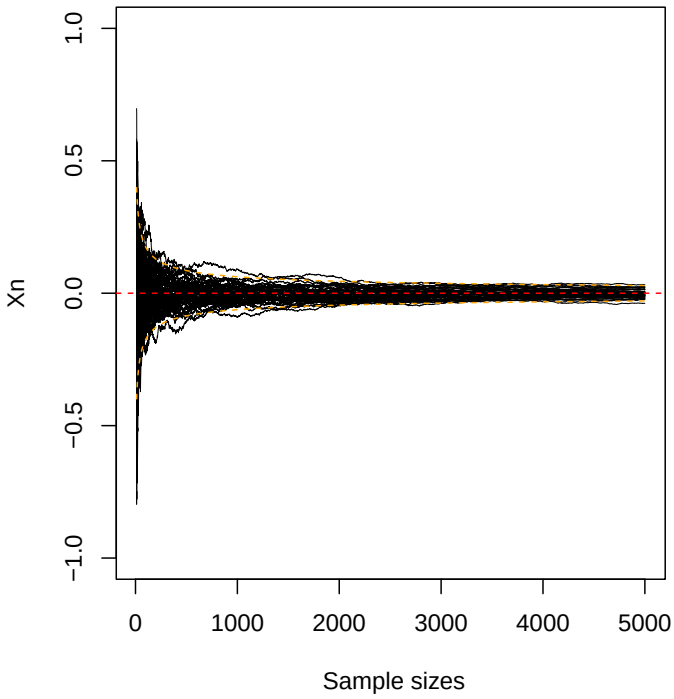


Figure 1.3: The 50 black curves represent 50 different sample paths $\{X_n(\omega)\}_{n=11}^{5000}$, one for each fixed ω. The red dashed line represents the true mean 0 of $N(0,1)$. The two orange dashed curves represent the convergence rate $\pm\sigma\cdot\sqrt{\frac{2\cdot\log(\log n)}{n}}$. The following `R` code produces these curves.

```r
n=5000
X_bar=vector()
for (itr in 1:50) {
  Z=rnorm(n)
  for (m in 11:n) { X_bar[m-10]=mean(Z[1:m]) }
  if(itr==1){
    plot(11:n, X_bar, type = "l", ylim = c(-1, 1), lwd=0.1,
         xlab = " Sample sizes", ylab = "Xn")
  }else{
    lines(11:n, X_bar, lwd=0.1)
  }
}
lines(11:n, sqrt((2/(11:n))*log(log(11:n))), col="orange", lty=2)
lines(11:n, -sqrt((2/(11:n))*log(log(11:n))), col="orange", lty=2)
abline(h=0, col="red", lty=2)
```

1.3.4 Law of the Iterated Logarithm

The LLN establishes convergence of the sample mean but does not describe the size of its pathwise fluctuations. For iid random variables with finite mean μ and finite, positive variance σ^2, the Hartman-Wintner law of the iterated logarithm (LIL) gives the precise almost-sure scale:

$$\mathbb{P}\left\{\omega \in \Omega \;\middle|\; \limsup_{n\to\infty} \frac{\left|\frac{1}{n}\sum_{i=1}^{n} X_i(\omega) - \mu\right|}{\sigma\sqrt{\frac{2\log(\log n)}{n}}} = 1\right\} = 1. \tag{1.3.2}$$

See Theorem 22.11 and Chapter 5.4 of Klenke (2020). Equation (1.3.2) is a limit-superior statement along almost every sample path; it is not a finite-sample probability bound.

Because the rigorous statement is beyond the scope of this course, we will use the following heuristic interpretation.

Theorem 1.3.2 (Heuristic version of the law of the iterated logarithm)**.** *Let $\{X_i\}_{i=1}^{\infty}$ be iid random variables with mean μ and variance $\sigma^2 \in (0,\infty)$. Define the sample-mean error*

$$e_n(\omega) := \left|\frac{1}{n}\sum_{i=1}^{n} X_i(\omega) - \mu\right|.$$

For large n, its almost-sure fluctuation scale is heuristically

$$\boxed{e_n(\omega) \;\lesssim\; \sigma\sqrt{\frac{2\log(\log n)}{n}}.} \tag{1.3.3}$$

The symbol $\lesssim$ emphasizes that this is a useful asymptotic heuristic, not an exact inequality for every n.

Remark: Etemadi's LLN requires only a finite mean, whereas this version of the LIL also requires a finite, positive variance.

The scale in Eq. (1.3.3) is informative in stochastic approximation (Chapter 2) because it indicates how slowly sample-mean errors may decay. The following visualization illustrates this heuristic envelope.

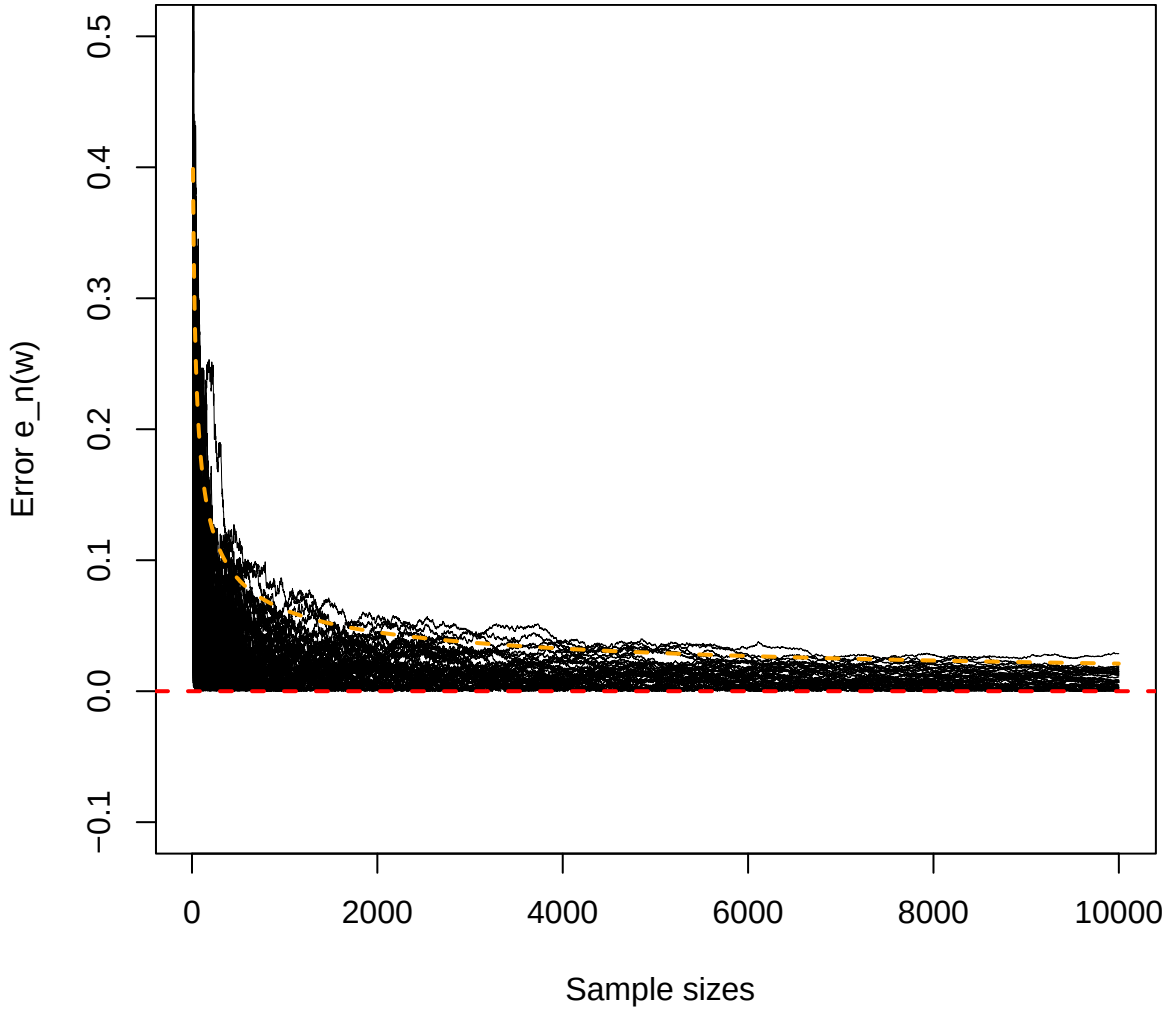


Figure 1.4: The approximation error $e_n(\omega) := |\frac{1}{n}\sum_{i=1}^{n} X_i(\omega) - \mathbb{E}X_1|$ for ten simulated sample paths. The dashed orange curve is $g(n) = \sqrt{\mathbb{V}(X_1)}\sqrt{\frac{2\log(\log n)}{n}}$, the almost-sure fluctuation scale suggested by the LIL. The `R` code used to produce the curves is provided below.

```r
n=10000

X_bar=vector()
for (itr in 1:10) {
  Z=rnorm(n)
  for (m in 11:n) { X_bar[m-10]=mean(Z[1:m]) }
  if(itr==1){
    plot(11:n, abs(X_bar), type = "l", ylim = c(-0.1, 0.5), lwd=0.1,
         xlab = " Sample sizes", ylab = "Error e_n(w)")
  }else{
    lines(11:n, abs(X_bar), lwd=0.1)
  }
}
abline(h=0, col="red", lty=2, lwd=2)
```

```
15
16 lines(11:n, sqrt((2/(11:n))*log(log(11:n))), col="orange", lty=2, lwd=2)
```

1.3.5 Central Limit Theorem

Whereas the LLN identifies the limiting value of a sample average, the central limit theorem describes its asymptotic fluctuations. A widely used version is the following.

Theorem 1.3.3. *(CLT for iid sequences) Let $X_1, X_2, \ldots$ be iid with $\mathbb{E}X_1 = \mu$ and $\mathbb{V}X_1 = \sigma^2 > 0$. Then,*

$$\sqrt{n} \cdot \frac{\left(\frac{1}{n}\sum_{i=1}^n X_i\right) - \mu}{\sigma} \xrightarrow{d} N(0,1), \tag{1.3.4}$$

where $\xrightarrow{d}$ denotes "convergence in distribution," i.e.,

$$\lim_{n\to\infty} \mathbb{P}\left\{\omega \in \Omega \middle| \sqrt{n} \cdot \frac{\left(\frac{1}{n}\sum_{i=1}^n X_i(\omega)\right) - \mu}{\sigma} \le t\right\} = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi}} e^{-\frac{\tau^2}{2}} d\tau = \Phi(t \,|\, \mu = 0, \sigma = 1),$$

where Φ is the normal CDF defined in Eq. (1.2.7).

A rigorous proof of Theorem 1.3.3 can be found in Chapter 3.4.1 of Durrett (2019). Another version of CLT more general than Theorem 1.3.3 is the Lindeberg-Feller CLT (see Theorem 3.4.5 of Durrett (2019)), which is out of the scope of this course and will be taught in APMA 2630/2640.

Figure 1.5 illustrates the CLT. We generate $X_1(\omega), \ldots, X_n(\omega) \overset{iid}{\sim} \text{Pois}(1)$, the Poisson distribution with parameter $\lambda = 1$, and compute the standardized quantity $\sqrt{n} \cdot \frac{\left(\frac{1}{n}\sum_{i=1}^n X_i(\omega)\right) - \mu}{\sigma}$. Repeating this procedure 1,000 times, once for each ω, produces 1,000 standardized quantities. Their histogram approximately matches the PDF of $N(0,1)$.

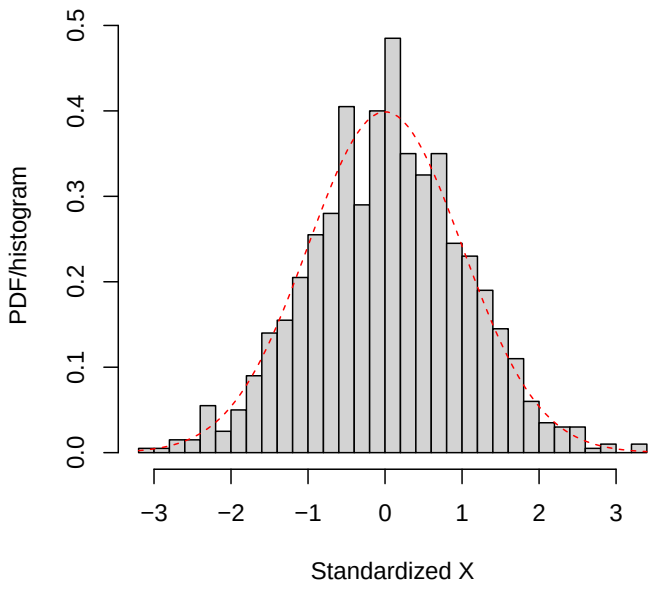


Figure 1.5: The gray bars show the histogram of the standardized quantities $\sqrt{n}\cdot\frac{\left(\frac{1}{n}\sum_{i=1}^{n}X_i(\omega)\right)-\mu}{\sigma}$, and the red dashed curve shows the PDF of $N(0,1)$. The following `R` code produces this visualization.

```r
n=10000
standardized_X=vector()
for (itr in 1:1000) {
  X=rpois(n, lambda = 1)
  standardized_X[itr]=sqrt(n)*(mean(X)-1)
}
hist(standardized_X, freq = FALSE, breaks = 30,
     xlab = "Standardized X", ylab = "PDF/histogram", main = "")
curve(dnorm, add = TRUE, col="red", lty=2)
```

1.3.6 Glivenko-Cantelli Theorem

The preceding results concern sample averages. The Glivenko-Cantelli theorem instead controls the entire empirical CDF and will later motivate one diagnostic for pseudorandom-number generators. To state it, we first introduce random functions, a generalization of the random variables in Definition 1.2.6.

Definition 1.3.1. *Let $(\Omega,\mathbb{P})$ be a probability space. Any bivariate function of the following form is called a* ***random function***

$$F:\ \Omega\times\mathbb{R}\to\mathbb{R},$$
$$(\omega,t)\mapsto F(\omega,t).$$

This function is also referred to as a ***continuous-time stochastic process*** *because its index set is $\mathbb{R}$. Here,*

continuous-time *describes the index set; it does not imply that the sample paths $t \mapsto F(\omega, t)$ are continuous.*

In applications, the underlying CDFs, which generate data of interest, are unknown. Suppose $X_1, \ldots, X_n \overset{iid}{\sim} F$, where F is the underlying CDF and unknown. Without any further assumptions, the following **empirical CDF** based on $\{X_i\}_{i=1}^n$ is a widely used estimator of the true CDF $F(t)$

$$F_n(\omega, t) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{X_i(\omega) \le t\} = \frac{\text{number of } \{X_i(\omega)\}_{i=1}^n \text{ such that } X_i(\omega) \le t}{n}, \quad \text{for all } t \in \mathbb{R}. \tag{1.3.5}$$

Although the true CDF F is a deterministic function, the estimator $F_n(\omega, t)$ defined in Eq. (1.3.5) is random as it depends on ω (see Definition 1.3.1).

Since X_i are iid random variables, for each fixed t, the indicators $\mathbb{1}\{X_i(\omega) \le t\}$ are binary-valued random variables which take values only in $\{0, 1\}$. Therefore, $\mathbb{1}\{X_i(\omega) \le t\} \overset{iid}{\sim} \text{Bernoulli}(p)$, where $p := \mathbb{E}(\mathbb{1}\{X_1(\omega) \le t\})$. In particular, their independence makes all covariance terms vanish when computing the variance of their sum. Thus, for each fixed t, we have

$$\begin{aligned}
&\mathbb{E}F_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\mathbb{1}\{X_i \le t\}\right] = \mathbb{E}\left[\mathbb{1}\{X_1 \le t\}\right] = \mathbb{P}(X_1 \le t) = \text{ the CDF of } X_1 \\
&\text{(the second ``=" is from that } X_i \text{ are identically distributed)}, \\
&\mathbb{V}F_n(t) = \frac{1}{n^2}\sum_{i=1}^{n} \mathbb{V}\left[\mathbb{1}\{X_i \le t\}\right] = \frac{1}{n}\mathbb{V}\left[\mathbb{1}\{X_1 \le t\}\right] = \frac{1}{n}F(t)\left[1 - F(t)\right].
\end{aligned} \tag{1.3.6}$$

We may apply the LLN (see Theorem 1.3.1) and Eq. (1.3.6) to derive the following asymptotic behavior of $F_n(t)$, for each fixed t, as $n \to \infty$.

Claim 1.3.4. *For each fixed t, we have $\lim_{n\to\infty} F_n(t) \to F(t)$ with probability one, i.e.,*

$$\mathbb{P}\left\{\omega \in \Omega : \lim_{n\to\infty} F_n(\omega, t) = F(t)\right\} = 1$$

Proof. Set $Z_i(\omega) := \mathbb{1}\{X_i(\omega) \le t\}$, then the desired result follows from Eq. (1.3.1). □

Claim 1.3.4 is a "t-wise" result, i.e., we have to fix t before taking limit $n \to \infty$. The following theorem provides the asymptotic behavior of F_n uniformly with respect to t.

Theorem 1.3.5 (Glivenko-Cantelli, 1933; see Theorem 5.23 of Klenke (2020))**.** *Let $F(t)$ be the CDF shared by iid random variables $X_1, \ldots, X_n, \ldots$, and $F_n(\omega, t)$ is the empirical CDF based on $\{X_i(\omega)\}_{i=1}^n$. Then, the empirical*

CDF converges to $F(t)$ uniformly with probability one, i.e.,[*]

$$\mathbb{P}\left\{\omega \in \Omega : \lim_{n\to\infty}\left(\sup_{t\in\mathbb{R}}|F_n(\omega,t)-F(t)|\right)=0\right\}=1.$$

One proof of Theorem 1.3.5 is left for homework. Please think about the following question: *Why does the proof of Claim 1.3.4 not work for Theorem 1.3.5?* Thinking about this question is not required in APMA 1690, while it is helpful if you will take APMA 2630/2640.

Theorem 1.3.5 is introduced in this course for the following purposes

- It guarantees that we can uncover the true underlying distribution if we have a sufficient amount of data.
- It motivates the definition of *pseudorandom number generator* (see Definition 1.5.1 in Section 1.5).

We conclude this section with a simulation study illustrating the Glivenko–Cantelli theorem. We generate iid $X_1,\ldots,X_n$ from $N(0,1)$ for $n \in \{10, 50, 100, 500, 1000, 10000\}$ and compute the empirical CDF F_n defined by Eq. (1.3.5). Figure 1.6 shows the empirical CDF approaching the true CDF as the displayed sample size increases. This finite-sample behavior is consistent with the theorem's uniform-convergence conclusion.

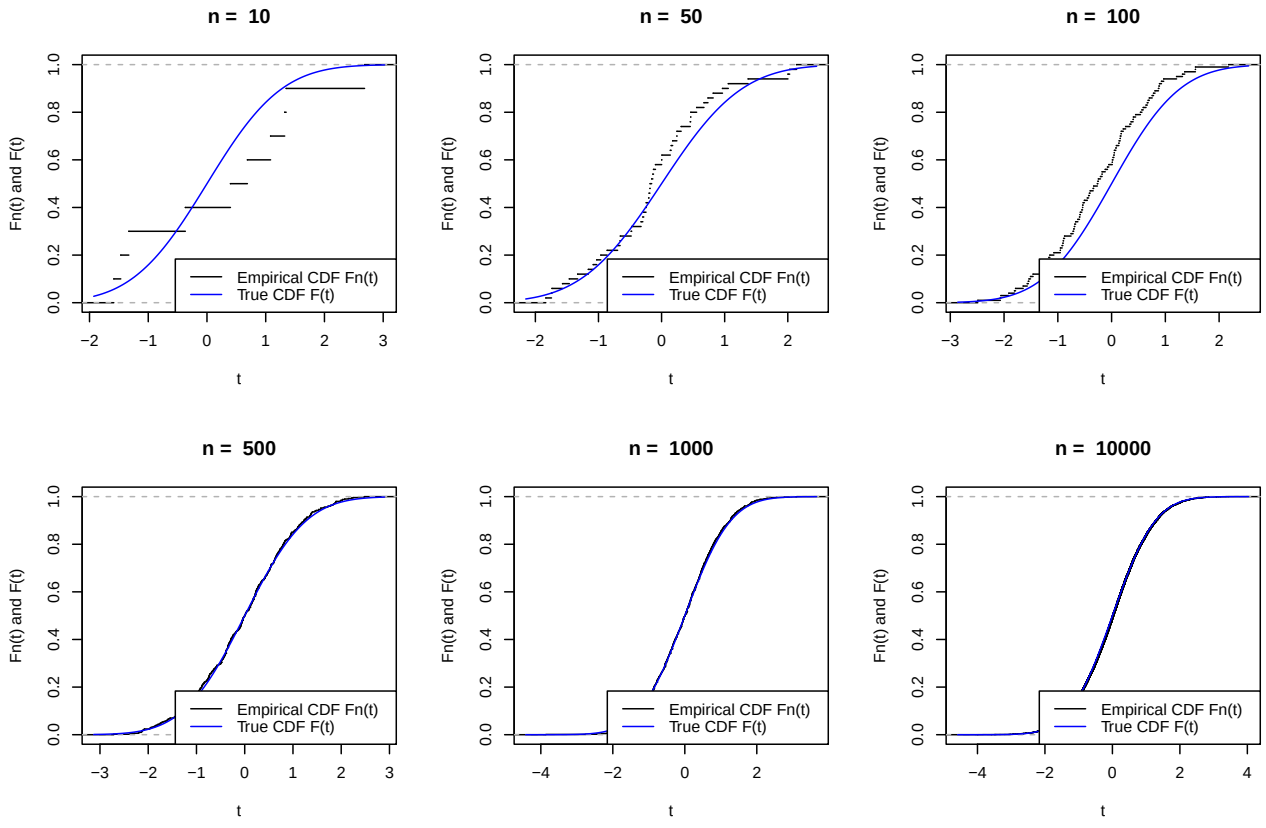


Figure 1.6: Comparison of the empirical CDF F_n (black) with the true CDF F of $N(0,1)$ (blue). The following code produces the simulation and visualization.

*: The "sup" denotes the "supremum." If you are not familiar with the "sup," please feel free to just view it as maximum, i.e., "max."

```r
# Empirical CDF and Glivenko Cantelli theorem
par(mfrow=c(2, 3))
for (n in c(10, 50, 100, 500, 1000, 10000)) {
  X_i=rnorm(n)
  plot(ecdf(X_i), pch = 16, cex = 0.2, xlab = "t", ylab = "Fn(t) and F(t)",
  main = paste("n = ", as.character(n)))
  curve(pnorm, add = TRUE, col="blue")
  legend("bottomright", c("Empirical CDF Fn(t)", "True CDF F(t)"),
  col = c("black", "blue"), lty = c(1,1))
}
```

1.4 True-Random Samples and Their Realizations

Before delving into pseudorandom numbers, we first distinguish a true-random sample, which is produced by a stochastic generating mechanism, from a fixed numerical realization of that sample.

Definition 1.4.1. *Let $(\Omega, \mathbb{P})$ be a probability space, and let $X_1, X_2, \ldots, X_n$ be iid real-valued random variables with common CDF† F. The random vector $(X_1, X_2, \ldots, X_n)$ is called a* ***true-random sample*** *of size n from F. For an outcome $\omega \in \Omega$, the fixed vector*

$$\big(X_1(\omega), X_2(\omega), \ldots, X_n(\omega)\big)$$

is called a ***realization*** *of this random sample. Thus, true randomness is a property of the stochastic generating mechanism represented by $X_1, X_2, \ldots, X_n$, not a property of the fixed numerical vector observed after an outcome is realized.*

1.5 Pseudorandom Number Generators (PRNGs)

> *"Anyone who considers arithmetical methods of producing random digits is, of course, in a state of sin."*
>
> — John von Neumann (von Neumann, 1951)

Simulating randomness—specifically, generating iid random variables following a known distribution of interest—is useful in many areas of science and engineering. We take stochastic approximation (see Chapter 2) as an example. Suppose we want to compute the integral $\int_0^1 f(x)dx$. However, the integrand $f(x)$ is so complicated that we cannot compute this integral analytically. One way to compute $\int_0^1 f(x)dx$ numerically is to apply the LLN (see Theorem 1.3.1). Suppose we can generate a large iid sample $X_1, \ldots, X_n$ from the uniform distribution on $[0, 1]$, whose PDF

†: Recall that the CDF of a real-valued random variable contains all the information about its distribution.

is $\mathbb{1}_{[0,1]}$. The LLN implies the following limit:

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} f(X_i) = \mathbb{E}[f(X_1)] = \int_0^1 f(x)dx. \tag{1.5.1}$$

Therefore, for large n, we may approximate the integral $\int_0^1 f(x)dx$ by

$$\int_0^1 f(x)dx \approx \frac{1}{n}\sum_{i=1}^{n} f(X_i).$$

The question is, *how can we generate a large number of random variables from a given distribution?* A hardware random number generator (HRNG) is a partial solution to this question. HRNGs generate random numbers from physical processes. However, HRNGs produce only a limited number of random numbers per second. Hence, generating many random numbers with an HRNG can be very time-consuming, making the approximation in Eq. (1.5.1) expensive. We need a faster random number generator: a pseudorandom number generator (PRNG).

Roughly speaking, a pseudorandom number generator (PRNG), denoted by g, generates a sequence of numbers $g(1), g(2), \ldots$. As indicated by the word "pseudo," the numbers generated by a PRNG are not truly random. They are generated by a deterministic mechanism but are designed to look like iid random numbers from a given distribution. We will make the phrase "look like" more precise in Section 1.9. Although PRNGs do not produce true randomness, they can generate numbers very quickly. We first introduce the general mathematical definition of PRNGs, which is partially motivated by the Glivenko-Cantelli theorem; please compare Theorem 1.3.5 with Definition 1.5.1.

Definition 1.5.1. *Suppose $F : \mathbb{R} \to [0,1]$ is the CDF of a given distribution of real-valued random variables. We call a function $g : \{1, 2, \cdots\} \to \mathbb{R}$ a pseudorandom number generator (PRNG) for CDF F if*

$$\sup_{-\infty<t<+\infty} \left| \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\Big(g(i) \le t\Big) - F(t) \right| \to 0, \quad \text{as } n \to \infty. \tag{1.5.2}$$

In addition, the numbers $g(1), g(2), \ldots, g(n), \ldots$ are called pseudo-random numbers.

Before going to any explanations, we keep the following conclusion/claim in our minds:

The real numbers $g(1), g(2), \ldots, g(n)$ look like (but are not) true-random numbers iid from F.

Suppose we know and are interested in the CDF F, and a PRNG g for F is available (constructing g is left for later sections). Imagine we have an underlying probability space $(\Omega, \mathbb{P})$, which can be any probability space and is

not important. For a sufficiently large sample size n, we define the following random variables

$$\begin{aligned} X_1(\omega) &= g(1) \quad \text{for all } \omega, \\ X_2(\omega) &= g(2) \quad \text{for all } \omega, \\ &\cdots \\ X_n(\omega) &= g(n) \quad \text{for all } \omega. \end{aligned} \tag{1.5.3}$$

The numbers $\{X_i(\omega)\}_{i=1}^n$ defined in Eq. (1.5.3) are viewed as random numbers following distribution F. One may notice the phrase "for all ω"—each $X_i(\omega)$ is a constant in ω, i.e., each X_i is deterministic instead of truly random.

1.6 PRNGs and the Glivenko-Cantelli Theorem

The Glivenko-Cantelli theorem motivates empirical-CDF convergence as one diagnostic for the marginal distribution of a PRNG's output. It does not, by itself, characterize iid sampling. Further evaluation of the performance of a PRNG will be provided in Section 1.9. Recall the Glivenko-Cantelli theorem as follows (also see Eq. (1.3.5)): if $X_1, X_2, \ldots$ are iid with common CDF F, then, with probability one,

$$\sup_{t\in\mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{X_i(\omega) \le t\} - F(t) \right| \to 0, \quad \text{as } n \to \infty. \tag{1.6.1}$$

For the deterministic variables $X_i(\omega) = g(i)$ in Eq. (1.5.3), the limit in Eq. (1.6.1) reduces to the empirical-CDF criterion in Definition 1.5.1.

What the criterion establishes: If the output sequence satisfies this criterion, its empirical marginal distribution converges to F.

What the criterion does not establish: The converse of the Glivenko-Cantelli implication is false. Empirical-CDF convergence does not imply that a sequence is independent or behaves like an iid sample; dependent and deterministic sequences can satisfy the same marginal limit. Consequently, this criterion must be supplemented by diagnostics that assess serial dependence and other aspects of the joint distribution.

1.7 Multiplicative Congruential Generator

We now give an explicit example of a PRNG: the multiplicative congruential generator (MCG). Its goal is to generate pseudorandom numbers that approximate draws from Unif$(0, 1)$. The construction has three steps:

i) Construct a PRNG for Unif$(\{1, 2, \ldots, m-1\})$.

ii) Construct a PRNG for $\text{Unif}(\{\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m-1}{m}\})$.

iii) Construct an approximate PRNG for $\text{Unif}(0, 1)$.

1.7.1 PRNG for $\text{Unif}(\{1, 2, \ldots, m-1\})$

Let X be a random variable defined on the probability space $(\Omega, \mathbb{P})$, and X takes values only in $\{1, 2, \ldots, m-1\}$, where m is a fixed positive integer. Furthermore, X satisfies the following

$$\mathbb{P}\{\omega \in \Omega \,|\, X(\omega) = i\} = \frac{1}{m-1}, \quad \text{for all } i = 1, 2, \ldots, m-1. \tag{1.7.1}$$

That is, $X \sim \text{Unif}(\{1, 2, \ldots, m-1\})$ = the uniform distribution on the set $\{1, 2, \ldots, m-1\}$. Then, it is easy to verify that the CDF of X is the following

$$F_X(t) = \frac{1}{m-1} \sum_{i=1}^{m-1} \mathbb{1}(i \le t). \tag{1.7.2}$$

The goal of the multiplicative congruential generator (MCG) is the following: construct a PRNG for the CDF in Eq. (1.7.2). The MCG is presented by the following algorithm

Algorithm 1 : Multiplicative Congruential Generator

Input: (i) positive integers m (see Eq. (1.7.2)) and a; (ii) a seed $s \in \{1, 2, \ldots, m-1\}$; (iii) sample size n.
Output: pseudo-random numbers $g(1), g(2), \ldots, g(n)$ intended to mimic draws from $\text{Unif}(\{1, 2, \ldots, m-1\})$.

1: Initialization: $g(1) \leftarrow s$.
2: **for all** $i = 1, 2, \ldots, n-1$, **do**
3: $\quad g(i+1) \leftarrow (a \cdot g(i) \bmod m)$ = the remainder of $\frac{a \cdot g(i)}{m}$.
4: **end for**

The phrase "properly chosen" can be made precise through the period of the deterministic recurrence. For example, suppose that m is prime and that a is a primitive root modulo m. Then, for every seed $s \in \{1, \ldots, m-1\}$, the sequence visits each nonzero residue modulo m exactly once before repeating. Its period is therefore $m-1$, and over every complete cycle,

$$\frac{1}{m-1} \sum_{i=1}^{m-1} \mathbb{1}\{g(i) = j\} = \frac{1}{m-1}, \qquad j = 1, \ldots, m-1.$$

Thus, the empirical distribution over a full cycle is exactly uniform, even though the sequence itself is deterministic. More generally, we have the following result.

Theorem 1.7.1. *For "properly chosen" positive integers m and a that yield a full-period orbit on $\{1, 2, \ldots, m-1\}$, the function g constructed in Algorithm 1 is a PRNG for the CDF in Eq. (1.7.2) for every valid seed $g(1) \in \{1, 2, \ldots, m-1\}$.*

Remark 1.7.1. *i) A complete characterization of full-period parameters uses number theory and is beyond our scope. One standard choice is $m = 2^{31} - 1$ and $a = 7^5 = 16807$, used in the Park–Miller "minimal standard" generator and in early mathematical software.*

ii) For valid nonzero seeds on a full-period orbit, the asymptotic full-cycle frequencies are the same. However, the phase and every finite output sequence depend on the seed. Invalid seeds can produce degenerate behavior; for example, the seed $g(1) = 0$ produces the constant sequence $g(i) = 0$ for all i.

iii) "Random seeds are often generated from the state of the computer system (such as the time), a cryptographically secure pseudorandom number generator or from a hardware random number generator" (see Wikipedia).

iv) The MCG is a special case of the linear congruential generator.

1.7.2 PRNG for $\mathrm{Unif}(\{\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m-1}{m}\})$

If the random variable $X(\omega)$ defined on the probability space $(\Omega, \mathbb{P})$ follows the distribution $\mathrm{Unif}(\{1, 2, \ldots, m-1\})$ (see Eq. (1.7.1)), then we have

$$Y(\omega) = \frac{X(\omega)}{m} \sim \mathrm{Unif}\left(\left\{\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m-1}{m}\right\}\right). \tag{1.7.3}$$

That is, the random variable Y defined above is a random variable defined on probability space $(\Omega, \mathbb{P})$ and taking values only in $\{\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m-1}{m}\}$, and

$$\mathbb{P}\left\{\omega \in \Omega \,\middle|\, Y(\omega) = \frac{i}{m}\right\} = \frac{1}{m-1}, \quad \text{for all } i = 1, 2, \ldots, m-1.$$

The CDF of the Y defined above is

$$F_Y(t) = \frac{1}{m-1} \sum_{i=1}^{m-1} \mathbb{1}\left(\frac{i}{m} \le t\right). \tag{1.7.4}$$

Let g denote the PRNG for the CDF in Eq. (1.7.2) (see Algorithm 1). Motivated by Eq. (1.7.3), we define the following function

$$\begin{aligned} h: \ & \{1, 2, \ldots\} \to \mathbb{R}, \\ & i \mapsto \frac{g(i)}{m}. \end{aligned} \tag{1.7.5}$$

Then, it is easy to show that the function h defined in Eq. (1.7.5) is a PRNG for the CDF in Eq. (1.7.4).

1.7.3 PRNG for Unif$(0,1)$

We have the following approximation when m is large

$$\frac{1}{m-1}\sum_{i=1}^{m-1}\mathbb{1}\left(\frac{i}{m}\le t\right)\approx\begin{cases}0 & \text{if } t<0\\ t & \text{if } 0\le t<1\\ 1 & \text{if } 1\le t.\end{cases} \tag{1.7.6}$$

Specifically, using the definition of Riemann integrals, we have the following

$$\lim_{m\to\infty}\frac{1}{m-1}\sum_{i=1}^{m-1}\mathbb{1}\left(\frac{i}{m}\le t\right)=\begin{cases}0 & \text{if } t<0\\ t & \text{if } 0\le t<1\\ 1 & \text{if } 1\le t.\end{cases}$$

In the approximation above, the left-hand side is in Eq. (1.7.4), and the right-hand side is the CDF of Unif$(0,1)$.

Therefore, when m is very large (e.g., $m=2^{31}-1$ used in an old version of MATLAB), we can approximately use the PRNG for the CDF in Eq. (1.7.4) as a PRNG for Unif$(0,1)$. That is, the function h defined in Eq. (1.7.5) is approximately a PRNG for Unif$(0,1)$.

Let us conclude the logic: **Algorithm 1 defines the function g; we define the function h using Eq. (1.7.5); then, this h is approximately a PRNG for** Unif$(0,1)$**.**

1.8 Basic Randomness and Its Transforms

"*I learned very early the difference between knowing the name of something and knowing something.*"

— Richard Feynman (1965 Nobel Laureate in Physics)

The preceding section constructed pseudorandom numbers intended to mimic draws from the uniform distribution on $[0,1]$. We call this distribution, denoted by Unif$(0,1)$, **basic randomness**, because every real-valued random variable can be represented as a transformation of a Unif$(0,1)$ random variable. **From this point onward, we assume that a PRNG intended to mimic iid** Unif$(0,1)$ **draws is available.** Its output closely resembles iid uniform draws in the diagnostics considered here, while its number-theoretic construction in Section 1.7 is largely independent of the remainder of the course.

Many programming languages have functions generating Unif$(0,1)$ (pseudo) random variables using PRNGs, i.e., `R` implements function `runif` as follows.

```
# n = the sample size of interest, e.g., n = 10000
# A PRNG for Unif([0,1])
n=10000
```

```r
U=runif(n)
hist(U, freq = FALSE)
```

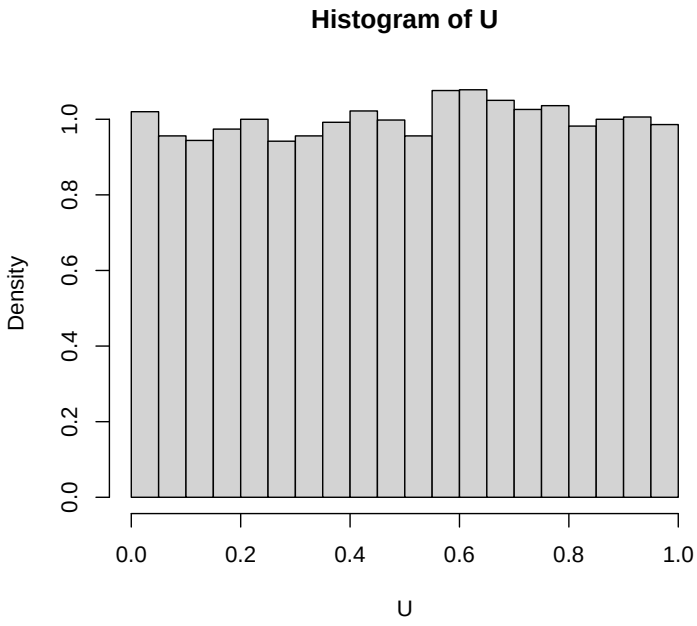


Figure 1.7: The histogram of the $n = 10000$ pseudo random numbers generated by the `R` function `runif`. This histogram looks similar to the density function $\mathbb{1}_{[0,1]}$ of $\text{Unif}(0,1)$.

As mentioned in the preceding paragraph, every real-valued random variable following any distribution can be viewed as a transform of a $\text{Unif}(0,1)$ random variable. A trivial transform is the following

Claim 1.8.1. *Suppose a and b are two real numbers with $a < b$. If $U \sim \text{Unif}(0,1)$, then $(b-a)\cdot U + a \sim \text{Unif}(a,b) =$ the uniform distribution on interval $[a,b]$.*

Generating random numbers from other distributions using $U \sim \text{Unif}(0,1)$ requires additional, nontrivial transformations. We study two of them: the *inverse CDF method* and the *rejection sampling method*. Two remarks apply to both methods:

- The inverse CDF method uses CDFs, and the rejection sampling method uses (un-normalized) PDFs.
- The inverse CDF method works for all distributions including but not limited to continuous and discrete distributions, and the rejection sampling method applies mainly for continuous distributions.

1.8.1 Inverse CDF Method

The inverse CDF method is based on the following theorem, which works for all random variables—continuous, discrete, and more complicated random variables.

Theorem 1.8.2. *Let X be any real-valued random variable, $U \sim \mathrm{Unif}(0,1)$, and F_X be the CDF of X. Define a new random variable Y by $Y := F_X^{-1,*}(U)$, where*

$$F_X^{-1,*}(t) := \inf\{x \in \mathbb{R} : F_X(x) \geq t\}, \qquad 0 < t < 1. \tag{1.8.1}$$

Then $X \overset{d}{=} Y$, i.e., X and Y share the same distribution/CDF F_X.

Remarks: (i) For any $u \in (0,1)$, the notation "$\{t \in \mathbb{R} : F(t) \geq u\}$" denotes the set of real numbers t for which $F(t) \geq u$.

(ii) The quantity $\inf\{t \in \mathbb{R} : F(t) \geq u\}$ is the greatest lower bound of this set. An infimum need not itself belong to the set, so it should not generally be described as its smallest element.

(iii) If F has an ordinary inverse, then the generalized inverse $F^{-1,*}$ agrees with F^{-1} on $(0,1)$.

We first make several remarks about the "formal inverse" $F_X^{-1,*}$. They rely on the following general result from calculus.

Claim 1.8.3. *If the function $F : \mathbb{R} \to \mathbb{R}$ is **strictly increasing and continuous**, then its inverse $F^{-1} : F(\mathbb{R}) \to \mathbb{R}$ **exists**; additionally, this inverse is **strictly increasing and continuous** as well.*

i) If the CDF F_X is continuous and strictly increasing, Claim 1.8.3 indicates that the inverse F_X^{-1} exists; additionally, F_X^{-1} is continuous and strictly increasing. The continuity and limits $\lim_{t\to-\infty} F_X(t) = 0$ and $\lim_{t\to\infty} F_X(t) = 1$ further indicate the following

$$\begin{aligned} &F_X : \quad \mathbb{R} \to (0,1) \text{ is continuous,} \\ &F_X^{-1} : (0,1) \to \mathbb{R} \text{ is continuous,} \\ &\lim_{u\downarrow 0} F_X^{-1}(u) = -\infty, \qquad \lim_{u\uparrow 1} F_X^{-1}(u) = +\infty. \end{aligned}$$

Since $U \sim \mathrm{Unif}(0,1)$ satisfies $\mathbb{P}\{0 < U < 1\} = 1$, the inverse only needs to be defined on $(0,1)$ for the inverse CDF method.

ii) In the discussion above, the reason we need the continuity of F_X and F_X^{-1} is to make the inverse F_X^{-1} well-defined on the entire open interval $(0,1)$. For example, the CDF presented in Figure 1.1 is strictly increasing but not continuous. Because the strict increasingness, the inverse of this CDF exists. However, the inverse is defined only on $(0,a) \cup (b,1)$ for some $a < b$. As a result, for $U \sim \mathrm{Unif}(0,1)$, the expression $F_X^{-1}(U)$ may not be well-defined if $a < U < b$. The motivation for using Eq. (1.8.1) to define the formal inverse $F_X^{-1,*}$ is to make such expression $F_X^{-1}(U)$ well-defined in some sense.

iii) One may verify that, when the CDF F_X is continuous and strictly increasing, $F_X^{-1}(u) = F_X^{-1,*}(u)$ for all $u \in (0,1)$ (see Eq. (1.8.1)), i.e., $F_X^{-1,*}$ is indeed the inverse of F_X. However, in many common scenarios,

CDFs are neither continuous nor strictly increasing. For example, the CDF of Bernoulli($\frac{1}{2}$) is $\frac{1}{2}\mathbb{1}_{[0,\infty)}+\frac{1}{2}\mathbb{1}_{[1,\infty)}$ (see Figure 1.8). In these scenarios, the true inverse F_X^{-1} does not exist, and we use the "formal inverse" $F_X^{-1,*}$ defined in Eq. (1.8.1).

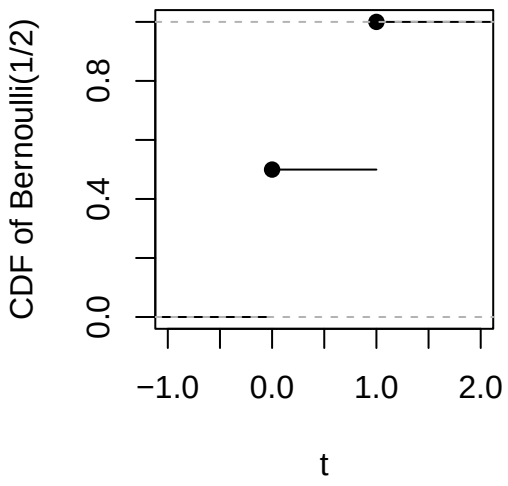


Figure 1.8: The CDF of distribution Bernoulli($\frac{1}{2}$).

Proof. (of Theorem 1.8.2) Let $G = F_X^{-1,*}$. Because every CDF is nondecreasing and right-continuous, the generalized inverse satisfies

$$G(u) \le t \quad \Longleftrightarrow \quad u \le F_X(t), \qquad u \in (0,1),\ t \in \mathbb{R}.$$

Indeed, if $u \le F_X(t)$, then t belongs to the set whose infimum defines $G(u)$, so $G(u) \le t$. Conversely, if $G(u) \le t$, monotonicity and right-continuity of F_X imply $F_X(t) \ge u$. Therefore,

$$\begin{aligned} F_Y(t) &= \mathbb{P}\{\omega \in \Omega : Y(\omega) \le t\} \\ &= \mathbb{P}\{\omega \in \Omega : G(U(\omega)) \le t\} \\ &= \mathbb{P}\{\omega \in \Omega : U(\omega) \le F_X(t)\} \\ &= F_X(t), \qquad t \in \mathbb{R}, \end{aligned}$$

where the last equality follows from $U \sim \text{Unif}(0,1)$. Hence, $F_Y = F_X$, so X and Y have the same distribution. □

The following algorithm gives the procedure for applying the inverse CDF method.

As an example, we apply the inverse CDF method (Algorithm 2) to generate random numbers from the exponential distribution Exp(1), whose CDF is $(1-e^{-t})\cdot\mathbb{1}(t>0)$. The CDF of Exp(1) is continuous and strictly increasing, hence, its inverse exists and is continuous and strictly increasing. Specifically, the inverse is $\log\frac{1}{1-u}$.

Algorithm 2 : Inverse CDF Method

Input: (i) The CDF F of interest; (ii) sample size n.
Output: A sequence of (pseudo) iid random numbers $x_1, \ldots, x_n$ following the distribution F.

1: Generate (pseudo) iid random variables $u_1, u_2, \ldots, u_n$ from Unif$(0, 1)$.
2: **for all** $i = 1, \ldots, n,$ **do**
3: Compute $x_i \leftarrow F^{-1,*}(u_i) = \inf\{x \in \mathbb{R} : F(x) \geq u_i\}$
4: **end for**

Therefore, we first generate random variables $U_1, \ldots, U_n$ from Unif$(0, 1)$, then we compute $X_i = \log \frac{1}{1-U_i}$ for all $i \in \{1, \ldots, n\}$. Figure 1.9 compares the empirical CDFs (black) of these generated $X_1, \ldots, X_n$ with the true Exp(1) CDF (blue), providing a finite-sample illustration of the inverse CDF method. In Section 1.9, we will evaluate the performance of these generated $X_1, \ldots, X_n$ using hypothesis testing.

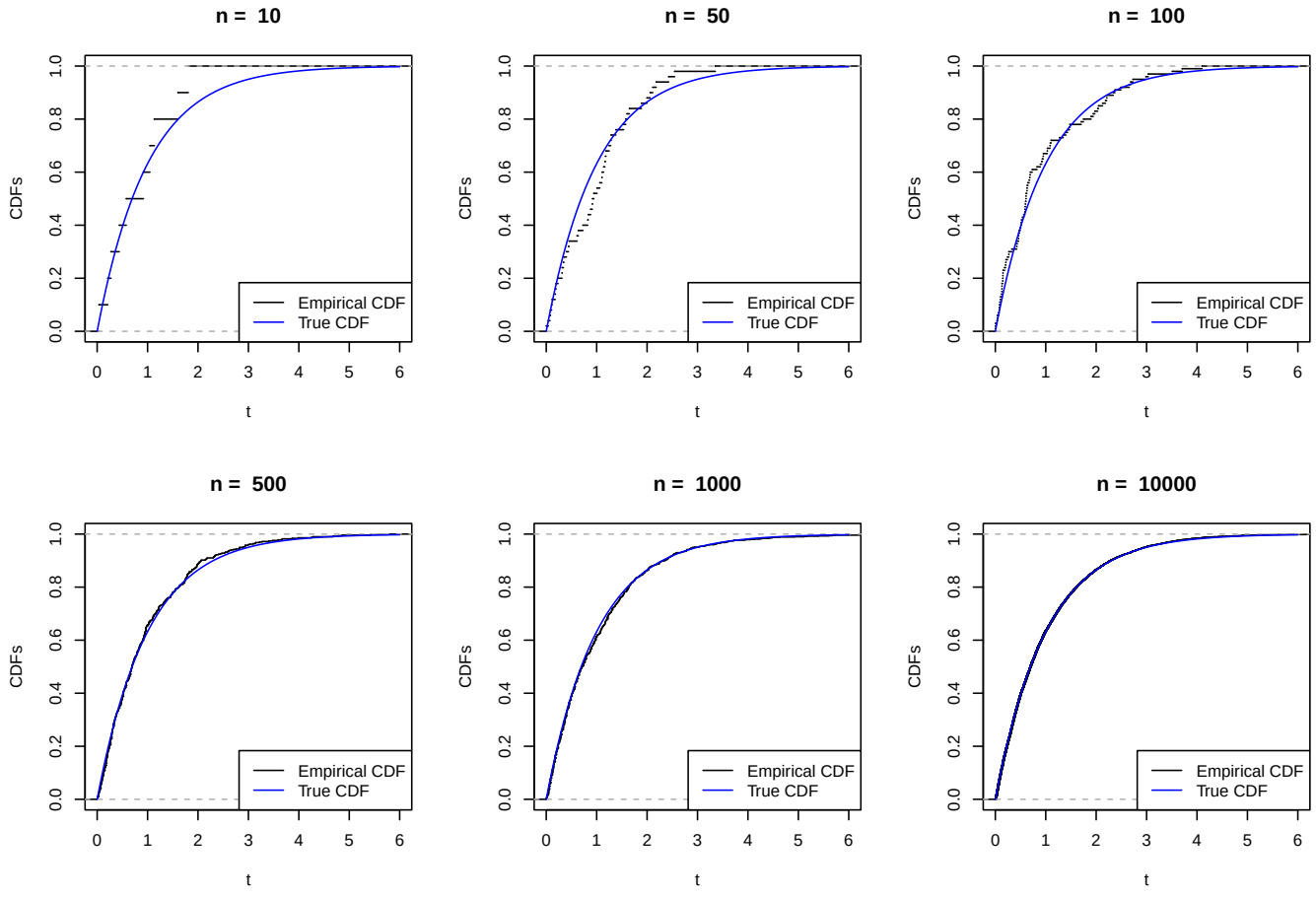


Figure 1.9: Let g denote the PRNG constructed in this example. The empirical CDF (black) of the generated numbers is $\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(g(i) \leq t)$. Across the six displayed sample sizes, the empirical CDF becomes closer to the true CDF F. This behavior is consistent with Eq. (1.5.2) in Definition 1.5.1, but the finite plots do not establish convergence. The following `R` code produces the simulation.

```r
# Inverse CDF example: exponential distribution with parameter rate = 1
par(mfrow=c(2, 3))
for (n in c(10, 50, 100, 500, 1000, 10000)) {
```

```
  U_i=runif(n)
  X_i=log(1/(1-U_i))
  plot(ecdf(X_i), pch = 16, cex = 0.2, xlab = "t", ylab = "CDFs",
       main = paste("n = ", as.character(n)), xlim = c(0,6))
  curve(pexp, add = TRUE, col="blue")
  legend("bottomright", c("Empirical CDF", "True CDF"),
         col = c("black", "blue"), lty = c(1,1))
}
```

1.8.2 Knothe–Rosenblatt Rearrangement*

The inverse CDF method generates one real-valued random variable from one uniform random variable. The **Knothe–Rosenblatt rearrangement** (KR rearrangement), introduced in Knothe (1957) and Rosenblatt (1952), extends this idea to random vectors by applying one-dimensional inverse CDF transforms sequentially. The presentation below follows (Sanz-Alonso and Al-Ghattas, 2024, Section 2.1.2).

We begin with two dimensions. Let $\boldsymbol{X} = (X_1, X_2)^\intercal$ have a strictly positive joint PDF f on $\mathbb{R}^2$, and let $U_1, U_2 \overset{iid}{\sim} \mathrm{Unif}(0,1)$. The density factorization

$$f(x_1, x_2) = f_{X_1}(x_1) f_{X_2|X_1=x_1}(x_2 \mid x_1)$$

suggests a sequential construction. First generate X_1 from its marginal distribution; then, after obtaining a value x_1, generate X_2 from its conditional distribution given $X_1 = x_1$. In terms of generalized inverse CDFs, define

$$T_1(u_1) := F_{X_1}^{-1,*}(u_1),$$
$$T_2(u_1, u_2) := F_{X_2|X_1=T_1(u_1)}^{-1,*}(u_2).$$

Then

$$\boldsymbol{T}(u_1, u_2) := \begin{pmatrix} T_1(u_1) \\ T_2(u_1, u_2) \end{pmatrix}$$

satisfies $\boldsymbol{T}(U_1, U_2) \overset{d}{=} \boldsymbol{X}$. Notice the order of the construction: the second component may depend on both uniforms, whereas the first component depends only on the first uniform.

The same idea extends recursively to d dimensions. For a vector $u = (u_1, \ldots, u_d)^\intercal$, write $u_{1:i} := (u_1, \ldots, u_i)^\intercal$. Similarly, $X_{1:i} := (X_1, \ldots, X_i)^\intercal$ and $T_{1:i} := (T_1, \ldots, T_i)^\intercal$ denote the first i components of the corresponding vectors.

Theorem 1.8.4 (Knothe–Rosenblatt rearrangement). *Let $\boldsymbol{X} = (X_1, \ldots, X_d)^\intercal$ have a strictly positive PDF on $\mathbb{R}^d$,*

and let $\boldsymbol{U} = (U_1, \dots, U_d)^\intercal$ *have iid* $\mathrm{Unif}(0,1)$ *components. Define*

$$\begin{aligned} T_1(u_1) &:= F_{X_1}^{-1,*}(u_1), \\ T_i(u_{1:i}) &:= F^{-1,*}_{X_i \mid X_{1:i-1} = T_{1:i-1}(u_{1:i-1})}(u_i), \qquad i = 2, \dots, d, \end{aligned}$$

and set

$$\boldsymbol{T}(u) := \begin{pmatrix} T_1(u_1) \\ T_2(u_{1:2}) \\ \vdots \\ T_d(u_{1:d}) \end{pmatrix}.$$

Then $\boldsymbol{T}(\boldsymbol{U}) \overset{d}{=} \boldsymbol{X}$*. Moreover,* T_i *is nondecreasing in* u_i*. Because its* i*-th component depends only on* $u_1, \dots, u_i$*,* $\boldsymbol{T}$ *is called a* ***triangular map****; wherever its Jacobian exists, that Jacobian is lower triangular.*

Proof. Theorem 1.8.2 gives $T_1(U_1) \overset{d}{=} X_1$. Suppose the first $i-1$ generated coordinates have the same joint distribution as $X_{1:i-1}$. Conditional on those coordinates, U_i remains uniform and independent of $U_1, \dots, U_{i-1}$. Another application of the inverse CDF method therefore gives the correct conditional distribution of X_i given $X_{1:i-1}$. Induction over $i = 1, \dots, d$ proves that the complete generated vector has the same joint distribution as $\boldsymbol{X}$. The monotonicity in u_i follows because every generalized inverse CDF is nondecreasing. □

Consequently, if $\boldsymbol{U}^{(1)}, \dots, \boldsymbol{U}^{(n)}$ are iid uniform random vectors on $(0,1)^d$, then

$$\boldsymbol{X}^{(k)} := \boldsymbol{T}\big(\boldsymbol{U}^{(k)}\big), \qquad k = 1, \dots, n,$$

are iid draws from the target PDF f. Operationally, each draw is generated coordinate by coordinate: evaluate the marginal inverse CDF for the first coordinate, then evaluate each conditional inverse CDF using the coordinates already generated.

Computational perspective. Once the map $\boldsymbol{T}$ is available, generating samples is fast. The difficult step is constructing or evaluating the required conditional CDFs and their generalized inverses. This step can be expensive in high dimensions. Conditional-independence structure can simplify some of these conditionals and make the triangular map sparse, which is one reason the KR rearrangement is useful in structured high-dimensional models.

1.8.3 Rejection Sampling Method*

Rejection sampling is also called "acceptance-rejection sampling." In many applications, a PDF is known only up to a normalizing constant. Specifically, the target PDF $f(x)$ can be written as

$$f(x) = \frac{1}{c} \cdot g(x), \tag{1.8.2}$$

and we only know $g(x)$. For example, the following function is a PDF

$$\frac{1}{c} \cdot g(x) = \frac{1}{c} \cdot \exp\left\{-(\sin x)^{1690}\right\} \cdot (\cos x)^{1690} \cdot \mathbb{1}_{[-1,1]},$$

where $c = \int_{-1}^{1} \exp\left\{-(\sin x)^{1690}\right\} \cdot (\cos x)^{1690}\, dx$ is the normalizing constant. However, it is not easy to know (or we are too lazy to know) the exact number of this constant. Fortunately, we will see that the rejection sampling method only needs the non-normalized version $g(x)$ (see Eq. (1.8.2)). This method is based on the following

Theorem 1.8.5. *Let B denote the region under the curve of g and above x-axis, i.e.,*

$$B = \left\{(x, y) \in \mathbb{R}^2 : 0 \le y \le g(x)\right\}, \tag{1.8.3}$$

where function $g(x)$ satisfies Eq. (1.8.2). Let (X, Y) be a 2-dimensional random vector following the uniform distribution on B (denoted as $\mathrm{Unif}(B)$, see Eq. (1.2.9)). Then, the marginal distribution of X is the one with PDF $f(x)$.

Proof. Recall Eq. (1.2.9), i.e., $\mathbb{P}\{\omega \in \Omega : X(\omega) \in A\} = \frac{\mathrm{Area}(A \cap B)}{\mathrm{Area}(B)}$ for all subsets $A \subseteq \mathbb{R}^2$. Let $t \in \mathbb{R}$ be fixed. We set $A = \{(x, y) : x \le t\}$, then $A \cap B = \{(x, y) : 0 \le y \le g(x) \text{ and } x \le t\}$. Using Eq. (1.2.9), we have

$$\begin{aligned} F_X(t) &= \mathbb{P}\{\omega \in \Omega : X(\omega) \le t\} \\ &= \mathbb{P}\{\omega \in \Omega : X(\omega) \in A\} \\ &= \frac{\mathrm{Area}(A \cap B)}{\mathrm{Area}(B)} \\ &= \frac{\int_{-\infty}^{t} g(x)dx}{\int_{-\infty}^{\infty} g(x)dx} \\ &= \frac{c \cdot \int_{-\infty}^{t} f(x)dx}{c \cdot \int_{-\infty}^{\infty} f(x)dx} \\ &= \int_{-\infty}^{t} f(x)dx. \end{aligned} \tag{1.8.4}$$

Using Eq. (1.2.4), the equation above implies that the PDF of random variable X is $f(x)$. The proof is completed. □

Remark 1.8.1. *In Eq. (1.8.4), the normalizing constant $\frac{1}{c}$ is canceled out. It is the reason we do not need normalized PDFs in rejection sampling.*

Based on Theorem 1.8.5, if we are able to generate a random vector (X, Y) from the uniform distribution on the region B defined by Eq. (1.8.3), the first component X of this random vector is our desired number. One remaining question is, *how to generate vector-valued random variables from* $\mathrm{Unif}(B)$*?* This question is solved by the following claim.

Claim 1.8.6. *Let E be a region such that $B \subseteq E$. Suppose $(\widetilde{X}, \widetilde{Y}) \sim \mathrm{Unif}(E)$. Then, conditioning on $(\widetilde{X}, \widetilde{Y}) \in B$, we have $(\widetilde{X}, \widetilde{Y}) \sim \mathrm{Unif}(B)$, i.e., $\left[(\widetilde{X}, \widetilde{Y}) \middle| (\widetilde{X}, \widetilde{Y}) \in B\right] \sim \mathrm{Unif}(B)$.*

Proof. Let the regions A, B, and E satisfy $A \subseteq B \subseteq E$. Then, the definition of conditional probability implies

$$\begin{aligned}
&\mathbb{P}\left\{\omega \in \Omega : \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in A \middle| \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in B\right\} \\
&= \frac{\mathbb{P}\left\{\omega \in \Omega : \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in A \cap B\right\}}{\mathbb{P}\left\{\omega \in \Omega : \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in B\right\}} \\
&= \frac{\mathbb{P}\left\{\omega \in \Omega : \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in A\right\}}{\mathbb{P}\left\{\omega \in \Omega : \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in B\right\}} \quad (\text{because } A \subseteq B) \\
&= \frac{\mathrm{Area}(A)/\,\mathrm{Area}(E)}{\mathrm{Area}(B)/\,\mathrm{Area}(E)} \\
&= \frac{\mathrm{Area}(A)}{\mathrm{Area}(B)},
\end{aligned}$$

which is the distribution of $\mathrm{Unif}(B)$. In the equation above, we may assume $\mathbb{P}\left\{\omega \in \Omega : \left(\widetilde{X}(\omega), \widetilde{Y}(\omega)\right) \in B\right\} > 0$ without loss of generality. The proof is completed. □

The region E in Claim 1.8.6 is called an **envelope** because it contains the "letter" B. For certain envelopes E, we can generate random variables from $\mathrm{Unif}(E)$. Under this assumption, Claim 1.8.6 leads to the following rejection sampling algorithm.

Algorithm 3 : General Rejection Sampling Method

Input: (i) Un-normalized PDF $g(x)$ (see Eq. (1.8.2)); (ii) An envelope region E; (iii) sample size n.
Output: A sequence of random numbers $X_1, \ldots, X_n$ following the distribution with PDF f.

```
for all i = 1, ..., n, do
    Generate 2-dimensional random vector (X̃_i, Ỹ_i) ~ Unif(E).
    while (X̃_i, Ỹ_i) ∉ B = {(x, y) ∈ R^2 : 0 ≤ y ≤ g(x)}, i.e., Ỹ_i > g(X̃_i) do
        Generate 2-dimensional random vector (X̃_i, Ỹ_i) ~ Unif(E).
    end while
    Set X_i ← X̃_i.
end for
```

Then, Algorithm 3 boils the rejection sampling method down to the following questions:

- *For a given region B defined in Eq. (1.8.3), what is a proper region E?*

- *How to generate random variables from* $\mathrm{Unif}(E)$*?*

Rectangle Envelopes

We first give a simple answer, which is generally not optimal. To avoid some mathematical techniques, we assume all PDFs are "supported in a bounded interval," i.e., $f(x) = 0$ whenever $x < a$ or $x > b$ for some real values a and b. This assumption is not true for many distributions, e.g., normal distributions. However, most mass of these distributions are concentrated in bounded intervals. For example, 99.7% of the mass of $N(0,1)$ is in $[-3,3]$. Hence, this assumption is approximately true. Suppose the region B determined by $g(x)$ (see Eq. (1.8.3)) is contained in a rectangle $[a,b] \times [0,M]$, i.e.,

$$B = \big\{(x,y) \in \mathbb{R}^2 : 0 \le y \le g(x)\big\} \subseteq [a,b] \times [0,M].$$

Set $E = [a,b] \times [0,M]$. Sampling from $\mathrm{Unif}(E)$ is straightforward: generate $X \sim \mathrm{Unif}(a,b)$ and $Y \sim \mathrm{Unif}([0,M])$ independently, so that $(X,Y) \sim \mathrm{Unif}(E)$. This observation and Algorithm 3 yield the following algorithm for generating random numbers from the PDF $f(x) = \frac{1}{c}g(x)$.

Algorithm 4 : Rejection Sampling with Rectangle Envelopes

Input: (i) Un-normalized PDF $g(x)$ (see Eq. (1.8.2)); (ii) rectangle envelope $[a,b] \times [0,M]$; (iii) sample size n.
Output: A sequence of random numbers $X_1, \dots, X_n$ following the distribution with PDF f.

1: **for all** $i = 1, \dots, n,$ **do**
2: Generate $\tilde{X}_i \sim \mathrm{Unif}(a,b)$ and $\tilde{Y}_i \sim \mathrm{Unif}([0,M])$.
3: **while** $(\tilde{X}_i, \tilde{Y}_i) \notin B = \big\{(x,y) \in \mathbb{R}^2 : 0 \le y \le g(x)\big\}$, i.e., $\tilde{Y}_i > g(\tilde{X}_i)$ **do**
4: Generate $\tilde{X}_i \sim \mathrm{Unif}(a,b)$ and $\tilde{Y}_i \sim \mathrm{Unif}([0,M])$.
5: **end while**
6: Set $X_i \leftarrow \tilde{X}_i$.
7: **end for**

As an example, we apply Algorithm 4 to generate random variable from the following PDF

$$f(x) = \frac{1}{c} \cdot g(x) = \frac{1}{c} \cdot \sqrt{1-x^2} \cdot \mathbb{1}_{[-1,1]}, \tag{1.8.5}$$

where the normalizing constant is actually $c = \frac{\pi}{2}$. Specifically, we set the inputs of Algorithm 4 to be $g(x) = \sqrt{1-x^2} \cdot \mathbb{1}_{[-1,1]}$, $[a,b] \times [0,M] = [-1,1] \times [0,1]$, and $n = 100000$. The result of this example is presented in Figure 1.10, which visually suggests that the generated numbers are from the PDF defined in Eq. (1.8.5).

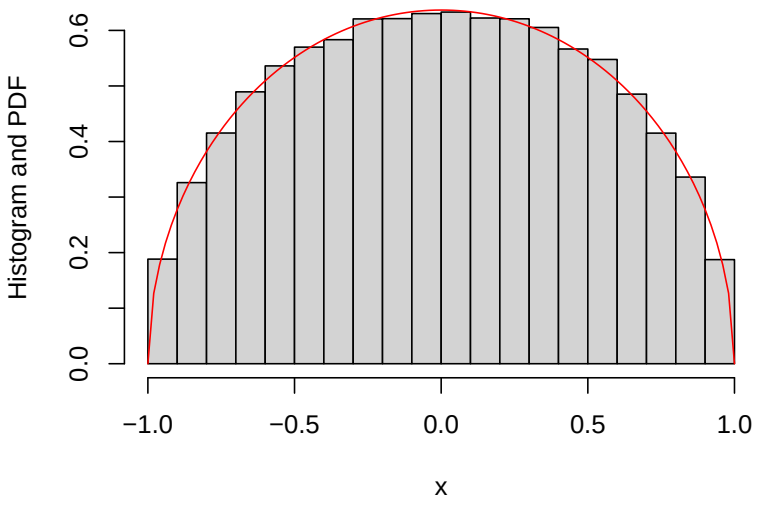


Figure 1.10: The red curve is the true PDF defined in Eq. (1.8.5), and the gray bars form a histogram of the values generated by Algorithm 4. The R code used for this simulation is provided below.

```r
# An Example of the Rejection Sampling with Rectangle Envelopes
g=function(x){ sqrt(1-x^2) }
f=function(x){ (2/pi)*g(x) }
X=vector()
n=100000
for (i in 1:n) {
  X_tilde=runif(1, min = -1, max = 1)
  Y_tilde=runif(1, min = 0, max = 1)
  while (Y_tilde>g(X_tilde)) {
    X_tilde=runif(1, min = -1, max = 1)
    Y_tilde=runif(1, min = 0, max = 1)
  }
  X[i]=X_tilde
}
hist(X, freq = FALSE, xlab = "x", ylab = "Histogram and PDF", main = "")
curve(f, add = TRUE, col = "red")
```

Envelopes Derived from Scaled PDFs

In Algorithm 4, only the generated $\tilde{X}_i$ satisfying $\tilde{Y}_i \le g(\tilde{X}_i)$ are retained. Other generated $\tilde{X}_i$ are actually wasted. The ratio of accepted $\tilde{X}_i$ is approximately

$$\mathbb{P}\{(\tilde{X}_i, \tilde{Y}_i) \in B\} = \frac{\text{area of } B}{\text{area of } E}.$$

In the example above (see Eq. (1.8.5)), the ratio of accepted $\tilde{X}_i$ is $\frac{\pi/2}{2} \approx 0.7854$. Hence, to generate $n = 100000$ desired X_i, we need to generate around 127324 pairs $(\tilde{X}_i, \tilde{Y}_i)$; equivalently, around 27324 pairs $(\tilde{X}_i, \tilde{Y}_i)$ are wasted. To increase the acceptance ratio $\frac{\text{area of } B}{\text{area of } E}$, we need a region E satisfying the following two conditions: (i) $B \subseteq E$ and (ii) B **and** E **are as similar as possible**. Hence, we use another way to construct the envelope E: deriving E from a **scaled PDF** $C \cdot h(x)$.

Suppose we know how to generate random numbers from a PDF $h(x)$. Additionally, this PDF satisfies the following: there exists a positive constant C such that

$$B = \big\{(x, y) \in \mathbb{R}^2 : 0 \le y \le g(x)\big\} \subseteq \big\{(x, y) \in \mathbb{R}^2 : 0 \le y \le C \cdot h(x)\big\} =: E \subseteq [a, b] \times [0, M]. \tag{1.8.6}$$

Because the newly defined envelope $E = \big\{(x, y) \in \mathbb{R}^2 : 0 \le y \le C \cdot h(x)\big\}$ is contained in the rectangle $[a, b] \times [0, M]$, its area is no larger than that of the rectangle. Consequently, the waste ratio for the newly defined envelope is no greater than that for the rectangle envelope $[a, b] \times [0, M]$. For example, for the un-normalized PDF $g(x) = \sqrt{1 - x^2} \cdot \mathbb{1}_{[-1,1]}$, we use the envelope $E = \big\{(x, y) \in \mathbb{R}^2 : 0 \le y \le C \cdot h(x)\big\}$ with the following

- $h(x) = \frac{1}{2} \text{Beta}(\frac{x+1}{2}; \alpha = 1.4, \beta = 1.4) \mathbb{1}_{[-1,1]}(x)$, where $\text{Beta}(z; \alpha, \beta)$ denotes the PDF of the Beta distribution. Thus, h is the PDF of $X = 2Z - 1$ when $Z \sim \text{Beta}(1.4, 1.4)$;
- $C = \frac{1}{h(0)} = \frac{2}{\text{Beta}(\frac{1}{2}; 1.4, 1.4)}$.

For $x \in [-1, 1]$, the scaled proposal satisfies

$$Ch(x) = (1 - x^2)^{0.4} \ge (1 - x^2)^{0.5} = g(x).$$

Thus, the condition in Eq. (1.8.6) is satisfied over the full support (see Figure 1.11).

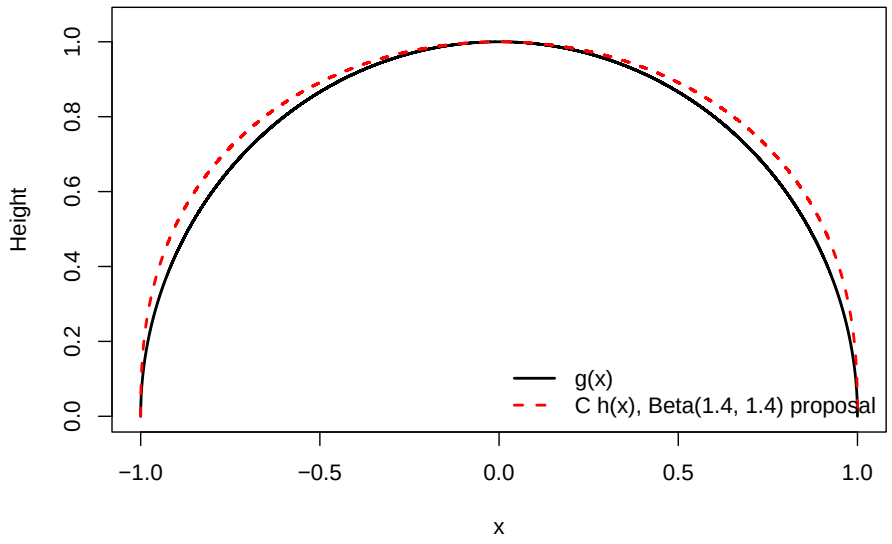


Figure 1.11: The solid (black) curve presents the un-normalized PDF $g(x) = \sqrt{1-x^2} \cdot \mathbb{1}_{[-1,1]}$, and the dashed (red) curve presents the **scaled PDF** $Ch(x)$, where h is the PDF of $2Z-1$ for $Z \sim \text{Beta}(1.4, 1.4)$ and $C = 1/h(0)$. Because $Ch(x) = (1-x^2)^{0.4} \geq g(x)$ on $[-1,1]$, Eq. (1.8.6) is satisfied with $a = -1$, $b = 1$, and $M = 1$.

To apply the rejection sampling method, we need to generate random variables from $\text{Unif}(E)$ with $E = \big\{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq C \cdot h(x)\big\}$, which implies the following theorem.

Theorem 1.8.7. *Let $\widetilde{X}$ be a random variable with PDF h. For any fixed x, conditional on $\widetilde{X} = x$, let $\widetilde{Y}$ have distribution $\text{Unif}([0, C \cdot h(x)])$, i.e., $(\widetilde{Y} \mid \widetilde{X} = x) \sim \text{Unif}([0, C \cdot h(x)])$. Then, $(\widetilde{X}, \widetilde{Y}) \sim \text{Unif}(E)$, where $E = \big\{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq C \cdot h(x)\big\}$.*

Before the proof of Theorem 1.8.7, we need the following **joint–marginal–conditional density factorization**, which will also be used in Section 1.10. This factorization underlies Bayes' rule and is widely used in Bayesian statistics (see Section 2.1). The result can be found in Definition 4.2.3 of Casella and Berger (2002), Example 5.1.4 of Durrett (2019), or Theorem 4.1 of Shao (2003); its proof is omitted herein.

Theorem 1.8.8 (Joint–marginal–conditional density factorization)**.** *Let $f(x,y)$ be the joint PDF of a continuous random vector (X,Y), and define the marginal PDF $f_X(x) := \int_{-\infty}^{\infty} f(x,y)dy$. For every x such that $f_X(x) > 0$, a conditional PDF of Y given $X = x$ is*

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}, \tag{1.8.7}$$

and hence

$$f(x,y) = f_{Y|X}(y|x)\, f_X(x).$$

Bayes' rule for densities follows by combining the two factorizations of the joint density and normalizing. Wherever the relevant marginal densities are positive,

$$f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y)f_Y(y)}{f_X(x)}.$$

Proof. (of Theorem 1.8.7) The proof combines the PDF of Unif(E) (see Eq. (1.2.11)) with the density factorization in Theorem 1.8.8, and it is left for homework. □

Based on Theorem 1.8.7 and the general Algorithm 3, we provide the following rejection-sampling algorithm using envelopes derived from scaled PDFs

Algorithm 5 : Rejection Sampling with Envelopes Derived from Scaled PDFs

Input: (i) Un-normalized PDF $g(x)$ (see Eq. (1.8.2)); (ii) scaled PDF $C \cdot h(x)$ such that $g(x) \le C \cdot h(x)$; (iii) sample size n.

Output: A sequence of random numbers $X_1, \ldots, X_n$ following the distribution with PDF f.

1: **for all** $i = 1, \ldots, n,$ **do**
2: Generate $\tilde{X}_i$ from PDF h; then, generate $U \sim \text{Unif}(0,1)$ and let $\tilde{Y}_i = C \cdot h(\tilde{X}_i) \cdot U$.
3: **while** $(\tilde{X}_i, \tilde{Y}_i) \notin B = \{(x,y) \in \mathbb{R}^2 : 0 \le y \le g(x)\}$, i.e., $\tilde{Y}_i > g(\tilde{X}_i)$ **do**
4: Generate $\tilde{X}_i$ from PDF h; then, generate $U \sim \text{Unif}(0,1)$ and let $\tilde{Y}_i = C \cdot h(\tilde{X}_i) \cdot U$.
5: **end while**
6: Set $X_i \leftarrow \tilde{X}_i$.
7: **end for**

To illustrate Algorithm 5, we generate random variables from the PDF in Eq. (1.8.5) using the scaled beta PDF envelope. Details of this example are provided in Figure 1.12, which heuristically suggests that Algorithm 5 works.

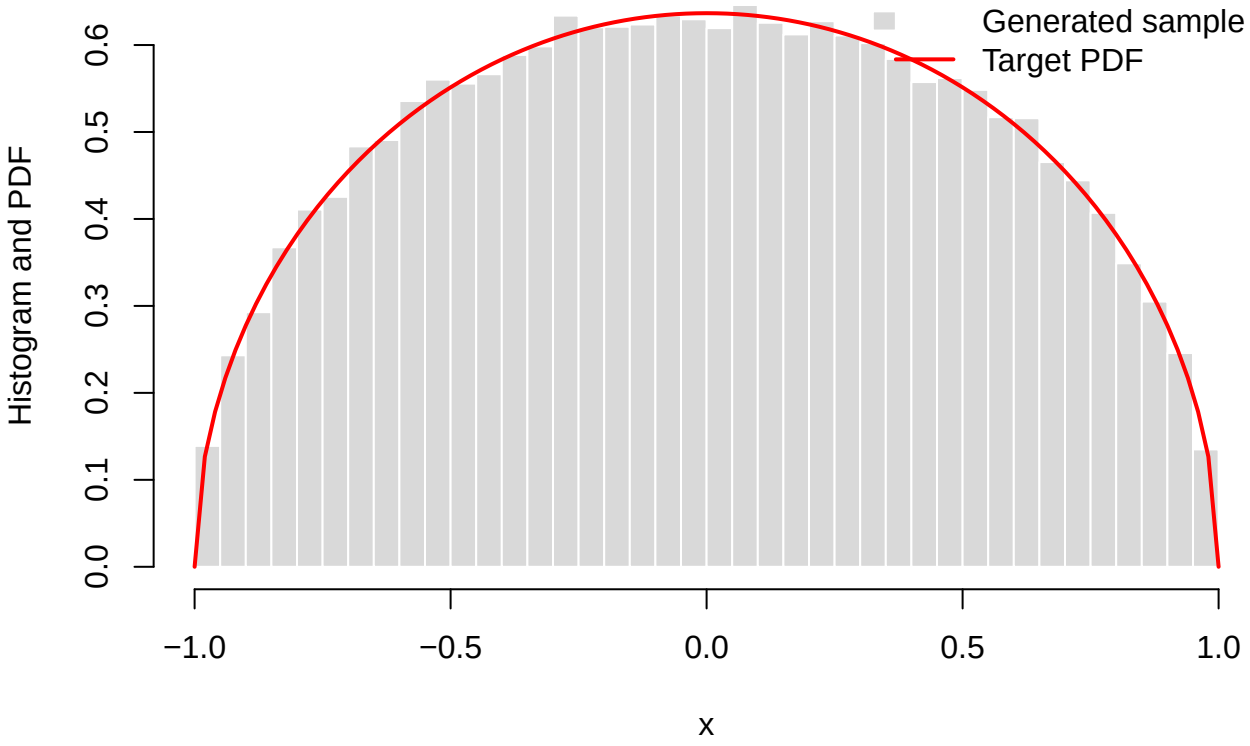


Figure 1.12: Algorithm 5 with (i) $g(x) = \sqrt{1-x^2} \cdot \mathbb{1}_{[-1,1]}$, (ii) the PDF h of $2Z-1$ for $Z \sim \text{Beta}(1.4, 1.4)$ and $C = 1/h(0)$, and (iii) $n = 100{,}000$. The gray bars form a histogram of the generated values, and the solid red curve is the true PDF $f(x)$ from Eq. (1.8.5). The R code used for this simulation is provided below.

```r
# An Example of the Rejection Sampling
# with Envelopes Derived from PDFs
set.seed(1690)
g=function(x){ sqrt(pmax(0,1-x^2)) }
f=function(x){ (2/pi)*g(x) }
shape_para=1.4
h=function(x){ 0.5*dbeta((x+1)/2, shape1 = shape_para, shape2 = shape_para) }
C=1/h(0)
grid=seq(-1,1,length.out=100001)
stopifnot(all(g(grid)<=C*h(grid)+sqrt(.Machine$double.eps)))
n=100000
X=numeric(n)
for (i in 1:n) {
  Z=rbeta(1, shape1 = shape_para, shape2 = shape_para)
  X_tilde=2*Z-1
```

```
  U=runif(1, min = 0, max = 1)
  Y_tilde=C*h(X_tilde)*U
  while (Y_tilde>g(X_tilde)) {
    Z=rbeta(1, shape1 = shape_para, shape2 = shape_para)
    X_tilde=2*Z-1
    U=runif(1, min = 0, max = 1)
    Y_tilde=C*h(X_tilde)*U
  }
  X[i]=X_tilde
}
hist(X, freq = FALSE, xlab = "x", ylab = "Histogram and PDF",
     main = "", breaks = 30, col = "gray85", border = "white")
curve(f, from = -1, to = 1, add = TRUE, col = "red", lwd = 2)
legend("topright", c("Generated sample", "Target PDF"),
       fill = c("gray85", NA), border = c("white", NA),
       lty = c(NA, 1), lwd = c(NA, 2), col = c("black", "red"), bty = "n")
```

1.9 Evaluation of PRNGs via Hypothesis Testing*

The preceding sections assessed simulated numbers primarily through empirical CDFs, density curves, and histograms. Such plots are useful diagnostics, but they do not provide a formal criterion for comparison with a target distribution. This section introduces hypothesis tests as a complementary diagnostic. As emphasized by von Neumann's remark at the beginning of Section 1.5, PRNG outputs are deterministic; the relevant question is how closely their empirical behavior agrees with that expected under a proposed random model. Section 1.9.1 first introduces significance levels and related terminology, after which we apply the framework to PRNG outputs.

As we mentioned in Section 1.5, one important advantage of PRNGs is the high speed of generating numbers. That is, we can make the sample size n of iid data $X_1, \ldots, X_n$ generated by a PRNG extremely large. Therefore, we mainly focus on the "large n" scenario—asymptotic hypothesis testing.

1.9.1 General Theory of Hypothesis Testing

Before the application of hypothesis testing to evaluate PRNGs, we give a brief overview of the general theory of hypothesis testing. The terminologies in this overview are based on Chapters 2.4.2 and 6.1 of Shao (2003). Additionally, this overview is designed specifically for evaluating PRNG.

Suppose we have the following two hypotheses, which **contradict** each other.

- H_0: a statement about the underlying CDF F generating iid data $\{X_i\}_{i=1}^n$
  (this statement is referred to as the **null** hypothesis);

- H_1: another statement about F contradicting H_0

  (this statement is referred to as the **alternative** hypothesis).

Concepts in the following definition are needed in the evaluation of PRNG.

Definition 1.9.1. *Suppose H_0 and H_1 are the null and alternative hypotheses of interest on the underlying distribution F. Data $X_1, \dots, X_n \overset{iid}{\sim} F$.*

i) Any random variable $T_n(\omega) = t(X_1(\omega), \dots, X_n(\omega))$ taking values in $\{0, 1\}$ is called a test for "H_0 vs. H_1"; $T_n = 0$ means "failing to reject H_0," and $T_n = 1$ means "rejecting H_0."

ii) When H_0 is true, $\mathbb{E}T_n = \mathbb{P}\{\omega \in \Omega : T_n(\omega) = 1\}$ is called the type I error rate of test T_n—the probability of incorrectly rejecting H_0. In this case, we write $\mathbb{E}(T_n|H_0) = \mathbb{P}\{\omega \in \Omega : T_n(\omega) = 1|H_0\}$.

iii) When H_1 is true, $\mathbb{E}T_n = \mathbb{P}\{\omega \in \Omega : T_n(\omega) = 1\}$ is called the power of test T_n—the probability of correctly rejecting H_0. In this case, we write $\mathbb{E}(T_n|H_1) = \mathbb{P}\{\omega \in \Omega : T_n(\omega) = 1|H_1\}$.

iv) If the limit $\lim_{n\to\infty} \mathbb{E}(T_n|H_0)$ exists, its value is called the asymptotic type I error rate—the approximate probability of (wrongly) rejecting H_0 when H_0 is true. (Of course, the practical use of this concept depends on large sample sizes n.) We can define the asymptotic power in the same way.

Remark 1.9.1.

- *In the expression $T_n(\omega) = t(X_1(\omega), \dots, X_n(\omega))$, the function $t(\cdot)$ is a deterministic n-variable function. However, data $X_1, \dots, X_n$ are random, which is emphasized by the variable ω. Hence, $T_n(\omega)$ is a composition of deterministic and random quantities and is itself random. An example is the following t-test for testing whether the mean of a normal distribution is zero*

$$T_n = t(X_1, \dots, X_n) = \mathbb{1}\left(\left|\sqrt{n} \cdot \frac{\frac{1}{n}\sum_{i=1}^n X_i}{\sqrt{\frac{1}{n-1}\sum_{i=1}^n \left(X_i - \frac{1}{n}\sum_{i'=1}^n X_{i'}\right)^2}}\right| > c\right) \in \{0, 1\},$$

  where t-statistic $\sqrt{n} \cdot \frac{\frac{1}{n}\sum_{i=1}^n X_i}{\sqrt{\frac{1}{n-1}\sum_{i=1}^n \left(X_i - \frac{1}{n}\sum_{i'=1}^n X_{i'}\right)^2}}$, under the corresponding null hypothesis, follows the t distribution with $n-1$ degrees of freedom.

- *To compute the exact value of the power and type I error rate, we need the exact distribution of test T_n. However, in many (actually most) applications, the exact distribution of T_n is not available. If the limit $\lim_{n\to\infty} \mathbb{E}(T_n|H_0)$ exists and known, we may let this limit to play the role of type I error rate/power when the sample size n is large. The word "asymptotic" indicates the scenario where n is large or the limit as $n \to \infty$. The asymptotic type I error rate will be used in this section to evaluate the performance of PRNG.*

- ***The interpretation of asymptotic type I error rate: When null hypothesis H_0 is true and sample size n is extremely large, the probability of making the type I error—wrongly rejecting the true H_0—is approximately equal to $\lim_{n\to\infty} \mathbb{E}(T_n|H_0)$.***

- * *To avoid the existence of the limit* $\lim_{n\to\infty}\mathbb{E}(T_n|H_0)$*, we may replace the limit with the limit superior* $\limsup_{n\to\infty}\mathbb{E}(T_n|H_0)$.

Definition 1.9.1 does not provide any criteria for choosing proper tests T_n. The most widely used criterion for choosing tests T_n is to control the type I error rates. For example, we only allow the type I error rates to be at most 5%. An explicit criterion for choosing tests T_n is in the following definition.

Definition 1.9.2. *For a fixed* $\alpha\in(0,1)$*, e.g.,* $\alpha=0.05$ *if we allow a type I error rate of at most* 5%*, any test* T_n *whose asymptotic type I error rate* $\lim_{n\to\infty}\mathbb{E}(T_n|H_0)$*, if it exists, is at most* α *is referred to as a* ***test with asymptotic significance level*** α.

To evaluate the performance of PRNGs, we introduce two ways of constructing test T_n—Kolmogorov–Smirnov and χ^2 goodness-of-fit tests. **The Kolmogorov–Smirnov test works for continuous distributions and the χ^2 goodness-of-fit test works for discrete scenarios**.

Throughout this section, numbers $\{X_i\}_{i=1}^n$ are generated by a PRNG for CDF F; additionally, we are interested in the following null and alternative hypotheses

$$H_0:\ X_1,\ldots,X_n\overset{iid}{\sim}F,\quad vs.\quad H_1:\ \text{otherwise.}$$

1.9.2 Kolmogorov–Smirnov Test

We first introduce the Kolmogorov–Smirnov test, which is closely relevant to the Glivenko-Cantelli theorem. One interesting historical fact is that the original papers of both the Kolmogorov–Smirnov test and Glivenko-Cantelli theorem were published in the same year and in the same volume of the same journal, i.e.,

- (Kolmogorov–Smirnov test, Kolmogorov (1933)) Kolmogorov A (1933). "Sulla determinazione empirica di una legge di distribuzione". Giorn. Ist. Ital. Attuari 4: 83–91.
- (Glivenko-Cantelli theorem, Glivenko (1933)) Glivenko, V. (1933). "Sulla determinazione empirica delle leggi di probabilità". Giorn. Ist. Ital. Attuari 4: 92-99.
- (Glivenko-Cantelli theorem, Cantelli (1933)) Cantelli, F. P. (1933). "Sulla determinazione empirica delle leggi di probabilità". Giorn. Ist. Ital. Attuari 4: 421-424.

Let $F_n(\omega,t):=\frac{1}{n}\sum_{i=1}^n\mathbb{1}\{X_i(\omega)\le t\}$ denote the empirical CDF derived from the data $\{X_i\}_{i=1}^n$ generated by a PRNG for CDF F. Heuristically (also see Theorem 1.3.5, i.e., the Glivenko-Cantelli theorem), if $X_1,\ldots,X_n\overset{iid}{\sim}F$, the true CDF F and empirical CDF F_n should be close to each other, i.e., the following difference is small

$$|F_n(\omega,t)-F(t)|=\left|\frac{1}{n}\sum_{i=1}^n\mathbb{1}\{X_i(\omega)\le t\}-F(t)\right|=\left|\frac{1}{n}\sum_{i=1}^n\Big[\mathbb{1}\{X_i(\omega)\le t\}-F(t)\Big]\right|.$$

Noticing that the $\frac{1}{n}\sum_{i=1}^{n}\left[\mathbb{1}\{X_i \le t\} - F(t)\right]$ above is of a "mean form," to mimic the central limit theorem (see Eq. (1.3.4)), we will focus on the following scaled mean form

$$\sqrt{n}\cdot|F_n(t) - F(t)| = \left|\sqrt{n}\cdot\frac{1}{n}\sum_{i=1}^{n}\left[\mathbb{1}\{X_i \le t\} - F(t)\right]\right|.$$

Noticing that the quantity $\sqrt{n}\cdot|F_n(t) - F(t)|$ depends on t. To get a test working uniformly for all $t \in \mathbb{R}$, we investigate the following

$$\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(t) - F(t)|.$$

Recall the randomness of $F_n(\omega, t) := \frac{1}{n}\sum_{i=1}^{n}\mathbb{1}\{X_i(\omega) \le t\}$ (the randomness of X_i), then $\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(t) - F(t)|$ is random as well.

If $\{X_i\}_{i=1}^{n}$ are truly iid from F, then $\sup_{t\in\mathbb{R}}|F_n(t) - F(t)| \approx 0$ (see Theorem 1.3.5, i.e., the Glivenko-Cantelli theorem). Therefore, the null hypothesis H_0 is likely untrue if $\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(t) - F(t)|$ is excessively large, i.e., $\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(t) - F(t)| > c$ for some threshold c. (The threshold c will be chosen to make the Kolmogorov–Smirnov test T_n^{KS} be with a predetermined significance level α.) The Kolmogorov–Smirnov test is defined as the following binary-valued random variable

$$T_n^{KS} = \mathbb{1}\left\{\left(\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(t) - F(t)|\right) > c\right\}, \text{ where} \tag{1.9.1}$$

$$T_n^{KS} = 1 \text{ indicates that the Kolmogorov–Smirnov test rejects } H_0, \tag{1.9.2}$$

$$T_n^{KS} = 0 \text{ indicates that the Kolmogorov–Smirnov test does not reject } H_0. \tag{1.9.3}$$

The Kolmogorov–Smirnov test—especially the threshold c making the Kolmogorov–Smirnov test T_n^{KS} be with a predetermined asymptotic significance level α—is based on the following asymptotic result.

Theorem 1.9.1. *Suppose the CDF F is **continuous**. When $X_1, \ldots, X_n \overset{iid}{\sim} F$, i.e., the null hypothesis H_0 is true, we have the following limit*

$$\lim_{n\to\infty}\mathbb{P}_{H_0}\left\{\omega\in\Omega\,\middle|\,\left(\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(\omega,t) - F(t)|\right) \le t\right\} = 1 - 2\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\exp\left(-2k^2t^2\right). \tag{1.9.4}$$

The proof of Theorem 1.9.1 is faaaaar beyond the scope of this course. One modern proof of this theorem can be found in Pitman and Yor (2001). One important fact is that **the right-hand side of Eq.(1.9.4) does not depend on CDF** F, and the proof of this fact is left for homework. In the proof of this fact, one can see why we need the continuity of CDF F in Theorem 1.9.1. In applications, the continuity of F is emphasized as well. For example, the `R` function `ks.test` emphasizes that "only continuous CDFs are valid;" please run the following `R` code and inspect the corresponding output.

```
?ks.test
```

Figure 1.13 provides a numerical check of the finite-sample behavior predicted by Theorem 1.9.1. Interpreting the study is left as homework.

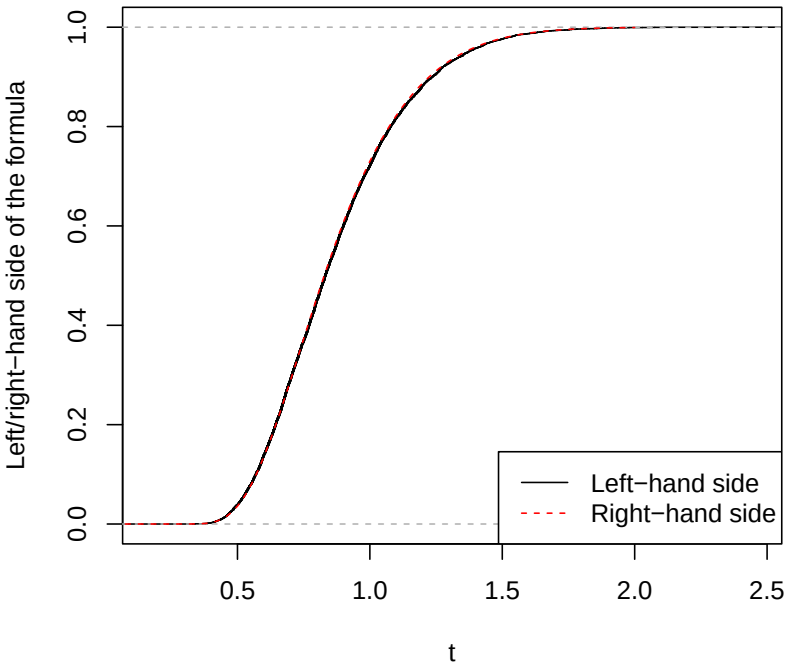


Figure 1.13: A numerical illustration of Theorem 1.9.1. The solid black curve is the left-hand side of Eq. (1.9.4) as a function of t, and the dashed red curve is the right-hand side. The code used to produce the figure is provided below.

```
# Left-hand side of the Kolmogorov-Smirnov formula
KS_data=vector()
for (i in 1:10000) {
  print(i)
  n=10000
  X=rnorm(n)
  ks_test_results=ks.test(X, pnorm, exact = FALSE)
  KS_data[i]=sqrt(n)*ks_test_results$statistic
}
quantile(KS_data, probs = 0.95)   # 95%-quantile = 1.355991
plot(ecdf(KS_data), xlab = "t", ylab = "Left/right-hand side of the formula", main = "")

# Right-hand side of the formula
KS_CDF=function(t){
  s=0
```

```
  for (j in 1:10000) { s=s+((-1)^(j-1))*exp(-2*(j*t)^2) }
  return(1-2*s)
}
curve(KS_CDF, from = 0, to = 2, add = TRUE, col="red", lty=2)
legend("bottomright", c("Left-hand side", "Right-hand side"),
       col = c("black", "red"), lty = c(1, 2))
```

Using Theorem 1.9.1, the asymptotic type I error rate of the test T_n^{KS} defined in Eq. (1.9.1) is the following

$$\begin{aligned}\lim_{n\to\infty}\mathbb{E}(T_n^{KS}|H_0) &= \lim_{n\to\infty}\mathbb{P}_{H_0}\left\{\omega\in\Omega\,\middle|\,\left(\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(\omega,t)-F(t)|\right)>c\right\}\\ &= 1-\lim_{n\to\infty}\mathbb{P}_{H_0}\left\{\omega\in\Omega\,\middle|\,\left(\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(\omega,t)-F(t)|\right)\le c\right\}\\ &= 2\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\exp\left(-2k^2c^2\right).\end{aligned}$$

Let c_α^* be the threshold c such that $2\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\exp\left(-2k^2c^2\right)=\alpha$, e.g., $c_\alpha^*\approx 1.356$ **when** $\alpha=0.05$. If we want the test T_n^{KS} to have asymptotic significance level at most α, we need to choose $c\ge c_\alpha^*$. Additionally, when H_0 is false, the limit $\lim_{n\to\infty}\mathbb{E}(T_n^{KS}|H_1)$ is the asymptotic power of T_n^{KS}—the probability of correctly rejecting the false H_0, and we want the power to be as large as possible. We notice that $\lim_{n\to\infty}\mathbb{E}(T_n^{KS}|H_1)$ is a non-increasing function of threshold c. It is straightforward that $c=c_\alpha^*$ makes the asymptotic power the largest under the constraint $c\ge c_\alpha^*$.

We conclude the Kolmogorov–Smirnov test in the following algorithm.

Algorithm 6 : Kolmogorov–Smirnov Test

Input: (i) Data $\{X_i\}_{i=1}^n$; (ii) the potential underlying CDF F of interest; (iii) significance level $\alpha\in[0,1]$.
Output: Decisions: "reject H_0 at asymptotic significance level α," or "fail to reject H_0."

1: Compute $M_n:=\sqrt{n}\cdot\sup_{t\in\mathbb{R}}|F_n(t)-F(t)|$ using inputs $\{X_i\}_{i=1}^n$ and F.
2: Compute the c_α^* by solving equation $2\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\exp\left(-2k^2c^2\right)=\alpha$.
3: **if** $M_n>c_\alpha^*$ **then** Adopt the decision "reject H_0 at asymptotic significance level α."
4: **else** Adopt the decision "fail to reject H_0."
5: **end if**

Many programming languages provide functions, such as `ks.test` in `R`, that automatically compute $M_n:=\sqrt{n}\sup_{t\in\mathbb{R}}|F_n(t)-F(t)|$. To illustrate Algorithm 6, we generate $n=10{,}000$ iid observations from Exp(1) by the inverse CDF method used in Figure 1.9. We then apply the algorithm with the exponential CDF and $\alpha=0.05$ and report the resulting decision. The `R` code is provided below.

```
# Generate one collection of Exp(1) random variables using the inverse CDF method
n=10000
U_i=runif(n)
X_i=log(1/(1-U_i))
# Compute the Kolmogorov-Smirnov statistic
ks_test_results=ks.test(X_i, pexp, rate = 1)
```

```
KS_statistic=sqrt(n)*ks_test_results$statistic

# H_0: the data X_i generated with the inverse CDF method follow the Exp(1) distribution.
# At the significance level 0.05 (i.e., allowing 0.05 type I error rate),
# we reject H_0 if KS_statistic > 1.356.
```

Since the F specified herein is the true CDF, we are under the null hypothesis. Then, the asymptotic type I error rate is expected to be around 0.05. To provide a numerical check of this approximation, we replicate the simulation study 10,000 times and compute the incorrect rejection rate across the 10,000 runs. The resulting value of `mean(T_KS)` can then be compared with 0.05; agreement is consistent with, but does not prove, the asymptotic result.

```
# Replicate the procedure above 10,000 times.
# For each replication, we do the Kolmogorov-Smirnov test with significance level 0.05.
# Under the null hypothesis, the expected type I error rate is 0.05.
T_KS=vector()
for (itr in 1:10000) {
  print(itr)
  U_i=runif(n)
  X_i=log(1/(1-U_i))
  ks_test_results=ks.test(X_i, pexp, rate = 1)
  T_KS[itr]=sqrt(n)*ks_test_results$statistic>1.356
}
mean(T_KS)  # which is approximately equal to 0.05, e.g., 0.0517.
```

A multidimensional version of the Kolmogorov–Smirnov test can be found in Fasano and Franceschini (1987).

1.9.3 χ^2 Goodness-of-Fit Test

In Section 1.9.2, we introduced the Kolmogorov-Smirnov test to evaluate PRNGs for continuous distributions. However, the Kolmogorov-Smirnov test does not theoretically work for discrete distributions—the ones generating random variables taking values in $\{1, 2, 3, \ldots\}$†. In this subsection, we introduce a method of evaluating PRNGs for discrete distributions.

For each discrete distribution, the PMF $\boldsymbol{p} = (p_1, p_2, \ldots)$ fully determines the distribution, where $\sum_{j=1}^{\infty} p_j = 1$ and $p_j \geq 0$ for all j. Because $\sum_{j=1}^{\infty} p_j$ converges, we have $\lim_{j\to\infty} p_j = 0$. Therefore, most of the probability mass of the distribution determined by $\boldsymbol{p} = (p_1, p_2, \ldots)$ is contained in finitely many p_j. For example, using the ϵ-N definition of convergence, there exists a sufficiently large k such that $\sum_{j=1}^{k} p_j \geq 0.99$, i.e., at least 99% of the probability mass is contained in the first k entries $\{p_j\}_{j=1}^{k}$. Hence, to investigate the discrete distribution determined by $\boldsymbol{p} = (p_1, p_2, \ldots)$, we can approximately study the **categorical distribution** with the following

† Generally, a discrete random variable takes values in a countable set $\{x_j\}_{j=1}^{\infty}$. We can always identify each x_j with integer j. Therefore, we can view all discrete random variables as taking values in $\{1, 2, 3, \ldots\}$.

probabilities:

$$\mathbb{P}\{X = j\} = \frac{p_j}{\sum_{j'=1}^{k} p_{j'}}, \quad \text{for all } j = 1, 2, \ldots, k.$$

Therefore, we focus herein on evaluating PRNGs for categorical distributions.

Suppose the distribution F of interest is a categorical distribution with probability vector $\boldsymbol{p} = (p_1, \ldots, p_k)$ with $\sum_{j=1}^{k} p_j = 1$. The χ^2 goodness-of-fit test is based on the following theorem.

Theorem 1.9.2. *Suppose* $X_1, \ldots, X_n \overset{iid}{\sim} \text{Categorical}(\boldsymbol{p})$. *Define*

$$\begin{aligned} Y_j &:= \#\{X_i = j\} = \text{ the number of } X_i \text{ satisfying } X_i = j, \\ S_n &:= \sum_{j=1}^{k} \frac{(Y_j - np_j)^2}{np_j} \end{aligned} \tag{1.9.5}$$

Then, we have $S_n \xrightarrow{d} \chi^2_{k-1} =$ ***the*** χ^2 ***distribution with*** $k-1$ ***degrees of freedom****, i.e.,*

$$\lim_{n\to\infty} \mathbb{P}\{\omega \in \Omega | S_n(\omega) \le t\} = \mathbb{P}\{\chi^2_{k-1} \le t\}.$$

The proof and a more detailed version of Theorem 1.9.2 can be found in Chapter 6.4.3 of Shao (2003). We omit the proof of Theorem 1.9.2 in this course‡. Based on Theorem 1.9.2, it is easy to verify that the following test is a test with asymptotic significance level α

$$T_n^{\chi^2} := \mathbb{1}\left(S_n > \chi^2_{k-1,1-\alpha}\right),$$

where S_n is defined in Eq.(1.9.5), and $\chi^2_{k-1,1-\alpha}$ is the $1-\alpha$ quantile of χ^2_{k-1}, i.e., $\mathbb{P}(\chi^2_{k-1} \le \chi^2_{k-1,1-\alpha}) = 1-\alpha$. The χ^2 goodness-of-fit test is concluded in the following algorithm.

Algorithm 7 : χ^2 Goodness-of-Fit Test

Input: (i) Data $\{X_i\}_{i=1}^n$; (ii) probability vector $\boldsymbol{p} = (p_1, \ldots, p_k)$; (iii) asymptotic significance level $\alpha \in [0, 1]$.
Output: Decisions: "reject H_0 at asymptotic significance level α," or "fail to reject H_0."

1: Compute Y_j for $j = 1, \ldots, k$ and S_n using Eq.(1.9.5).
2: Compute $T_n^{\chi^2} = \mathbb{1}(S_n > \chi^2_{k-1,1-\alpha})$.
3: **if** $T_n^{\chi^2} = 1$ **then** Adopt the decision "reject H_0 at asymptotic significance level α."
4: **else** Adopt the decision "fail to reject H_0."
5: **end if**

We provide a simulation study to illustrate the finite-sample performance of Algorithm 7. We set $\boldsymbol{p} = (0.4, 0.3, 0.2, 0.1)$ and generate $\{X_i\}_{i=1}^n \overset{iid}{\sim} \text{Categorical}(\boldsymbol{p})$ with $n = 10000$. Then, we apply Algorithm 7 to $\{X_i\}_{i=1}^n$, $\boldsymbol{p}$, and $\alpha = 0.05$. The `R` code for this simulation study is the following

‡ It is a half-page proof involving the *spectral decomposition of projection matrices*. Although the mathematical tool is within the context of undergraduate level linear algebra, it is distractive and will not be used in the rest of this course. Therefore, we omit this proof. This proof can be found in Chapter 6.4.3 of Shao (2003)

```r
# An example of the chisq goodness-of-fit test
n=10000
p=c(4,3,2,1)
p_0=p/sum(p)
X_i=rmultinom(n, size = 1, prob = p_0)
Y_j=apply(X_i, 1, sum)
as.numeric( sum((Y_j-n*p_0)^2/(n*p_0))>qchisq(0.95, df=length(p)-1) )
```

To provide a numerical check of the asymptotic χ^2_{k-1} approximation in Theorem 1.9.2, we repeat the simulation study above 10,000 times and obtain 10,000 values of S_n (see Eq. (1.9.5)). In Figure 1.14, the solid black curve is the empirical CDF of these values, and the dashed red curve is the CDF of the χ^2_{k-1} distribution. Their close agreement over the plotted range is consistent with the theorem but does not establish the asymptotic result.

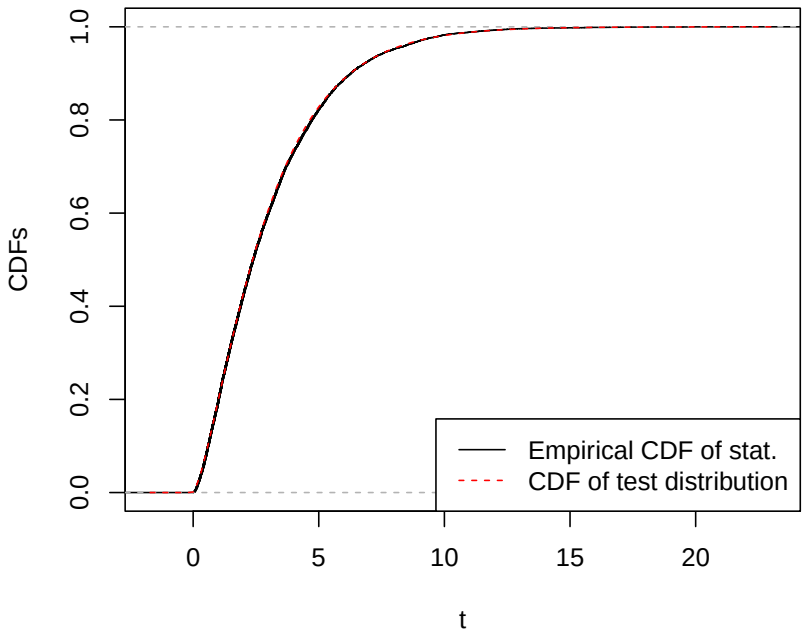


Figure 1.14: The solid black curve is the empirical CDF of S_n, and the dashed red curve is the CDF of its asymptotic distribution as $n \to \infty$. The R code used to produce this figure is provided below.

```r
# Replicate the procedure above 10,000 times.
# For each replication, we do the chi-square goodness-of-fit test with significance level 0.05.
# Under the null hypothesis, the expected type I error rate is 0.05.
number_replicates=10000
chisq_statistic=vector()
for (itr in 1:number_replicates) {
  print(itr)
  X_i=rmultinom(n, size = 1, prob = p_0)
```

```
  Y_j=apply(X_i, 1, sum)
  chisq_statistic[itr]=sum((Y_j-n*p_0)^2/(n*p_0))
}
mean(chisq_statistic>qchisq(0.95, df=length(p)-1))  # which is approximately 0.05
plot(ecdf(chisq_statistic), xlab = "t", ylab = "CDFs", main = "")
test_distribution_cdf=function(x){ return(pchisq(x, df=length(p)-1))}
curve(test_distribution_cdf, col="red", add = TRUE, lty=2)
legend("bottomright", c("Empirical CDF of stat.", "CDF of test distribution"),
       lty = c(1, 2), col = c("black", "red"))
```

1.10 Generating Random Vectors

The preceding methods generate one real-valued random variable at a time. We now extend the same ideas to random vectors by exploiting a joint–marginal–conditional factorization. To keep the notation transparent, we treat continuous bivariate distributions; the same sequential principle extends to discrete and higher-dimensional settings.

Suppose we want to generate a random vector (X, Y) with bivariate PDF $f(x, y)$. The marginal PDF of X is $f_X(x) := \int f(x, y)\, dy$. For each x such that $f_X(x) > 0$, let $f_{Y|X}(y|x)$ denote the conditional PDF of Y given $X = x$. The joint–marginal–conditional density factorization (Theorem 1.8.8) then gives

$$f(x, y) = f_{Y|X}(y|x) f_X(x). \tag{1.10.1}$$

Equation (1.10.1) suggests the following two-step procedure for generating (X, Y):

i) generate X from the marginal PDF f_X;

ii) conditional on the generated value of X, generate Y from $f_{Y|X}(\cdot \mid X)$.

The resulting pair has joint PDF f.

For example, consider the decomposition

$$\text{If } f(x, y) = e^{-y} \cdot \mathbb{1}(0 < x < y < \infty) \text{ with } f_X(x) = e^{-x} \cdot \mathbb{1}(x > 0) \text{ and } f_{Y|X}(y|x) = e^{x-y} \cdot \mathbb{1}(y > x). \tag{1.10.2}$$

Verifying Eq. (1.10.2) and implementing the corresponding simulation are left as homework. To generate (X, Y) with PDF $f(x, y) = e^{-y} \cdot \mathbb{1}(0 < x < y < \infty)$, proceed as follows:

i) generate X from the marginal distribution $f_X(x) = e^{-x} \cdot \mathbb{1}(x > 0)$;

ii) conditional on the generated value of X, generate Y from $f_{Y|X}(y|X) = e^{X-y} \cdot \mathbb{1}(y > X)$.

In high-dimensional spaces, sequential simulation based on Eq. (1.10.1) can be cumbersome. Markov chain Monte Carlo provides a widely used alternative when direct simulation is difficult; see Chapter 4.

1.11 Homework for Chapter 1

Problem 1.11.1. *These problems are meant to jog your memory about probability. If most of them seem completely foreign, or very difficult, then you may not have the necessary probability background to do this course. Please feel free to come talk to me about it.*

i) A biased die has the following probabilities of landing on each face:

Face value	*1*	*2*	*3*	*4*	*5*	*6*
Probability	*.1*	*.1*	*.2*	*.2*	*.4*	*0*

I win if the die shows even. What is the probability that I win? Is this better or worse than a fair die?

ii) I flip a coin 3 times. What is the sample space?

iii) Suppose $P(\text{snow today}) = 30\%$, $P(\text{snow tomorrow}) = 60\%$, and $P(\text{snow today and tomorrow}) = 25\%$. Given that it snows today, what is the chance that it will snow tomorrow?

iv) Give a formula for $P(G|H^c)$ in terms of $P(G)$, $P(H)$ and $P(G \cap H)$ only.

v) A box contains w white balls and b black balls. A ball is chosen at random.

- *If the chosen ball is white, we add d white balls to the box, that is, now there are $w + d$ white balls and b black balls.*
- *If the chosen ball is black, we add d black balls to the box, that is, now there are w white balls and $b + d$ black balls.*

After adding the d balls, another ball is drawn at random from the box. Show that the probability that the second chosen ball is white does not depend on d.

vi) Roll two fair five-sided dice with sides labeled $\{-2, -1, 0, 1, 2\}$.

(a) What is the pmf for the product of the two dice?

(b) What is the cdf for the product of the two dice?

(c) Let Y be 1 for a positive product, -1 for a negative product and 0 for a zero product. What is the pmf for Y?

vii) *X has pdf $f_X(x) = \lambda e^{-\lambda x} 1_{\{x \geq 0\}}$ for some $\lambda > 0$. Compute*

- *(a) the cdf of X.*
- *(b) the mean of X.*
- *(c) the variance of X.*
- *(d) the probability that X is greater than its mean.*

viii) *A coin with $P(H) = p$ is tossed repeatedly. Given that the second heads occurs at the 10th toss, what is the (conditional) pmf of the time of the first heads?*

Problem 1.11.2. *Verify that the pair $(\Omega, \mathbb{P})$ defined in Example 1.2.1 is a probability space.*

Problem 1.11.3. *Prove parts (iii), (iv), (vi), and (vii) of Theorem 1.2.1. You may use Definition 1.2.4 and parts (i) and (ii), which were proved above.*

Problem 1.11.4. *Let $\Omega = [0, 1]$. For every event $A \subseteq \Omega$, define*

$$\mathbb{P}(A) := \int_0^1 \mathbb{1}_A(x)\, dx,$$

where $\mathbb{1}_A$ is the indicator function from Definition 1.2.5. You may assume that $\mathbb{P}$ is a probability. Define the random variable X on Ω by $X(\omega) = \omega + 1$.

i) *Let $A = \{\omega \in \Omega : X(\omega) = 1.5\}$. Compute $\mathbb{P}(A)$. Is A an impossible event?*

ii) *Let $B = \{\omega \in \Omega : X(\omega) = 0.5\}$. Compute $\mathbb{P}(B)$. Is B an impossible event?*

iii) *Derive the CDF $F_X(t)$ for every $t \in \mathbb{R}$.*

Problem 1.11.5. *Show that all CDFs are nondecreasing functions. (Hint: apply Theorem 1.2.1 and Eq. (1.2.3).)*

Problem 1.11.6. *Suppose X is a random variable defined on probability space $(\Omega, \mathbb{P})$, and it takes values in positive integers $\{1, 2, \ldots\}$ and satisfies $p_j = \mathbb{P}\{\omega \in \Omega : X(\omega) = j\}$ for all $j = 1, 2, \ldots$. Show that the CDF of X is $F_X(t) = \sum_{j=1}^{\infty} p_j \cdot \mathbb{1}(j \leq t)$.*

Problem 1.11.7. *i)* *Derive an analytic formula of the CDF F_X of the random variable X defined in Eq. (1.2.12). This analytic formula can be expressed using the CDF $\Phi(t \,|\, \mu, \sigma)$ defined in Eq. (1.2.7).*

ii) *Prove the result in Eq. (1.2.14).*

Problem 1.11.8. *One way of proving Theorem 1.3.5 is to use the following two results.*

- *(Dvoretzky–Kiefer–Wolfowitz inequality, 1956, see Dvoretzky et al. (1956) or Lemma 5.1 of Shao (2003)) Suppose* $X_1, \ldots, X_n \overset{iid}{\sim} F$. *Let* F_n *be the empirical CDF based on* $\{X_i\}_{i=1}^n$ *(see Eq. (1.3.5)). There exists a positive constant* C *(not depending on* F*) such that*

$$\mathbb{P}\left\{\omega \in \Omega \,\middle|\, \sup_{t\in\mathbb{R}} |F_n(\omega, t) - F(t)| > \epsilon\right\} \leq Ce^{-2n\epsilon^2}.$$

  (Massart (1990) proved that the sharp constant C *in the inequality above is* $C = 2$*.)*

- *Let* $Z(\omega), Z_1(\omega), \ldots, Z_n(\omega), \cdots$ *be random variables. If, for any* $z > 0$*, we have*

$$\sum_{n=1}^{\infty} \mathbb{P}\{\omega \in \Omega : |Z_n(\omega) - Z(\omega)| \geq z\} < \infty,$$

  then Z_n *converges to* Z *with probability one, i.e.,* $\mathbb{P}\{\omega \in \Omega : \lim_{n\to\infty} |Z_n(\omega) - Z(\omega)| = 0\} = 1$.

We assume the two results above are correct. (They are actually correct, although their proofs involve a lot of advanced machinery.) Using the two results above, prove Theorem 1.3.5.

Problem 1.11.9. *Look up the definitions of "pairwise independence" and "mutual independence" and explain the difference between these two definitions.*

Problem 1.11.10. *Consider the infinite sequence of independent fair-coin indicators* $X_1, X_2, \ldots$ *constructed in Section 1.3.2. Define*

$$\Omega_1 := \left\{\omega \in \Omega : \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i(\omega) = \frac{1}{2}\right\},$$
$$\Omega_0 := \Omega \setminus \Omega_1.$$

i) Determine whether each of the following outcomes belongs to Ω_1 *or* Ω_0*, and explain your answer:*

$$\omega^{(1)} = (1, 0, 0, 1, 0, 0, \ldots),$$
$$\omega^{(2)} = (1, 0, 1, 0, 1, 0, \ldots).$$

ii) Use the LLN to prove that $\mathbb{P}(\Omega_1) = 1$ *and* $\mathbb{P}(\Omega_0) = 0$.

Problem 1.11.11. *Use a programming language of your choice to investigate the heuristic LIL envelope in Theorem 1.3.2.*

i) Generate 10,000 *iid observations* $X_1, \ldots, X_{10000} \sim$ Bernoulli(1/2).

ii) For $n = 100, \dots, 10{,}000$, compute

$$e_n := \left| \frac{1}{n} \sum_{i=1}^{n} X_i - \frac{1}{2} \right|.$$

iii) Repeat the preceding steps ten times and overlay the ten curves of e_n against n.

iv) On the same plot, add

$$g(n) := \frac{1}{2} \sqrt{\frac{2 \log(\log n)}{n}}.$$

Submit the plot and your code, and explain what the comparison illustrates. In particular, explain why $g(n)$ should be interpreted as an asymptotic fluctuation scale rather than a finite-sample bound.

Problem 1.11.12. *Let X have the standard Cauchy PDF*

$$p_X(x) = \frac{1}{\pi(1 + x^2)}, \qquad x \in \mathbb{R}.$$

i) For every $M > 0$, evaluate $\int_{-M}^{M} x p_X(x)\, dx$ and then compute its limit as $M \to \infty$.

ii) Does $\mathbb{E}X$ exist under Definition 1.2.13? Prove your answer. Explain why the symmetric limit in the preceding part does not settle this question.

iii) For $n = 1, \dots, 1000$, generate iid observations $X_1, \dots, X_n$ from the standard Cauchy distribution and plot the running averages $\overline{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Repeat the experiment many times, overlay the trajectories, and submit both the plot and your code.

iv) Based on the LLN assumptions and your simulation, explain whether one should expect $\overline{X}_n$ to converge almost surely.

Problem 1.11.13. *This problem examines the multiplicative congruential generator in Algorithm 1.*

i) Write a function that takes a sample size n, a modulus m, a multiplier a, and a seed $s \in \{1, \dots, m-1\}$ as inputs and returns $g(1), \dots, g(n)$.

ii) Set $m = 2^{31} - 1$, $a = 7^5$, and $g(1) = 1690$. Report $g(1), \dots, g(10)$ and submit your code.

iii) Generate $g(1), \dots, g(10000)$ and plot a histogram of $u_i = g(i)/m$. Submit the histogram and your code.

iv) Explain how the histogram is related to the uniform-distribution approximation in Eqs. (1.7.4)–(1.7.5). Why does a uniform-looking histogram not imply that the sequence is genuinely random or independent?

v) Use $u_1, \dots, u_{10000}$ to construct a Monte Carlo estimate of

$$\int_0^\infty x e^{-10x}\, dx.$$

Explain your estimator, report its value, and compare it with the exact integral. (Hint: use the inverse CDF method to set

$$X_i = -\frac{1}{10}\log(1-u_i),$$

so that the X_i behave like draws from an exponential distribution with rate 10. Since its density is $10e^{-10x}$,

$$\int_0^\infty x e^{-10x}\, dx = \frac{1}{10}\mathbb{E}[X] = \frac{1}{100},$$

Thus, with $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$, estimate the integral using

$$\widehat{I}_n = \frac{1}{10}\overline{X}_n = \frac{1}{10n}\sum_{i=1}^n X_i.$$

)

Problem 1.11.14. *For each positive integer n, let $X_n \sim \mathrm{Unif}(\{1, \dots, n\})$ and define $Y_n = X_n/n$.*

i) Show that the CDF of Y_n is

$$F_n(t) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\left\{\frac{i}{n} \le t\right\}, \qquad t \in \mathbb{R}.$$

ii) Use Riemann sums to prove that $F_n(t)$ converges pointwise to the CDF of $\mathrm{Unif}(0,1)$ as $n \to \infty$. Treat the cases $t < 0$, $0 \le t < 1$, and $t \ge 1$ explicitly.

Problem 1.11.15. *Let $X \sim \mathrm{Bernoulli}(1/3)$, where $\mathbb{P}(X = 1) = 1/3$.*

i) Derive an explicit formula for the generalized inverse

$$G(u) = \inf\{t \in \mathbb{R} : F_X(t) \ge u\}, \qquad 0 < u < 1.$$

ii) Explain why the same formula at $u = 0$ does not define a finite real number. How does this observation justify the domain $(0,1)$ in Eq. (1.8.1)?

iii) Apply G to the first ten normalized MCG outputs $u_i = g(i)/m$ from the preceding MCG problem and report the resulting ten Bernoulli pseudo-random numbers.

iv) Apply G to the first 10,000 normalized MCG outputs, plot the empirical CDF of the resulting Bernoulli pseudo-random numbers, and compare it with the true CDF F_X. Submit your plot and code.

Problem 1.11.16. *Prove Theorem 1.8.7. Hint: combine the PDF of* $\mathrm{Unif}(E)$ *(see Eq. (1.2.11)) and the density factorization in Theorem 1.8.8.*

Problem 1.11.17. *i) Let X be a real-valued random variable with* ***continuous*** *CDF F_X. Then $U := F_X(X) \sim \mathrm{Unif}(0,1)$. (This result can be viewed as an "inverse version" of Theorem 1.8.2.)*

ii) Suppose $X_1, \ldots, X_n \overset{iid}{\sim} F$, where F is continuous. Then the distribution of $D_n := \sup_{t\in\mathbb{R}} |F_n(t) - F(t)| = \sup_{t\in\mathbb{R}} \left| \frac{1}{n}\sum_{i=1}^n \left[\mathbb{1}\{X_i \le t\} - F(t)\right]\right|$ does not depend on the CDF F. (Hint: first show that the distribution of $D_n^+ := \sup_{t\in\mathbb{R}} (F_n(t) - F(t))$ does not depend on F, then show that the distribution of $D_n^- := \sup_{t\in\mathbb{R}} (F(t) - F_n(t))$ does not depend on F. The desired result follows from $D_n = \max\{D_n^+, D_n^-\}$.)

Problem 1.11.18. *Read and explain the simulation study presented in Figure 1.13, which provides a numerical check of the finite-sample behavior predicted by Theorem 1.9.1.*

Problem 1.11.19. *i) If $f(x,y) = e^{-y} \cdot \mathbb{1}(0 < x < y < \infty)$, then $f_X(x) = e^{-x} \cdot \mathbb{1}(x > 0)$ and $f_{Y|X}(y|x) = e^{x-y} \cdot \mathbb{1}(y > x)$.*

ii) Using the inverse CDF method, generate $n = 10000$ random vectors from this PDF $f(x,y)$.

iii) Using the rejection sampling method, generate $n = 10000$ random vectors from this PDF $f(x,y)$.

iv) Plot the generated $\{(X_i, Y_i)\}_{i=1}^n$ on the x-y plane.

Chapter 2

Stochastic Approximation

"If you can talk with crowds and keep your virtue, Or walk with Kings—nor lose the common touch..."

— Rudyard Kipling

Many difficult integrals can be rewritten as expectations and then approximated using simulated draws. This chapter develops that idea, beginning with ordinary Monte Carlo integration and its uncertainty, and then turning to importance sampling for problems in which direct simulation is inefficient. For example, for any function $h : [0,1] \to \mathbb{R}$ with $\int_0^1 |h(x)|dx < \infty$, we have

$$\int_0^1 h(x)dx = \mathbb{E}[h(U)], \quad \text{where } U \sim \text{Unif}(0,1). \tag{2.0.1}$$

When $h(x)$ is very complicated, analytically computing the exact value of $\int_0^1 h(x)\,dx$ may be extremely difficult. The expectation representation in Eq. (2.0.1) instead permits a Monte Carlo approximation based on iid draws $U_1, \ldots, U_n \sim \text{Unif}(0,1)$. The LLN gives consistency of the sample mean $n^{-1}\sum_{i=1}^n h(U_i)$ for the integral. Under the additional finite-variance condition $\mathbb{E}[h(U)^2] < \infty$, the CLT provides an asymptotic normal approximation for its sampling uncertainty.

2.1 Motivation: Bayes Estimators*

"Bayes or not Bayes, is this the question?"

— the title of paper Hackenberger (2019)

Let θ be an unknown parameter and suppose that, conditional on θ, the data are iid with density $f(\cdot|\theta)$; that is, $X_1, \ldots, X_n \mid \theta \overset{iid}{\sim} f(\cdot|\theta)$. Suppose we are interested in estimating θ. For the estimation of θ, there are two philosophies—"frequentist" and "Bayes."

- People holding the frequentist philosophy view the parameter θ as fixed and deterministic. A popular frequentist estimator of θ is the following **maximum likelihood estimation** (MLE), which will be mentioned in APMA 1740/2610 from the information theory perspective

$$\widehat{\theta}_{MLE}(\omega) := \underset{\theta}{\operatorname{argmax}} \left\{ \prod_{i=1}^{n} f\left(X_i(\omega)|\theta\right) \right\}.$$

- People holding the Bayesian philosophy **view the parameter θ as a random variable**. Under the conditional iid assumption $X_1, \ldots, X_n \mid \theta \overset{iid}{\sim} f(\cdot|\theta)$, the conditional joint PDF of $(X_1, \ldots, X_n)$ given θ is $\prod_{i=1}^{n} f(x_i|\theta)$. Suppose $f(\theta)$ is the PDF of θ, which is called the **prior** of θ. Then, the density factorization in Theorem 1.8.8 implies that the PDF of the joint distribution of $(X_1, \ldots, X_n, \theta)$ is

$$f(x_1, \cdots, x_n, \theta) := \prod_{i=1}^{n} f(x_i|\theta) f(\theta),$$

and the distribution of θ conditioning on $X_1 = x_1, \ldots, X_n = x_n$ is the following

$$f(\theta|x_1, \ldots, x_n) = \frac{\prod_{i=1}^{n} f(x_i|\theta) f(\theta)}{\int \left[\prod_{i=1}^{n} f(x_i|\theta') f(\theta')\right] d\theta'},$$

i.e., $(\theta|X_1 = x_1, \ldots, X_n = x_n) \sim f(\theta|x_1, \ldots, x_n)$. The PDF $f(\theta|x_1, \ldots, x_n)$ is called the **posterior** of θ given data $X_1 = x_1, \ldots, X_n = x_n$. The **Bayes estimator** of θ is defined as follows (see Section 7.2.3 of Casella and Berger (2002))

$$\begin{aligned} \widehat{\theta}_{Bayes} &:= \mathbb{E}(\theta|X_1 = x_1, \ldots, X_n = x_n) \\ &= \int \theta \cdot \left(\frac{\prod_{i=1}^{n} f(x_i|\theta) f(\theta)}{\int \left[\prod_{i=1}^{n} f(x_i|\theta') f(\theta')\right] d\theta'} \right) d\theta. \end{aligned} \tag{2.1.1}$$

The integral in Eq. (2.1.1) is extremely complicated, and it also involves another difficult integral. In general, we cannot compute these integrals analytically. Stochastic approximation methods are needed to compute the integral in Eq. (2.1.1), especially when the parameter θ is a high-dimensional vector†.

2.2 Monte Carlo Integration

"Monte Carlo methods, or Monte Carlo experiments, are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. The underlying concept is to use randomness to solve problems that might be deterministic in principle."

— from the Wikipedia page for the "Monte Carlo method."

† Markov chain Monte Carlo is a widely used approach when direct integration or direct sampling is difficult; see Chapter 4.

The Bayes estimator is one instance of a more general computational problem: approximating an integral represented as an expectation. We now develop the Monte Carlo framework for that general problem.

2.2.1 Estimators

Suppose the integral of interest can be represented as

$$v := \int_{-\infty}^{\infty} H(x) \cdot f(x) dx, \tag{2.2.1}$$

where $f(x)$ is a PDF and $\int_{-\infty}^{\infty} |H(x)| \cdot f(x) dx < \infty$. Then $v = \mathbb{E}[H(X)]$, where $X \sim f$. Suppose we generate iid $X_1, \ldots, X_n \overset{iid}{\sim} f$ (using a PRNG, which can deal with large sample sizes n), the LLN (Theorem 1.3.1) indicates

$$\lim_{n\to\infty} \left[\frac{1}{n} \sum_{i=1}^{n} H(X_i) \right] = \int_{-\infty}^{\infty} H(x) \cdot f(x) dx = v, \quad \text{with probability one.}$$

Therefore, when the sample size n is extremely large (PRNGs can easily guarantee this fact), we may adopt the following approximation to estimate the integral in Eq. (2.2.1)

$$\widehat{v}_n := \frac{1}{n} \sum_{i=1}^{n} H(X_i) \approx \int_{-\infty}^{\infty} H(x) \cdot f(x) dx = v. \tag{2.2.2}$$

We take several examples in Table 2.1 to illustrate this approximation. Details of these examples are presented in the caption of Table 2.1. In these examples, the Monte Carlo mean approximation in Eq. (2.2.2) performs well in terms of estimating the integrals of interest.

Table 2.1: Examples illustrating the Monte Carlo approximation in Eq. (2.2.2). We take $f(x) = \mathbb{1}_{[0,1]}(x) =$ the PDF of Unif$(0,1)$, $H(x) =$ the corresponding functions in the table, and $n = 100000$. For each scenario, we run the approximation three times and present all the estimates. The `R` code for producing this table is provided right after this table.

$H(x)$	True values	1st run	2nd run	3rd run
x	$1/2 = 0.5000$	0.4994	0.5010	0.5003
x^5	$1/6 \approx 0.1667$	0.1659	0.1677	0.1665
$\sin(x) \cdot \log(1+x)$	≈ 0.2265 (via `WolframAlpha`)	0.2260	0.2272	0.2266

```r
# Examples for Monte Carlo Integration
n=10000
for (itr in 1:3) {
  print(paste("Run No.: ", as.character(itr)))
  U=runif(n)
  v_1_hat=mean(U)
  v_2_hat=mean(U^5)
  v_3_hat=mean(sin(U)*log(1+U))
  print(paste("v_1_hat = ", as.character(round(v_1_hat, 4))))
```

```
  print(paste("v_2_hat = ", as.character(round(v_2_hat, 4))))
  print(paste("v_3_hat = ", as.character(round(v_3_hat, 4))))
}
```

2.2.2 Estimation Error

The LLN describes the limiting behavior of the Monte Carlo estimator, but every finite simulation still has approximation error. Define

$$e_n = \left| \frac{1}{n} \sum_{i=1}^{n} H(X_i) - \int_{-\infty}^{\infty} H(x) \cdot f(x) dx \right|.$$

The law of the iterated logarithm (see Section 1.3.4) implies

$$e_n \leq \sqrt{\mathrm{Var}\left(H(X_1)\right)} \cdot \sqrt{\frac{2\log(\log n)}{n}},$$

which holds approximately (rather than exactly). We estimate the population variance $\mathrm{Var}\left(H(X_1)\right)$ by the following sample variance

$$\widehat{\sigma}_n^2 := \frac{1}{n-1} \sum_{i=1}^{n} \left(H(X_i) - \widehat{v}_n\right)^2.$$

Then, $\sqrt{\mathrm{Var}\left(H(X_1)\right)}$ is estimated by the standard deviation

$$\sqrt{\widehat{\sigma}_n^2}.$$

Therefore, in applications, we use the following as an approximate upper bound for the approximation error when computing $\widehat{v}_n$

$$e_n \leq \sqrt{\widehat{\sigma}_n^2} \cdot \sqrt{\frac{2\log(\log n)}{n}}.$$

2.2.3 Asymptotic Confidence Intervals*

A complementary way to quantify finite-sample uncertainty is through an asymptotic confidence interval. Because $X_1, \ldots, X_n$ are random, the estimator $\widehat{v}_n$ in Eq. (2.2.2) varies from one run to another. Writing the dependence on the outcome ω explicitly,

$$\widehat{v}_n(\omega) = \frac{1}{n} \sum_{i=1}^{n} H\left(X_i(\omega)\right).$$

The randomness of $\widehat{v}_n(\omega)$ creates uncertainty in the integral estimate, which may be quantified through the variance of $\widehat{v}_n(\omega)$. For this subsection, assume $\mathbb{E}[H(X_1)^2] < \infty$ and let $\sigma^2 := \mathbb{V}[H(X_1)] > 0$. The finite-second-moment assumption ensures that the CLT applies and that the sample variance below is consistent. If $\sigma^2 = 0$, then $H(X_1) = v$ almost surely, so $\widehat{v}_n = v$ almost surely and the confidence interval degenerates to $[\widehat{v}_n, \widehat{v}_n]$. The remainder of this discussion concerns the nondegenerate case $\sigma^2 > 0$. We have

$$\mathbb{V}(\widehat{v}_n) = \mathbb{V}\left[\frac{1}{n}\sum_{i=1}^{n} H\left(X_i\right)\right] = \frac{1}{n^2}\cdot\mathbb{V}\left[\sum_{i=1}^{n} H\left(X_i\right)\right] = \frac{1}{n^2}\cdot\sum_{i=1}^{n}\mathbb{V}\left[H\left(X_i\right)\right] = \frac{\sigma^2}{n} \to 0, \quad \text{as } n\to\infty;$$

Moreover, the CLT (Theorem 1.3.3) implies

$$\sqrt{n}\cdot\frac{\widehat{v}_n - v}{\sigma} = \sqrt{n}\cdot\frac{\frac{1}{n}\sum_{i=1}^{n} H\left(X_i\right) - v}{\sigma} \xrightarrow{d} N(0,1), \quad \text{as } n\to\infty.$$

This asymptotic normality leads to confidence intervals. For any $0 < \alpha < 1$, we have

$$1-\alpha \approx \mathbb{P}\left(\left|\sqrt{n}\cdot\frac{\widehat{v}_n - v}{\sigma}\right| \le z_{1-\alpha/2}\right) = \mathbb{P}\left(\widehat{v}_n - z_{1-\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} \le v \le \widehat{v}_n + z_{1-\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right),$$

where Φ is the CDF of $N(0,1)$ (see Eq. (1.2.7)), and $z_{1-\alpha/2}$ is the $1-\alpha/2$ quantile of $N(0,1)$, i.e., $\Phi(z_{1-\alpha/2}) = 1-\alpha/2$; for example, $z_{0.975} \approx 1.96$ ($\alpha = 0.05$). Therefore, the asymptotic $1-\alpha$ confidence interval of integral $v = \int_{-\infty}^{\infty} H(x)\cdot f(x)dx$ is the following

$$\left(\widehat{v}_n - z_{1-\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\,,\, \widehat{v}_n + z_{1-\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right).$$

Since the true variance $\sigma^2 = \mathbb{V}[H\left(X_1\right)]$ is unknown, we may estimate σ using the following sample variance

$$\widehat{\sigma}_n^2 := \frac{1}{n-1}\sum_{i=1}^{n}\left(H(X_i) - \widehat{v}_n\right)^2;$$

the LLN implies that $\widehat{\sigma}_n^2 \approx \sigma^2$ when n is sufficiently large. Thus, an asymptotic $1-\alpha$ confidence interval for v is

$$\left(\widehat{v}_n - z_{1-\alpha/2}\cdot\frac{\widehat{\sigma}}{\sqrt{n}}\,,\, \widehat{v}_n + z_{1-\alpha/2}\cdot\frac{\widehat{\sigma}}{\sqrt{n}}\right). \tag{2.2.3}$$

The following algorithm summarizes the estimator in Eq. (2.2.2) and the confidence interval in Eq. (2.2.3).

Algorithm 8 : Monte Carlo Integration

Input: (i) function $H(x)$; (ii) PDF $f(x)$; (iii) sample size n; (iv) asymptotic significance level $0 < \alpha < 1$ (equivalently, asymptotic confidence level $1 - \alpha$).

Output: (i) an estimator $\widehat{v}_n$ of the integral $v = \int_{-\infty}^{\infty} H(x) \cdot f(x)dx$; (ii) the $1 - \alpha$ asymptotic confidence interval of v.

1: Generate $X_1, \ldots, X_n \overset{iid}{\sim} f$.

2: $\widehat{v}_n \leftarrow \frac{1}{n}\sum_{i=1}^{n} H(X_i)$.

3: $\widehat{\sigma}_n^2 \leftarrow \frac{1}{n-1}\sum_{i=1}^{n} \left(H(X_i) - \widehat{v}_n\right)^2$ and $\widehat{\sigma}_n \leftarrow \sqrt{\widehat{\sigma}_n^2}$.

4: Return the estimator $\widehat{v}_n$ and the confidence interval $\left(\widehat{v}_n - z_{1-\alpha/2} \cdot \frac{\widehat{\sigma}}{\sqrt{n}}\,,\, \widehat{v}_n + z_{1-\alpha/2} \cdot \frac{\widehat{\sigma}}{\sqrt{n}}\right)$.

Algorithm 8 also applies to random vectors. Specifically, if $\boldsymbol{X} = (X^{(1)}, \ldots, X^{(d)})$ is an $\mathbb{R}^d$-valued random variable and $H : \mathbb{R}^d \to \mathbb{R}$, then $H(\boldsymbol{X})$ is the real-valued random variable defined by $H(\boldsymbol{X})(\omega) = H(\boldsymbol{X}(\omega))$. Thus, the same sample-average estimator applies when $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ are iid random vectors from a multivariate PDF.

2.3 Monte Carlo Integration versus Riemann Sums

"*More than cleverness, we need kindness and gentleness.*"

— a sentence in the final speech of the movie *The Great Dictator*

Monte Carlo is not the only numerical approach to integration. To clarify when it is useful, we first compare it with the deterministic Riemann-sum method. Recall that if a function $H : [0, 1] \to \mathbb{R}$ is Riemann integrable, then

$$\text{Riemann integral } \int_0^1 H(x)dx = \lim_{n\to\infty} \left\{ \frac{1}{n} \sum_{i=1}^{n} H\left(\frac{i}{n}\right) \right\}.$$

One may ask why we do not apply the following approximation to estimate integrals:

$$\int_0^1 H(x)dx \approx \frac{1}{n} \sum_{i=1}^{n} H\left(\frac{i}{n}\right). \tag{2.3.1}$$

The Riemann sum approximation above is deterministic and avoids the need to generate random numbers. Table 2.2 compares it with Monte Carlo integration. For these selected low-dimensional examples, the Riemann sum estimates are highly accurate. Why, then, use Monte Carlo integration? Its principal advantage emerges in high dimensions.

Table 2.2: Performance of the Riemann sum approximation in Eq. (2.3.1) with n = 100,000 grid points. The `R` code used to produce the table appears below.

$H(x)$	True values	$\frac{1}{n}\sum_{i=1}^{n} H\left(\frac{i}{n}\right)$
x	$1/2 = 0.5000$	0.5000
x^5	$1/6 \approx 0.1667$	0.1667
$\sin(x) \cdot \log(1 + x)$	≈ 0.2265 (via `WolframAlpha`)	0.2265

```r
par(mfrow=c(1,3))
n=100000
grids=1:n/n
print( round(mean(grids), 4) )
print( round(mean(grids^5), 4) )
print( round(mean(sin(grids)*log(1+grids)), 4) )
```

Consider the following high-dimensional integral:

$$v = \int_0^1 \cdots \int_0^1 \left(\sum_{i=1}^{d} i \cdot x_i \right)^2 dx_1 \cdots dx_d = \frac{1}{16} d^4 + \frac{11}{72} d^3 + \frac{5}{48} d^2 + \frac{1}{72} d.$$

For $d = 100$, $v = 6403820.833333333954215049743652343 75$. Let $H(x_1, \ldots, x_d) = (\sum_{i=1}^{d} i \cdot x_i)^2$. Then

$$v = \mathbb{E}[H(\boldsymbol{X})], \quad \text{where } \boldsymbol{X} = (X_1, \ldots, X_d)^T \sim \text{Unif}([0,1]^d),$$
$$\text{equivalently, } X_1, \ldots, X_d \overset{iid}{\sim} \text{Unif}(0,1).$$

Applying Algorithm 8 with sample size $n = 10^6$ and confidence level 95% ($\alpha = 0.05$) gives

$$(6402597.78 \pm 1661.41),$$

which contains the true value of v. The `R` code used to compute this interval is provided below.

```r
d=100
H=function(x){ return( (sum((1:d)*x))^2 )}

n=10^(6)
U=runif(n*d)
U_matrix=matrix(U, ncol = n, nrow = d)
v_n=mean(apply(U_matrix, 2, H))
semi_length_CI=1.96*sd(apply(U_matrix, 2, H))/sqrt(n)
print(paste("( ", as.character(v_n), " + or - ", as.character(semi_length_CI), " )"))
```

Now consider approximating v by a Riemann sum. Dividing each interval $[0, 1]$ into only 10 pieces already produces $10^d = 10^{100}$ grid points when $d = 100$—a googol.‡ Storing, much less evaluating, so many grid points is infeasible. Thus, in this example, the Riemann sum approximation is computationally impractical.

‡ The number of grains of sand required to fill the known universe is estimated to be approximately 10^{63}, far smaller than 10^{100}.

2.4 Importance Sampling

"*Bitterness for one, and also loneliness.....These great artists, these geniuses, can also suffer. Finding inspiration in the bitterness of coffee is really interesting.*"

— some words I heard somewhere

The preceding examples used uniform sampling, with f equal to the density of $\text{Unif}(0,1)$ or, in several dimensions, $\text{Unif}([0,1]^d)$. Uniform sampling is natural when the integrand is important throughout the domain, but it can be extremely inefficient when most of the integral is concentrated in a small region. The following example makes this limitation precise and motivates importance sampling.

2.4.1 Curse of Dimensionality: A Motivating Example

The following example illustrates a limitation of uniform sampling. Consider the integral below. Although its domain is $[-1,1]^d$ rather than $[0,1]^d$, the method is unchanged:

$$\begin{aligned} v_d &= \int_{-1}^{1} \cdots \int_{-1}^{1} \mathbb{1}\{x_1^2 + \cdots + x_d^2 < 1\} dx_1 \cdots dx_d \\ &= \text{the volume of the unit ball in } d\text{-dimensional space in } \mathbb{R}^d. \\ &= \begin{cases} \dfrac{\pi^{d/2}}{(d/2)!}, & \text{if } d \text{ is even,} \\ \dfrac{2^d(\sqrt{\pi})^{d-1}}{d!} \cdot \left(\dfrac{d-1}{2}\right)!, & \text{if } d \text{ is odd.} \end{cases} \end{aligned} \tag{2.4.1}$$

Set $H(x_1, \cdots, x_d) = 2^d \cdot \mathbb{1}\{x_1^2 + \cdots + x_d^2 < 1\}$ and let $f(x_1, \cdots, x_d) = 2^{-d}\mathbb{1}_{[-1,1]^d}$ be the PDF of $\text{Unif}([-1,1]^d)$. Then $v_d = \mathbb{E}[H(U_1, \cdots, U_d)]$, where $(U_1, \ldots, U_d) \sim \text{Unif}([-1,1]^d)$. We apply Algorithm 8 for several dimensions. Figure 2.1 illustrates two limitations of uniform sampling.

- **Failure 1**: The estimator can have an extremely large relative variance. Let $I_d := \mathbb{1}\{U_1^2 + \cdots + U_d^2 < 1\}$, where $U_1, \ldots, U_d \overset{iid}{\sim} \text{Unif}([-1,1])$. Since $\mathbb{P}(I_d = 1) = v_d/2^d$, the absolute variance of one Monte Carlo summand is

$$\sigma_d^2 := \mathbb{V}(2^d I_d) = 2^d v_d - v_d^2.$$

  This absolute variance does not increase monotonically with d and eventually decreases. In contrast, its variance relative to the squared target is

$$\frac{\sigma_d^2}{v_d^2} = \frac{2^d}{v_d} - 1 \approx \frac{2^d}{v_d},$$

which becomes enormous as d increases. Consequently, the relative variance of the sample-mean estimator $\widehat{v}_{d,n}$ is $\frac{1}{n}(\frac{2^d}{v_d} - 1)$. Increasing the sample size reduces its standard error only at rate $O(1/\sqrt{n})$; reducing the relative standard error by a factor of 10 requires increasing n by a factor of 100.

- **Failure 2**: When d is large, the estimates may be exactly zero. This failure is more severe than the first. Indeed, $v_d \to 0$ as $d \to \infty$, whereas the volume of $[-1, 1]^d$ is 2^d and grows exponentially. Thus, the unit ball occupies a negligible fraction of the sampling region in high dimensions. As d increases, fewer sampled points fall inside the unit ball, so $\mathbb{1}\{U_1^2 + \cdots + U_d^2 < 1\}$ equals zero for nearly all observed points.

Both failures—large relative variance and the negligible fraction of the cube occupied by the unit ball—reflect the curse of dimensionality. They motivate importance sampling, which places more probability in important regions and can also reduce variance.

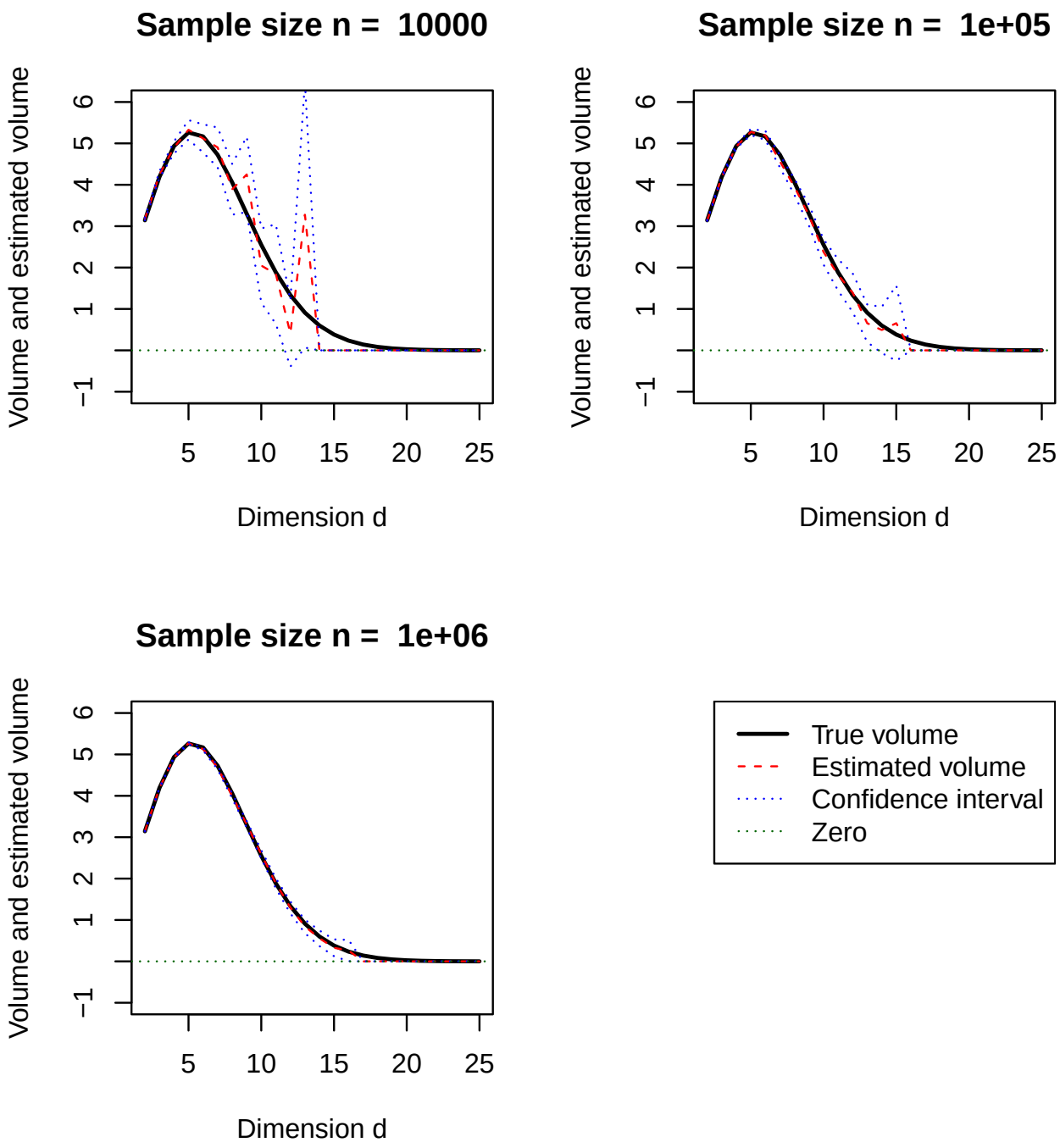


Figure 2.1: Results from Algorithm 8 with sample sizes $n \in \{10^4, 10^5, 10^6\}$. For each $d \in \{2, 3, \ldots, 25\}$, the solid black curve gives the true volume v_d, the dashed red curve gives the estimate $\widehat{v}_{d,n}$, and the dotted blue curves give the 95% confidence interval. The `R` code used to produce the figure is provided below.

```r
vol=function(d){
  if(d%%2==0){
    return( pi^(d/2)/(factorial(d/2)) )
  }else{
    return( (2^d)*((sqrt(pi))^(d-1))*factorial((d-1)/2)/factorial(d) )
  }
}
```

```r
# vol(d) = the volume of the unit ball in d-dimensional space.


par(mfrow=c(2,2))
for (p in 4:6) {

  n=10^p
  v_d_hat=vector()
  semi_length_CI=vector()
  v_d=vector()
  for (d in 2:25) {
    print(d)
    H=function(x){ return( (2^d)*as.numeric(sum(x^2)<1) ) }
    U=runif(n*d, min=-1, max=1)
    U_matrix=matrix(U, ncol = n, nrow = d)
    H_i=apply(U_matrix, 2, H)
    v_d_hat[d-1]=mean(H_i)
    semi_length_CI[d-1]=1.96*sd(H_i)/sqrt(n)
    v_d[d-1]=vol(d)
  }

  plot(2:25, v_d, type = "l", xlab = "Dimension d", ylab = "Volume and estimated volume",
       lwd=2, main = paste("Sample size n = ", as.character(10^p)), ylim = c(-1, 6))
  lines(2:25, v_d_hat, col="red", lty=2)
  lines(2:25, v_d_hat-semi_length_CI, col="blue", lty=3)
  lines(2:25, v_d_hat+semi_length_CI, col="blue", lty=3)
  abline(h=0, col="darkgreen", lty=3)
  if(p==6){
    plot.new()
    legend("topright", c("True volume", "Estimated volume", "Confidence interval", "Zero"),
           lty = c(1,2,3,3), col = c("black", "red", "blue", "darkgreen"), lwd = c(2,1,1,1))
  }

}
```

2.4.2 Algorithm

The preceding examples identify two related goals: place more sample points in important regions and reduce the estimator's variance. Importance sampling addresses both by introducing a proposal density f and writing

$$\int H(\boldsymbol{x})d\boldsymbol{x} = \int \frac{H(\boldsymbol{x})}{f(\boldsymbol{x})} \cdot f(\boldsymbol{x})d\boldsymbol{x},$$

where $f(\boldsymbol{x})$ is the PDF of a d-dimensional random vector. This identity requires $H(\boldsymbol{x}) = 0$ almost everywhere on $\{\boldsymbol{x} : f(\boldsymbol{x}) = 0\}$; we define $H(\boldsymbol{x})/f(\boldsymbol{x}) = 0$ when $f(\boldsymbol{x}) = 0$. **The art of importance sampling is the game of choosing the PDF** $f(\boldsymbol{x})$. Let random vectors $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \overset{iid}{\sim} f(\boldsymbol{x})$. If

$$\int_{\{f>0\}} \frac{|H(\boldsymbol{x})|^2}{f(\boldsymbol{x})} d\boldsymbol{x} < \infty,$$

then the importance-sampling summand has a finite second moment (and hence a finite first moment). Define the importance weights, their sample mean, and their sample variance by

$$W_i := \frac{H(\boldsymbol{X}_i)}{f(\boldsymbol{X}_i)}, \qquad \widehat{v}_{d,n} := \frac{1}{n}\sum_{i=1}^{n} W_i, \qquad \widehat{\Sigma}_n^2 := \frac{1}{n-1}\sum_{i=1}^{n} (W_i - \widehat{v}_{d,n})^2 . \tag{2.4.2}$$

Let

$$\Sigma_d^2 := \mathbb{V}(W_1) = \mathbb{V}\left[\frac{H(\boldsymbol{X}_1)}{f(\boldsymbol{X}_1)}\right].$$

If $0 < \Sigma_d^2 < \infty$, the strong law of large numbers and the Hartman–Wintner law of the iterated logarithm give

$$\widehat{v}_{d,n} \longrightarrow v_d := \int H(\boldsymbol{x}) d\boldsymbol{x} \quad \text{almost surely,}$$
$$\limsup_{n\to\infty} \sqrt{\frac{n}{2\log\log n}}\, |\widehat{v}_{d,n} - v_d| = \Sigma_d \quad \text{almost surely.}$$

The second statement describes an asymptotic fluctuation scale, not a finite-sample bound. The central limit theorem, together with the consistency of $\widehat{\Sigma}_n$, yields the asymptotic $(1-\alpha)$ confidence interval

$$\left(\widehat{v}_{d,n} - z_{1-\alpha/2}\frac{\widehat{\Sigma}_n}{\sqrt{n}},\ \widehat{v}_{d,n} + z_{1-\alpha/2}\frac{\widehat{\Sigma}_n}{\sqrt{n}}\right).$$

To make $\widehat{v}_{d,n}$ a good approximation to the integral $\int H(\boldsymbol{x})d\boldsymbol{x}$, we need to choose a PDF $f(\boldsymbol{x})$ satisfying the following conditions.

i) First, the choice of $f(\boldsymbol{x})$ should make $\mathbb{V}(W_1)$ as small as possible. If $H \geq 0$ and $0 < v_d := \int H(\boldsymbol{x})d\boldsymbol{x} < \infty$, the formal zero-variance density is $f^*(\boldsymbol{x}) = H(\boldsymbol{x})/v_d$, for which $W_1 = v_d$ almost surely. It is unavailable in practice because its normalizing constant v_d is exactly the integral being estimated. More generally, among admissible densities, a density proportional to $|H|$ minimizes the second moment of the importance weights; zero variance occurs only when H does not change sign.

ii) Secondly, $H(\boldsymbol{x})$ must equal zero almost everywhere on $\{\boldsymbol{x} : f(\boldsymbol{x}) = 0\}$. An isolated point at which $H(\boldsymbol{x}) \neq 0$

and $f(\boldsymbol{x}) = 0$ does not affect the integral. In addition, finite variance requires

$$\int_{\{f>0\}} \frac{|H(\boldsymbol{x})|^2}{f(\boldsymbol{x})} d\boldsymbol{x} < \infty.$$

iii) Finally, $f(\boldsymbol{x})$ should place substantial probability where $|H(\boldsymbol{x})|$ is large. Merely covering the support of H is necessary for unbiasedness but does not by itself guarantee a low-variance estimator.

Remark 2.4.1. *Requirements (i) and (iii) are theoretically aligned: placing sufficient probability where $|H|$ is large helps reduce the second moment of the importance weights, and the second-moment-optimal proposal is proportional to $|H|$. Nevertheless, a finite-sample variance estimate can be highly misleading when the sample misses a rare but important region. For example, in the first panel of Figure 2.1, with $n = 10000$, none of the sampled points enters the unit ball. Consequently, all observed importance weights and their sample variance are zero, even though the estimator misses the positive integral and the proposal does not have zero true variance. Thus, an empirical variance of zero in this setting reflects the absence of observed rare events, not an optimal proposal.*

Finding the optimal $f(\boldsymbol{x})$ among **all** possible PDFs is generally infeasible in practice, and a detailed study of proposal design is beyond the scope of this course. The following example shows how a suitable choice can nevertheless improve the finite-sample performance of Algorithm 8.

Suppose we are still interested in the integral $\int_{-1}^{1} \cdots \int_{-1}^{1} \mathbb{1}\{x_1^2 + \cdots + x_d^2 < 1\} dx_1 \cdots dx_d$ in Eq. (2.4.1). A simple volume ratio explains why the boundary is important in high dimensions. For any fixed $\epsilon \in (0, 1)$, the fraction of the unit-ball volume lying in the outer shell $1 - \epsilon < \|\boldsymbol{x}\| < 1$ is

$$\frac{v_d - (1-\epsilon)^d v_d}{v_d} = 1 - (1-\epsilon)^d \longrightarrow 1 \qquad \text{as } d \to \infty.$$

Thus, most of the volume lies near the unit sphere when d is large.

In the first panel of Figure 2.1, the uniform-distribution method fails when $d = 15$ and $n = 10000$. We instead set $H(\boldsymbol{x}) = \mathbb{1}\{\|\boldsymbol{x}\| < 1\}$ and choose a proposal from the following family of centered multivariate normal densities:

$$f_\sigma(\boldsymbol{x}) := (2\pi\sigma^2)^{-\frac{d}{2}} \exp\left\{-\frac{x_1^2 + \cdots + x_d^2}{2\sigma^2}\right\}, \quad \text{for } \sigma > 0. \tag{2.4.3}$$

If $\boldsymbol{X} \sim f_\sigma$, then its coordinates are independent $N(0, \sigma^2)$ random variables, and hence

$$\mathbb{E}\|\boldsymbol{X}\|^2 = \sum_{j=1}^{d} \mathbb{E}\left(X^{(j)}\right)^2 = d\sigma^2.$$

Matching this expected squared radius to the unit sphere gives

$$\sigma^* = \frac{1}{\sqrt{d}}. \tag{2.4.4}$$

For $d = 15$, this is $\sigma^* = 1/\sqrt{15} \approx 0.2582$, the value used below. Radial matching is a useful heuristic rather than a claim of optimality: it places proposal mass near the region where the high-dimensional unit ball has most of its volume.

We generate $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \overset{iid}{\sim} f$ = the PDF defined in Eq. (2.4.3) with $\sigma = 0.2582$, where $n = 10000$. Then, we compute the approximation $\widehat{v}_{d,n} = \frac{1}{n}\sum_{i=1}^n \frac{H(\boldsymbol{X}_i)}{f(\boldsymbol{X}_i)}$, the 95% asymptotic confidence interval $(\widehat{v}_{d,n} \pm 1.96 \cdot \frac{\widehat{\Sigma}_n}{\sqrt{n}})$ using the importance-weight variance estimate in Eq. (2.4.2), and the LIL fluctuation scale $\left(v_d \pm \widehat{\Sigma}_n \cdot \sqrt{\frac{2\log(\log n)}{n}}\right)$. The estimation results of this example are presented in Figure 2.2. Please compare this figure with the first panel of Figure 2.1. The importance sampling method—specifically, choosing $f = f_{0.2582}$ instead of a uniform distribution—performs much better than applying Algorithm 8 with sample points from uniform distributions.

We conclude the importance sampling method in the following algorithm, which is essentially a special case of Algorithm 8

Algorithm 9 : Importance Sampling Integration

Input: (i) integrand $H(\boldsymbol{x})$; (ii) a properly chosen PDF $f(\boldsymbol{x})$; (iii) sample size n; (iv) asymptotic significance level α (equivalently, asymptotic confidence level $1-\alpha$).

Output: (i) an estimator $\widehat{v}_n$ of the integral $v = \int_{\mathbb{R}^d} H(\boldsymbol{x})d\boldsymbol{x}$; (ii) the $1-\alpha$ asymptotic confidence interval of v.

1: Generate $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \overset{iid}{\sim} f(\boldsymbol{x})$ and set $W_i = H(\boldsymbol{X}_i)/f(\boldsymbol{X}_i)$.

2: Compute $\widehat{v}_n = \frac{1}{n}\sum_{i=1}^n W_i$ and $\widehat{\Sigma}_n^2 = \frac{1}{n-1}\sum_{i=1}^n (W_i - \widehat{v}_n)^2$.

3: Return $\widehat{v}_n$ and the $1-\alpha$ asymptotic confidence interval $(\widehat{v}_n \pm z_{1-\alpha/2} \cdot \frac{\widehat{\Sigma}_n}{\sqrt{n}})$.

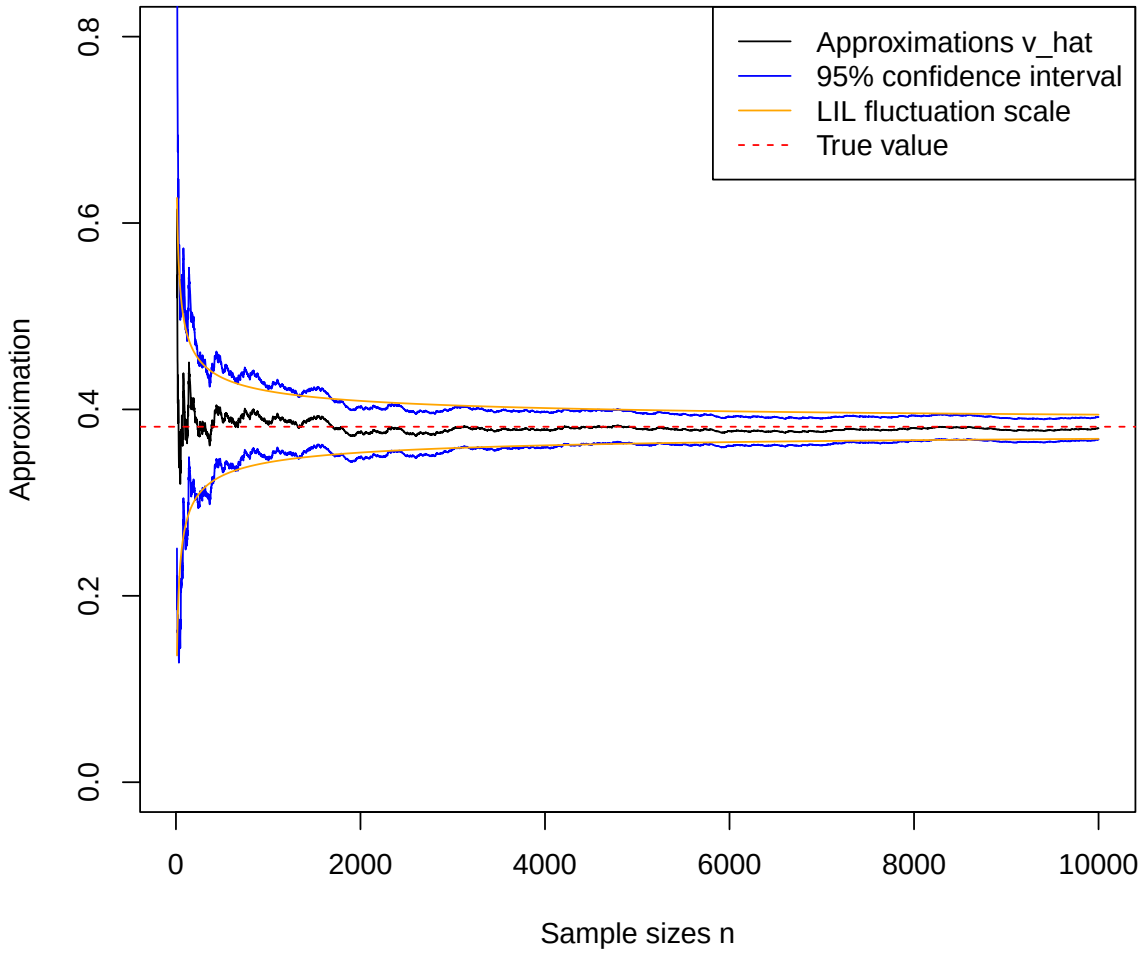


Figure 2.2: Approximation of integral $\int_{-1}^{1}\cdots\int_{-1}^{1}\mathbb{1}\{x_1^2+\cdots+x_d^2<1\}dx_1\cdots dx_d$ using importance sampling, where $d=15$. The approximations $\widehat{v}_{d,n}$, for $n\in\{11,12,\cdots,10000\}$, are presented by the black solid curve, the true value v_d is indicated by the red dashed line, the 95% asymptotic confidence intervals are presented by the blue solid curves, and the LIL fluctuation scale is indicated by the orange solid curves.

```
library("mvtnorm")
set.seed(1690)  # Make the numerical illustration reproducible.
# We need the function "rmvnorm" in package "mvtnorm" to generate
# random vectors from multivariate normal distributions.

d=15
true_vol=(2^d)*((sqrt(pi))^(d-1))*factorial((d-1)/2)/factorial(d)
n=10000
H=function(x){ return( as.numeric(sum(x^2)<1) ) }

sig=0.2582  # A good choice of the parameter sigma.
```

```
X_matrix=rmvnorm(n, mean = rep(0,d), sigma = diag(rep(sig^2, d)))
H_i=apply(X_matrix, 1, H)
f=function(x){ return(dmvnorm(x, mean = rep(0,d), sigma = diag(rep(sig^2, d)))) }
f_i=apply(X_matrix, 1, f)
mean(H_i/f_i)

true_sigma_est=sd(H_i/f_i)

v_hat=vector()
semi_CI_width=vector()
Iterated_logarithm_rate=vector()
for (m in 11:n) {
  v_hat[m-10]=mean(H_i[1:m]/f_i[1:m])
  semi_CI_width[m-10]=1.96*true_sigma_est/sqrt(m)
  Iterated_logarithm_rate[m-10]=true_sigma_est*sqrt(2*log(log(m))/m)
}

plot(11:n, v_hat, type = "l", ylim = c(0, 0.8), xlab = "Sample sizes n", ylab = "Approximation")
abline(h=true_vol, col="red", lty=2)
lines(11:n, v_hat+semi_CI_width, col="blue")
lines(11:n, v_hat-semi_CI_width, col="blue")
lines(11:n, true_vol+Iterated_logarithm_rate, col="orange")
lines(11:n, true_vol-Iterated_logarithm_rate, col="orange")
legend("topright", col = c("black", "blue", "orange", "red"), lty = c(1, 1, 1, 2),
       c("Approximations v_hat", "95% confidence interval", "LIL fluctuation scale", "True value"
    ))
```

2.5 Homework for Chapter 2

Problem 2.5.1. *Let $X_1, \dots, X_n$ be generated from the normal distribution $N(\theta, \sigma^2)$. Suppose the prior of θ is the PDF of normal distribution $N(\mu, \tau^2)$, where σ^2, μ, and τ^2 are known. Derive the Bayes estimator $\mathbb{E}(\theta | X_1 = x_1, \dots, X_n = x_n)$; see Eq. (2.1.1). Express the estimator in terms of σ^2, μ, τ^2, and the data.*

Problem 2.5.2. *i) Let v_d denote the volume of the d-dimensional unit ball $\{\boldsymbol{x} \in \mathbb{R}^d : \|\boldsymbol{x}\| \le 1\}$. For a fixed $\epsilon \in (0,1)$, compute the fraction of its volume contained in the shell $1 - \epsilon < \|\boldsymbol{x}\| \le 1$ and explain why most of the volume concentrates near the unit sphere as $d \to \infty$.*

ii) Let f_σ be the d-dimensional normal density in Eq. (2.4.3). Show that $\mathbb{E}\|\boldsymbol{X}\|^2 = d\sigma^2$ when $\boldsymbol{X} \sim f_\sigma$, and hence derive the radial-matching choice $\sigma^* = 1/\sqrt{d}$ in Eq. (2.4.4).

iii) Set $d = 100$, $\sigma^* = 1/10$, and $n = 100000$. Use Algorithm 9 with proposal f_{σ^*} to estimate v_{100}, the volume of the 100-dimensional unit ball. Report the estimator, its estimated standard error, and your code. Compare the

estimate with the exact value $v_{100} = \pi^{50}/50!$. *To avoid numerical underflow or overflow, you may compute the normal density and importance weights on the logarithmic scale.*

Problem 2.5.3. *Use either Algorithm 8 or Algorithm 9 to approximately compute the following integrals—report the estimators and the corresponding 95% asymptotic confidence intervals. Increase the sample sizes* n *and choose appropriate proposal densities* f *in Algorithm 9, justifying your choices, to make your approximations as accurate as possible.*

i) $\int_0^1 |\cos(\pi x)|^{2.5} \cdot e^{-10x} dx$,

ii) $\int_0^1 \int_0^1 |\cos(x_1 \cdot x_2)| \cdot e^{-x_1^2} dx_1 dx_2$.

iii) $\int_0^\infty \int_0^2 \int_0^\infty \{\cos(x_1 \cdot x_2)\} \cdot e^{-x_1 - x_3^2(x_1+x_2)} dx_1 dx_2 dx_3$.

The integral in part (iii) is absolutely convergent. Indeed, after bounding $|\cos(x_1 x_2)|$ *by* 1 *and integrating with respect to* x_3, *its absolute integral is at most*

$$\frac{\sqrt{\pi}}{2} \int_0^\infty e^{-x_1} \int_0^2 \frac{1}{\sqrt{x_1 + x_2}} dx_2 dx_1 \le \sqrt{2\pi} < \infty.$$

Chapter 3

Markov Chains

"*Given the present, the future does not depend on the past.*"

— a heuristic description of Markov property

Chapters 1 and 2 relied primarily on independent and identically distributed samples. Chapters 3 and 4 replace independence with a structured form of dependence: the Markov property. An iid sequence is itself a Markov chain, whereas a stationary Markov chain is identically distributed over time but is generally dependent.

This chapter develops the definitions and asymptotic results needed for Markov chain Monte Carlo (MCMC) in Chapter 4. To see why this theory is needed, suppose that we want to compute an integral represented as

$$\mathbb{E}[H(\boldsymbol{X})], \quad \text{where } \boldsymbol{X} \sim \text{ a distribution } \pi.$$

Chapter 2 approximated such an expectation using iid draws $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \sim \pi$. Direct iid sampling can be difficult, however, especially when π is high-dimensional; in the Ising model of Section 4.1, for example, the dimension can be several thousand or more. MCMC instead constructs a Markov chain $\{\boldsymbol{X}_n\}_{n=0}^{\infty}$ for which

$$\lim_{n\to\infty} \left[\frac{1}{n} \sum_{i=1}^{n} H(\boldsymbol{X}_i) \right] = \mathbb{E}[H(\boldsymbol{X})] \quad \text{with probability one.}$$

This limit resembles the LLN, but the iid version of that theorem does not apply directly because MCMC samples are dependent. The remainder of this chapter therefore develops the Markov-chain machinery, especially asymptotic behavior, that justifies the approximation.

3.1 Transition Probabilities and the Markov Property

"*And yet don't look too good, nor talk too wise.*"

— Rudyard Kipling

3.1.1 Definition of Markov Chains

Roughly speaking, a Markov chain is a sequence of random variables $\{X_n\}_{n=0}^{\infty}$ satisfying the Markov property.† Throughout Chapters 3 and 4, we assume that all random variables take values in a generic countable set $\mathcal{X}$, i.e., $\mathcal{X}$ can be expressed as $\mathcal{X} = \{x_0, x_1, \ldots, x_n, \ldots\}$. For example, $\mathcal{X} = \mathbb{N} = \{0, 1, 2, \ldots\}$. The countability assumption on $\mathcal{X}$ substantially simplifies the theory of Markov chains. The following is the definition‡ of Markov chains.

Definition 3.1.1. *i) A sequence of random variables $\{X_n\}_{n=0}^{\infty}$ taking values in $\mathcal{X}$ is called a **Markov chain** if this sequence satisfies*

$$\mathbb{P}(X_{n+1} = y \,|\, X_n = x, X_{n-1} = \xi_{n-1}, \ldots, X_0 = \xi_0) = \mathbb{P}(X_{n+1} = y \,|\, X_n = x) \tag{3.1.1}$$

*for all $n = 0, 1, \ldots$, all $y \in \mathcal{X}$, and all the $\xi_0, \xi_1, \ldots, \xi_{n-1}, x \in \mathcal{X}$ such that $\mathbb{P}(X_n = x, X_{n-1} = \xi_{n-1}, \ldots, X_0 = \xi_0) > 0$. The property in Eq. (3.1.1) is called the **Markov property**.*

ii) Furthermore, if there exists a bivariate function $p : \mathcal{X} \times \mathcal{X} \to [0, 1]$ such that

$$p(x, y) = \mathbb{P}(X_{n+1} = y \,|\, X_n = x) \quad \textit{(this function does not depend on } n\textit{)},$$

*for all $n = 0, 1, \ldots$, then $\{X_n\}_{n=0}^{\infty}$ is called a **homogeneous** Markov chain, and the function $p(\cdot, \cdot)$ is called the **transition probability** of this Markov chain.*

*iii) In the literature, $\mathcal{X}$ is usually referred to as the **state space** of the Markov chain $\{X_n\}_{n=0}^{\infty}$; each element in $\mathcal{X}$ is referred to as a **state**.*

Throughout this course, all the Markov chains will be homogeneous. Hence, **we will omit the word "homogeneous" hereafter**. In Eq. (3.1.1), if we call X_n "the present," X_{n+1} "the future," and $X_{n-1}, \ldots, X_0$ "the past," the Markov property means, "given the present, the future does not depend on the past"—the condition $X_{n-1} = x_{n-1}, \ldots, X_0 = x_0$ in Eq. (3.1.1) does not play any role!

The value $p(x, y)$ is the probability of moving from x to y and is therefore called a transition probability. For each fixed $x \in \mathcal{X}$, the function $p(x, \cdot) : y \mapsto p(x, y)$ is a PMF. Together with the distribution of the initial variable X_0, the transition function p determines the distribution of the entire Markov chain $\{X_n\}_{n=0}^{\infty}$, as formalized in the following theorem.

† For Markov chains in this course, we always let the index n go from 0 instead of 1. This is just a convention.

‡ The definition herein works only for the scenario where a Markov chain takes values in a countable space $\mathcal{X} = \{x_0, x_1, \ldots, x_n, \ldots\}$. This is one reason we assume $\mathcal{X}$ is countable. For the general definition of Markov chains and the relevant details, see Definition 17.1 and Remark 17.2 of Klenke (2020). Since the material on general Markov chains requires substantial real analysis background (see APMA 2110), we skip general Markov chains in this course.

Theorem 3.1.1. *Suppose the PMF of the initial random variable X_0 is μ. Then, we have the following formula*

$$\mathbb{P}(X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) = \mu(x_0) \cdot p(x_0, x_1) \cdot p(x_1, x_2) \dots p(x_{n-1}, x_n), \quad \textit{for all } n = 1, 2, \dots.$$

The proof of Theorem 3.1.1 follows by applying the chain rule for joint probabilities and then using the Markov property in Eq. (3.1.1) to simplify each conditional factor. The details are left for homework.

At the end of this section, we investigate a special scenario of Markov chains—$\mathcal{X}$ is a finite set $\{x_1, x_2, \dots, x_S\}$. This scenario is important as the computation of probabilities becomes a linear algebra problem. Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in $\mathcal{X} = \{x_1, x_2, \dots, x_S\}$ and having transition probability p. We define the following **transition matrix**

$$\boldsymbol{P} := \begin{pmatrix} p(x_1, x_1) & p(x_1, x_2) & \cdots & p(x_1, x_S) \\ p(x_2, x_1) & p(x_2, x_2) & \cdots & p(x_2, x_S) \\ \vdots & \vdots & \ddots & \vdots \\ p(x_S, x_1) & p(x_S, x_2) & \cdots & p(x_S, x_S) \end{pmatrix}. \tag{3.1.2}$$

Because $\sum_{k=1}^{S} p(x_i, x_k) = 1$ for any fixed i, the sum of each row of $\boldsymbol{P}$ is equal to 1[‡]. We adopt the following notations

$$\boldsymbol{P}^n := \underbrace{\boldsymbol{P}\boldsymbol{P}\cdots\boldsymbol{P}}_{n \text{ times}}, \quad \text{for any positive integers } n,$$

$$(\boldsymbol{P}^n)_{ij} = \text{ the entry at the intersection of the } i\text{-th row and } j\text{-th column of matrix } \boldsymbol{P}^n.$$

Then, we have the following useful formula

$$\mathbb{P}(X_n = x_j | X_0 = x_i) = (\boldsymbol{P}^n)_{ij} \tag{3.1.3}$$

Proof. (of Eq. (3.1.3)) When $n = 1$, Eq. (3.1.3) is just the definition of transition probability p. When $n = 2$, the

‡ The matrix $\boldsymbol{P}$ defined in Eq. (3.1.2) satisfies the following properties

i) All the entries of $\boldsymbol{P}$ are nonnegative.

ii) The sum of each row is equal to one, i.e., $\sum_{j=1}^{S} \boldsymbol{P}_{ij} = 1$ for each i.

In the literature, any matrices satisfying these two properties are called stochastic matrices.

law of total probability implies

$$\begin{aligned}\mathbb{P}(X_2 = x_j | X_0 = x_i) &= \sum_{k=1}^{S} \mathbb{P}(X_2 = x_j | X_1 = x_k, X_0 = x_i) \cdot \mathbb{P}(X_1 = x_k | X_0 = x_i) \\ &= \sum_{k=1}^{S} \mathbb{P}(X_2 = x_j | X_1 = x_k) \cdot \mathbb{P}(X_1 = x_k | X_0 = x_i) \\ &= \sum_{k=1}^{S} \boldsymbol{P}_{kj} \cdot \boldsymbol{P}_{ik} \\ &= (\boldsymbol{P}^2)_{ij},\end{aligned}$$

where the second equality comes from the Markov property, and the last equality comes from the definition of matrix multiplication. Suppose the desired result is true for $n-1$, then we have

$$\begin{aligned}\mathbb{P}(X_n = x_j | X_0 = x_i) &= \sum_{k=1}^{S} \mathbb{P}(X_n = x_j | X_{n-1} = x_k, X_0 = x_i) \cdot \mathbb{P}(X_{n-1} = x_k | X_0 = x_i) \\ &= \sum_{k=1}^{S} \mathbb{P}(X_n = x_j | X_{n-1} = x_k) \cdot \mathbb{P}(X_{n-1} = x_k | X_0 = x_i) \\ &= \sum_{k=1}^{S} \boldsymbol{P}_{kj} \cdot (\boldsymbol{P}^{n-1})_{ik} \\ &= (\boldsymbol{P}^n)_{ij}.\end{aligned}$$

Eq. (3.1.3) follows from the mathematical induction. □

Suppose the PMF of the initial random variable X_0 is μ. We define the vector $\boldsymbol{\mu} := (\mu(x_1), \ldots, \mu(x_S))^\intercal$. Then, Eq. (3.1.3) implies the following

$$\begin{aligned}\mathbb{P}(X_n = x_i) &= \sum_{k=1}^{S} \mathbb{P}(X_n = x_i | X_0 = x_k) \mathbb{P}(X_0 = x_k) \\ &= \sum_{k=1}^{S} (\boldsymbol{P}^n)_{ki} \cdot \mu(x_k) \\ &= (\boldsymbol{\mu}^\intercal \boldsymbol{P}^n)_i \\ &= \text{ the } i\text{-th entry of the row vector } \boldsymbol{\mu}^\intercal \boldsymbol{P}^n,\end{aligned}$$

that is, $(\mathbb{P}(X_n = x_1), \ldots, \mathbb{P}(X_n = x_S)) = \boldsymbol{\mu}^\intercal \boldsymbol{P}^n$. We may interpret the formula as follows: the initial distribution $\boldsymbol{\mu}$ is updated by the transition matrix $\boldsymbol{P}$ through the matrix right-multiplication.

3.1.2 Construction of Markov Chains

The joint-distribution formula above explains what law a Markov chain must have. The following algorithm gives the corresponding sequential construction from an initial distribution μ and transition probability p.

Algorithm 10 : Generating Markov Chains (I)

Input: (i) initial distribution PMF μ, (ii) transition probability p, and (iii) desired number of transitions n^*.
Output: the n^*+1 states $X_0, \dots, X_{n^*}$ of a Markov chain with initial distribution μ and transition probability p.

1: Generate $X_0 \sim \mu$.
2: Generate X_1 from the PMF $p(X_0, \cdot)$.
3: **for all** $i = 1, 2, \dots, n^* - 1$ **do**
4: Generate X_{i+1} from the PMF $p(X_i, \cdot)$.
5: **end for**

Algorithm 10 assumes we know how to generate random variables from the PMF $p(X_i, \cdot)$ derived from transition probability p. However, this assumption might not be true, and generating random variables from a general PMF might not be easy—this is the reason we devote the entire Chapter 1 to generating random numbers. When we do not know how to generate random numbers from $p(X_i, \cdot)$ directly, we can still generate Markov chains from given transition probability and initial distribution—we still need to know how to generate the basic randomness $\mathrm{Unif}(0,1)$ (see Section 1.7). Consider the finite state space $\mathcal{X} = \{x_1, \dots, x_S\}$. The aim is to simulate a Markov chain $\{X_n\}_{n=0}^{\infty}$ with a given transition probability p. Suppose that the computer provides a (pseudo) random number generator for $\mathrm{Unif}(0,1)$. For each $i \in \{1, \dots, S\}$, define

$$r(i,0) := 0, \quad r(i,j) := \sum_{k=1}^{j} p(x_i, x_k), \quad \text{for all } j = 1, \dots, S.$$

Suppose we have generated a sequence $U_1, U_2, \dots \overset{iid}{\sim} \mathrm{Unif}(0,1)$. For each $i, j \in \{1, \dots, S\}$, define

$$I_{ij} := \begin{cases} [r(i, j-1), r(i,j)), & j = 1, \dots, S-1, \\ [r(i, S-1), 1], & j = S. \end{cases}$$

The sets $I_{i1}, \dots, I_{iS}$ form a partition of $[0,1]$, and the length of I_{ij} is $p(x_i, x_j)$. Define a sequence of random functions $\{R_n\}_{n=1}^{\infty}$ on $\mathcal{X}$ by

$$R_n(\omega; x_i) := x_j \quad \text{when} \quad U_n(\omega) \in I_{ij}, \qquad i, j \in \{1, \dots, S\}. \tag{3.1.4}$$

Here, ω is written explicitly to emphasize that the randomness of R_n comes from U_n. Because the variables $U_1, U_2, \dots$ are independent, the random functions $R_1, R_2, \dots$ are independent as well. Using the random functions

defined in Eq. (3.1.4), define a sequence of random variables $\{X_n\}_{n=0}^{\infty}$ through the following procedure:

$$\begin{aligned} &X_0 \sim \mu = \text{ a PMF on } \mathcal{X} = \{x_1, \dots, x_S\}, \\ &X_0 \text{ is independent of } \{U_n\}_{n=1}^{\infty}, \\ &X_n(\omega) := R_n\Big(\omega; X_{n-1}(\omega)\Big), \quad \text{for all } n = 1, 2, \dots, \end{aligned} \tag{3.1.5}$$

where the random function $R_n(\omega; \cdot)$ is defined in Eq. (3.1.4).

Theorem 3.1.2. *The sequence of random variables $\{X_n\}_{n=0}^{\infty}$ defined by Eq. (3.1.5) is a homogeneous Markov chain on $\mathcal{X} = \{x_1, \dots, x_S\}$ with initial distribution μ and transition probability p.*

Proof. The initial distribution of X_0 is μ by construction. For every $k \le n$, the random variable X_k is a function of $X_0, U_1, \dots, U_k$. Consequently, the event

$$H_n := \{X_0 = x_{i_0}, X_1 = x_{i_1}, \dots, X_n = x_{i_n}\}$$

is determined by $X_0, U_1, \dots, U_n$. The random variable U_{n+1} is independent of these variables and hence independent of H_n.

Now fix a history H_n with $\mathbb{P}(H_n) > 0$. On this event, $X_n = x_{i_n}$, so Eq. (3.1.5) and the definition of R_{n+1} give

$$\begin{aligned} \mathbb{P}(X_{n+1} = x_j \mid H_n) &= \mathbb{P}\big(R_{n+1}(x_{i_n}) = x_j \mid H_n\big) \\ &= \mathbb{P}\big(U_{n+1} \in I_{i_n j} \mid H_n\big) \\ &= \mathbb{P}\big(U_{n+1} \in I_{i_n j}\big) \\ &= \text{length}(I_{i_n j}) \\ &= p(x_{i_n}, x_j). \end{aligned}$$

The same calculation conditioned only on $X_n = x_{i_n}$ shows that

$$\mathbb{P}(X_{n+1} = x_j \mid X_n = x_{i_n}) = p(x_{i_n}, x_j)$$

whenever $\mathbb{P}(X_n = x_{i_n}) > 0$. Therefore,

$$\mathbb{P}(X_{n+1} = x_j \mid H_n) = \mathbb{P}(X_{n+1} = x_j \mid X_n = x_{i_n}),$$

and the common conditional probability $p(x_{i_n}, x_j)$ does not depend on n. Thus, $\{X_n\}_{n=0}^{\infty}$ is a homogeneous Markov chain with transition probability p. □

The procedure in Eq. (3.1.5) and Theorem 3.1.2 are summarized in the following algorithm.

Algorithm 11 : Generating Markov Chains (II)

Input: (i) initial distribution PMF μ, (ii) transition probability p, and (iii) desired number of transitions n^*.
Output: the n^*+1 states $X_0, \ldots, X_{n^*}$ of a Markov chain with initial distribution μ and transition probability p.

1: Generate $X_0 \sim \mu$.
2: Generate $\{U_n\}_{n=1}^{n^*} \overset{iid}{\sim} \text{Unif}(0,1)$ and define $R_n(x_i) := x_j$ whenever $U_n \in I_{ij}$, for $i, j = 1, \ldots, S$ and $n = 1, \ldots, n^*$.
3: **for all** $n = 1, 2, \ldots, n^*$ **do**
4: $\quad X_n = R_n(X_{n-1})$
5: **end for**

You will be asked to write programming code of Algorithm 11 in homework.

3.1.3 Random Walks

The abstract Markov property becomes more concrete in random walks, where each new state is obtained by adding an independent increment to the current state. We first define integer-valued random walks directly and then identify their transition probabilities.†.

Definition 3.1.2. *Let $\xi_1, \xi_2, \ldots$ be iid random variables taking values in $\mathbb{Z}^d$, and let $x_0 \in \mathbb{Z}^d$ be fixed. Define*

$$X_0 := x_0, \qquad X_n := x_0 + \sum_{i=1}^{n} \xi_i, \quad n = 1, 2, \ldots.$$

*Then, the sequence of $\mathbb{Z}^d$-valued random variables $\{X_n\}_{n=0}^{\infty}$ is called a **random walk starting from** x_0. Unless stated otherwise, our examples use $x_0 = \mathbf{0}$.*

An integer-valued random walk has a simple gambling interpretation. Flip a fair coin in each round: earn one dollar for heads and lose one dollar for tails. Let ξ_i denote the gain in round i, so $\mathbb{P}(\xi_i = -1) = \mathbb{P}(\xi_i = 1) = 0.5$. After n rounds, the net gain is $X_n = \sum_{i=1}^{n} \xi_i$, and $\{X_n\}_{n=0}^{\infty}$ is an integer-valued random walk.

To visualize a $\mathbb{Z}^2$-valued random walk, let $\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \ldots$ be iid $\mathbb{Z}^2$-valued random variables satisfying

$$\mathbb{P}\left(\boldsymbol{\xi}_i = (1,0)^\intercal\right) = \mathbb{P}\left(\boldsymbol{\xi}_i = (0,1)^\intercal\right) = \mathbb{P}\left(\boldsymbol{\xi}_i = (-1,0)^\intercal\right) = \mathbb{P}\left(\boldsymbol{\xi}_i = (0,-1)^\intercal\right) = \frac{1}{4}.$$

Then, $\{\boldsymbol{X}_n = \sum_{i=1}^{n} \boldsymbol{\xi}_i\}_{n=0}^{\infty}$ is a random walk. Figure 3.1 shows unscaled paths of this random walk, i.e., collections of line segments $\{\overline{\boldsymbol{X}_0\boldsymbol{X}_1}, \overline{\boldsymbol{X}_1\boldsymbol{X}_2}, \ldots, \overline{\boldsymbol{X}_{n-1}\boldsymbol{X}_n}\}$, for $n = 100, 1000, 5000, 10000$. A rigorous connection with Brownian motion requires rescaling both time and space. Specifically, define the linearly interpolated process

$$\boldsymbol{W}_n(t) := \frac{1}{\sqrt{n}} \left[\boldsymbol{X}_{\lfloor nt \rfloor} + (nt - \lfloor nt \rfloor)\left(\boldsymbol{X}_{\lfloor nt \rfloor + 1} - \boldsymbol{X}_{\lfloor nt \rfloor}\right)\right], \qquad 0 \le t \le 1.$$

† Generally, random walks can be defined on continuous spaces, e.g., $\mathbb{R}^d$, instead of integers. More generally, random walks can be defined on topological Abelian groups. The Abelian group structure enables us to define the addition in $X_n = \sum_{i=1}^{n} \xi_i$. To be consistent with the countable set $\mathcal{X}$ assumption in our discussion of Markov chains, in APMA 1690, we only focus on integer-valued (or $\mathbb{Z}^d$-valued) random walks.

Donsker's theorem states that $\boldsymbol{W}_n$ converges in distribution to a two-dimensional Brownian motion $\boldsymbol{B}$ satisfying $\mathrm{Cov}(\boldsymbol{B}(t)) = (t/2)\boldsymbol{I}_2$. The unscaled lattice paths themselves do not converge to Brownian motion. This result is studied in APMA 2640.

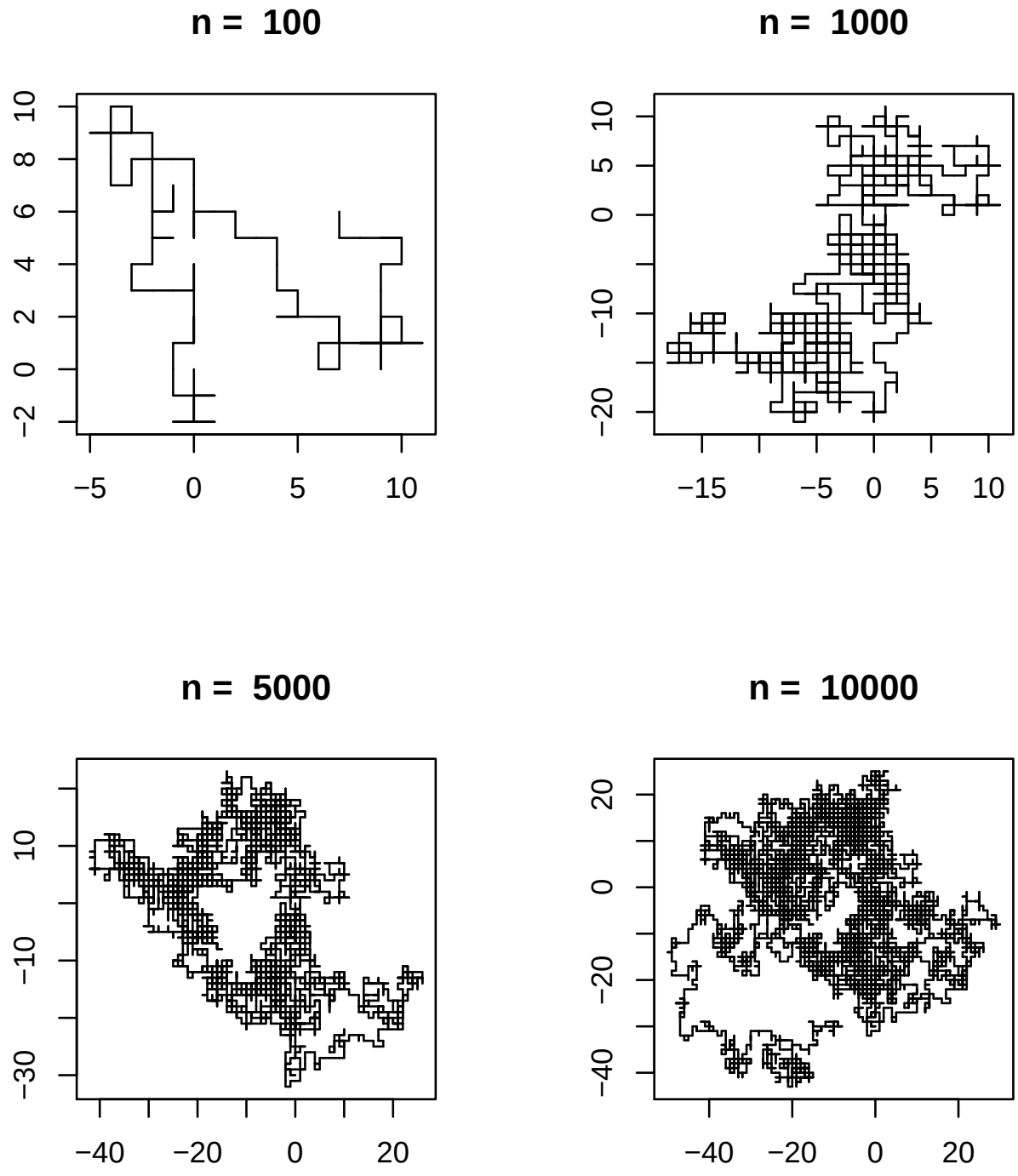


Figure 3.1: Unscaled paths of the random walk $\boldsymbol{X}_n = \sum_{i=1}^{n} \boldsymbol{\xi}_i$, shown as collections of line segments $\{\overline{\boldsymbol{X}_0\boldsymbol{X}_1}, \overline{\boldsymbol{X}_1\boldsymbol{X}_2}, \ldots, \overline{\boldsymbol{X}_{n-1}\boldsymbol{X}_n}\}$ for $n = 100, 1000, 5000, 10000$. The connection to Brownian motion requires the time-and-space rescaling described in the text. The `R` code used to produce these visualizations is provided below.

```r
par(mfrow=c(2, 2))
n=10000

```

```
Xn=matrix(NA, ncol = 2, nrow = n+1)
Xn[1,]=c(0, 0)
for (i in 1:n) {
  direction_ind=sample(1:4, size = 1, prob = rep(1, 4))
  if(direction_ind==1){ xi=c(1, 0) }
  if(direction_ind==2){ xi=c(0, 1) }
  if(direction_ind==3){ xi=c(-1, 0) }
  if(direction_ind==4){ xi=c(0, -1) }
  Xn[i+1,]=Xn[i,]+xi
}

sample_sizes=c(101, 1001, 5001, 10001)
for (k in 1:4) {
  plot(c(0, 0), xlim = range(Xn[1:sample_sizes[k],1]), ylim = range(Xn[1:sample_sizes[k],2]),
       col="white", xlab = "", ylab = "",
       main = paste("n = ", as.character(sample_sizes[k]-1)))
  for (i in 1:(sample_sizes[k]-1)) {
    segments(x0=Xn[i,1], y0=Xn[i,2], x1=Xn[i+1,1], y1=Xn[i+1,2])
  }
}
```

Definition 3.1.2 indicates that all random walks are Markov chains, which follows from the following equation derived from the definition of random walks $\{X_n\}_{n=0}^{\infty}$

$$X_{n+1} = X_n + \xi_{n+1}.$$

We present this result by the following theorem.

Theorem 3.1.3. *The random walks $\{X_n\}_{n=0}^{\infty}$ defined in Definition 3.1.2 are Markov chains.*

Proof. For any fixed n, we assume $\mathbb{P}(X_n = x, \ldots, X_0 = x_0) > 0$. Then, the definition of random walk $\{X_n\}_n$ implies

$$\begin{aligned}
\mathbb{P}(X_{n+1} = y \,|\, X_n = x, X_{n-1} = x_{n-1}, \ldots, X_0 = x_0) &= \frac{\mathbb{P}(X_{n+1} = y, X_n = x, \ldots, X_0 = x_0)}{\mathbb{P}(X_n = x, \ldots, X_0 = x_0)} \\
&= \frac{\mathbb{P}(\xi_{n+1} = y - x, X_n = x, \ldots, X_0 = x_0)}{\mathbb{P}(X_n = x, \ldots, X_0 = x_0)} \\
&\overset{(1)}{=} \frac{\mathbb{P}(\xi_{n+1} = y - x) \cdot \mathbb{P}(X_n = x, \ldots, X_0 = x_0)}{\mathbb{P}(X_n = x, \ldots, X_0 = x_0)} \\
&= \mathbb{P}(\xi_{n+1} = y - x),
\end{aligned}$$

where the equality "(1)" follows from the fact that ξ_{n+1} is independent of $\{\xi_1, \ldots, \xi_n\}$ (hence, ξ_{n+1} is independent

of $X_m = x_0 + \sum_{i=1}^{m} \xi_i$ for all $1 \le m \le n$). On the other hand, using the same independence argument, we have

$$\begin{aligned}
\mathbb{P}(X_{n+1} = y \,|\, X_n = x) &= \frac{\mathbb{P}(X_{n+1} = y, X_n = x)}{\mathbb{P}(X_n = x)} \\
&= \frac{\mathbb{P}(\xi_{n+1} = y - x, X_n = x)}{\mathbb{P}(X_n = x)} \\
&= \frac{\mathbb{P}(\xi_{n+1} = y - x) \cdot \mathbb{P}(X_n = x)}{\mathbb{P}(X_n = x)} \\
&= \mathbb{P}(\xi_{n+1} = y - x).
\end{aligned}$$

Therefore, the Markov property in Eq. (3.1.1) is satisfied. □

Since random walks are Markov chains (see Theorem 3.1.3), we can determine a random walk by specifying its initial and transition probability. Specifically, we have the following

Theorem 3.1.4. *Let $\{X_n\}_{n=0}^{\infty}$ be the random walk in Definition 3.1.2, with $\xi_1, \xi_2, \ldots \overset{iid}{\sim} \mu$, where μ is a PMF on $\mathbb{Z}^d$. Then, $\{X_n\}_{n=0}^{\infty}$ is a homogeneous Markov chain whose initial distribution is concentrated at x_0 and whose transition probability is*

$$p(x, y) = \mu(y - x), \qquad x, y \in \mathbb{Z}^d.$$

The proof of Theorem 3.1.4 is left for homework.

3.2 Recurrence and Transience

"*A drunk man will eventually find his way home, but a drunk bird may get lost forever.*"

— Shizuo Kakutani

Having constructed several Markov chains, we now begin their long-run analysis. The first question is whether a chain returns to a given state. Recurrence corresponds to an eventual return with probability one, whereas transience allows the chain to leave and never return. In the random-walk metaphor, this distinction asks whether a person walking at random will eventually find the way home. The material in this section is based on Chapter 6.4 of Durrett (2019).

3.2.1 Definition of Recurrence

Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in the countable set $\mathcal{X}$. For this Markov chain and each element $y \in \mathcal{X}$, we define the following random variable

$$\begin{aligned} &T_y(\omega) := \min\{n > 0 \mid X_n(\omega) = y\} = \text{ the time at which the sequence } \{X_n(\omega)\}_{n=1}^{\infty} \text{ first visits } y, \\ &T_y^1(\omega) = T_y(\omega), \quad \text{and} \\ &T_y^k(\omega) := \min\left\{n > T_y^{k-1}(\omega) \mid X_n(\omega) = y\right\} = \text{ the time of the } k\text{-th visit to } y. \end{aligned}$$

Here we use the convention that the minimum of the empty set is ∞; thus, $T_y = \infty$ precisely when the chain never visits y at a positive time. We explicitly write down ω to emphasize that T_y^k are random variables. We denote the following probability

$$\begin{aligned} \rho_{xy} :&= \mathbb{P}(T_y < \infty \mid X_0 = x) \\ &= \text{the conditional probability that } \{X_n\}_{n=1}^{\infty} \text{ will visit } y \text{ at least once, given that it starts from } x. \end{aligned}$$

With the notation $\{\rho_{xy}\}_{x,y\in\mathcal{X}}$, we can define recurrence and transience as follows.

Definition 3.2.1. *Suppose we have a Markov chain $\{X_n\}_{n=0}^{\infty}$. We provide the following two versions of the same definition.*

*i) (Rigorous/mathematical version) A state $y \in \mathcal{X}$ is said to be **recurrent** for this Markov chain if $\rho_{yy} = 1$; otherwise (i.e., $\rho_{yy} < 1$), the state y is said to be **transient** for this Markov chain.*

ii) (Heuristic version) If the Markov chain starting from y (i.e., $X_0 = y$) will almost surely (i.e., with probability one) return to y, the state y is said to be recurrent for this Markov chain; otherwise, y is said to be transient.

To illustrate recurrence, Shizuo Kakutani famously remarked that "a drunk man will eventually find his way home, but a drunk bird may get lost forever." Let y represent home and let $\{X_n\}_{n=0}^{\infty}$ represent the path. Then T_y is the first time the path reaches home: $T_y < \infty$ means that it eventually returns, whereas $T_y = \infty$ means that it never does. Thus, the recurrence condition $\rho_{yy} = \mathbb{P}(T_y < \infty \mid X_0 = y) = 1$ says that a path starting at home returns there with probability one. The distinction between the two-dimensional walk of the "drunk man" and the three-dimensional walk of the "drunk bird" illustrates how strongly dimension can affect probabilistic behavior. The corresponding mathematical justification is left as homework.

The following theorem relates the first visit time T_y with the k-th visit time T_y^k

Theorem 3.2.1 (Theorem 6.4.1 of Durrett (2019))**.** $\mathbb{P}(T_y^k < \infty \mid X_0 = x) = \rho_{xy} \cdot \rho_{yy}^{k-1}$.

The rigorous proof of Theorem 3.2.1 needs the so-called "strong Markov property" (Durrett, 2019, Theorem 6.3.4), which is outside the scope of this course. Instead of providing a cheating proof, we heuristically explain Theorem

3.2.1 with $k = 2$. That is, *the probability that the Markov chain visits y at least twice is equal to the probability that the Markov chain visits y at least once times the probability that this chain returns to y at least once after the first visit.*

The following identity is useful in investigating the recurrence of Markov chains.

$$\begin{aligned} N_y(\omega) &:= \sum_{n=1}^{\infty} \mathbb{1}\left(X_n(\omega) = y\right) = \text{ the number of visits to } y \text{ at positive times,} \\ \mathbb{E}(N_y \,|\, X_0 = x) &= \sum_{n=1}^{\infty} \mathbb{P}(X_n = y \,|\, X_0 = x) = \rho_{xy} \cdot \sum_{k=1}^{\infty} \rho_{yy}^{k-1}, \end{aligned} \tag{3.2.1}$$

where the infinite series is allowed to be ∞. Additionally, we would emphasize that the sum is over all positive times, not including $n = 0$. The proof of Eq. (3.2.1) is left for homework. Eq. (3.2.1) implies the following useful result.

Theorem 3.2.2. *Let $\{X_n\}_{n=0}^{\infty}$ be a homogeneous Markov chain taking values in the discrete state space $\mathcal{X}$. For any $y \in \mathcal{X}$, we have the following*

i) The state y is recurrent for the Markov chain if and only if

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n = y \,|\, X_0 = y) = \infty.$$

ii) If y is not recurrent, we have the following for all $x \in \mathcal{X}$

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n = y \,|\, X_0 = x) < \infty.$$

To see why this criterion follows from Eq. (3.2.1), first set $x = y$. If y is recurrent, then $\rho_{yy} = 1$ and the geometric series $\sum_{k=1}^{\infty} \rho_{yy}^{k-1}$ diverges. If y is transient, then $\rho_{yy} < 1$, so for every starting state x,

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n = y \mid X_0 = x) = \frac{\rho_{xy}}{1 - \rho_{yy}} < \infty.$$

Thus, recurrence can be recognized through an infinite expected number of returns, whereas a transient state has a finite expected number of visits from every starting state.

3.2.2 Properties of Recurrence

The following theorem shows that recurrence is "contagious."

Theorem 3.2.3. *If x is recurrent (i.e., $\rho_{xx} = 1$) and $\rho_{xy} > 0$, then we have*

i) $\rho_{yy} = 1$, i.e., the state y is recurrent;

ii) $\rho_{xy} = \rho_{yx} = 1$.

That is, $\rho_{xx} = 1$ *and* $\rho_{xy} > 0$ *implies* $\rho_{xx} = \rho_{yy} = \rho_{xy} = \rho_{yx} = 1$.

The proof of Theorem 3.2.3 is provided below. You are not required to remember this proof; please just try to feel the elegance of this proof. Reading this proof helps you have an idea of what the **average** homework questions of APMA 2630/2640 look like.

Proof. If $x = y$, then there is nothing to prove. Hereafter, we assume $x \neq y$.

Step 1: showing $\rho_{yx} = 1$. We have

$$\begin{aligned} 0 < \rho_{xy} &= \mathbb{P}(T_y < \infty \,|\, X_0 = x) \\ &= \mathbb{P}(\text{ there exists at least one } n \geq 1 \text{ such that } X_n = y \,|\, X_0 = x) \\ &= \mathbb{P}\left(\bigcup_{n=1}^{\infty} \{\omega \in \Omega : X_n(\omega) = y\} \,\middle|\, X_0 = x \right) \\ &\leq \sum_{n=1}^{\infty} \mathbb{P}(X_n = y \,|\, X_0 = x), \end{aligned} \tag{3.2.2}$$

where the "$\leq$" follows from Theorem 1.2.1 vii). Eq. (3.2.2) implies that there exists at least one $k \geq 1$ such that $\mathbb{P}(X_k = y \,|\, X_0 = x) > 0$; hence, the collection $\{k : \mathbb{P}(X_k = y \,|\, X_0 = x) > 0\}$ is not empty. Then,

$$K := \min\{k : \mathbb{P}(X_k = y \,|\, X_0 = x) > 0\} < \infty.$$

Because of the following positive marginal probability

$$\mathbb{P}(X_K = y \,|\, X_0 = x) = \sum_{y_1', \ldots, y_{K-1}' \in \mathcal{X}} \mathbb{P}(X_K = y, X_{K-1} = y_{K-1}', \ldots, X_1 = y_1' \,|\, X_0 = x) > 0,$$

there is a sequence $y_1, \ldots, y_{K-1}$ such that $\mathbb{P}(X_K = y, X_{K-1} = y_{K-1}, \ldots, X_1 = y_1 \,|\, X_0 = x) > 0$. The definition of K implies that $y_i \neq x$ for $1 \leq i \leq K-1$. Otherwise, if there exists $k' < K$ such that $y_{k'} = x$, we have

$$\begin{aligned} 0 &< \mathbb{P}(X_K = y, X_{K-1} = y_{K-1}, \ldots, X_1 = y_1 \,|\, X_0 = x) \\ &\leq \mathbb{P}(X_K = y, X_{k'} = x \,|\, X_0 = x) \\ &= \mathbb{P}(X_K = y \,|\, X_{k'} = x, X_0 = x) \cdot \mathbb{P}(X_{k'} = x \,|\, X_0 = x) \\ &\overset{(*)}{=} \mathbb{P}(X_{K-k'} = y \,|\, X_0 = x) \cdot \mathbb{P}(X_{k'} = x \,|\, X_0 = x), \end{aligned}$$

where $\overset{(*)}{=}$ follows from the Markov property.[†] Then $\mathbb{P}(X_{K-k'} = y \,|\, X_0 = x) > 0$, which implies the contradiction $K \leq K - k'$.

If $\rho_{yx} < 1$, the condition $\rho_{xx} = 1$ implies

$$
\begin{aligned}
0 &= 1 - \rho_{xx} \\
&= \mathbb{P}(T_x = \infty \,|\, X_0 = x) \\
&= \mathbb{P}(X_n \neq x \text{ for all } n \,|\, X_0 = x) \\
&\geq \mathbb{P}(X_1 = y_1, \dots X_{K-1} = y_{K-1}, X_K = y, \text{ and } X_n \neq x \text{ for all } n > K \,|\, X_0 = x) \\
&\overset{(1)}{=} \mathbb{P}(X_n \neq x \text{ for all } n > K \,|\, X_0 = x, X_1 = y_1, \dots X_{K-1} = y_{K-1}, X_K = y) \cdot \mathbb{P}(X_1 = y_1, \dots X_{K-1} = y_{K-1}, X_K = y \,|\, X_0 = x) \\
&\overset{(2)}{=} \mathbb{P}(X_n \neq x \text{ for all } n > K \,|\, X_K = y) \cdot \mathbb{P}(X_1 = y_1, \dots X_{K-1} = y_{K-1}, X_K = y \,|\, X_0 = x) \\
&\overset{(3)}{=} \mathbb{P}(X_n \neq x \text{ for all } n > 0 \,|\, X_0 = y) \cdot \mathbb{P}(X_1 = y_1, \dots X_{K-1} = y_{K-1}, X_K = y \,|\, X_0 = x) \\
&= (1 - \rho_{yx}) \cdot \mathbb{P}(X_1 = y_1, \dots X_{K-1} = y_{K-1}, X_K = y \,|\, X_0 = x) \\
&\overset{(4)}{>} 0,
\end{aligned}
$$

where $\overset{(1)}{=}$ comes from the law of total probability, $\overset{(2)}{=}$ and $\overset{(3)}{=}$ follow from the Markov property†, and $\overset{(4)}{>}$ follows from the assumption $\rho_{yx} < 1$ and contradicts $\rho_{xx} = 1$. Therefore, we have $\rho_{yx} = 1$.

Step 2: showing the recurrence of y. Because of the following

$$
\begin{aligned}
0 < \rho_{yx} &= \mathbb{P}(T_x < \infty \,|\, X_0 = y) \\
&= \mathbb{P}(\text{ there exists at least one } n \geq 1 \text{ such that } X_n = x \,|\, X_0 = y) \\
&= \mathbb{P}\left(\bigcup_{n=1}^{\infty} \{\omega \in \Omega : X_n(\omega) = x\} \,\middle|\, X_0 = y \right) \\
&\leq \sum_{n=1}^{\infty} \mathbb{P}(X_n = x \,|\, X_0 = y),
\end{aligned}
$$

† Actually, they follow from the following equation derived from the Markov property

$$\mathbb{E}\left(h(X_n, X_{n+1}, \dots) \,|\, X_n = x_n, \dots, X_1 = x_1, X_0 = x_0\right) = \mathbb{E}(h(X_0, X_1, \dots) \,|\, X_0 = x_n),$$

for any bounded functions $h : \mathcal{X}^{\mathbb{N}} \to \mathbb{R}$. The proof of this equation needs the $\pi - \lambda$ theorem (see Theorem 2.1.2 of Durrett (2019)) that is outside the scope of this course. The proof of this equation can be found in Chapter 6.3 of Durrett (2019).

there exists an L such that $\mathbb{P}(X_L = x \,|\, X_0 = y) > 0$. From the following

$$
\begin{aligned}
&\mathbb{P}(X_{L+n+K} = y \,|\, X_0 = y) \\
&\geq \mathbb{P}(X_L = x, X_{L+n} = x, X_{L+n+K} = y \,|\, X_0 = y) \\
&= \mathbb{P}(X_{L+n+K} = y \,|\, X_0 = y, X_L = x, X_{L+n} = x) \cdot \mathbb{P}(X_L = x, X_{L+n} = x \,|\, X_0 = y) \\
&= \mathbb{P}(X_{L+n+K} = y \,|\, X_0 = y, X_L = x, X_{L+n} = x) \cdot \mathbb{P}(X_{L+n} = x \,|\, X_0 = y, X_L = x) \cdot \mathbb{P}(X_L = x \,|\, X_0 = y) \\
&\overset{(5)}{=} \mathbb{P}(X_{L+n+K} = y \,|\, X_{L+n} = x) \cdot \mathbb{P}(X_{L+n} = x \,|\, X_L = x) \cdot \mathbb{P}(X_L = x \,|\, X_0 = y) \\
&\overset{(6)}{=} \mathbb{P}(X_K = y \,|\, X_0 = x) \cdot \mathbb{P}(X_n = x \,|\, X_0 = x) \cdot \mathbb{P}(X_L = x \,|\, X_0 = y),
\end{aligned}
$$

where $\overset{(5)}{=}$ and $\overset{(6)}{=}$ follow from the Markov property[†], we have

$$
\begin{aligned}
\sum_{m=1}^{\infty} \mathbb{P}(X_m = y \,|\, X_0 = y) &= \sum_{l=1}^{L+K} \mathbb{P}(X_l = y \,|\, X_0 = y) + \sum_{n=1}^{\infty} \mathbb{P}(X_{L+n+K} = y \,|\, X_0 = y) \\
&\geq \sum_{l=1}^{L+K} \mathbb{P}(X_l = y \,|\, X_0 = y) \\
&\quad + \mathbb{P}(X_K = y \,|\, X_0 = x) \cdot \mathbb{P}(X_L = x \,|\, X_0 = y) \cdot \left[\sum_{n=1}^{\infty} \mathbb{P}(X_n = x \,|\, X_0 = x) \right].
\end{aligned}
$$

The recurrence of x and Theorem 3.2.2 implies $\sum_{n=1}^{\infty} \mathbb{P}(X_n = x \,|\, X_0 = x) = \infty$. Then, we have $\sum_{m=1}^{\infty} \mathbb{P}(X_m = y \,|\, X_0 = y) = \infty$. Applying Theorem 3.2.2 again, we have that y is recurrent. The proof is completed. □

The following definition is important for the MCMC theory in Chapter 4.

Definition 3.2.2. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in the countable space $\mathcal{X}$ and having transition probability p.*

*i) The Markov chain $\{X_n\}_{n=0}^{\infty}$ or transition probability p is said to be **recurrent** if all states of $\mathcal{X}$ are recurrent for this Markov chain.*

*ii) The Markov chain $\{X_n\}_{n=0}^{\infty}$ or transition probability p is said to be **irreducible** if $\rho_{xy} > 0$ for all $x, y \in \mathcal{X}$, i.e., starting from any state x, the chain can reach any state y with positive probability in some finite number of steps.*

Because of the following theorem, the scenario where the state space $\mathcal{X}$ is finite is important in the Markov chain theory.

Theorem 3.2.4. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in the state space $\mathcal{X}$. If $\mathcal{X}$ is finite (i.e., $\#\mathcal{X} < \infty$), we have*

i) $\mathcal{X}$ has at least one state that is recurrent for $\{X_n\}_{n=0}^{\infty}$;

ii) furthermore, if $\{X_n\}_{n=0}^{\infty}$ *is irreducible,* $\{X_n\}_{n=0}^{\infty}$ *is recurrent, i.e., all states in* $\mathcal{X}$ *are recurrent for* $\{X_n\}_{n=0}^{\infty}$.

Proof. In view of Theorem 3.2.3, it suffices to prove result i). Suppose result i) is false. Then, for all $y \in \mathcal{X}$, we have $\rho_{yy} < 1$, and Eq. (3.2.1) implies $\mathbb{E}(N_y \,|\, X_0 = x) = \sum_{n=1}^{\infty} \mathbb{P}(X_n = y \,|\, X_0 = x) = \frac{\rho_{xy}}{1-\rho_{yy}}$. This is impossible because it implies

$$\begin{aligned}
\infty &\overset{(1)}{>} \sum_{y\in\mathcal{X}} \mathbb{E}(N_y \,|\, X_0 = x) \\
&= \sum_{y\in\mathcal{X}} \sum_{n=1}^{\infty} \mathbb{P}(X_n = y \,|\, X_0 = x) \\
&\overset{(2)}{=} \sum_{n=1}^{\infty} \sum_{y\in\mathcal{X}} \mathbb{P}(X_n = y \,|\, X_0 = x) \\
&\overset{(3)}{=} \sum_{n=1}^{\infty} 1 = \infty,
\end{aligned}$$

where $\overset{(1)}{>}$ follows from the finiteness of $\mathcal{X}$, and $\overset{(2)}{=}$ follows from the same fact. The equality $\overset{(3)}{=}$ follows from $\sum_{y\in\mathcal{X}} \mathbb{P}(X_n = y \,|\, X_0 = x) = 1$ for any fixed x. Therefore, at least one state y must be recurrent for $\{X_n\}_{n=0}^{\infty}$. The proof is complete. □

We use the following example to illustrate the use of Theorems 3.2.3 and 3.2.4.

Example 3.2.1. *(A seven-state chain) Consider the following transition probability matrix*

$$\boldsymbol{P} = \begin{pmatrix}
0.3 & 0 & 0 & 0 & 0.7 & 0 & 0 \\
0.1 & 0.2 & 0.3 & 0.4 & 0 & 0 & 0 \\
0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\
0 & 0 & 0 & 0.5 & 0 & 0.5 & 0 \\
0.6 & 0 & 0 & 0 & 0.4 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.2 & 0.8 \\
0 & 0 & 0 & 1 & 0 & 0 & 0
\end{pmatrix}. \tag{3.2.3}$$

We draw a directed graph in Figure 3.2 that contains an arrow from x_i *to* x_j *if* $p(x_i, x_j) = \boldsymbol{P}_{ij} > 0$. *We have the following conclusions*

i) Figure 3.2 shows that there is a positive probability path from x_2 *to* x_1, *then* $\rho_{x_2x_1} > 0$. *However, there is no positive probability path getting into node* x_2, *then* $\rho_{x_1x_2} = 0$. *Then, we have that* x_2 *is not recurrent for the Markov chain. Otherwise, Theorem 3.2.3 will imply that* $\rho_{x_1x_2} = \rho_{x_2x_1} = 1$.

ii) Although the state space of this Markov chain is finite, we cannot apply Theorem 3.2.4 to show that all states are recurrent—this recurrence result is actually false. The theorem does not apply because this Markov chain

is not irreducible.

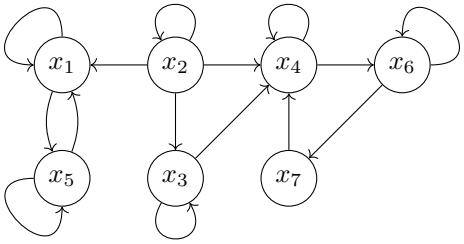


Figure 3.2: The directed graph has an arrow from x_i to x_j if $p(x_i, x_j) = \boldsymbol{P}_{ij} > 0$.

Motivated by Example 3.2.1, we provide a result that helps us to visually identify the irreducibility of Markov chains taking values in finite state spaces.

Definition 3.2.3. *For the transition probability matrix $\boldsymbol{P}$ of a Markov chain taking values in $\mathcal{X} = \{x_1, \dots, x_S\}$, we define the **directed graph** $G(\boldsymbol{P}) = (V, E)$ **for** $\boldsymbol{P}$ as follows*

i) $V := \mathcal{X} = \{x_1, \dots, x_S\}$;

ii) the directed edge $(x_i \to x_j) \in E$ if $\boldsymbol{P}_{ij} > 0$.

For example, the directed graph of the transition probability matrix in Eq. (3.2.3) is the one in Figure 3.2.

Definition 3.2.4. *Let $G(\boldsymbol{P})$ be the directed graph defined in Definition 3.2.3. For two given vertices $x, y \in V$, we say that there exists a directed path from x to y if there are finitely many vertices, say $\xi^{(1)}, \xi^{(2)}, \dots, \xi^{(l)}$, such that the following conditions are satisfied:*

- $\xi^{(1)}, \xi^{(2)}, \dots, \xi^{(l)} \in V$;
- *the directed edges $(x \to \xi^{(1)}), (\xi^{(1)} \to \xi^{(2)}), (\xi^{(2)} \to \xi^{(3)}), \dots, (\xi^{(l-1)} \to \xi^{(l)}), (\xi^{(l)} \to y)$ belong to E.*

The collection $\{(x \to \xi^{(1)}), (\xi^{(1)} \to \xi^{(2)}), (\xi^{(2)} \to \xi^{(3)}), \dots, (\xi^{(l-1)} \to \xi^{(l)}), (\xi^{(l)} \to y)\}$ of directed edges is called a path from x to y.

For example, in Figure 3.2, $\{(x_2 \to x_3), (x_3 \to x_4)\}$ is a directed path going from x_2 to x_4.

Claim 3.2.5. *Let $\boldsymbol{P}$ be the transition probability matrix of a homogeneous Markov chain taking values in $\mathcal{X} = \{x_1, \dots, x_S\}$. The Markov chain is irreducible if and only if $G(\boldsymbol{P})$ satisfies the following: for each pair of vertices x_i and x_j, there exists at least one directed path going from x_i to x_j and one directed path going from x_j to x_i.*

The proof of Claim 3.2.5 is conceptual and easy. It is left for homework. Using Claim 3.2.5 and the directed graph in Figure 3.2, the Markov chain with the transition probability in Eq. (3.2.3) is not irreducible.

3.2.3 Recurrence of Simple Random Walks

At the end of this section, we go back to the drunk man/bird example and translate it into mathematics.

Definition 3.2.5. *The random walk $X_n = x_0 + \sum_{i=1}^n \xi_i$ taking values in $\mathbb{Z}^d$ is called a* ***simple symmetric random walk*** *if the iid $\mathbb{Z}^d$-valued increments $\xi_1, \xi_2, \dots$ satisfy*

$$\mathbb{P}(\xi_i = \boldsymbol{e}_k) = \mathbb{P}(\xi_i = -\boldsymbol{e}_k) = \frac{1}{2d}, \quad \text{for all } k = 1, \dots, d, \tag{3.2.4}$$

where $\boldsymbol{e}_k$ is the k-th axis of $\mathbb{R}^d$, i.e.,

$$\boldsymbol{e}_k = \quad (0, \dots, 0, \underset{\substack{\uparrow \\ k^{th}}}{1}, 0, \dots, 0).$$

Each random variable ξ_i can be viewed as one step of the drunk man ($d = 2$) or the drunk bird ($d = 3$), while x_0 is the starting point. The accumulated increments form the walking or flying path. By Theorem 3.1.4, this process is a homogeneous Markov chain with transition probability

$$p(x, y) = \begin{cases} \dfrac{1}{2d}, & y - x \in \{\boldsymbol{e}_1, -\boldsymbol{e}_1, \dots, \boldsymbol{e}_d, -\boldsymbol{e}_d\}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.2.5}$$

Thus, from every state, the walk moves to one of its $2d$ nearest lattice neighbors with equal probability.

The preceding distinction between dimensions $d = 2$ and $d = 3$ is formalized by the following theorem.

Theorem 3.2.6. *A simple random walk is recurrent when $d \leq 2$ and transient when $d \geq 3$.*

The proof of Theorem 3.2.6 involves some tricks in combinatorics and is skipped in this course. One can find this proof in Chapter 4.2 of Durrett (2019).

Let $\mathbf{0} = (0, 0, \dots, 0)^\intercal \in \mathbb{Z}^d$ and $\{X_n\}_{n=0}^\infty$ be a simple random walk. Then, a more detailed version of Theorem 3.2.6 is provided in Table 3.1 (also see Table 17.1 of Klenke (2020)). The numbers in Table 3.1 are essentially derived from the Chung-Fuchs theorem (see Theorem 17.42 of Klenke (2020)) proposed in 1951. The theoretical computation of the numbers in Table 3.1 involves some advanced mathematics and is skipped in this course. You will be asked to numerically estimate some of these numbers in homework (see the code for producing Figure 3.1).

Table 3.1: Recurrence of Simple Random Walks in Different Dimensions

d	$\rho_{\mathbf{00}} = \mathbb{P}(T_{\mathbf{0}} < \infty \mid X_0 = \mathbf{0})$	Recurrence
1	1	Yes
2	1	Yes
3	≈ 0.34053732955	No
4	≈ 0.19320167322	No
5	≈ 0.13517860982	No
6	≈ 0.10471549562	No
7	≈ 0.08584493411	No
8	≈ 0.07291264996	No

3.3 Stationary Distributions

"*In addition to saying something useful, the Perron–Frobenius theory is elegant. It is a testament to the fact that beautiful mathematics eventually tends to be useful, and useful mathematics eventually tends to be beautiful.*"

— Carl D. Meyer, Matrix Analysis and Applied Linear Algebra (2000)

Before providing the definition of stationary distributions, we explain the motivation for introducing this concept. One of our ultimate goals is to investigate the asymptotic distribution of a Markov chain $\{X_n\}_{n=0}^{\infty}$. For the moment, suppose that the state space $\mathcal{X}$ is finite and that the limits $\pi(x) := \lim_{n\to\infty} \mathbb{P}(X_n = x)$ exist for all $x \in \mathcal{X}$. The law of total probability implies

$$\mathbb{P}(X_n = x) = \sum_{y\in\mathcal{X}} \mathbb{P}(X_n = x \mid X_{n-1} = y) \cdot \mathbb{P}(X_{n-1} = y), \tag{3.3.1}$$

Taking limits on both sides of Eq. (3.3.1) and interchanging the limit with the finite sum, we obtain

$$\begin{aligned} \pi(x) &= \lim_{n\to\infty} \mathbb{P}(X_n = x) \\ &= \sum_{y\in\mathcal{X}} \Big[\lim_{n\to\infty} \mathbb{P}(X_{n-1} = y)\Big] \cdot \mathbb{P}(X_n = x \mid X_{n-1} = y) \\ &= \sum_{y\in\mathcal{X}} \pi(y) \cdot p(y, x), \end{aligned} \tag{3.3.2}$$

where p is the transition probability of $\{X_n\}_{n=0}^{\infty}$. Thus, whenever a limiting distribution exists, it must satisfy $\pi(x) = \sum_y \pi(y)p(y, x)$. The converse is not automatic: a stationary distribution need not be the limiting distribution from every starting state. Periodicity provides a simple counterexample, as we will see in Example 3.4.1.

Definition 3.3.1. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in the state space $\mathcal{X}$ and having transition*

probability p. If a PMF π defined on $\mathcal{X}$ satisfies the following equation,

$$\pi(x) = \sum_{y \in \mathcal{X}} \pi(y) \cdot p(y, x), \quad \text{for all } x \in \mathcal{X}, \tag{3.3.3}$$

the PMF π is called a ***stationary distribution*** *or* ***invariant distribution*** *for p or $\{X_n\}_{n=0}^{\infty}$.*

The term "stationary" has the following direct interpretation. If $X_0 \sim \pi$, then Eq. (3.3.3) gives $X_1 \sim \pi$; induction then gives $X_n \sim \pi$ for every $n \geq 0$. In matrix notation for a finite state space, this condition is $\boldsymbol{\pi}^\intercal \boldsymbol{P} = \boldsymbol{\pi}^\intercal$.

For a general state space, neither existence nor uniqueness of a stationary distribution is automatic. For example, simple random walks on $\mathbb{Z}^d$ do not have stationary probability distributions. The general theory is developed in Chapter 6.5 of Durrett (2019). Every Markov chain on a finite state space has at least one stationary distribution; the Brouwer fixed-point argument in the homework establishes this fact. Irreducibility strengthens the conclusion to uniqueness and strict positivity, as stated below. This result also follows from the irreducible nonnegative-matrix version of the Perron–Frobenius theorem; see Chapter 8.3 of Meyer (2000).

Theorem 3.3.1. *Let $\boldsymbol{P} = (p(x_i, x_j))_{1 \leq i,j \leq S}$ be the transition probability matrix of a Markov chain $\{X_n\}_{n=0}^{\infty}$ taking values in a finite state space $\mathcal{X} = \{x_1, x_2, \ldots, x_S\}$. If the Markov chain $\{X_n\}_{n=0}^{\infty}$ is* ***irreducible****, this chain has a unique stationary distribution on $\mathcal{X}$, i.e., $\pi(x) = \sum_{y \in \mathcal{X}} \pi(y) \cdot p(y, x)$, for all $x \in \mathcal{X}$; equivalently,*

$$\boldsymbol{\pi}^\intercal = \boldsymbol{\pi}^\intercal \boldsymbol{P}, \tag{3.3.4}$$

where $\boldsymbol{\pi} := (\pi(x_1), \ldots, \pi(x_S))^\intercal$. Furthermore, $\pi(x_i) > 0$ for all $i = 1, \ldots, S$.

Theorem 3.3.1, especially the matrix form in Eq. (3.3.4), is equivalent to that the matrix $\boldsymbol{P}^\intercal$ (instead of $\boldsymbol{P}$) has 1 as its eigenvalue and the corresponding eigenvector provides a PMF on $\mathcal{X}$. The proof is a simple application of the "irreducible non-negative matrices version" of the Perron-Frobenius theorem. In the homework, you will be asked to prove a special case of Theorem 3.3.1 partially. In this homework problem, you will see how algebraic topology (MATH 2410), linear algebra, and probability theory interact with one another.

3.4 Asymptotic Behavior

Having defined Markov chains, recurrence, and stationary distributions, we now study how a chain behaves over long time horizons. These asymptotic results provide the bridge from Markov-chain theory to the computational methods of Chapter 4.

As mentioned at the beginning of Section 3.3, if the limiting distribution $\pi(x) = \lim_{n \to \infty} \mathbb{P}(X_n = x)$ of a Markov chain $\{X_n\}_n$ exists, it must be a stationary distribution of $\{X_n\}_n$. In general, however, a stationary distribution

need not be unique, and the distribution of X_n need not converge to it. To avoid these complications, we focus on finite irreducible chains, for which Theorem 3.3.1 guarantees a unique stationary distribution. Furthermore, Theorem 3.2.4 and irreducibility imply that every state is recurrent (see Theorem 3.2.3).

We present two complementary results. The first is an analogue of the LLN for empirical averages along a Markov chain; the second concerns convergence of the distribution of X_n itself. Complete proofs require more machinery than we develop here and may be found in Chapter 6.6 of Durrett (2019). Section 3.5 later gives a partial proof from a spectral viewpoint.

Theorem 3.4.1. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in a finite state space $\mathcal{X}$. If this chain is **irreducible**, the empirical distribution of X_n converges with probability one to the stationary distribution of this chain as $n \to \infty$, i.e., for each fixed $x \in \mathcal{X}$,*

$$\mathbb{P}\left\{\omega \in \Omega \,\middle|\, \lim_{n\to\infty} \widehat{\pi}_n(\omega; x) = \pi(x)\right\} = 1,$$
$$\text{where} \quad \widehat{\pi}_n(\omega; x) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\left(X_i(\omega) = x\right),$$

where π is the stationary distribution of $\{X_n\}_{n=0}^{\infty}$ (Theorem 3.3.1 guarantees the uniqueness of π).

The second asymptotic result is on the limit $\lim_{n\to\infty} \mathbb{P}(X_n = x)$. Because of the following

$$\begin{aligned} \lim_{n\to\infty} \mathbb{P}(X_n = x) &= \lim_{n\to\infty} \left[\sum_{y\in\mathcal{X}} \mathbb{P}(X_n = x \,|\, X_0 = y) \cdot \mathbb{P}(X_0 = y)\right] \\ &= \sum_{y\in\mathcal{X}} \left[\lim_{n\to\infty} \mathbb{P}(X_n = x \,|\, X_0 = y)\right] \cdot \mathbb{P}(X_0 = y), \end{aligned} \tag{3.4.1}$$

it suffices to investigate $\lim_{n\to\infty} \mathbb{P}(X_n = x \,|\, X_0 = y)$ for each fixed $y \in \mathcal{X}$. Let $\mathcal{X} = \{x_1, \ldots, x_S\}$, then $\mathbb{P}(X_n = x_j \,|\, X_0 = x_i) = (\boldsymbol{P}^n)_{ij}$. Hence, it suffices to investigate the limiting matrix

$$\lim_{n\to\infty} \boldsymbol{P}^n \quad \text{(if it exists)}.$$

Unfortunately, the limit above does not exist in general even if the Markov chain is irreducible. We have the following example.

Example 3.4.1. *(Periodicity) The following matrix is a stochastic matrix—the transition matrix of Markov chain taking values in a two-point set. By Claim 3.2.5, it is straightforward that this Markov chain is irreducible.*

$$\boldsymbol{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then, we have

$$\boldsymbol{P}^n = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{if } n \text{ is even}; \qquad \boldsymbol{P}^n = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{if } n \text{ is odd}.$$

Thus, $\boldsymbol{P}^n$ alternates with period 2, and the limit $\lim_{n\to\infty} \boldsymbol{P}^n$ does not exist.

We will show that the "periodicity" in Example 3.4.1 is the only thing that can prevent the convergence of $\boldsymbol{P}^n$ as $n \to \infty$. First, we need the following definition to characterize the "periodicity" in Example 3.4.1.

Definition 3.4.1. *Let $\boldsymbol{P} = (p(x_i, x_j))_{1\le i,j\le S}$ be the transition probability matrix of a Markov chain $\{X_n\}_{n=0}^{\infty}$ taking values in a finite state space $\mathcal{X} = \{x_1, \dots, x_S\}$. Suppose the Markov chain is irreducible. Let $\mathcal{I}_i := \{n \ge 1 : (\boldsymbol{P}^n)_{ii} > 0\}$, and let d_i be the **greatest common divisor** of $\mathcal{I}_i$. The positive integer d_i is called the **period** of state x_i.*

When the state space $\mathcal{X}$ is finite and the Markov chain is irreducible, Theorem 3.2.4 implies that all states are recurrent. Then, Theorem 3.2.2 implies that $\sum_{n=1}^{\infty} \mathbb{P}(X_n = x_i \,|\, X_0 = x_i) = \sum_{n=1}^{\infty} (\boldsymbol{P}^n)_{ii} = \infty$ for all $i = 1, \dots, S$. Therefore, the collection $\mathcal{I}_i = \{n \ge 1 : (\boldsymbol{P}^n)_{ii} > 0\}$ is not empty, and the greatest common divisor of $\mathcal{I}_i$ is well-defined. In the scenario where the state space is finite and Markov chain is irreducible, the following theorem allows us remove the index i in d_i.

Theorem 3.4.2. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in a finite state space $\mathcal{X} = \{x_1, \dots, x_S\}$. If this chain is irreducible, $d_i = d_j$ for all $i, j = 1, \dots, S$. In this scenario, we call the value shared by all d_i as the **period** of this Markov chain.*

This theorem is essentially the Lemma 6.6.2 of Durrett (2019); the proof can be found therein. Based on the theorem above, we define the following concept.

Definition 3.4.2. *Let $\{X_n\}_{n=0}^{\infty}$ be an irreducible Markov chain taking values in a finite state space $\mathcal{X} = \{x_1, \dots, x_S\}$. If its period is equal to 1, this Markov chain is said to be **aperiodic**.*

At the end of this chapter, we arrive at the result of the asymptotic behaviors of Markov chains.

Theorem 3.4.3. *Let $\{X_n\}_{n=0}^{\infty}$ be a homogeneous Markov chain taking values in a finite state space $\mathcal{X} = \{x_1, \dots, x_S\}$. If this Markov chain is **irreducible** and **aperiodic**, we have*

$$\begin{aligned} &\lim_{n\to\infty} \mathbb{P}(X_n = x_j \,|\, X_0 = x_i) = \pi(x_j) \quad \text{for all } i, j \in \{1, \dots, S\}, \quad \text{equivalently} \\ &\lim_{n\to\infty} (\boldsymbol{P}^n)_{ij} = \pi(x_j) \quad \text{for all } i, j \in \{1, \dots, S\}, \end{aligned} \tag{3.4.2}$$

where π is the unique stationary distribution of the Markov chain.

The existence and uniqueness of the stationary distribution π in Eq. (3.4.2) are guaranteed by Theorem 3.3.1. The proof and a more detailed version of Theorem 3.4.3 can be found in Chapter 18.2 of Klenke (2020) (Theorem 18.13 therein). Combining Eq. (3.4.1) and Eq. (3.4.2), we have

$$\begin{aligned}\lim_{n\to\infty}\mathbb{P}(X_n=x_j)&=\sum_{i=1}^{S}\left[\lim_{n\to\infty}\mathbb{P}(X_n=x_j\,|\,X_0=x_i)\right]\cdot\mathbb{P}(X_0=x_i)\\&=\pi(x_j)\cdot\sum_{i=1}^{S}\mathbb{P}(X_0=x_i)\\&=\pi(x_j),\quad\text{for all } j=1,\dots,S,\end{aligned}$$

that is, X_n looks like a π-distributed random variable when n is sufficiently large.

3.5 Spectral Theory of Transition Probability Matrices*

The preceding probabilistic results describe the long-run behavior of a Markov chain. On a finite state space, linear algebra provides a complementary explanation. The spectrum of a matrix is the set of its eigenvalues; by studying the eigenvalues and eigenvectors of a transition matrix, we can prove a special case of Theorem 3.4.3 and obtain a basic estimate of the convergence rate.

3.5.1 Review of Eigenvalues and Eigenvectors

This section provides a brief review of eigenvalues and eigenvectors and is based on Wikipedia. Because this review is specifically devoted to the analysis of transition probability matrices, we will not provide the general versions of the introduced results.

Definition 3.5.1. *Let $\boldsymbol{A}$ be an S-by-S complex-valued matrix. If there exists a complex number $\lambda\in\mathbb{C}$ and a complex-valued vector $\boldsymbol{v}\neq\boldsymbol{0}:=(0,0,\dots,0)^\intercal$ such that $\boldsymbol{A}\boldsymbol{v}=\lambda\boldsymbol{v}$, we say λ is an eigenvalue of matrix $\boldsymbol{A}$ and $\boldsymbol{v}$ is the eigenvector corresponding to λ.*

Eigenvalues of $\boldsymbol{A}$ are derived from the polynomial equation

$$\det(\boldsymbol{A}-\lambda\boldsymbol{I})=0.$$

The fundamental theorem of algebra indicates that the characteristic polynomial $\det(\boldsymbol{A}-\lambda\boldsymbol{I})$ has exactly S complex roots counted with algebraic multiplicity. These roots are the eigenvalues of matrix $\boldsymbol{A}$. The fundamental theorem of algebra works for the field of complex numbers $\mathbb{C}$, and this is the reason we focus on $\mathbb{C}$ in Definition 3.5.1.

Additionally, because

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \det(\boldsymbol{A}^\intercal - \lambda \boldsymbol{I}),$$

matrix $\boldsymbol{A}$ and its transpose $\boldsymbol{A}^\intercal$ share the same eigenvalues. Given a matrix, we can derive its eigenvalues and eigenspaces using the following algorithm.

Algorithm 12 Derivation of Eigenvalues and Eigenspaces

Input: An S-by-S matrix $\boldsymbol{A}$

Output: (i) The eigenvalues of $\boldsymbol{A}$, counted with algebraic multiplicity; (ii) a basis for the eigenspace of each distinct eigenvalue; (iii) whether $\boldsymbol{A}$ is diagonalizable.

1: Solve $\det(\boldsymbol{A}-\lambda \boldsymbol{I}) = 0$. Denote the distinct roots by $\mu_1, \dots, \mu_r$ and their algebraic multiplicities by $m_1, \dots, m_r$, where $\sum_{j=1}^r m_j = S$. (In applications, numerical methods are generally needed.)
2: **for all** $j = 1, \dots, r$ **do**
3: Compute a basis $\mathcal{B}_j$ for the eigenspace $E_{\mu_j} := \ker(\boldsymbol{A} - \mu_j \boldsymbol{I})$ and record its geometric multiplicity $g_j := \dim(E_{\mu_j})$.
4: **end for**
5: If $\sum_{j=1}^r g_j = S$, then $\boldsymbol{A}$ is diagonalizable and $\bigcup_{j=1}^r \mathcal{B}_j$ is an eigenbasis of $\mathbb{C}^S$. Otherwise, $\boldsymbol{A}$ is defective and has fewer than S linearly independent eigenvectors.

Remark 3.5.1. *For each eigenvalue μ_j, the eigenspace E_{μ_j} contains infinitely many vectors, including the zero vector; its nonzero elements are the eigenvectors corresponding to μ_j. Its geometric multiplicity satisfies $1 \le g_j \le m_j$.* ***If all entries of $\boldsymbol{A}$ and the eigenvalue μ_j are real-valued, then E_{μ_j} has a basis consisting of real-valued vectors.***

Example 3.5.1. *Let $\boldsymbol{P}$ be a transition probability matrix with $\boldsymbol{P}_{ij} = p(x_i, x_j)$.*

i) Let $\mathbf{1} := (1, 1, \dots, 1)^\intercal$. Since $\sum_{j=1}^S p(x_i, x_j) = 1$ for every fixed i, we have

$$\boldsymbol{P}\mathbf{1} = \mathbf{1}. \tag{3.5.1}$$

Therefore, $\mathbf{1}$ is an eigenvector of $\boldsymbol{P}$ corresponding to the eigenvalue 1.

ii) Eq. (3.3.4) implies

$$\boldsymbol{P}^\intercal \boldsymbol{\pi} = \boldsymbol{\pi},$$

where $\boldsymbol{\pi} = (\pi(x_1), \dots, \pi(x_S))^\intercal$, and π is the stationary distribution of the transition probability p. Therefore, $\boldsymbol{\pi}$ is an eigenvector of the transpose $\boldsymbol{P}^\intercal$ corresponding to the eigenvalue 1.

The following theorem is one of the most important theorems in linear algebra.

Theorem 3.5.1. *Let $\boldsymbol{A}$ be a real-valued symmetric S-by-S matrix.*

i) All eigenvalues $\lambda_1, \ldots, \lambda_S$ of $\boldsymbol{A}$ are real numbers. Without loss of generality, we always assume $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_S$.

ii) There exists an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_S\}$ of $\mathbb{R}^S$ consisting of eigenvectors of $\boldsymbol{A}$, with $\boldsymbol{A}\boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i$. Equivalently, within each eigenspace corresponding to a repeated eigenvalue, choose an orthonormal basis; eigenspaces corresponding to distinct eigenvalues are orthogonal. Arbitrary unit eigenvectors from the same eigenspace need not be orthogonal.

iii) Let $\boldsymbol{U}$ be the matrix with $\boldsymbol{u}_i$ as its i-th column, for $i = 1, \ldots, S$. Then, we have $\boldsymbol{A}\boldsymbol{U} = \boldsymbol{U}\,\mathrm{diag}\{\lambda_1, \ldots, \lambda_S\}$, i.e.,

$$\boldsymbol{A} = \boldsymbol{U}\,\mathrm{diag}\{\lambda_1, \ldots, \lambda_S\}\boldsymbol{U}^\intercal.$$

Remark 3.5.2.

- $\mathrm{diag}\{\lambda_1, \ldots, \lambda_S\}$ *denotes the following diagonal matrix*

$$\begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_S \end{pmatrix}.$$

- *Theorem 3.5.1 ii) implies that $\boldsymbol{U}$ is an orthogonal matrix, i.e., $\boldsymbol{U}^\intercal = \boldsymbol{U}^{-1}$.*
- *This theorem will play a critical role in the theory of **principal component analysis** (see Chapter 6).*
- *Wikipedia claims that Theorem 3.5.1 was first proved by Augustin-Louis Cauchy (you are not mistaken, "the Cauchy").*
- *(not required*) Theorem 3.5.1 is a special case of the spectral resolution of self-adjoint operators due to von Neumann (see Chapter 32.1 of Lax (2014)). In an infinite-dimensional space, a self-adjoint operator is different from one that is merely symmetric. In applications, for most people, the difference between self-adjointness and symmetry is not interesting. The following "historical notes" can be found at the end of Chapter 33 of Lax (2014). "In the 1960s, Friedrichs met Heisenberg, and used the occasion to express to him the deep gratitude of the community of mathematicians for having created quantum mechanics, which gave birth to the beautiful theory of operators in Hilbert space. Heisenberg allowed that this was so; Friedrichs then added that the mathematicians have, in some measure, returned the favor. Heisenberg looked noncommittal, so Friedrichs pointed out that it was a mathematician, von Neumann, who clarified the difference between a self-adjoint operator and one that is merely symmetric. 'What's the difference,' said Heisenberg."*

3.5.2 Properties of Transition Probability Matrices

The materials in this section are based on the lecture notes "Markov chains and mixing times" by Professor James R. Lee.

Throughout Section 3.5, we assume that π is a **reversible distribution** of the transition probability p (see Example 6.5.4 of Durrett (2019) or Eq. (18.8) of Klenke (2020)). i.e.,

$$\pi(x)p(x,y) = \pi(y)p(y,x), \quad \text{for all } x, y \in \mathcal{X} = \{x_1, \ldots, x_S\}. \tag{3.5.2}$$

Because $\sum_{x\in\mathcal{X}} \pi(x)p(x,y) = \sum_{x\in\mathcal{X}} \pi(y)p(y,x) = \pi(y)\sum_{x\in\mathcal{X}} p(y,x) = \pi(y)$, **a reversible distribution for p is necessarily a stationary distribution for p.** Although the asymptotic results in Section 3.4 do not need π to be a reversible distribution—just a stationary distribution, we assume Eq. (3.5.2) in this section for the following reasons

- The reversible assumption in Eq. (3.5.2) extremely simplifies the proof of the results in Section 3.4.
- As mentioned, the chapter on Markov chains is a preparation for the MCMC theory, where we only introduce the Metropolis-Hastings algorithm and Gibbs sampler. The Metropolis-Hastings algorithm satisfies the reversible assumption.

This section develops properties of reversible transition matrices that will be used to analyze the corresponding Markov chains.

Theorem 3.5.2. *Suppose $\boldsymbol{P} = (p(x_i, x_j))_{1\le i,j\le S}$ is a transition probability matrix satisfying Eq. (3.5.2) with $\pi(x_i) > 0$ for all $i = 1, \ldots, S$.*

i) 1 is an eigenvalue of $\boldsymbol{P}^\intercal$.

ii) All eigenvalues of $\boldsymbol{P}^\intercal$ are real-valued.

iii) All eigenvalues of $\boldsymbol{P}^\intercal$ are in the real-valued closed interval $[-1, 1]$.

The proof of Theorem 3.5.2 depends on the following lemma, which is the essential reason we need the reversible assumption in Eq. (3.5.2). The proof of this lemma is straightforward and left for homework.

Lemma 3.5.1. *Suppose $\boldsymbol{P} = (p(x_i, x_j))_{1\le i,j\le S}$ is a transition probability matrix satisfying Eq. (3.5.2) with $\pi(x_i) > 0$ for all $i = 1, \ldots, S$. Then, $\boldsymbol{D}^{-1/2}\boldsymbol{P}^\intercal\boldsymbol{D}^{1/2}$ is a symmetric matrix, where*

$$\begin{aligned}
\boldsymbol{D} &:= \operatorname{diag}\{\pi(x_1), \ldots, \pi(x_S)\}, \\
\boldsymbol{D}^{1/2} &:= \operatorname{diag}\left\{\sqrt{\pi(x_1)}, \ldots, \sqrt{\pi(x_S)}\right\}, \\
\boldsymbol{D}^{-1/2} &:= \operatorname{diag}\left\{\frac{1}{\sqrt{\pi(x_1)}}, \ldots, \frac{1}{\sqrt{\pi(x_S)}}\right\}.
\end{aligned}$$

(Since $\pi(x_i) > 0$ for all $i = 1, \dots, S$, the matrix $\boldsymbol{D}^{-1/2}$ is well-defined.)

Proof. (of Theorem 3.5.2)

i). Eq. (3.5.1) indicates that 1 is an eigenvalue of $\boldsymbol{P}$. Since $\boldsymbol{P}$ and $\boldsymbol{P}^\intercal$ share the same eigenvalues, 1 is an eigenvalue of $\boldsymbol{P}^\intercal$ as well.

ii). Let λ_i be an eigenvalue of $\boldsymbol{P}^\intercal$ corresponding to the eigenvector $\boldsymbol{v}_i$, i.e., $\boldsymbol{P}^\intercal \boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$, which further implies

$$\boldsymbol{P}^\intercal \boldsymbol{D}^{1/2} \left(\boldsymbol{D}^{-1/2} \boldsymbol{v}_i \right) = \lambda_i \boldsymbol{D}^{1/2} \left(\boldsymbol{D}^{-1/2} \boldsymbol{v}_i \right) \quad \Longrightarrow \quad \boldsymbol{D}^{-1/2} \boldsymbol{P}^\intercal \boldsymbol{D}^{1/2} \left(\boldsymbol{D}^{-1/2} \boldsymbol{v}_i \right) = \lambda_i \left(\boldsymbol{D}^{-1/2} \boldsymbol{v}_i \right).$$

Then λ_i is an eigenvalue of $\boldsymbol{D}^{-1/2} \boldsymbol{P}^\intercal \boldsymbol{D}^{1/2}$. Theorem 3.5.1, together with Lemma 3.5.1, implies that λ_i is real. Indeed, $\boldsymbol{D}^{-1/2} \boldsymbol{P}^\intercal \boldsymbol{D}^{1/2}$ and $\boldsymbol{P}^\intercal$ are similar, so they have the same eigenvalues,‡ while the symmetry of $\boldsymbol{D}^{-1/2} \boldsymbol{P}^\intercal \boldsymbol{D}^{1/2}$ guarantees that these eigenvalues are real.

iii). Suppose λ is an eigenvalue of $\boldsymbol{P}^\intercal$, and $\boldsymbol{v} := (v_1, \dots, v_S)^\intercal \neq \boldsymbol{0}$ is the corresponding eigenvector, i.e., they satisfy the equation $\boldsymbol{P}^\intercal \boldsymbol{v} = \lambda \boldsymbol{v}$. Then,

$$\begin{aligned} |\lambda| \left(\sum_{i=1}^S |v_i| \right) &= \sum_{i=1}^S |\lambda v_i| \\ &= \sum_{i=1}^S \left| \sum_{j=1}^S \boldsymbol{P}_{ji} v_j \right| \\ &\leq \sum_{j=1}^S \left(\sum_{i=1}^S \boldsymbol{P}_{ji} \right) |v_j| \\ &= \sum_{j=1}^S |v_j|. \end{aligned}$$

Since $\sum_{i=1}^S |v_i| > 0$, we have $|\lambda| \leq 1$. The proof is completed. □

The following theorem illustrates the irreducibility of transition probability matrices from the perspective of eigenvalues.

Theorem 3.5.3. *Suppose $\boldsymbol{P} = (p(x_i, x_j))_{1 \leq i,j \leq S}$ is a transition probability matrix satisfying Eq. (3.5.2) with $\pi(x_i) > 0$ for all $i = 1, \dots, S$. Furthermore, if $\boldsymbol{P}$ is **irreducible**, then $\boldsymbol{\pi} = (\pi(x_1), \dots, \pi(x_S))^\intercal$ is the unique eigenvector of $\boldsymbol{P}^\intercal$ corresponding to 1 whose entries are nonnegative and sum to 1.*

Proof. Let $\boldsymbol{v} = (v_1, \dots, v_S)^\intercal$ be a real-valued eigenvector of $\boldsymbol{P}^\intercal$ corresponding to the eigenvalue 1, i.e., $\boldsymbol{P}^\intercal \boldsymbol{v} = \boldsymbol{v}$. Define the matrix† $\boldsymbol{L} := \boldsymbol{D} - \boldsymbol{P}^\intercal \boldsymbol{D}$ and vector $\boldsymbol{w} = (w_1, \dots, w_S)^\intercal := \boldsymbol{D}^{-1} \boldsymbol{v}$. Using the reversible assumption in

‡ Two matrices are similar if and only if they represent the same linear operator with respect to possibly different bases.

† You will learn—perhaps you have already learned—that $\boldsymbol{L}$ is a Laplacian operator in some sense.

Eq. (3.5.2), it is easy to verify the following identity, which is left for homework

$$\boldsymbol{w}^\intercal \boldsymbol{L}\boldsymbol{w} = \frac{1}{2}\sum_{i,j=1}^{S} \pi(x_i)\cdot p(x_i,x_j)\cdot (w_i - w_j)^2 . \tag{3.5.3}$$

Additionally, we have the following derivation

$$\boldsymbol{P}^\intercal \boldsymbol{v} = \boldsymbol{v} \implies \boldsymbol{P}^\intercal \boldsymbol{D}\boldsymbol{D}^{-1}\boldsymbol{v} = \boldsymbol{D}\boldsymbol{D}^{-1}\boldsymbol{v} \implies \boldsymbol{D}\boldsymbol{w} - \boldsymbol{P}^\intercal \boldsymbol{D}\boldsymbol{w} = 0 \implies \boldsymbol{L}\boldsymbol{w} = 0 \implies \boldsymbol{w}^\intercal \boldsymbol{L}\boldsymbol{w} = 0.$$

Since $\pi(x_i) > 0$ for all i, the derivation above indicates that $p(x_i,x_j)\cdot(w_i - w_j)^2 = 0$ for all $i,j \in \{1,\dots,S\}$. For any pair $x_i, x_j \in \mathcal{X}$, since the Markov chain of interest is **irreducible**, it is easy to verify that there exists a sequence of states $x_{l_0}, x_{l_1}, \dots, x_{l_K} \in \mathcal{X}$ such that $x_{l_0} = x_i$, $x_{l_K} = x_j$, and $p(x_{l_{k-1}}, x_{l_k}) > 0$ for all $k \in \{1,\dots,K\}$ (the proof of this fact is left for homework). Therefore, $p(x_{l_{k-1}}, x_{l_k}) > 0$ for all $k \in \{1,\dots,K\}$ implies

$$w_i = w_{l_0} = w_{l_1} = \cdots = w_{l_{K-1}} = w_{l_K} = w_j.$$

Hence, $\boldsymbol{w}$ must be of the form $(\alpha,\alpha,\dots,\alpha)^\intercal$ for some $\alpha \in \mathbb{R}$. Then, $\boldsymbol{v} = \boldsymbol{D}\boldsymbol{w} = \alpha(\pi(x_1),\dots,\pi(x_S))^\intercal = \alpha\boldsymbol{\pi}$, i.e., every eigenvector of $\boldsymbol{P}^\intercal$ corresponding to the eigenvalue 1 is a scalar multiple of $\boldsymbol{\pi}$. Among all vectors in $\{\alpha\boldsymbol{\pi} \,|\, \alpha \in \mathbb{R}\}$, $\boldsymbol{\pi}$ is the unique one whose entries are nonnegative and sum to 1. □

The following theorem illustrates the aperiodicity from the perspective of eigenvalues. The proof of this is left for homework.

Theorem 3.5.4. *Suppose $\boldsymbol{P} = (p(x_i,x_j))_{1\le i,j\le S}$ is a transition probability matrix satisfying Eq. (3.5.2) with $\pi(x_i) > 0$ for all $i = 1,\dots,S$, and $\boldsymbol{P}$ is irreducible. Furthermore, if $\boldsymbol{X}$ is aperiodic, -1 is not an eigenvalue of $\boldsymbol{P}^\intercal$.*

Let us recall the following periodic transition probability matrix we mentioned in Example 3.4.1

$$\boldsymbol{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

From the spectral viewpoint, the essential reason that $\boldsymbol{P}$ is periodic is that -1 is an eigenvalue of it. See the following `R` code

```
P=rbind(c(0,1), c(1,0))
eigen(P)

# Output:
# eigen() decomposition
# $values
# [1]  1 -1
```

```
8
9  # $vectors
10 # [,1]       [,2]
11 # [1,] 0.7071068 -0.7071068
12 # [2,] 0.7071068  0.7071068
```

3.5.3 Convergence and Its Speed

Under the reversibility assumption in Eq. (3.5.2), we now give a simple proof of Theorem 3.4.3 and a rough estimate of the rate at which $(\boldsymbol{P}^n)_{ij}$ converges to $\pi(x_j)$.

Proof of Theorem 3.4.3 Under the Reversible Assumption

Theorem 3.5.2 shows that $\boldsymbol{P}^\intercal$ has real-valued eigenvalues $1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_S \geq -1$, and these eigenvalues are also eigenvalues of the real symmetric matrix $\boldsymbol{D}^{-1/2}\boldsymbol{P}^\intercal\boldsymbol{D}^{1/2}$. Theorem 3.5.3 implies that $\boldsymbol{\pi}$ is an eigenvector corresponding to $\lambda_1 = 1$, and $1 = \lambda_1 > \lambda_2$. Additionally, Theorem 3.5.4 implies $\lambda_S > -1$. Let $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_S$ be the eigenvectors of $\boldsymbol{D}^{-1/2}\boldsymbol{P}^\intercal\boldsymbol{D}^{1/2}$ forming an orthonormal basis. Then,

$$\boldsymbol{D}^{-1/2}\boldsymbol{P}^\intercal\boldsymbol{D}^{1/2}\boldsymbol{u}_i = \lambda_i\boldsymbol{u}_i \quad \Longrightarrow \quad \boldsymbol{P}^\intercal\left(\boldsymbol{D}^{1/2}\boldsymbol{u}_i\right) = \lambda_i\left(\boldsymbol{D}^{1/2}\boldsymbol{u}_i\right).$$

The collection $\{\boldsymbol{D}^{1/2}\boldsymbol{u}_1, \ldots, \boldsymbol{D}^{1/2}\boldsymbol{u}_S\}$ forms a linear basis of $\mathbb{R}^S$, as $\boldsymbol{D}^{1/2}\boldsymbol{U}$ has rank S. For any initial distribution $\boldsymbol{\mu} = (\mu(x_1), \ldots, \mu(x_S))^\intercal$ with $\mu(x_i) = \mathbb{P}(X_0 = x_i)$ for all i, we have $\boldsymbol{\mu} = \sum_{i=1}^S \alpha_i\boldsymbol{v}_i$, where $\boldsymbol{v}_i := \boldsymbol{D}^{1/2}\boldsymbol{u}_i$. Then,

$$(\boldsymbol{P}^\intercal)^n\boldsymbol{\mu} = \sum_{i=1}^S \alpha_i(\boldsymbol{P}^\intercal)^n\boldsymbol{v}_i = \sum_{i=1}^S \alpha_i\lambda_i^n\boldsymbol{v}_i. \tag{3.5.4}$$

This equation implies

$$\lim_{n\to\infty}(\boldsymbol{P}^\intercal)^n\boldsymbol{\mu} = \sum_{i=1}^S \alpha_i \cdot \lim_{n\to\infty}(\boldsymbol{P}^\intercal)^n\boldsymbol{v}_i = \sum_{i=1}^S \alpha_i\left(\lim_{n\to\infty}\lambda_i^n\right)\boldsymbol{v}_i = \alpha_1\boldsymbol{v}_1,$$

where $\lim_{n\to\infty}\lambda_i^n = 0$ for all $i \geq 2$ as $|\lambda_i| < 1$ for all $i \geq 2$. Since all entries of $(\boldsymbol{P}^\intercal)^n\boldsymbol{\mu}$ are nonnegative, all entries of $\alpha_1\boldsymbol{v}_1$ are nonnegative. Additionally,

$$1 = \lim_{n\to\infty}\sum_{i=1}^S \mathbb{P}(X_n = x_i) = \lim_{n\to\infty}\sum_{i=1}^S \left((\boldsymbol{P}^\intercal)^n\boldsymbol{\mu}\right)_i = \sum_{i=1}^S \left(\lim_{n\to\infty}(\boldsymbol{P}^\intercal)^n\boldsymbol{\mu}\right)_i = \sum_{i=1}^S \left(\alpha_1\boldsymbol{v}_1\right)_i.$$

That is, $\alpha_1 \boldsymbol{v}_1$ is an eigenvector of $\boldsymbol{P}^\intercal$ corresponding to eigenvalue 1, and the sum of its entries is equal to 1. Theorem 3.5.3 indicates $\alpha_1 \boldsymbol{v}_1 = \boldsymbol{\pi}$. Hence,

$$\lim_{n\to\infty} (\boldsymbol{P}^\intercal)^n \boldsymbol{\mu} = \boldsymbol{\pi},$$

for any PMF $\boldsymbol{\mu} = (\mu(x_1), \dots, \mu(x_S))^\intercal$. □

Speed of Convergence

The speed of convergence of Markov chains is a deep topic. Here, we only provide a very rough estimation of the convergence rate to give you an idea of what influences the speed of convergence.

In the proof above, we showed $\alpha_1 \boldsymbol{v}_1 = \boldsymbol{\pi}$. Then we have $\boldsymbol{\mu} = \boldsymbol{\pi} + \sum_{i=2}^S \alpha_i \boldsymbol{v}_i$ and

$$(\boldsymbol{P}^\intercal)^n \boldsymbol{\mu} = \boldsymbol{\pi} + \sum_{i=2}^S \lambda_i^n \cdot \alpha_i \boldsymbol{v}_i.$$

Additionally, irreducibility and aperiodicity imply $1 > \lambda_2 \geq \cdots \geq \lambda_S > -1$; then, we have $|\lambda_i| \leq \max\{|\lambda_2|, |\lambda_S|\}$. For the Euclidean norm $\|\cdot\|$ on $\mathbb{R}^S$, defined by $\|\boldsymbol{\mu}\| = \sqrt{\sum_{i=1}^S \mu_i^2}$, we have

$$\begin{aligned} \|(\boldsymbol{P}^\intercal)^n \boldsymbol{\mu} - \boldsymbol{\pi}\| &= \left\| \sum_{i=2}^S \lambda_i^n \cdot \alpha_i \boldsymbol{v}_i \right\| \\ &\leq \left[\sum_{i=2}^S |\alpha_i| \cdot \|\boldsymbol{v}_i\| \right] \cdot \max\{|\lambda_2|^n, |\lambda_S|^n\}. \end{aligned}$$

The inequality above indicates that the convergence rate depends on the eigenvalues of $\boldsymbol{P}$.

3.6 Homework for Chapter 3

Problem 3.6.1. *Prove Theorem 3.1.1. (Hint: first expand the joint probability using the chain rule*

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mathbb{P}(X_0 = x_0) \prod_{k=1}^n \mathbb{P}(X_k = x_k \mid X_0 = x_0, \dots, X_{k-1} = x_{k-1}),$$

and then apply the Markov property to every factor.)

Problem 3.6.2. *For the transition matrix $\boldsymbol{P}$ in Eq. (3.2.3) and the initial distribution $\mu = \mathrm{Unif}(\{x_1, \dots, x_7\})$,*

i) write the programming code of Algorithm 11;

ii) generate 10 paths of length 10000 from the Markov chain corresponding to $\boldsymbol{P}$ and μ;

iii) for each generated path, summarize the frequencies of x_i for $i = 1, \dots, 7$ in a table.

Problem 3.6.3. *Prove Theorem 3.1.4. As a special case, derive Eq. (3.2.5). Explain why, when $d = 1$, the walk moves from x to each of $x - 1$ and $x + 1$ with probability $1/2$.*

Problem 3.6.4. *Let $X_0 = 0$ and $X_n = \sum_{i=1}^{n} \xi_i$, where $\xi_1, \xi_2, \ldots$ are iid and satisfy $\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = 1/2$.*

i) Show that $\mathbb{P}(X_{2m+1} = 0) = 0$ and

$$\mathbb{P}(X_{2m} = 0) = \binom{2m}{m} 2^{-2m}, \qquad m = 1, 2, \ldots.$$

ii) Use Stirling's formula to show that

$$\mathbb{P}(X_{2m} = 0) \sim \frac{1}{\sqrt{\pi m}} \qquad \text{as } m \to \infty.$$

iii) Deduce that $\sum_{n=1}^{\infty} \mathbb{P}(X_n = 0 \mid X_0 = 0) = \infty$, and use Theorem 3.2.2 to conclude that the one-dimensional simple symmetric random walk is recurrent.

Problem 3.6.5. *Prove the identity $\mathbb{E}(N_y \mid X_0 = x) = \sum_{n=1}^{\infty} \mathbb{P}(X_n = y \mid X_0 = x) = \rho_{xy} \cdot \sum_{k=1}^{\infty} \rho_{yy}^{k-1}$ in Eq. (3.2.1). In your proof, you may assume that the order of $\mathbb{E}$ and $\sum$ can be interchanged; this assumption is justified by Fubini's theorem.*

Problem 3.6.6. *Please design a numerical method for approximately computing the probability $\mathbb{P}(T_{\mathbf{0}} < \infty \mid X_0 = \mathbf{0})$, for $d = 2$ and 3, in Table 3.1. Please provide the description and justification of your design, the programming code, and your estimation results. In addition, use your approximation result to justify the claim, "A drunk man will eventually find his way home, but a drunk bird may get lost forever."*

Problem 3.6.7. *Prove Claim 3.2.5.*

Problem 3.6.8. *In this question, you will be asked to prove a special case of Theorem 3.3.1. In solving this problem, you will see how algebraic topology (MATH 2410), linear algebra, and probability theory interact with each other.*

We first state a generalized version of the Brouwer fixed-point theorem: ***Let $K \subseteq \mathbb{R}^S$ be convex, bounded, and closed. Every continuous function $f : K \to K$ has a fixed point, i.e., there exists $x \in K$ such that $f(x) = x$.*** *This theorem is a result in algebraic topology.*

(Notice: To use the Brouwer fixed-point theorem, you have to verify that $f(K) \subseteq K$, i.e., $f(\boldsymbol{\xi}) \in K$ for all $\boldsymbol{\xi} \in K$.)

We assume that the Brouwer fixed-point theorem is true and apply it to show the following: ***Let $\boldsymbol{P} = (p(x_i, x_j))_{1 \le i,j \le S}$ be the transition probability matrix of a homogeneous Markov chain $\{X_n\}_{n=0}^{\infty}$ taking values in a finite state space $\mathcal{X} = \{x_1, x_2, \ldots, x_S\}$. Then, there exists $\boldsymbol{\pi} \in \Delta$ such that $\boldsymbol{P}^{\intercal}\boldsymbol{\pi} = \boldsymbol{\pi}$, where***

$\Delta = \{\boldsymbol{\xi} = (\xi_1, \ldots, \xi_S)^\intercal \in \mathbb{R}^S \mid \sum_{i=1}^S \xi_i = 1$ ***and*** $\xi_i \geq 0$ ***for all*** $i\}$***. Furthermore, if all entries of*** $\boldsymbol{P}$ ***are positive, i.e.,*** $p(x_i, x_j) > 0$ ***for all*** i ***and*** j***, all entries of the vector*** $\boldsymbol{\pi}$ ***are positive.***

Hint: Apply the Brouwer fixed-point theorem with setting $K = \Delta$ *and* $f(\boldsymbol{\xi}) = \boldsymbol{P}^\intercal \boldsymbol{\xi}$*. In your proof, you do not need to show that* $K = \Delta$ *is bounded and closed, which might be outside the scope of APMA 1690. But you need to show that* $K = \Delta$ *is convex (see the definition of convex sets by clicking the link).*

Remark: *The Brouwer fixed-point argument proves that every transition matrix on a finite state space has at least one stationary distribution; irreducibility is not needed for existence. The assumption that all entries of* $\boldsymbol{P}$ *are strictly positive is used here only to show that every entry of the resulting stationary vector* $\boldsymbol{\pi}$ *is positive. More generally, irreducibility guarantees both strict positivity and uniqueness of the stationary distribution. The set* Δ *defined above is usually referred to as the* ***probability simplex****, which is a fundamental building block of the simplicial homology theory in algebraic topology.*

Problem 3.6.9. *Prove Lemma 3.5.1.*

Problem 3.6.10. *Verify the following steps in the proof of Theorem 3.5.3.*

i) Let $\{X_n\}_{n=0}^\infty$ *be a Markov chain taking values in the finite state space* $\mathcal{X} = \{x_1, \ldots, x_S\}$*, and* p *is its transition probability. Please prove the following fact: if* p *is irreducible, for any pair* $x_i, x_j \in \mathcal{X}$*, there exists a sequence of states* $y_0, y_1, \ldots, y_K$ *such that* $y_0 = x_i$*,* $y_K = x_j$*, and* $p(y_{k-1}, y_k) > 0$ *for all* $k \in \{1, \ldots, K\}$*.*

ii) Eq.(3.5.3). (Hint: use the reversible assumption in Eq. (3.5.2).)

Problem 3.6.11. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain on $\mathcal{X} = \{x_1, x_2, x_3\}$ with transition matrix*

$$\boldsymbol{P} = \begin{pmatrix} 0 & 1/3 & 2/3 \\ 1/3 & 0 & 2/3 \\ 1 & 0 & 0 \end{pmatrix}.$$

i) Prove that the chain is irreducible.

ii) Prove that the chain is aperiodic.

iii) Compute its stationary distribution π.

iv) Define $f(x_k) = k^2$ for $k = 1, 2, 3$. Compute $\sum_{x \in \mathcal{X}} f(x)\pi(x)$.

v) Compute $\lim_{n\to\infty} \boldsymbol{P}^n$ and explain how your answer follows from irreducibility, aperiodicity, and the stationary distribution.

Chapter 4

Markov Chain Monte Carlo

".....data, signals, images. The role of mathematics is in developing new representations for those data."

— David Donoho

Markov chain Monte Carlo (MCMC) is designed for expectations under distributions from which direct sampling is difficult. One important example is a posterior distribution in Bayesian statistics, especially when computing the following Bayes estimator (see also Eq. (2.1.1)):

$$\widehat{\theta}_{Bayes} := \mathbb{E}(\theta | X_1 = x_1, \ldots, X_n = x_n) = \int \theta \cdot p(\theta | x_1, \ldots, x_n) d\theta,$$
$$\text{where} \quad p(\theta | x_1, \ldots, x_n) := \frac{\prod_{i=1}^n f(x_i|\theta) f(\theta)}{\int [\prod_{i=1}^n f(x_i|\theta') f(\theta')] \, d\theta'}.$$

Chapter 2 introduced this Bayesian motivation; further discussion appears in Gelman et al. (2013) and Tierney (1994). The present chapter develops the general MCMC framework and its principal algorithms. We begin with a second motivating example from statistical mechanics, the Ising model, which will also serve as a running computational example.

4.1 Motivation: The Ising Model

The Ising model† provides both a motivation for MCMC and a concrete target on which to apply its algorithms. Its local interaction structure will also lead naturally to the Gibbs random fields developed in Chapter 5.

See Donoho (2018).

† German pronunciation is like the English word "easing," see the remark in Example 18.16 of Klenke (2020); the author of this textbook—Professor Achim Klenke—is a Professor für Mathematische Stochastik Johannes Gutenberg-Universität Mainz; so the remark on the German pronunciation is quite trustable. Also, see Wikipedia for the German pronunciation.

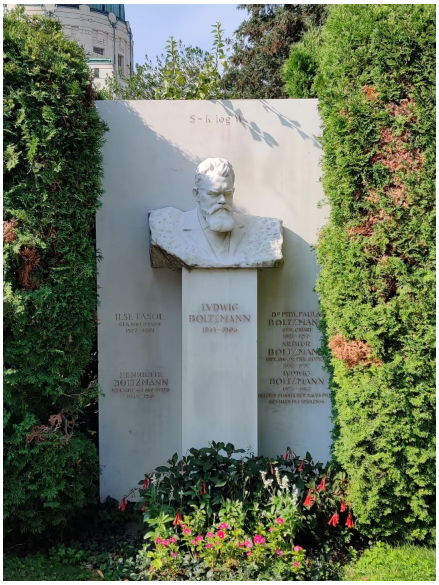


Figure 4.1: A photo of Boltzmann's Grave. This photo was taken by Lecheng Ren in 2022. Lecheng is a Ph.D. student in the Department of Physics at Brown University.

4.1.1 Boltzmann Distributions

In statistical mechanics, a **Boltzmann distribution** is a probability distribution that gives the probability that an equilibrium system will be in a certain state $\boldsymbol{s}$ as a function of that state's energy, say $H(\boldsymbol{s})$, and the temperature T of the system. The distribution is expressed in the form

$$p(\boldsymbol{s}) \propto \exp\left\{-\frac{H(\boldsymbol{s})}{k_B \cdot T}\right\}, \tag{4.1.1}$$

where $\boldsymbol{s}$ denotes a state, k_B is the Boltzmann constant, and T denotes the thermodynamic temperature.

4.1.2 Ising Model

The Ising model gives a model for the energy function $H(\boldsymbol{s})$ in Eq. (4.1.1).

Periodic Lattices

Suppose atoms are placed at the sites of a d-dimensional lattice Λ_N^d of size N (i.e., when $d = 2$, Λ_N^2 is an N-by-N lattice) with the periodic boundary condition, i.e.,

$$\Lambda_N^d = (\mathbb{Z}_N)^d,$$
$$\mathbb{Z}_N = \mathbb{Z}/\sim, \text{ where } n_1 \sim n_2 \text{ if and only if } n_1 \equiv n_2 \bmod N.$$

The structure $\mathbb{Z}_N$ is the quotient ring of integers modulo N; under addition, it is the additive cyclic group of integers modulo N. We identify $\mathbb{Z}_N$ with $\{0, 1, \ldots, N-1\}$ and perform coordinate arithmetic modulo N. In

particular, for $N \geq 3$, a site $(r, c) \in \Lambda_N^2$ has the four distinct neighbors

$$\mathcal{N}(r, c) = \big\{(r+1, c), (r-1, c), (r, c+1), (r, c-1)\big\} \pmod{N}. \tag{4.1.2}$$

Thus, the periodic boundary condition removes boundary and corner cases: moving past one side of the lattice returns on the opposite side. As an example, the 2D lattice Λ_3^2 of size 3 with periodic boundary condition is illustrated in Figure 4.2. Illustrations of lattices and the concept of **neighbors** are presented in Figures 4.3 and 4.4. The number of sites of Λ_N^d is N^d.

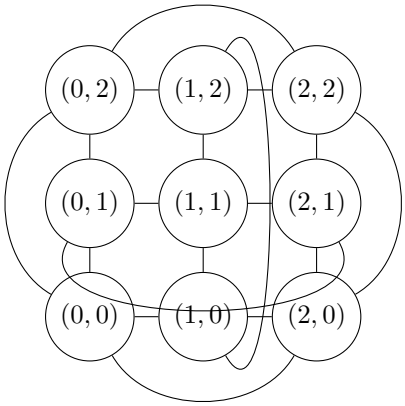


Figure 4.2: The lattice Λ_3^2 for a finite-volume 2D Ising model. The periodic boundary condition is also illustrated in this figure.

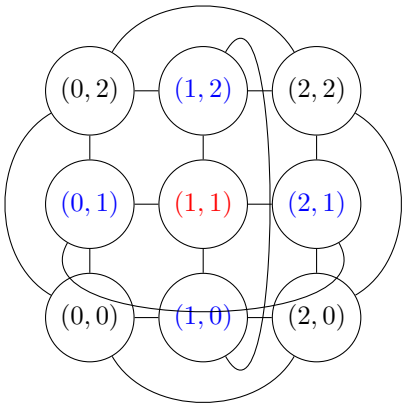


Figure 4.3: The lattice Λ_3^2 for a finite-volume 2D Ising model. The **neighbors** of site $(1, 1)$ (red) are presented in blue.

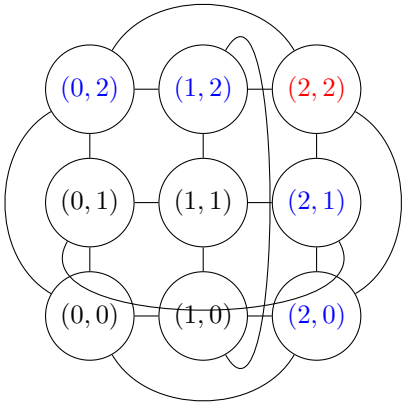


Figure 4.4: The lattice Λ_3^2 for a finite-volume 2D Ising model. The **neighbors** of site $(2, 2)$ (red) are presented in blue.

Energy Function

At each lattice site $k \in \Lambda_N^d$, there exists an atom with spin $s_k \in \{-1, 1\}$. A **spin configuration** $\boldsymbol{s} = (s_k)_{k \in \Lambda_N^d}$ assigns a spin value to each lattice site. The configuration space is $\{-1, 1\}^{\Lambda_N^d}$ and contains 2^{N^d} states (in two dimensions, 2^{N^2} states). Because of thermal fluctuations, $\boldsymbol{s}$ is random. Each spin configuration corresponds to an energy value $H(\boldsymbol{s})$. The Ising model for $H(\boldsymbol{s})$ is based on the following ideas

- Neighboring spins that agree have a lower energy than those that disagree.
- The system tends to the lowest energy but heat disturbs this tendency, thus creating the possibility of different structural phases.

Based on these ideas, the energy function $H(\boldsymbol{s})$ is modeled as follows

$$H_N^d(\boldsymbol{s}) := -\sum_{\langle i,j \rangle} J_{ij} s_i s_j - \mu \sum_j h_j s_j, \tag{4.1.3}$$

where $\langle i, j \rangle$ indicates that lattice sites i and j are neighbors and each undirected edge is counted once, J_{ij} measures the interaction strength between sites i and j, $\{h_j\}_{j \in \Lambda_N^d}$ is an external magnetic field, and the magnetic moment is given by μ. To avoid complications in our discussion, we assume that there is no external magnetic field, i.e., $h_j = 0$ for all j, and all neighbors have the same ferromagnetic interaction strength, i.e., $J_{ij} \equiv J > 0$. Then, the model in Eq. (4.1.3) becomes

$$H_N^d(\boldsymbol{s}) := -J \sum_{\langle i,j \rangle} s_i s_j. \tag{4.1.4}$$

Let $\beta_{\text{phys}} = (k_B T)^{-1}$ be the physical inverse temperature and define the dimensionless coupling

$$\beta := \beta_{\text{phys}} J = \frac{J}{k_B T}.$$

From this point onward, we measure energy in units of J, replace H_N^d/J by H_N^d, and use β for this dimensionless coupling. Thus, in the adopted convention, $J = 1$ and $H_N^d(\boldsymbol{s}) = -\sum_{\langle i,j \rangle} s_i s_j$.

As mentioned, because of the perturbation from heat, spin configuration $\boldsymbol{s}$ is a random variable taking values in $\{-1, 1\}^{\Lambda_N^d}$. Based on Eq. (4.1.1), the probability of the spin configuration $\boldsymbol{s}$ is the following

$$\pi_{N,\beta}^d(\boldsymbol{s}) = \frac{\exp\left\{-\beta H_N^d(\boldsymbol{s})\right\}}{Z_{N,\beta}} = \frac{\exp\left\{\beta \sum_{\langle i,j \rangle} s_i s_j\right\}}{Z_{N,\beta}}, \tag{4.1.5}$$

where

$$Z_{N,\beta} := \sum_{\boldsymbol{s} \in \{-1,1\}^{\Lambda_N^d}} \exp\left\{\beta \sum_{\langle i,j \rangle} s_i s_j\right\}$$

is the partition function. The two minus signs—one in the Boltzmann factor and one in the ferromagnetic energy—cancel. Reversing this sign changes the model from ferromagnetic to antiferromagnetic. The letter Z stands for the German word *Zustandssumme*, "sum over states."

4.1.3 Curie Temperature

The Curie temperature is the one above which certain materials lose their permanent magnetic properties. The Curie temperature is named after Pierre Curie. In this section, we describe the Curie temperature using the Ising model. The discussion in this section provides a nontrivial question that can be solved by MCMC algorithms.

Macroscopically, the individual spins cannot be observed, but the following average magnetization can

$$m_N^d(\beta) := \sum_{\boldsymbol{s} \in \{-1,1\}^{\Lambda_N^d}} \pi_{N,\beta}^d(\boldsymbol{s}) \left| \frac{\sum_{i \in \Lambda_N^d} s_i}{N^d} \right| = \mathbb{E}_{\pi_{N,\beta}^d} \left| \frac{\sum_{i \in \Lambda_N^d} s_i}{N^d} \right|. \tag{4.1.6}$$

For a very large system, the relevant quantity is the thermodynamic limit (when it exists)

$$m^d(\beta) := \lim_{N \to \infty} m_N^d(\beta). \tag{4.1.7}$$

The quantity $m^d(\beta)$ is an order parameter: $m^d(\beta) > 0$ indicates macroscopic magnetic order, whereas $m^d(\beta) = 0$ indicates its absence. The phase transition is a property of the thermodynamic limit, not of any fixed finite lattice. For the two-dimensional square-lattice model, the Peierls argument establishes the existence of an ordered low-temperature phase (Peierls, 1936); see also Bonati (2014). Thus, there is a critical dimensionless inverse temperature

β_C such that

$$m^2(\beta) = \begin{cases} 0, & 0 < \beta \le \beta_C, \\ > 0, & \beta > \beta_C. \end{cases}$$

Equivalently,

$$\beta_C := \inf\{\beta > 0 : m^2(\beta) > 0\}, \qquad T_C = \frac{J}{k_B \beta_C}. \tag{4.1.8}$$

In the energy units $J = 1$ adopted above, this becomes $T_C = (k_B \beta_C)^{-1}$. Onsager's exact solution identifies the critical point through $\sinh(2\beta_C) = 1$ (Onsager, 1944), so

$$\boxed{\beta_C = \frac{1}{2}\log(1 + \sqrt{2}) \approx 0.4406868.} \tag{4.1.9}$$

Yang's spontaneous-magnetization formula (Yang, 1952) gives the sharper description

$$m^2(\beta) = \begin{cases} 0, & 0 < \beta \le \beta_C, \\ \left[1 - \sinh^{-4}(2\beta)\right]^{1/8}, & \beta > \beta_C. \end{cases} \tag{4.1.10}$$

The exponent $1/8$ describes the behavior just below the Curie temperature. **Below T_C (equivalently, above β_C), the model is magnetically ordered; above T_C, it is disordered.** Two representative spin configurations are presented in Figure 4.5. For every fixed finite N, $m_N^2(\beta)$ varies smoothly and is positive because of the absolute value in Eq. (4.1.6); a sharp phase transition emerges only as $N \to \infty$. The exact zero-field solution and critical point are due to Onsager (1944). The first published derivation of the spontaneous magnetization in Eq. (4.1.10) is due to Yang (1952). Professor Yang completed this work in 1951 and published it in 1952.

Exact Curie temperatures are difficult to derive in dimensions $d \ge 2$. In his 1924 Ph.D. thesis, Ernst Ising solved the $d = 1$ model, for which $T_C = 0$ in an appropriate sense, or equivalently no positive Curie temperature exists. The MCMC algorithms developed later in this chapter provide a numerical route: approximate the limiting expectation $m^d(\beta)$ in Eq. (4.1.7) over a range of $\beta > 0$ and examine the resulting transition behavior.

4.2 General Theory

"*Perfection is not just about control. It's also about letting go.*"

— a line in the movie *Black Swan*

With the Ising model as motivation, we now state the general MCMC problem. Chapter 3 began with a known

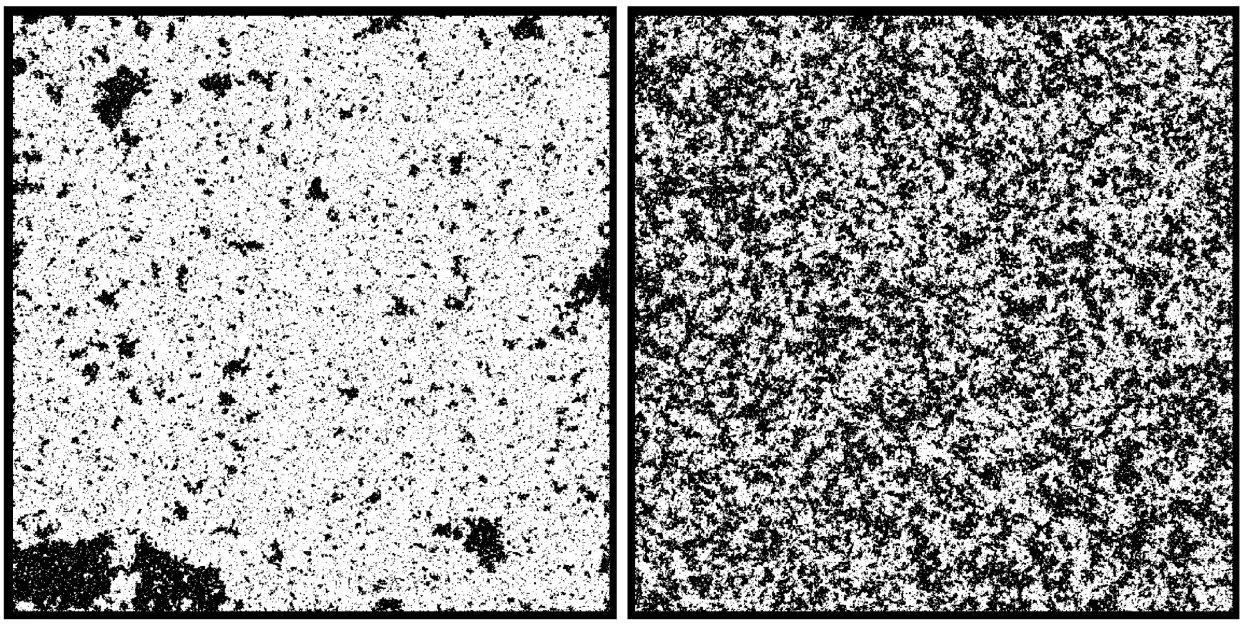
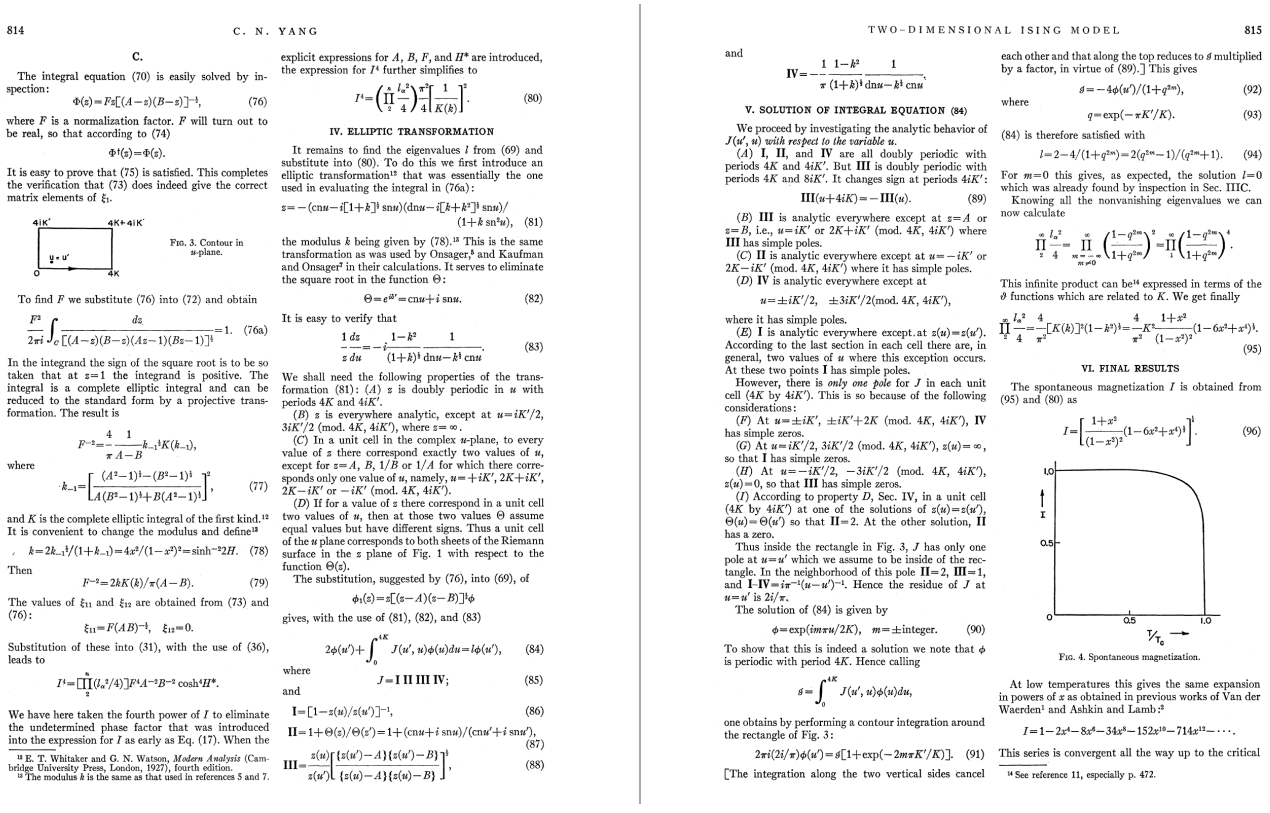


Figure 4.5: Equilibrium states of the Ising model on a 2-dimensional 800×800 lattice Λ_{800}^2 (black dot = spin $+1$). The two panels present spin configurations below (left) and above (right) the Curie temperature. This is Figure 18.5 of Klenke (2020). At temperatures above the Curie temperature, thermal fluctuations disrupt the low-temperature ordered configuration shown in the left panel, producing a disordered mixture of spins $+1$ (black) and -1 (white).



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C.

The integral equation (70) is easily solved by inspection:

$$\Phi(z) = Fz[(A-z)(B-z)]^{-\frac{1}{2}}, \tag{76}$$

where F is a normalization factor. F will turn out to be real, so that according to (74)

$$\Phi^\dagger(z) = \Phi(z).$$

It is easy to prove that (75) is satisfied. This completes the verification that (73) does indeed give the correct matrix elements of ξ_1.

FIG. 3. Contour in u-plane.

To find F we substitute (76) into (72) and obtain

$$\frac{F^2}{2\pi i}\int_C \frac{dz}{[(A-z)(B-z)(Az-1)(Bz-1)]^{\frac{1}{2}}} = 1. \tag{76a}$$

In the integrand the sign of the square root is to be so taken that at $z=1$ the integrand is positive. The integral is a complete elliptic integral and can be reduced to the standard form by a projective transformation. The result is

$$F^{-2} = -\frac{4}{\pi}\frac{1}{A-B}k_{-1}^{\frac{1}{2}}K(k_{-1}),$$

where

$$k_{-1} = \left[\frac{(A^2-1)^{\frac{1}{2}} - (B^2-1)^{\frac{1}{2}}}{A(B^2-1)^{\frac{1}{2}} + B(A^2-1)^{\frac{1}{2}}}\right]^2, \tag{77}$$

and K is the complete elliptic integral of the first kind.[12] It is convenient to change the modulus and define[13]

$$k = 2k_{-1}^{\frac{1}{2}}/(1+k_{-1}) = 4x^2/(1-x^2)^2 = \sinh^{-2}2H. \tag{78}$$

Then

$$F^{-2} = 2kK(k)/\pi(A-B). \tag{79}$$

The values of ξ_{11} and ξ_{12} are obtained from (73) and (76):

$$\xi_{11} = F(AB)^{-\frac{1}{2}}, \quad \xi_{12} = 0.$$

Substitution of these into (31), with the use of (36), leads to

$$I^4 = [\prod_1^n (l_n^2/4)]^2 A^{-2}B^{-2}\cosh^4 H^*.$$

We have here taken the fourth power of I to eliminate the undetermined phase factor that was introduced into the expression for I as early as Eq. (17). When the explicit expressions for A, B, F, and H^* are introduced, the expression for I^4 further simplifies to

$$I^4 = \left(\prod_1^\infty \frac{l_n^2}{4}\right)^2 \left[\frac{1}{K(k)}\right]^2. \tag{80}$$

IV. ELLIPTIC TRANSFORMATION

It remains to find the eigenvalues l from (69) and substitute into (80). To do this we first introduce an elliptic transformation[13] that was essentially the one used in evaluating the integral in (76a):

$$z = -(\mathrm{cn}u - i[1+k]^{\frac{1}{2}}\,\mathrm{sn}u)(\mathrm{dn}u - i[k+k^2]^{\frac{1}{2}}\,\mathrm{sn}u)/(1+k\,\mathrm{sn}^2 u), \tag{81}$$

the modulus k being given by (78).[13] This is the same transformation as was used by Onsager,[5] and Kaufman and Onsager[7] in their calculations. It serves to eliminate the square root in the function Θ:

$$\Theta = e^{iH'} = \mathrm{cn}u + i\,\mathrm{sn}u. \tag{82}$$

It is easy to verify that

$$\frac{1}{z}\frac{dz}{du} = -i\frac{1-k^2}{(1+k)^{\frac{1}{2}}}\frac{1}{\mathrm{dn}u - k^{\frac{1}{2}}\,\mathrm{cn}u}. \tag{83}$$

We shall need the following properties of the transformation (81): (A) z is doubly periodic in u with periods $4K$ and $4iK'$.

(B) z is everywhere analytic, except at $u = iK'/2$, $3iK'/2$ (mod. $4K$, $4iK'$), where $z = \infty$.

(C) In a unit cell in the complex u-plane, to every value of z there correspond exactly two values of u, except for $z = A$, B, $1/B$ or $1/A$ for which there corresponds only one value of u, namely, $u = +iK'$, $2K+iK'$, $2K-iK'$ or $-iK'$ (mod. $4K$, $4iK'$).

(D) If for a value of z there correspond in a unit cell two values of u, then at those two values Θ assume equal values but have different signs. Thus a unit cell of the u plane corresponds to both sheets of the Riemann surface in the z plane of Fig. 1 with respect to the function $\Theta(z)$.

The substitution, suggested by (76), into (69), of

$$\phi_1(z) = z[(z-A)(z-B)]^{\frac{1}{2}}\phi$$

gives, with the use of (81), (82), and (83)

$$2\phi(u') + \int_0^{4K} J(u',u)\phi(u)du = l\phi(u'), \tag{84}$$

where

$$J = \mathbf{I}\,\mathbf{II}\,\mathbf{III}\,\mathbf{IV}; \tag{85}$$

and

$$\mathbf{I} = [1 - z(u)/z(u')]^{-1}, \tag{86}$$

$$\mathbf{II} = 1 + \Theta(z)/\Theta(z') = 1 + (\mathrm{cn}u + i\,\mathrm{sn}u)/(\mathrm{cn}u' + i\,\mathrm{sn}u'), \tag{87}$$

$$\mathbf{III} = \frac{z(u)}{z(u')}\left[\frac{\{z(u')-A\}\{z(u')-B\}}{\{z(u)-A\}\{z(u)-B\}}\right]^{\frac{1}{2}}, \tag{88}$$

[12] E. T. Whittaker and G. N. Watson, *Modern Analysis* (Cambridge University Press, London, 1927), fourth edition.
[13] The modulus k is the same as that used in references 5 and 7.

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and

$$\mathbf{IV} = -\frac{1}{\pi}\frac{1-k^2}{(1+k)^{\frac{1}{2}}}\frac{1}{\mathrm{dn}u - k^{\frac{1}{2}}\,\mathrm{cn}u}.$$

V. SOLUTION OF INTEGRAL EQUATION (84)

We proceed by investigating the analytic behavior of $J(u', u)$ *with respect to the variable* u.

(A) **I**, **II**, and **IV** are all doubly periodic with periods $4K$ and $4iK'$. But **III** is doubly periodic with periods $4K$ and $8iK'$. It changes sign at periods $4iK'$:

$$\mathbf{III}(u + 4iK) = -\mathbf{III}(u). \tag{89}$$

(B) **III** is analytic everywhere except at $z = A$ or $z = B$, i.e., $u = iK'$ or $2K + iK'$ (mod. $4K$, $4iK'$) where **III** has simple poles.

(C) **II** is analytic everywhere except at $u = -iK'$ or $2K - iK'$ (mod. $4K$, $4iK'$) where it has simple poles.

(D) **IV** is analytic everywhere except at

$$u = \pm iK'/2, \quad \pm 3iK'/2 (\mathrm{mod.}\ 4K, 4iK'),$$

where it has simple poles.

(E) **I** is analytic everywhere except at $z(u) = z(u')$. According to the last section in each cell there are, in general, two values of u where this exception occurs. At these two points **I** has simple poles.

However, there is *only one pole* for J in each unit cell ($4K$ by $4iK'$). This is so because of the following considerations:

(F) At $u = \pm iK'$, $\pm iK' + 2K$ (mod. $4K$, $4iK'$), **IV** has simple zeros.

(G) At $u = iK'/2$, $3iK'/2$ (mod. $4K$, $4iK'$), $z(u) = \infty$, so that **I** has simple zeros.

(H) At $u = -iK'/2$, $-3iK'/2$ (mod. $4K$, $4iK'$), $z(u) = 0$, so that **III** has simple zeros.

(I) According to property D, Sec. IV, in a unit cell ($4K$ by $4iK'$) at one of the solutions of $z(u) = z(u')$, $\Theta(u) = \Theta(u')$ so that $\mathbf{II} = 2$. At the other solution, **II** has a zero.

Thus inside the rectangle in Fig. 3, J has only one pole at $u = u'$ which we assume to be inside of the rectangle. In the neighborhood of this pole $\mathbf{II} = 2$, $\mathbf{III} = 1$, and $\mathbf{I}\text{-}\mathbf{IV} = i\pi^{-1}(u - u')^{-1}$. Hence the residue of J at $u = u'$ is $2i/\pi$.

The solution of (84) is given by

$$\phi = \exp(im\pi u/2K), \quad m = \pm\text{integer}. \tag{90}$$

To show that this is indeed a solution we note that ϕ is periodic with period $4K$. Hence calling

$$s = \int_0^{4K} J(u', u)\phi(u)du,$$

one obtains by performing a contour integration around the rectangle of Fig. 3:

$$2\pi i(2i/\pi)\phi(u') = s[1 + \exp(-2m\pi K'/K)]. \tag{91}$$

[The integration along the two vertical sides cancel each other and that along the top reduces to s multiplied by a factor, in virtue of (89).] This gives

$$s = -4\phi(u')/(1+q^{2m}), \tag{92}$$

where

$$q = \exp(-\pi K'/K). \tag{93}$$

(84) is therefore satisfied with

$$l = 2 - 4/(1+q^{2m}) = 2(q^{2m}-1)/(q^{2m}+1). \tag{94}$$

For $m = 0$ this gives, as expected, the solution $l = 0$ which was already found by inspection in Sec. IIIC.

Knowing all the nonvanishing eigenvalues we can now calculate

$$\prod_1^\infty \frac{l_n^2}{4} = \prod_{m=-\infty,\, m\neq 0}^{\infty}\left(\frac{1-q^{2m}}{1+q^{2m}}\right)^2 = \prod_1^\infty\left(\frac{1-q^{2m}}{1+q^{2m}}\right)^4.$$

This infinite product can be[14] expressed in terms of the ϑ functions which are related to K. We get finally

$$\prod_1^\infty \frac{l_n^2}{4} = \frac{4}{\pi^2}[K(k)]^2(1-k^2)^{\frac{1}{2}} = \frac{4}{\pi^2}k^2\frac{1+x^2}{(1-x^2)^2}(1-6x^2+x^4)^{\frac{1}{2}}. \tag{95}$$

VI. FINAL RESULTS

The spontaneous magnetization I is obtained from (95) and (80) as

$$I = \left[\frac{1+x^2}{(1-x^2)^2}(1-6x^2+x^4)^{\frac{1}{2}}\right]^{\frac{1}{4}}. \tag{96}$$

FIG. 4. Spontaneous magnetization.

At low temperatures this gives the same expansion in powers of x as obtained in previous works of Van der Waerden[1] and Ashkin and Lamb:[2]

$$I = 1 - 2x^4 - 8x^8 - 34x^{12} - 152x^{16} - 714x^{20} - \cdots.$$

This series is convergent all the way up to the critical

[14] See reference 11, especially p. 472.

Figure 4.6: Two pages of Yang (1952).

transition probability p and studied its stationary distribution π. MCMC reverses this direction: it begins with a target distribution π and constructs a transition probability p for which π is stationary. Once p is available, we can generate a Markov chain $\{X_n\}_{n=0}^{\infty}$ (see Algorithms 10 and 11). If this chain is irreducible and aperiodic, Theorem 3.4.3 gives

$$\lim_{n\to\infty} \mathbb{P}(X_n = x) = \pi(x), \quad \text{for all } x \in \mathcal{X}. \tag{4.2.1}$$

Hence, X_n may be treated as approximately distributed according to π only when a quantitative convergence bound or an appropriate diagnostic indicates that the discrepancy between $\mathcal{L}(X_n)$ and π is sufficiently small. The required value of n depends on the initial state and the mixing properties of the chain; there is no universal "extremely large" threshold. This is the main theme of MCMC. The MCMC algorithms we will introduce in this chapter help us do the following two tasks

i) Generate a Markov chain whose marginal distribution approaches π, and obtain approximate draws from π after an appropriate convergence assessment.

ii) Compute the expected value $\mathbb{E}_\pi[f(X)]$, where $\mathbb{E}_\pi$ indicates that $X \sim \pi$. The average magnetization in Eq. (4.1.6) is of this expected value form.

It is useful to remember the reversal between the two subjects:

$$\begin{aligned} \text{Markov-chain analysis:} &\quad p \longrightarrow \text{identify or study } \pi, \\ \text{MCMC design:} &\quad \pi \longrightarrow \text{construct } p \longrightarrow \text{simulate a chain with stationary distribution } \pi. \end{aligned}$$

In the second line, constructing p is not enough by itself: one must also verify conditions such as irreducibility and aperiodicity before using the convergence result in Eq. (4.2.1).

This section outlines the theoretical foundation of MCMC for three types of state spaces: finite spaces, such as $\mathcal{X} = \{x_1, \ldots, x_S\}$; countable spaces, such as $\mathcal{X} = \{x_i\}_{i=1}^{\infty}$; and general spaces, such as $\mathcal{X} = \mathbb{R}^d$. We focus on finite state spaces, where the results of Section 3.4 apply directly, and briefly summarize the countable and general cases. The deeper theory for countable and general state spaces belongs to graduate-level probability courses such as APMA 2630/2640.

4.2.1 Scenario I: Finite State Spaces

The theoretical foundation for the first task is Eq. (4.2.1), equivalently Theorem 3.4.3. Specifically, for a given distribution π, if we can construct the transition probability p of an irreducible and aperiodic Markov chain such that π is the corresponding stationary distribution, then any X_n with sufficiently large n can be viewed approximately as a random variable from π.

The theoretical foundation for the second task comes from Theorem 3.4.1, i.e.,

$$\lim_{n\to\infty} \widehat{\pi}_n(x) = \pi(x), \text{ with probability one, for all } x \in \mathcal{X}, \quad \text{where } \widehat{\pi}_n(x) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(X_i = x).$$

Specifically, we have the following theorem, which is an analog of LLN for Markov chains (instead of iid sequences).

Theorem 4.2.1. *Let $\{X_n\}_{n=0}^{\infty}$ be a Markov chain taking values in a finite state space $\mathcal{X}$. If this chain is irreducible, for any function $f : \mathcal{X} \to \mathbb{R}$, we have the following analog of LLN*

$$\lim_{n\to\infty} \left\{ \frac{1}{n}\sum_{i=1}^{n} f(X_i) \right\} = \mathbb{E}_\pi[f(X)] \stackrel{\text{def}}{=} \sum_{x\in\mathcal{X}} \pi(x)\cdot f(x) \quad \textit{with probability one,}$$

where π is the stationary distribution of $\{X_n\}_{n=0}^{\infty}$; Theorem 3.3.1 guarantees the uniqueness of π, so we may use the definite article "the."

Proof. Notice the following representation of the average

$$\frac{1}{n}\sum_{i=1}^{n} f(X_i) = \mathbb{E}_{\widehat{\pi}_n}\left[f(X)|X_1,\ldots,X_n\right] = \sum_{x\in\mathcal{X}} f(x)\cdot\widehat{\pi}_n(x).$$

Then, we have

$$\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_\pi[f(X)]\right| = \left|\sum_{x\in\mathcal{X}} f(x)\cdot\widehat{\pi}_n(x) - \sum_{x\in\mathcal{X}} f(x)\cdot\pi(x)\right| \le \sum_{x\in\mathcal{X}} |f(x)|\cdot|\widehat{\pi}_n(x)-\pi(x)|;$$

with the finiteness of $\mathcal{X}$, we have the following limit with probability one

$$\lim_{n\to\infty}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_\pi[f(X)]\right| \le \sum_{x\in\mathcal{X}} |f(x)|\cdot\lim_{n\to\infty}|\widehat{\pi}_n(x)-\pi(x)| = 0,$$

with probability one. The proof is completed. □

Based on Theorem 4.2.1, to estimate the expected value $\mathbb{E}_\pi[f(X)]$, we may construct an irreducible Markov chain $\{X_n\}_{n=0}^{\infty}$ with π as its stationary distribution and use the approximation $\frac{1}{n}\sum_{i=1}^{n} f(X_i) \approx \mathbb{E}_\pi[f(X)]$. In applications, the first half, or some other initial portion, of the sequence $\{X_i\}_{i=1}^{n}$ is usually treated as not yet sufficiently close to the target distribution (Gelman et al., 2013, Section 11.4). This initial segment is called the burn-in period. After discarding the first half, one may use $\frac{2}{n}\sum_{i=\frac{n}{2}+1}^{n} f(X_i)$ as the approximation.

The remaining question is: ***given a distribution π, how can we construct an irreducible and aperiodic Markov chain with stationary distribution π?*** This chapter develops two widely used answers: the Metropolis–Hastings algorithm and the Gibbs sampler.

Throughout this chapter, we assume $\pi(x) > 0$ for all $x \in \mathcal{X}$. If $\pi(x') = 0$ for some $x' \in \mathcal{X}$, we can just remove x' from $\mathcal{X}$ and update $\mathcal{X} - \{x'\}$ as the new $\mathcal{X}$.

4.2.2 Scenario II: Countable State Spaces

The scenario where the state space is countable is very similar to the scenario where the state space is finite. The asymptotic results for the two scenarios are almost the same, but the proof in Scenario II is much more complicated than the proof in Scenario I. We briefly describe the results for Scenario II as follows.

- Suppose that the transition probability p defined on $\mathcal{X} \times \mathcal{X}$, with $\mathcal{X} = \{x_i\}_{i=1}^{\infty}$, is **irreducible and positive recurrent**. Equivalently, for an irreducible countable-state chain, assume that p admits a stationary probability distribution π. Then we have the following (total variation) **mean ergodicity**

$$\sup_{C \subseteq \mathcal{X}} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{P}(X_i \in C | X_0 = x) - \pi(C) \right| \to 0, \quad \text{as } n \to \infty,$$

  for all $x \in \mathcal{X}$, where $\{X_n\}_{n=0}^{\infty}$ is the Markov chain starting at x with transition probability p; in addition, for any function f defined on $\mathcal{X}$ with $\sum_{x \in \mathcal{X}} |f(x)| \cdot \pi(x) < \infty$, we have

$$\lim_{n \to \infty} \left\{ \frac{1}{n} \sum_{i=1}^{n} f(X_i) \right\} = \sum_{x \in \mathcal{X}} f(x) \cdot \pi(x) = \mathbb{E}_{\pi}[f(X)] \qquad \text{with probability one.}$$

- Furthermore, if p is **aperiodic**, we have the following (total variation) **ergodicity**

$$\sup_{C \subseteq \mathcal{X}} \left| \mathbb{P}(X_n \in C | X_0 = x) - \pi(C) \right| \to 0, \quad \text{as } n \to \infty,$$

  for all $x \in \mathcal{X}$.

These results can be found on the second page of Athreya et al. (1996) or in Chapter 6.6 of Durrett (2019).

4.2.3 Scenario III: General State Spaces

The finite and countable settings above make recurrence and aperiodicity comparatively concrete. On a general state space, their measure-theoretic counterparts are substantially more involved, so we state only a representative convergence result. Figure 4.7 reproduces the main result of Athreya et al. (1996), which gives a self-contained proof under concrete sufficient conditions. Its key condition, Harris recurrence, can be difficult to verify in applications and is stronger than irreducibility. Further treatments of the general theory include Tierney (1994)[‡] and Tierney (1998).

In addition, the paper Gidas (1995)[§] describes the mathematical frameworks and properties of Metropolis-type Monte Carlo simulation algorithms.

‡ The paper Tierney (1994) had been cited more than 5150 times by July 2022.

§ Professor Basilis Gidas is an applied mathematician at Brown University. In addition to his achievement in applied mathematics, Professor Gidas has also done widely recognized work in pure mathematics. His collaborations with Luis Caffarelli, Wei-Ming Ni, Louis Nirenberg, and Joel Spruck resulted in six papers in pure mathematics, five of which rank among the most cited in the field of elliptic partial differential equations. The collaboration of Gidas, Ni, and Nirenberg in particular was cited in the awarding of the Abel Prize to Nirenberg.

THEOREM 1. *Suppose that the Markov chain $\{X_n\}$ with transition function $P(x, C)$ has an invariant probability measure π, that is,* (1.1) *holds. Suppose that there is a set $A \in \mathscr{B}$, a probability measure ρ with $\rho(A) = 1$, a constant $\varepsilon > 0$ and an integer $n_0 \geq 1$ such that*

$$\pi\{x\colon P_x(T(A) < \infty) > 0\} = 1, \tag{1.4}$$

and

$$P^{n_0}(x, \cdot) \geq \varepsilon\rho(\cdot) \quad \text{for each } x \in A. \tag{1.5}$$

Suppose further that

$$\text{g.c.d.}\{m\colon \text{there is an } \varepsilon_m > 0 \text{ such that } P^m(x, \cdot) \geq \varepsilon_m \rho(\cdot) \text{ for each } x \in A\} = 1. \tag{1.6}$$

Then there is a set D such that

$$\pi(D) = 1 \quad \text{and} \quad \sup_{C \in \mathscr{B}} |P^n(x, C) - \pi(C)| \to 0 \quad \text{for each } x \in D. \tag{1.7}$$

THEOREM 2. *Suppose that the Markov chain $\{X_n\}$ with transition function $P(x, C)$ satisfies conditions* (1.1), (1.4) *and* (1.5). *Then*

$$\sup_{C \in \mathscr{B}} \left| \frac{1}{n_0} \sum_{r=0}^{n_0-1} P^{mn_0+r}(x, C) - \pi(C) \right| \to 0 \tag{1.8}$$

as $m \to \infty$ for $[\pi]$-almost all x,

and hence

$$\sup_{C \in \mathscr{B}} \left| \frac{1}{n} \sum_{j=1}^{n} P^j(x, C) - \pi(C) \right| \to 0 \tag{1.9}$$

as $n \to \infty$ for $[\pi]$-almost all x.

Let $f(x)$ be a measurable function on $(\mathscr{X}, \mathscr{B})$ such that $\int \pi(dy)|f(y)| < \infty$. Then

$$P_x\left\{ \frac{1}{n} \sum_{j=1}^{n} f(X_j) \to \int \pi(dy) f(y) \right\} = 1 \quad \text{for } [\pi]\text{-almost all } x \tag{1.10}$$

and

$$\frac{1}{n} \sum_{j=1}^{n} E_x\big(f(X_j)\big) \to \int \pi(dy) f(y) \quad \text{for } [\pi]\text{-almost all } x. \tag{1.11}$$

Figure 4.7: Screenshot of the main result in Athreya et al. (1996) on the convergence of the MCMC method in general state spaces. The Eq. (1.1) referred to in the screenshot is "$\pi(C) = \int P(x, C)\pi(dx)$ for all C," which is a general definition of stationary distributions $\pi(dx)$. We avoid the explanation of the result in our lecture notes. The proof of this result is long and involves advanced probability theory knowledge.

4.3 Metropolis–Hastings Algorithm

Metropolis and Ulam (1949) and Metropolis et al. (1953) were the first to describe Markov chain simulation of probability distributions. Their method became known as the Metropolis algorithm, which Hastings (1970) later generalized. We begin with the Metropolis algorithm to develop intuition from an optimization viewpoint and then present the more general Metropolis–Hastings algorithm.

The five authors of the 1953 paper include two married couples: the second and third authors, Arianna W. Rosenbluth and Marshall N. Rosenbluth, were married, as were the fourth and fifth authors, Augusta H. Teller and Edward Teller; see this historical account from the American Physical Society. Edward Teller is also a character in Christopher Nolan's 2023 film *Oppenheimer*, in which he is portrayed by Benny Safdie.

4.3.1 Metropolis Algorithm and Its Relation to Optimization

We provide the mathematical and algorithmic versions of the Metropolis algorithm.

Mathematical Version

To sample random variables from a given distribution π, the Metropolis algorithm generates a Markov chain using the following transition probability†

$$p(x,y) = \begin{cases} q(x,y) \cdot \min\left\{1, \dfrac{\pi(y)}{\pi(x)}\right\}, & \text{if } x \neq y, \\ 1 - \displaystyle\sum_{z:z\neq x} p(x,z), & \text{if } x = y. \end{cases} \tag{4.3.1}$$

where $q(x,y)$ is a **symmetric** transition probability, i.e., $q(x,y) = q(y,x)$, of an irreducible and aperiodic Markov chain. The transition probability q is referred to as the **jumping distribution**. Choosing q well requires care. The jumping distribution used in Eq. (4.3.1) has to satisfy the following conditions:

- $q(x,y)$ is a symmetric transition probability, i.e., $q(x,y) = q(y,x)$ for all $x, y \in \mathcal{X}$;
- $q(x,y)$ is the transition probability of an irreducible and aperiodic homogeneous Markov chain;
- For each fixed $x \in \mathcal{X}$, we know how to efficiently generate a random variable/vector from $q(x,\cdot)$.

At this point, we assume that q has been chosen. The following two observations explain why the construction works.

Claim 4.3.1. *The transition probability p in Eq. (4.3.1) is irreducible and aperiodic whenever q is irreducible and aperiodic.*

Proof. Because π is strictly positive, for $x \neq y$ we have $p(x,y) > 0$ if and only if $q(x,y) > 0$. Consequently, every positive-probability path under q is also a positive-probability path under p, so irreducibility of q implies irreducibility of p. The same observation shows that every possible return time under q is also a possible return time under p. Since the greatest common divisor of the return times under an aperiodic q is one, the greatest common divisor for p is also one. Thus, p is aperiodic. □

† You may ask, *why is $p(x,x)$ defined separately?* Answer: If we simply define $p(x,x) = q(x,x)$, then p would not be a transition probability. The transition probability assumption requires $1 = \sum_{y\in\mathcal{X}} p(x,y)$. However, $\frac{\pi(y')}{\pi(x)}$ can be strictly smaller than 1 for some $y' \neq x$. In this scenario, if we did not define $p(x,x)$ separately—that is, we define $p(x,x) = q(x,x)$, we will have

$$1 = \sum_{y\in\mathcal{X}} p(x,y) = \sum_{y\in\mathcal{X}} q(x,y) \cdot \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\} = q(x,x) + \sum_{y\neq x,y'} q(x,y) \cdot \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\} + q(x,y') \cdot \min\left\{1, \frac{\pi(y')}{\pi(x)}\right\} < \sum_{y\in\mathcal{X}} q(x,y) = 1,$$

which is a contradiction. Hence, we need to define $p(x,x)$ separately to correct for the "loss of mass" in the term $q(x,y') \cdot \min\left\{1, \frac{\pi(y')}{\pi(x)}\right\}$.

Claim 4.3.2. *The target distribution π is reversible for p; that is,*

$$\pi(x)p(x,y) = \pi(y)p(y,x), \quad \textit{for all } x, y \in \mathcal{X}.$$

Proof. For $x \neq y$, symmetry of q gives

$$\begin{aligned}\pi(x)p(x,y) &= q(x,y)\min\{\pi(x),\pi(y)\} \\ &= q(y,x)\min\{\pi(y),\pi(x)\} = \pi(y)p(y,x).\end{aligned}$$

The identity is automatic when $x = y$. Summing the detailed-balance identity over x yields

$$\sum_{x\in\mathcal{X}} \pi(x)p(x,y) = \pi(y)\sum_{x\in\mathcal{X}} p(y,x) = \pi(y),$$

so π is stationary for p. □

Algorithmic Version

The mathematical version is somewhat abstract. The algorithmic version below is easier to translate into code and makes the optimization-based intuition behind the acceptance rule transparent.

Algorithm 13 : Metropolis Algorithm

Input: (i) the distribution π of interest satisfying $\pi(x) > 0$ for all $x \in \mathcal{X}$; (ii) a jumping distribution—a symmetric transition probability q of an irreducible and aperiodic Markov chain; (iii) an initial starting point x_0; (iv) a large integer n^*.

Output: n^* transitions and the resulting n^*+1 states $X_0, X_1, \ldots, X_{n^*}$ of a Markov chain with π as its stationary distribution.

1: Initialize $X_0 \leftarrow x_0$.
2: **for all** $n = 1, 2, \ldots, n^*$ **do**
3: Sample a proposal X^* from the distribution $q(X_{n-1}, \cdot)$.
4: Compute the ratio $r \leftarrow \frac{\pi(X^*)}{\pi(X_{n-1})}$. (Remark: Since we assume that $\pi(x) > 0$ for all $x \in \mathcal{X}$, the ratio r is always well-defined.)
5: Generate $Y \sim$ Bernoulli$(\min\{r, 1\})$, i.e., $\mathbb{P}(Y = 1) = \min\{r, 1\}$.
6: **if** $Y = 1$ **then**
7: $X_n \leftarrow X^*$.
8: **else**
9: $X_n \leftarrow X_{n-1}$.
10: **end if**
11: **end for**

The acceptance step of Algorithm 13 follows this rule:

- If the X^* generated from $q(X_{i-1}, \cdot)$ **increases** π, i.e., $\frac{\pi(X^*)}{\pi(X_{i-1})} \geq 1$(equivalently, $\min\{1, \frac{\pi(X^*)}{\pi(X_{i-1})}\} = 1$), we set $X_i = X^*$ with probability one.
- If X^* **decreases** π, i.e., $r = \frac{\pi(X^*)}{\pi(X_{i-1})} < 1$, we set $X_i = X^*$ only with probability $r < 1$.

This acceptance/rejection rule shows that the Metropolis algorithm resembles a stochastic stepwise-maximization algorithm: it **always accepts proposals that increase the value of π, but accepts downward proposals only probabilistically**.

Under the conditions of Theorem 3.4.3, the limiting distribution does not depend on the starting point x_0. For a finite run, however, the bias and the chain length required for an adequate approximation can depend strongly on both the starting point and the mixing rate. Thus, "sufficiently large" has no universal meaning and should be assessed for the chain under study. The following simulation illustrates this finite-run behavior; we return to convergence assessment at the end of this chapter.

Here we sample random vectors from a bivariate normal distribution as an example of the Metropolis algorithm (Algorithm 13). In this example, the target distribution π of interest is

$$N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & 0.8\\0.8 & 1\end{pmatrix}\right).$$

The jumping distribution q is chosen as $q(\boldsymbol{x}, \cdot) = N(\boldsymbol{x}, 0.2^2\boldsymbol{I})$, i.e., $q(\boldsymbol{x}, \boldsymbol{y}) = \frac{25}{2\pi} \cdot \exp\left\{-\frac{25\|\boldsymbol{x}-\boldsymbol{y}\|^2}{2}\right\}$; this q is symmetric. We set the output chain length n^* to be 5000. We sample five chains with different starting points, and the sampled chains are shown in Figure 4.8. In these five finite runs, the trajectories after the warm-up period look similar despite their different starting points. This observation is consistent with convergence toward a common target distribution, but the plots alone do not establish asymptotic convergence.

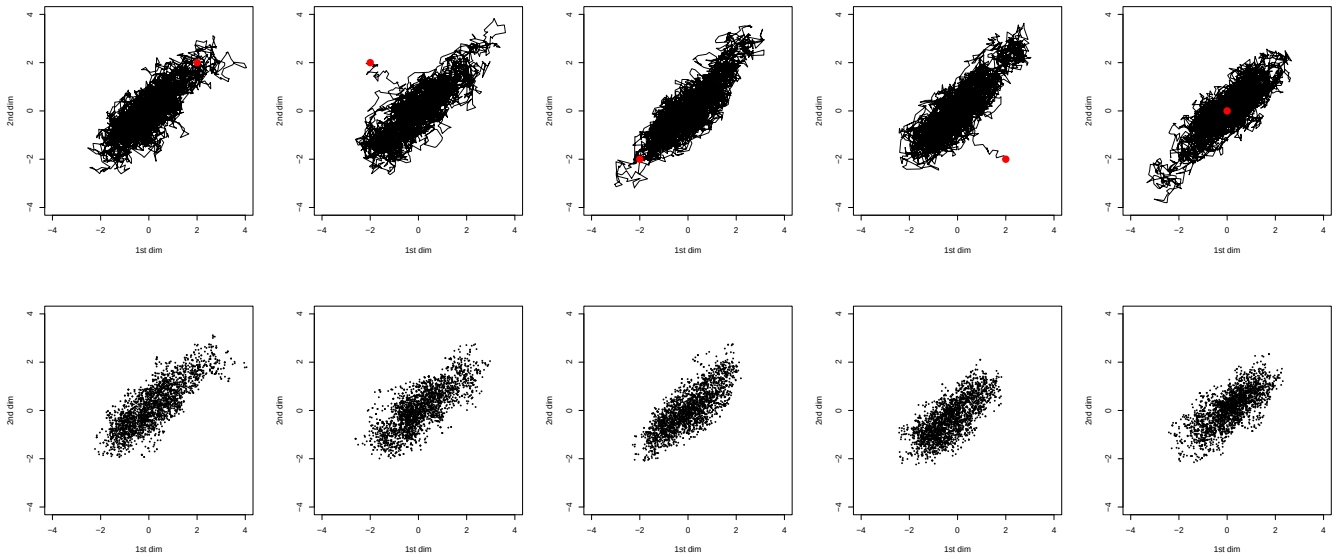


Figure 4.8: The first row shows five sampled paths starting at $x_0 = (2,2)^\intercal$, $(-2,2)^\intercal$, $(-2,-2)^\intercal$, $(2,-2)^\intercal$, and $(0,0)^\intercal$, respectively; red dots mark the starting points. Each panel in the second row shows the second half of the corresponding chain. The first half may still belong to the warm-up period, whereas the final 2,500 states resemble draws from the target distribution in the diagnostics shown; compare Figure 4.9. The R code used to produce the panels is provided below.

```r
install.packages("mvtnorm")
library("mvtnorm")
# We need the functions "rmvnorm" and "dmvnorm" in this package.

sigma_sq=0.2^2
n_star=5000

V=rbind(c(1, 0.8), c(0.8,1))
pi_dist=function(x){ return(dmvnorm(x, mean = rep(0,2), sigma = V)) }

X_list=list()
X_0_starting=rbind(c(2,2), c(-2,2), c(-2,-2), c(2,-2), c(0,0))
for (i in 1:5) {
  X_list[[i]]=matrix(NA, ncol = 2, nrow = n_star+1)
}

for (j in 1:5) {
  X_list[[j]][1,]=X_0_starting[j,]
  for (k in 1:(n_star)) {

    X_star=rmvnorm(1, mean = X_list[[j]][k,], sigma = sigma_sq*diag(2))
    r=min(1, pi_dist(X_star)/pi_dist(X_list[[j]][k,]))
```

```
    Y=rbinom(1, 1, prob = r)
    if(Y==1){
      X_list[[j]][k+1,]=X_star
    }else{
      X_list[[j]][k+1,]=X_list[[j]][k,]
    }

  }
}

par(mfrow=c(2,5))
for (j in 1:5) {
  plot(X_list[[j]], type = "l", xlim = c(-4,4), ylim = c(-4,4),
       xlab = "1st dim", ylab = "2nd dim")
  lines(X_0_starting[j,1], X_0_starting[j,2], col="red", type = "p", pch=16, cex=2)
}
for (j in 1:5) {
  plot(X_list[[j]][(floor(n_star/2)+1):n_star,], type = "p", xlim = c(-4,4), ylim = c(-4,4),
       xlab = "1st dim", ylab = "2nd dim", pch = 16, cex = 0.5, col = "black")
}
```

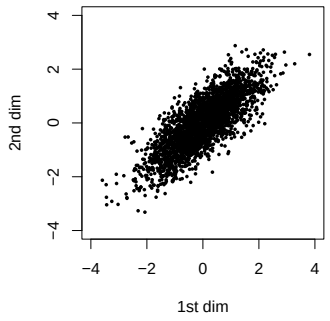


Figure 4.9: A sample of 2,500 iid draws from the target distribution, provided for comparison with the second row of Figure 4.8. The R code used to produce this figure is provided below.

```
install.packages("mvtnorm")
library("mvtnorm")
# We need the function "rmvnorm" in this package.

V=rbind(c(1, 0.8), c(0.8,1))
X=rmvnorm(2500, mean = rep(0,2), sigma = V)
plot(X, pch = 16, cex = 0.5, xlim = c(-4,4), ylim = c(-4,4),
     xlab = "1st dim", ylab = "2nd dim")
```

Choosing the proposal scale. For a random-walk proposal $q(\boldsymbol{x}, \cdot) = N(\boldsymbol{x}, \sigma^2 \boldsymbol{I})$, the value of σ controls a fundamental tradeoff. If σ is extremely small, most proposals are accepted, but consecutive states are nearly identical and the chain explores the target distribution very slowly. If σ is extremely large, proposals often land in regions where π is much smaller than at the current state, making the acceptance ratio and acceptance probability close to zero; the chain then remains at the same state for many iterations. A useful proposal scale balances movement size against acceptance probability. This balance depends on the target distribution, so it should be assessed rather than assumed.

4.3.2 Metropolis–Hastings Algorithm

The Metropolis algorithm requires the proposal transition $q(x, y)$ to be symmetric, i.e., $q(x, y) = q(y, x)$. This symmetry makes the target distribution π reversible, and hence stationary, for the transition probability p in Eq. (4.3.1). The Metropolis-Hastings algorithm removes this restriction by correcting for proposal asymmetry. Specifically, it replaces the ratio $r = \frac{\pi(X^*)}{\pi(X_{i-1})}$ in Algorithm 13 with

$$\frac{\pi(X^*) \cdot q(X^*, X_{i-1})}{\pi(X_{i-1}) \cdot q(X_{i-1}, X^*)}.$$

As we did in the preceding section, we present the Metropolis-Hastings algorithm in mathematical and algorithmic versions.

Mathematical Version

The Metropolis-Hastings algorithm generates Markov chains using the following transition probability to draw random variables from the target distribution π (see Definition 18.14 of Klenke (2020))

$$p(x, y) := \begin{cases} q(x, y) \cdot \min\left\{1, \dfrac{\pi(y) \cdot q(y, x)}{\pi(x) \cdot q(x, y)}\right\}, & \text{if } x \neq y \text{ and } q(x, y) > 0, \\ 0, & \text{if } x \neq y \text{ and } q(x, y) = 0, \\ 1 - \displaystyle\sum_{z : z \neq x} p(x, z), & \text{if } x = y, \end{cases} \tag{4.3.2}$$

where q is the transition probability of a Markov chain. If q is symmetric, Eq. (4.3.2) reduces to Eq. (4.3.1). The Metropolis–Hastings algorithm does not generally require symmetry. The following result, Theorem 18.15 of Klenke (2020), states suitable assumptions on q and provides the theoretical foundation for the algorithm.

Theorem 4.3.3. *Assume that q is irreducible and that for any $x, y \in \mathcal{X}$, we have $q(x, y) > 0$ if and only if $q(y, x) > 0$. Then the transition probability p defined in Eq. (4.3.2) is irreducible and has the unique stationary distribution π. If, in addition, q is aperiodic, then p is aperiodic as well.*

Algorithm 14 : Metropolis-Hastings Algorithm

Input: (i) the distribution π of interest; (ii) a jumping distribution—a transition probability q of an irreducible and aperiodic Markov chain; (iii) an initial starting point x_0; (iv) a (sufficiently large) chain length n^*.

Output: n^* transitions and the resulting n^*+1 states $X_0, X_1, \ldots, X_{n^*}$ of a Markov chain with π as its stationary distribution.

Set $X_0 = x_0$.
for all $i = 1, 2, \ldots, n^*$ **do**
  Sample a proposal X^* from the distribution $q(X_{i-1}, \cdot)$.
  Compute the ratio $r \leftarrow \frac{\pi(X^*)\cdot q(X^*, X_{i-1})}{\pi(X_{i-1})\cdot q(X_{i-1}, X^*)}$. (Remark: If $q(X_{i-1}, X^*) = 0$, then it is almost impossible to sample X^* in the preceding step. So, the ratio r is well-defined with probability one.)
  Generate $Y \sim \text{Bernoulli}(\min\{r, 1\})$.
  if $Y = 1$ **then**
    $X_i \leftarrow X^*$.
  else
    $X_i \leftarrow X_{i-1}$.
  end if
end for

4.4 Gibbs Sampler

"*Gibbs sampling is named after the physicist Josiah Willard Gibbs, in reference to an analogy between the sampling algorithm and statistical physics. The algorithm was described by brothers Stuart and Donald Geman in 1984, some eight decades after the death of Gibbs.*"

— *Wikipedia*†

The Gibbs sampler was first so named by Geman and Geman (1984)‡ in a discussion of applications to image processing. We first describe its transition probability mathematically, then translate that construction into an algorithm, and finally apply it to the Ising model from Section 4.1.

4.4.1 Mathematical Version

For simplicity, we first discuss two-dimensional distributions. The higher-dimensional construction follows the same coordinate-by-coordinate principle.

Let $\pi(\boldsymbol{x}) = \pi(\xi_1, \xi_2)$ be a strictly positive joint PMF on a finite product state space $\mathcal{X} = \mathcal{X}_1 \times \mathcal{X}_2$; that is, $\pi(\boldsymbol{x}) > 0$ for every $\boldsymbol{x} \in \mathcal{X}$. We define the following marginal distributions

$$\pi_1(\xi_1) := \sum_{\xi_2 \in \mathcal{X}_2} \pi(\xi_1, \xi_2), \quad \pi_2(\xi_2) := \sum_{\xi_1 \in \mathcal{X}_1} \pi(\xi_1, \xi_2).$$

† Professor Stuart Geman has been on the faculty at Brown University since 1977 and is the James Manning Professor of Applied Mathematics. Coincidentally, just as I began writing this section in Overleaf, he stopped by my office to say hello.

‡ According to Google Scholar, Geman and Geman (1984) had been cited 29,233 times as of July 22, 2026.

Then, we have the following conditional distributions

$$\pi_{1|2}(\xi_1|\xi_2) := \frac{\pi(\xi_1,\xi_2)}{\pi_2(\xi_2)}, \text{ which is a function of } \xi_1, \text{ and } \xi_2 \text{ is viewed as a fixed parameter;}$$
$$\pi_{2|1}(\xi_2|\xi_1) := \frac{\pi(\xi_1,\xi_2)}{\pi_1(\xi_1)}, \text{ which is a function of } \xi_2, \text{ and } \xi_1 \text{ is viewed as a fixed parameter.}$$

With these marginal and conditional distributions, the transition probability of the Gibbs sampler from state $\boldsymbol{x} = (\xi_1,\xi_2)^\intercal$ to $\boldsymbol{y} = (\eta_1,\eta_2)^\intercal$ is the following

$$p(\boldsymbol{x},\boldsymbol{y}) := \pi_{1|2}(\eta_1|\xi_2)\cdot\pi_{2|1}(\eta_2|\eta_1). \tag{4.4.1}$$

Systematic-scan Gibbs sampling is generally not reversible, because the order of the two coordinate updates matters.

Theorem 4.4.1. *Suppose that π is strictly positive on the finite product state space $\mathcal{X}_1\times\mathcal{X}_2$. The transition probability p in Eq. (4.4.1) is irreducible and aperiodic, and π is its unique stationary distribution.*

Proof. Strict positivity of π implies that both full conditional PMFs are strictly positive. Hence, $p(\boldsymbol{x},\boldsymbol{y}) > 0$ for every $\boldsymbol{x},\boldsymbol{y}\in\mathcal{X}$. In particular, every state can reach every other state in one step, so the chain is irreducible. Moreover, $p(\boldsymbol{x},\boldsymbol{x}) > 0$ for every $\boldsymbol{x}$, so the chain is aperiodic.

It remains to verify stationarity. For any $\boldsymbol{y} = (\eta_1,\eta_2)^\intercal$,

$$\begin{aligned}
\sum_{\boldsymbol{x}\in\mathcal{X}}\pi(\boldsymbol{x})p(\boldsymbol{x},\boldsymbol{y}) &= \sum_{\xi_1\in\mathcal{X}_1,\,\xi_2\in\mathcal{X}_2}\pi(\xi_1,\xi_2)\cdot\pi_{1|2}(\eta_1|\xi_2)\cdot\pi_{2|1}(\eta_2|\eta_1)\\
&= \sum_{\xi_1\in\mathcal{X}_1,\,\xi_2\in\mathcal{X}_2}\pi(\xi_1,\xi_2)\cdot\frac{\pi(\eta_1,\xi_2)}{\pi_2(\xi_2)}\cdot\frac{\pi(\eta_1,\eta_2)}{\pi_1(\eta_1)}\\
&= \sum_{\xi_2\in\mathcal{X}_2}\left(\sum_{\xi_1\in\mathcal{X}_1}\pi(\xi_1,\xi_2)\right)\cdot\frac{\pi(\eta_1,\xi_2)}{\pi_2(\xi_2)}\cdot\frac{\pi(\eta_1,\eta_2)}{\pi_1(\eta_1)}\\
&= \sum_{\xi_2\in\mathcal{X}_2}\pi_2(\xi_2)\cdot\frac{\pi(\eta_1,\xi_2)}{\pi_2(\xi_2)}\cdot\frac{\pi(\eta_1,\eta_2)}{\pi_1(\eta_1)}\\
&= \sum_{\xi_2\in\mathcal{X}_2}\pi(\eta_1,\xi_2)\cdot\frac{\pi(\eta_1,\eta_2)}{\pi_1(\eta_1)}\\
&= \pi_1(\eta_1)\cdot\frac{\pi(\eta_1,\eta_2)}{\pi_1(\eta_1)}\\
&= \pi(\eta_1,\eta_2)\\
&= \pi(\boldsymbol{y}).
\end{aligned}$$

Thus, π is stationary. Because the state space is finite and the chain is irreducible, the stationary distribution is unique by Theorem 3.3.1. □

4.4.2 Algorithmic Version

The preceding argument establishes why a coordinate-wise Gibbs update preserves the target distribution. We now assemble those updates into an implementable systematic scan. For a d-dimensional distribution on the product space $\mathcal{X} = \prod_{i=1}^{d} \mathcal{X}_i$, define the following marginal and conditional distributions for $i \in \{1, 2, \dots, d\}$:

$$\pi_{-i}(\xi_1, \dots, \xi_{i-1}, \xi_{i+1}, \dots, \xi_d) := \sum_{\xi_i \in \mathcal{X}_i} \pi(\xi_1, \dots, \xi_i, \dots, \xi_d),$$
$$\pi_{i|-i}(\xi_i \,|\, \xi_1, \dots, \xi_{i-1}, \xi_{i+1}, \dots, \xi_d) := \frac{\pi(\xi_1, \dots, \xi_i, \dots, \xi_d)}{\pi_{-i}(\xi_1, \dots, \xi_{i-1}, \xi_{i+1}, \dots, \xi_d)}$$

Algorithm 15 : Gibbs Sampling

Input: (i) the target distribution $\pi(\xi_1, \dots, \xi_d)$; (ii) a starting point $\boldsymbol{x}^{(0)} = (\xi_1^{(0)}, \dots, \xi_d^{(0)})^\intercal$; (iii) a (sufficiently large) chain length n^*.

Output: n^* transitions and the resulting $n^* + 1$ states $\boldsymbol{X}^{(0)}, \boldsymbol{X}^{(1)}, \dots, \boldsymbol{X}^{(n^*)}$ of a Markov chain with π as its stationary distribution.

1: Set $\boldsymbol{X}^{(0)} \leftarrow \boldsymbol{x}^{(0)}$.
2: **for all** $i = 1, \dots, n^*$ **do**
3: Sample $\xi_1^{(i)} \sim \pi_{1|-1}(\xi_1 | \xi_2^{(i-1)}, \dots, \xi_d^{(i-1)})$
4: **for all** $j = 2, \dots, d$ **do**
5: Sample $\xi_j^{(i)} \sim \pi_{j|-j}(\xi_j | \xi_1^{(i)}, \dots, \xi_{j-1}^{(i)}, \xi_{j+1}^{(i-1)}, \dots, \xi_d^{(i-1)})$
6: **end for**
7: Set $\boldsymbol{X}^{(i)} \leftarrow (\xi_1^{(i)}, \dots, \xi_d^{(i)})^\intercal$.
8: **end for**

We take the scenario $d = 2$ as an example:

- Step 1: Sample $\xi_1^{(1)} \sim \pi_{1|-1}(\xi_1 | \xi_2^{(0)})$.
- Step 2: Sample $\xi_2^{(1)} \sim \pi_{2|-2}(\xi_2 | \xi_1^{(1)})$; then set $\boldsymbol{X}^{(1)} = (\xi_1^{(1)}, \xi_2^{(1)})^\intercal$.
- Repeat these two coordinate updates for subsequent sweeps.

4.5 Homework for Chapter 4

Problem 4.5.1. *Let q be a symmetric transition probability of an irreducible and aperiodic Markov chain on $\mathcal{X}$, and let p be the Metropolis transition probability defined in Eq. (4.3.1). Prove Claim 4.3.1 directly from the definitions of irreducibility and aperiodicity.*

Problem 4.5.2. *Under the assumptions of the preceding problem, prove the detailed-balance identity*

$$\pi(x)p(x, y) = \pi(y)p(y, x), \qquad x, y \in \mathcal{X},$$

and use it to prove that π is stationary for p (Claim 4.3.2).

Problem 4.5.3 (Effect of the proposal scale in the Metropolis algorithm)**.** *Let the target density π be the density of*

$$N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & 0.8\\0.8 & 1\end{pmatrix}\right).$$

For each fixed $\boldsymbol{x} = (x_1, x_2)^\intercal$, let the proposal distribution $q(\boldsymbol{x}, \cdot)$ be

$$N\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}, \begin{pmatrix}\sigma^2 & 0\\0 & \sigma^2\end{pmatrix}\right).$$

This proposal is symmetric, so Algorithm 13 applies.

i) Set $\sigma = 0.7$ and $\boldsymbol{x}_0 = (-2, 2)^\intercal$. Generate 20,000 Metropolis transitions. Plot the states $\boldsymbol{X}_{10001}, \ldots, \boldsymbol{X}_{20000}$ and provide your code.

ii) Replace the starting point by $\boldsymbol{x}_0 = (2, 2)^\intercal$ and repeat the preceding part. Compare the two plots and comment on the effect of the starting point after the warm-up period.

iii) Generate 20,000 iid observations directly from the target distribution π. Plot them and provide your code.

iv) Set $\sigma = 0.07$, return to $\boldsymbol{x}_0 = (-2, 2)^\intercal$, and repeat the first part. Compare the result with the iid sample from the preceding part. If the plots differ, explain why.

v) Set $\sigma = 70$ and repeat the first part. Again compare the result with the iid sample and explain any discrepancy.

For the final two parts, support your explanation by considering both the distance between consecutive states and the acceptance ratio

$$r = \frac{\pi(\boldsymbol{X}^*)}{\pi(\boldsymbol{X}_{n-1})}.$$

You may also report the empirical acceptance rate for each value of σ.

Problem 4.5.4 (Metropolis–Hastings sampling from a Poisson distribution)**.** *Let the target PMF be the Poisson distribution with mean 10,*

$$\pi(x) = e^{-10}\frac{10^x}{x!}, \qquad x \in \{0, 1, 2, \ldots\}.$$

Define the proposal transition probability by

$$q(x,y)=\begin{cases}\frac{1}{2}, & x\geq 1 \text{ and } y=x-1 \text{ or } y=x+1,\\ \frac{1}{2}, & x=0 \text{ and } y\in\{0,1\},\\ 0, & \text{otherwise.}\end{cases}$$

Although the state space is countably infinite, the Metropolis–Hastings construction still applies. Starting from $X_0=0$*, generate* 10,000 *transitions using Algorithm 14. Plot histograms of the four collections*

$$\{X_{25},\dots,X_{50}\},\qquad \{X_{50},\dots,X_{100}\},$$
$$\{X_{500},\dots,X_{1000}\},\qquad \{X_{5000},\dots,X_{10000}\}.$$

Overlay or otherwise compare each histogram with the exact Poisson(10) *PMF. Submit the four plots and the code used to generate the chain, and explain how the approximation changes as more post-warm-up states are retained.*

Problem 4.5.5 (A three-dimensional Gibbs transition probability)**.** *Let* $\pi(\xi_1,\xi_2,\xi_3)$ *be a strictly positive PMF on a finite product state space. For* $\boldsymbol{x}=(\xi_1,\xi_2,\xi_3)^\intercal$ *and* $\boldsymbol{y}=(\eta_1,\eta_2,\eta_3)^\intercal$*, construct an explicit transition probability* $p(\boldsymbol{x},\boldsymbol{y})$ *corresponding to one systematic Gibbs sweep through coordinates* $1,2,3$*. Prove directly that*

$$\sum_{\boldsymbol{x}}\pi(\boldsymbol{x})p(\boldsymbol{x},\boldsymbol{y})=\pi(\boldsymbol{y}).$$

Thus, prove that π *is stationary for your transition probability. More than one valid construction is possible; provide only one. Use Eq.* (4.4.1) *and Algorithm 15 as guides.*

Chapter 5

Gibbs Random Fields

"In your life, you will inevitably misspeak, trust the wrong person, underreact, overreact, hurt the people who didn't deserve it, overthink, not think at all, self-sabotage, create a reality where only your experience exists, ruin perfectly good moments for yourself and others, deny any wrongdoing, not take steps to make it right, feel very guilty, let the guilt eat you, hit rock bottom, finally address the pain you caused, try to do better next time, rinse, repeat. And I'm not gonna lie. These mistakes will cause you to lose things."

— Taylor Swift (NYU 2022 commencement speech)

Chapter 4 already introduced a high-dimensional state: a spin configuration on the 100-by-100 lattice Λ_{100}^2 has 10,000 coordinates. This chapter makes the structure of such distributions explicit. Beginning with the Ising model, we show how graph-based factorizations turn a high-dimensional joint distribution into local conditional computations. We then formalize this idea through Gibbs random fields (GRFs), relate it to the Markov property, and use the resulting locality in Gibbs updates.

5.1 Motivation: The Ising Model Revisited

5.1.1 The Two-Dimensional Model and Its Critical Point

We first collect the ingredients of the two-dimensional Ising model that will be used throughout this chapter. This recap makes the connection between the statistical-mechanics formulation in Chapter 4 and the graphical-model computations below explicit.

Periodic square lattice. For $N \geq 3$, let

$$\Lambda_N^2 = (\mathbb{Z}/N\mathbb{Z})^2$$

be an $N \times N$ square lattice with periodic boundary conditions. We identify each coordinate with $\{0, 1, \ldots, N-1\}$ and perform coordinate arithmetic modulo N. Thus, a site (r, c) has the four distinct neighbors

$$\mathcal{N}(r, c) = \{(r+1, c), (r-1, c), (r, c+1), (r, c-1)\} \pmod{N}. \tag{5.1.1}$$

Moving beyond one side of the displayed square therefore returns on the opposite side. This convention avoids separate edge and corner cases in both the mathematics and the code. Figures 4.2–4.4 illustrate the toroidal lattice and its neighborhoods.

Spin configurations and probability mass function. At each site $i \in \Lambda_N^2$ there is a spin $s_i \in \{-1, 1\}$. A spin configuration is

$$\boldsymbol{s} = (s_i)_{i \in \Lambda_N^2} \in \mathcal{X}_N := \{-1, 1\}^{\Lambda_N^2}, \qquad |\mathcal{X}_N| = 2^{N^2}.$$

For the ferromagnetic, zero-field model, the energy and Boltzmann PMF are

$$H_N(\boldsymbol{s}) = -\sum_{\langle i,j \rangle} s_i s_j, \tag{5.1.2}$$

$$\pi_{N,\beta}(\boldsymbol{s}) = \frac{1}{Z_{N,\beta}} \exp\left\{\beta \sum_{\langle i,j \rangle} s_i s_j\right\}, \tag{5.1.3}$$

where $\langle i, j \rangle$ ranges over undirected neighboring pairs and counts every edge once, and $Z_{N,\beta}$ normalizes the PMF. We use energy units in which the interaction strength and Boltzmann constant are both one, so the dimensionless inverse temperature is $\beta = 1/T$. In physical units, $\beta = J/(k_B T)$. The positive sign in Eq. (5.1.3) favors aligned neighboring spins; reversing it gives the antiferromagnetic model.

Magnetization and the Curie point. The finite-volume average absolute magnetization is

$$m_N(\beta) := \mathbb{E}_{\pi_{N,\beta}} \left| \frac{1}{N^2} \sum_{i \in \Lambda_N^2} s_i \right|. \tag{5.1.4}$$

The phase transition concerns the thermodynamic limit

$$m(\beta) := \lim_{N \to \infty} m_N(\beta), \tag{5.1.5}$$

For integers a and b, the notation $a \equiv b \pmod{N}$ means that N divides $a - b$; equivalently, there exists an integer k (not necessarily positive) such that $a - b = kN$. Coordinates that differ by a multiple of N therefore represent the same lattice position. For example, $-1 \equiv N - 1 \pmod{N}$ and $N \equiv 0 \pmod{N}$. See also modular arithmetic.

when this limit exists. The model is disordered when $m(\beta) = 0$ and magnetically ordered when $m(\beta) > 0$. Peierls' argument establishes an ordered low-temperature phase, and Onsager's exact solution identifies the critical inverse temperature (Peierls, 1936; Onsager, 1944). Thus,

$$m(\beta) = 0 \quad (0 < \beta \le \beta_C), \qquad m(\beta) > 0 \quad (\beta > \beta_C), \qquad \boxed{\beta_C = \frac{1}{2}\log(1+\sqrt{2}) \approx 0.4406868.} \tag{5.1.6}$$

Accordingly, $\beta_C = \inf\{\beta > 0 : m(\beta) > 0\}$ and, in the units used here, $T_C = 1/\beta_C$. No finite lattice has a sharp phase transition: $m_N(\beta)$ changes smoothly with β, and the sharp transition appears only in the limit $N \to \infty$.

5.1.2 The Computational Problem

For a generic high-dimensional target distribution, evaluating the single-coordinate conditional distribution

$$\pi_{i|-i}(\xi_i \,|\, \xi_1, \ldots, \xi_{i-1}, \xi_{i+1}, \ldots, \xi_d) := \frac{\pi(\xi_1, \ldots, \xi_i, \ldots, \xi_d)}{\pi_{-i}(\xi_1, \ldots, \xi_{i-1}, \xi_{i+1}, \ldots, \xi_d)} = \frac{\pi(\xi_1, \ldots, \xi_i, \ldots, \xi_d)}{\sum_{u \in \mathcal{X}_i} \pi(\xi_1, \ldots, \xi_{i-1}, u, \xi_{i+1}, \ldots, \xi_d)}.$$

may require repeated evaluation of a high-dimensional mass function. For the Ising model, a naive evaluation recomputes an energy involving all lattice edges, even though changing one spin affects only its neighboring edges. The graphical structure removes this redundancy.

The partition function

$$Z_{N,\beta} := \sum_{\boldsymbol{s} \in \{-1,1\}^{\Lambda_N^d}} \exp\left\{-\beta H_N^d(\boldsymbol{s})\right\} \tag{5.1.7}$$

contains 2^{N^d} terms and is generally infeasible to evaluate directly for a large lattice. Fortunately, it cancels from every Gibbs full conditional. We therefore focus on efficient local Gibbs updates; exact computation of $Z_{N,\beta}$ is outside the scope of this chapter.

5.1.3 Systematic-Scan Gibbs Sampling (Recap)

For a target PMF $\pi(\xi_1, \ldots, \xi_d)$ on a finite product space, write $\pi_{j|-j}$ for the conditional PMF of coordinate j given all other coordinates. One systematic Gibbs sweep updates the coordinates in order, using the already updated values for coordinates $1, \ldots, j-1$ and the previous-sweep values for coordinates $j+1, \ldots, d$.

Professor Lars Onsager was an instructor at Brown University from 1928 to 1933. While at Brown, he taught statistical mechanics and published the reciprocal relations for which he later became the sole recipient of the 1968 Nobel Prize in Chemistry. During the Great Depression, Brown did not renew his appointment because the university could not support a faculty member regarded as much stronger in research than in teaching; Onsager subsequently joined Yale University.

Algorithm 16 : Systematic-Scan Gibbs Sampling (recap)

Input: A target PMF $\pi(\xi_1, \ldots, \xi_d)$; an initial state $\boldsymbol{x}^{(0)}$; and a number of sweeps n^*.
Output: The states $\boldsymbol{X}^{(0)}, \boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(n^*)}$ of a Markov chain having π as a stationary distribution.

Set $\boldsymbol{X}^{(0)} \leftarrow \boldsymbol{x}^{(0)}$.
for all $n = 1, \ldots, n^*$ **do**
  Sample $\xi_1^{(n)} \sim \pi_{1|-1}(\cdot \mid \xi_2^{(n-1)}, \ldots, \xi_d^{(n-1)})$.
  for all $j = 2, \ldots, d$ **do**
    Sample $\xi_j^{(n)} \sim \pi_{j|-j}(\cdot \mid \xi_1^{(n)}, \ldots, \xi_{j-1}^{(n)}, \xi_{j+1}^{(n-1)}, \ldots, \xi_d^{(n-1)})$.
  end for
  Set $\boldsymbol{X}^{(n)} \leftarrow (\xi_1^{(n)}, \ldots, \xi_d^{(n)})^\intercal$.
end for

This is the general algorithm specialized below to the N^2 spin coordinates. Its stationary-distribution claim does not by itself guarantee rapid convergence; burn-in and run length must be assessed for the particular target and parameter values.

5.1.4 Gibbs Sampling Using the Graphical Structure of the Ising Model

We now exploit the local graphical structure of the Ising model to simplify each Gibbs update. Consider the model on the d-dimensional periodic lattice of size N, $\Lambda_N^d = (\mathbb{Z}_N)^d$. For each site $i \in \Lambda_N^d$, let

$$\mathcal{N}(i) := \{ \text{ the neighbors of site } i\};$$

and adopt the convention that $\mathcal{N}(i)$ does not include i. In two dimensions, $\mathcal{N}(i)$ consists of the four modular-coordinate neighbors in Eq. (4.1.2); this periodic-boundary structure is encoded by the modulo operations in the R code below. We focus on the following Ising model:

$$\pi_{N,\beta}^d(\boldsymbol{s}) = \frac{1}{Z_{N,\beta}} \exp\left\{\beta \sum_{\langle i,j\rangle} s_i s_j\right\} \tag{5.1.8}$$

$$= \frac{1}{Z_{N,\beta}} \exp\left\{\beta \sum_{\text{all edges } (i,j) \text{ on } \Lambda_N^d} s_i s_j\right\}. \tag{5.1.9}$$

where $\langle i, j\rangle$ indicates that sites i and j are neighbors and each undirected edge is counted once. As in Chapter 4, $\beta = J/(k_B T)$ is the dimensionless coupling.

Suppose we are interested in the spin s_{i^*} at site i^*. The graphical structure allows us to write the Boltzmann distribution $\pi_{N,\beta}^d(\boldsymbol{s})$ as

$$\pi_{N,\beta}^d(\boldsymbol{s}) = \frac{1}{Z_{N,\beta}} \exp\left\{\beta \left[\sum_{j \in \mathcal{N}(i^*)} s_{i^*} s_j\right] + \beta \left[\sum_{\langle i',j'\rangle \text{ and } i',j' \neq i^*} s_{i'} s_{j'}\right]\right\}. \tag{5.1.10}$$

A key quantity in Gibbs sampling is the conditional distribution of s_{i^*} given the other coordinates (see Algorithm

15), denoted by $\pi^d_{N,\beta}(s_{i^*}|\boldsymbol{s}_{-i^*})$. Using Eq. (5.1.10), this conditional distribution is

$$\begin{aligned}
\pi^d_{N,\beta}(s_{i^*}|\boldsymbol{s}_{-i^*}) &= \frac{\pi^d_{N,\beta}(\boldsymbol{s})}{\sum_{u\in\{-1,1\}}\pi^d_{N,\beta}(s_{i^*}=u,\boldsymbol{s}_{-i^*})} \\
&= \frac{\frac{1}{Z_{N,\beta}}\exp\left\{\beta\left[\sum_{\langle i',j'\rangle \text{ and } i',j'\neq i^*} s_{i'}s_{j'}\right]\right\}\exp\left\{\beta\left[\sum_{j\in\mathcal{N}(i^*)} s_{i^*}s_j\right]\right\}}{\frac{1}{Z_{N,\beta}}\exp\left\{\beta\left[\sum_{\langle i',j'\rangle \text{ and } i',j'\neq i^*} s_{i'}s_{j'}\right]\right\}\sum_{u\in\{-1,1\}}\exp\left\{\beta\left[\sum_{j\in\mathcal{N}(i^*)} us_j\right]\right\}} \\
&= \frac{\exp\left\{\beta\left[\sum_{j\in\mathcal{N}(i^*)} s_{i^*}s_j\right]\right\}}{\sum_{u\in\{-1,1\}}\exp\left\{\beta\left[\sum_{j\in\mathcal{N}(i^*)} us_j\right]\right\}} \\
&= \frac{\exp\left\{\beta\left[\sum_{j\in\mathcal{N}(i^*)} s_{i^*}s_j\right]\right\}}{\exp\left\{-\beta\left[\sum_{j\in\mathcal{N}(i^*)} s_j\right]\right\}+\exp\left\{\beta\left[\sum_{j\in\mathcal{N}(i^*)} s_j\right]\right\}},
\end{aligned} \tag{5.1.11}$$

Define the **local field** at i^* by $h_{i^*}(\boldsymbol{s}) := \sum_{j\in\mathcal{N}(i^*)} s_j$. Then the two conditional probabilities have the logistic form

$$\pi^d_{N,\beta}(1|\boldsymbol{s}_{-i^*}) = \frac{e^{\beta h_{i^*}}}{e^{-\beta h_{i^*}}+e^{\beta h_{i^*}}} = \frac{1}{1+e^{-2\beta h_{i^*}}}, \tag{5.1.12}$$

$$\pi^d_{N,\beta}(-1|\boldsymbol{s}_{-i^*}) = \frac{e^{-\beta h_{i^*}}}{e^{-\beta h_{i^*}}+e^{\beta h_{i^*}}} = 1-\pi^d_{N,\beta}(1|\boldsymbol{s}_{-i^*}). \tag{5.1.13}$$

Consequently, a single-site Gibbs update requires only the following Bernoulli draw:

i) Compute $h_{i^*} = \sum_{j\in\mathcal{N}(i^*)} s_j$ and $p_{i^*} = (1+e^{-2\beta h_{i^*}})^{-1}$.

ii) Draw $B \sim \text{Bernoulli}(p_{i^*})$ and set $s_{i^*} \leftarrow 2B-1$.

Equivalently, as in the direct conditional-sampling construction, one may compute

$$a = e^{\beta h_{i^*}}, \qquad b = e^{-\beta h_{i^*}}, \qquad p_{i^*} = \frac{a}{a+b},$$

then draw $B \sim \text{Bernoulli}(p_{i^*})$ and return $2B-1$. The logistic expression above is numerically preferable because a stable implementation such as `plogis` avoids separately exponentiating a and b. For every finite β, both conditional probabilities lie strictly between zero and one. A full systematic sweep can therefore reach any spin configuration with positive probability, and it can leave the current configuration unchanged with positive probability; the finite-state sweep chain is irreducible and aperiodic and has $\pi^d_{N,\beta}$ as its stationary distribution.

The factor $\frac{1}{Z_{N,\beta}}\exp\left\{\beta\left[\sum_{\langle i',j'\rangle \text{ and } i',j'\neq i^*} s_{i'}s_{j'}\right]\right\}$ in Eq. (5.1.11) cancels. This cancellation is important because it removes both the unknown partition function and all terms not involving s_{i^*}. We explain the computational benefit as follows.

- A naive evaluation of the second line recomputes an energy sum over dN^d undirected lattice edges, so its cost grows with the lattice size.

- The third line of Eq. (5.1.11) uses only the $2d$ neighboring spins (four spins when $d = 2$). Its cost is $O(d)$,† independent of the lattice size N.

Using the third line of Eq. (5.1.11), we can substantially improve the efficiency of code for simulating random spin configurations of the Ising model.

Estimating the Critical Inverse Temperature

Let $\boldsymbol{S}^{(n)} = (S_i^{(n)})_{i\in\Lambda_N^2}$ denote the configuration after the n-th complete sweep through the lattice. After discarding B burn-in sweeps, the finite-volume magnetization can be estimated by

$$\widehat{m}_{N,M,B}(\beta) := \frac{1}{M-B}\sum_{n=B+1}^{M}\left|\frac{1}{N^2}\sum_{i\in\Lambda_N^2} S_i^{(n)}\right|. \tag{5.1.14}$$

For fixed N, the Markov-chain ergodic theorem gives $\widehat{m}_{N,M,B}(\beta) \to m_N(\beta)$ almost surely as the number of retained sweeps grows. Thus, for a large lattice and a sufficiently long run,

$$m(\beta) \approx m_N(\beta) \approx \widehat{m}_{N,M,B}(\beta). \tag{5.1.15}$$

Approximating the thermodynamic quantity $m(\beta)$ involves two distinct errors:

i) **finite-size error**, from replacing $m(\beta)$ by $m_N(\beta)$;

ii) **Monte Carlo error**, from replacing $m_N(\beta)$ by Eq. (5.1.14) after a finite run.

Both errors are especially important near β_C, where finite-size rounding and slow mixing make the apparent transition less sharp. Burn-in length and retained sample size should therefore be supported by convergence diagnostics rather than chosen as universal constants.

The supplied HW 9 numerical experiment uses a 100×100 periodic lattice and starts each chain from the all-minus configuration. Figure 5.1 displays $\boldsymbol{S}^{(10000)}$ at 16 equally spaced values of β from 0.3 to 0.5. The growing regions of aligned spins illustrate the transition from disorder to order as β crosses the neighborhood of β_C.

† The notation $O(d)$ means that the number of operations is bounded by $C \cdot d$ for some constant C.

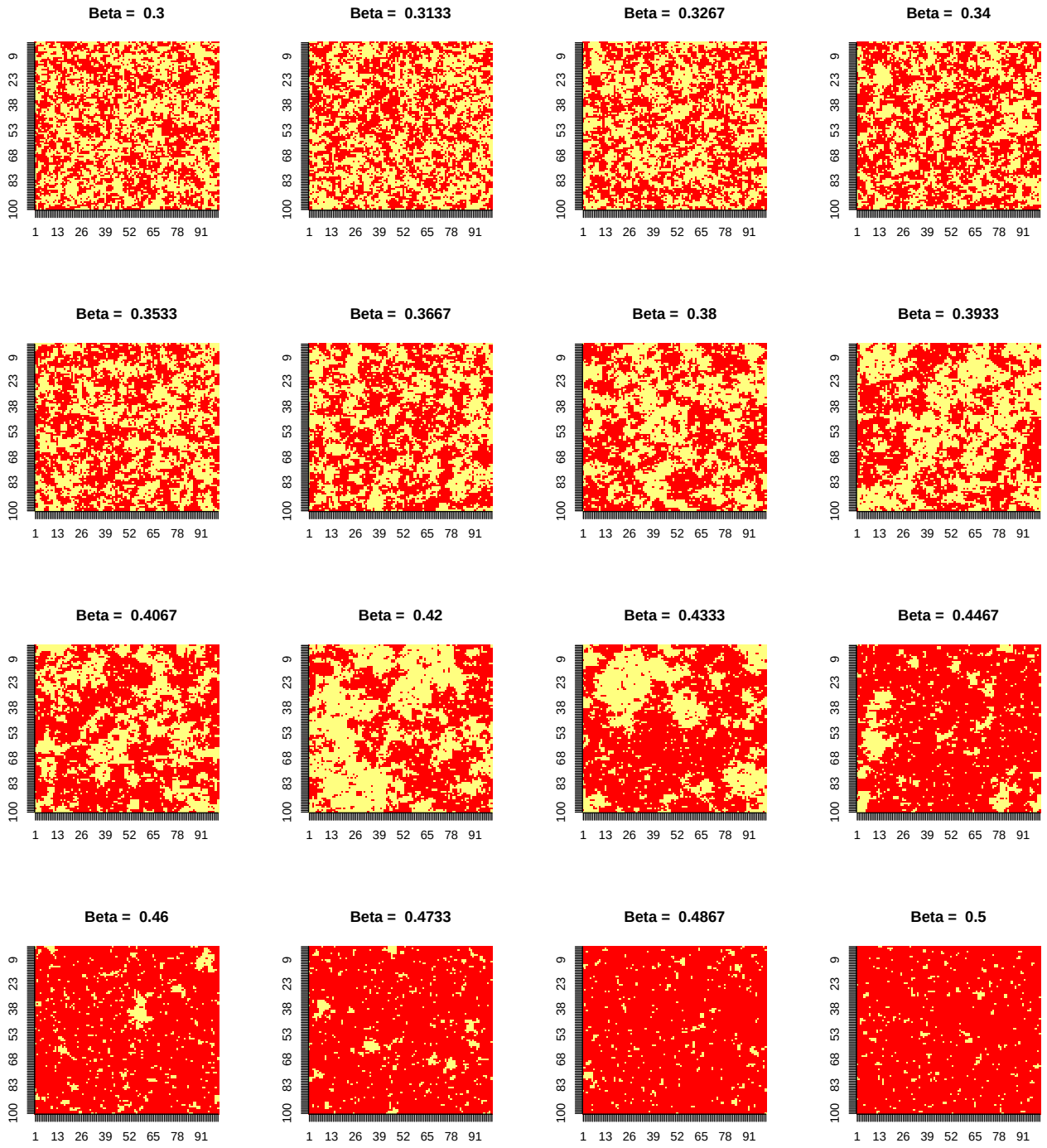


Figure 5.1: Final configurations after 10,000 systematic Gibbs sweeps on a 100×100 periodic lattice at 16 values of β. The exact critical inverse temperature is $\beta_C = \frac{1}{2}\log(1+\sqrt{2}) \approx 0.4407$. A final configuration alone is a qualitative illustration, not a convergence diagnostic.

For the accompanying magnetization experiment, take 20 equally spaced values of β between 0.2 and 0.6, set

$N = 100$ and $M = 10{,}000$, and compute Eq. (5.1.14). The supplied run used $B = 0$, so its plotted statistic is

$$\widehat{m}_{N,M,0}(\beta) := \frac{1}{M}\sum_{n=1}^{M}\left|\frac{1}{N^2}\sum_{i\in\Lambda_N^2} S_i^{(n)}\right|. \tag{5.1.16}$$

Figure 5.2 shows the resulting curve and the exact value of β_C. Its steep increase is consistent with Eq. (5.1.6). In a formal analysis, one should discard a justified burn-in period, estimate Monte Carlo uncertainty, and compare several lattice sizes before using the curve to estimate the thermodynamic critical point.

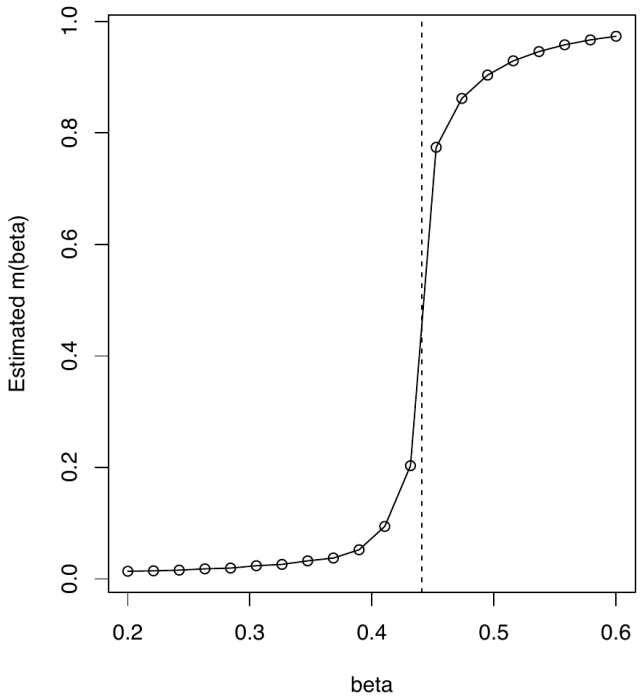


Figure 5.2: The HW 9 plot of $\widehat{m}_{N,M,0}(\beta)$ against β for $N = 100$ and $M = 10{,}000$. The vertical dashed line is the exact critical value $\beta_C \approx 0.4407$.

The implementation retains the four-neighbor lookup-table idea but records only the magnetization from each sweep, rather than storing all 10,000 full 100×100 configurations.

```r
set.seed(1690)

N <- 100L
n_sweeps <- 10000L
burn_in <- 0L  # Matches the supplied run; justify B > 0 in applications.
beta_grid <- seq(0.2, 0.6, length.out = 20L)

estimate_magnetization <- function(beta, N, n_sweeps, burn_in = 0L) {
```

```r
  X <- matrix(-1L, nrow = N, ncol = N)
  magnetization <- numeric(n_sweeps)

  # Lookup table: index 1 represents spin -1 and index 2 spin +1.
  p_plus <- array(0, dim = c(2L, 2L, 2L, 2L))
  for (s1 in 1:2) for (s2 in 1:2)
    for (s3 in 1:2) for (s4 in 1:2) {
      h <- 2 * (s1 + s2 + s3 + s4) - 12
      p_plus[s1, s2, s3, s4] <- plogis(2 * beta * h)
    }

  # One outer iteration is one complete systematic Gibbs sweep.
  for (sweep in seq_len(n_sweeps)) {
    for (j in seq_len(N)) {
      jp1 <- (j %% N) + 1L
      jm1 <- ((j - 2L) %% N) + 1L
      for (i in seq_len(N)) {
        ip1 <- (i %% N) + 1L
        im1 <- ((i - 2L) %% N) + 1L

        index <- (c(X[ip1, j], X[im1, j],
                    X[i, jp1], X[i, jm1]) + 3L) / 2L
        p <- p_plus[index[1], index[2], index[3], index[4]]
        X[i, j] <- if (runif(1) < p) 1L else -1L
      }
    }
    magnetization[sweep] <- abs(mean(X))
  }

  mean(magnetization[(burn_in + 1L):n_sweeps])
}

estimated_m <- vapply(
  beta_grid, estimate_magnetization, numeric(1),
  N = N, n_sweeps = n_sweeps, burn_in = burn_in
)

plot(beta_grid, estimated_m, type = "b",
     xlab = expression(beta), ylab = "Estimated m(beta)")
abline(v = log(1 + sqrt(2)) / 2, lty = 2)
```

5.2 Gibbs Random Fields and Markov Random Fields

"*He was too simple to wonder when he had attained humility. But he knew he had attained it, and he knew it was not disgraceful and it carried no loss of true pride.*"

— sentences from *The Old Man and the Sea*, by Ernest Hemingway

Section 5.1.4 showed that an Ising update depends only on neighboring spins. We now formalize that local dependence using graphs and extend it beyond the Ising model. After introducing the necessary graph terminology, we define Gibbs random fields and relate their factorization to the Markov property.

5.2.1 Basic Concepts in Graph Theory

We list the concepts needed for graphical models as follows.

- A **simple undirected graph** is an ordered pair $G = (V, E)$ consisting of:

  i) a set V of **vertices**, whose elements are usually indexed by positive integers;

  ii) a set

  $$E \subseteq \big\{\{i, j\} \;\big|\; i, j \in V \text{ and } i \neq j\big\}$$

  of **edges**, where each edge $\{i, j\}$ is a two-element, and hence unordered, subset of V.

- For a given graph $G = (V, E)$, two vertices i and j are called **adjacent** if $\{i, j\} \in E$.

- For a given graph $G = (V, E)$ and a vertex $i \in V$, the **neighborhood** of i is the set of vertices adjacent to i, i.e.,

  $$\mathcal{N}(i) := \big\{j \in V \;\big|\; \{i, j\} \in E\big\}.$$

  Because the graph is simple and therefore has no self-loops, $i \notin \mathcal{N}(i)$.

- For a given graph $G = (V, E)$, a set of vertices (i.e., a subset $c \subseteq V$) is called a **clique** if every pair of vertices in this set is adjacent. We denote $\mathscr{C}(G) := \{$ all cliques of graph $G\}$. We also adopt the convention that each individual vertex is a clique, i.e., $\{i\} \in \mathscr{C}(G)$ for all $i \in V$.

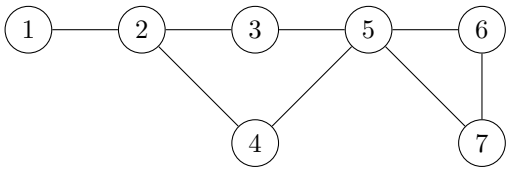


Figure 5.3: A graph $G = (V, E)$ with $V = \{1, 2, \ldots, 7\}$ and $E = \{\{1,2\}, \{2,3\}, \{2,4\}, \{3,5\}, \{4,5\}, \{5,6\}, \{5,7\}, \{6,7\}\}$.

Figure 5.3 provides an example of a graph. In this graph, for example, vertices 1 and 2 are adjacent, and vertices 2 and 3 are adjacent. Additionally, the neighborhood of vertex 5 is $\mathcal{N}(5) = \{3, 4, 6, 7\}$. We list all the elements of $\mathscr{C}(G)$ as follows:

$$\begin{aligned} &\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\}, \\ &\{1,2\}, \{2,3\}, \{2,4\}, \{3,5\}, \{4,5\}, \{5,6\}, \{5,7\}, \{6,7\}, \\ &\{5,6,7\}. \end{aligned} \tag{5.2.1}$$

We also adopt the following notation

- Let $G = (V, E)$ be a graph with $V = \{1, 2, \ldots, d\}$ and $\boldsymbol{x} = (x_1, x_2, \ldots, x_d)^\intercal$ a vector in the product space $\mathcal{X} = \mathcal{X}_1 \times \cdots \mathcal{X}_d$.
- For any subset $c = \{i_1, \ldots, i_k\} \subseteq V$ with $i_1 < i_2 < \cdots < i_k$, we denote $x_c := (x_{i_1}, x_{i_2}, \ldots, x_{i_k})^\intercal$ and $\mathcal{X}_c := \mathcal{X}_{i_1} \times \cdots \times \mathcal{X}_{i_k}$. For example, let $V = \{1, \ldots, 7\}$ and $c = \{1, 3, 5\}$, then $x_c = (x_1, x_3, x_5)^\intercal$ and $\mathcal{X}_c = \mathcal{X}_1 \times \mathcal{X}_3 \times \mathcal{X}_5$.

5.2.2 Gibbs Random Fields

The Ising calculation in Eq. (5.1.11) shows that factors unrelated to the updated spin cancel from its full conditional distribution. Gibbs random fields abstract precisely this local factorization. We first define a GRF, with the Ising model as a special case, and then show that the same cancellation yields a local Markov property.

Definition 5.2.1. *Let $\boldsymbol{X} = (X_1, \ldots, X_d)^\intercal$ be a random vector taking values in $\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_d$. Denote its PMF† by $p_{\boldsymbol{X}}(\boldsymbol{x})$. Let $G = (V, E)$ be a graph with $V = \{1, \ldots, d\}$. We say $\boldsymbol{X}$ is a Gibbs random field with respect to the graph G if there exists a collection $\{\varphi_c(x_c)\}_{c \in \mathscr{C}(G)}$ of functions such that*

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = p_{\boldsymbol{X}}(x_1, \ldots, x_d) = \frac{1}{Z} \prod_{c \in \mathscr{C}(G)} \varphi_c(x_c), \tag{5.2.2}$$

where φ_c are positive-valued functions for all $c \in \mathscr{C}(G)$, and Z is the normalizing constant defined as follows

$$Z := \sum_{x_1 \in \mathcal{X}_1} \cdots \sum_{x_d \in \mathcal{X}_d} \left\{ \prod_{c \in \mathscr{C}(G)} \varphi_c(x_c) \right\}. \tag{5.2.3}$$

Remark: Throughout this chapter all PMFs $p_{\boldsymbol{X}}(\boldsymbol{x})$ are assumed to be strictly positive. The functions $\varphi_c(x_c)$ in Eq. (5.2.2) are not necessarily PMFs; they only need to be positive. In addition, the factorization

† We focus on the PMF scenario for simplicity. This framework also works for the PDF or more general scenarios.

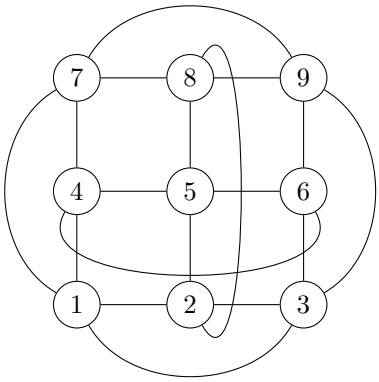


Figure 5.4: The graph for the 2D Ising model with lattice size of 3.

in Eq. (5.2.2) is not unique. For example, we can have

$$\varphi_c(x_c)\cdot\varphi_{c'}(x_{c'}) = \left(\frac{1}{1690}\cdot\varphi_c(x_c)\right)\cdot(1690\cdot\varphi_{c'}(x_{c'})) = \left(\frac{1}{1690}\cdot\varphi_c(x_c)\right)\cdot(1690\cdot\varphi_{c'}(x_{c'}))\cdot\varphi_{c''}(x_{c''}),$$

where $\varphi_{c''}(x_{c''})\equiv 1$. All we need is **to find one factorization**.

The random spin configuration $\boldsymbol{s}$ from the Ising model in Eq. (5.1.8) is a GRF because its distribution factors as

$$\pi^d_{N,\beta}(\boldsymbol{s}) = \frac{1}{Z_{N,\beta}}\prod_{\langle i,j\rangle}\exp\{\beta s_i s_j\}, \tag{5.2.4}$$

where $\langle i,j\rangle$ denotes an undirected neighboring pair, counted once. We take the scenario $d=2, N=3$ as an example and investigate the graph in Figure 5.4. Because each row and column of the 3×3 periodic lattice forms a 3-cycle, its cliques comprise the singletons, the neighboring pairs, and the following six three-vertex cliques:

$$\{1,2,3\},\ \{4,5,6\},\ \{7,8,9\},\ \{1,4,7\},\ \{2,5,8\},\ \{3,6,9\}.$$

Define $\varphi_{\{i,j\}}(s_i,s_j) = \exp\{\beta s_i s_j\}$ for neighboring pairs and set the potentials of all other cliques equal to one. Then, Eq. (5.2.4) satisfies the definition of a GRF with respect to the graph in Figure 5.4. (For $N\geq 4$, the square torus has no three-vertex cliques.)

5.2.3 Markov Property and Markov Random Fields

The GRF factorization now yields its central consequence: a coordinate's conditional distribution depends only on factors involving that coordinate, and hence only on its graph neighbors. This is the Markov property. The derivation generalizes the cancellation used for the Ising model in Eq. (5.1.11).

Let $\boldsymbol{X} = (X_1, \ldots, X_d)^\intercal$ be a GRF with respect to the graph $G = (V, E)$ and factorization

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{Z} \prod_{c \in \mathscr{C}(G)} \varphi_c(x_c).$$

Suppose we are interested in the i-th component of $\boldsymbol{X}$ and would like to compute the conditional distribution $(X_i | X_{-i})$ (perhaps for the purpose of conducting Gibbs sampling). Then,

$$\begin{aligned} p(x_i | x_{-i}) &= \frac{p_{\boldsymbol{X}}(\boldsymbol{x})}{\sum_{x_i \in \mathcal{X}_i} p_{\boldsymbol{X}}(\boldsymbol{x})} \\ &= \frac{\prod_{c \in \mathscr{C}(G)} \varphi_c(x_c)}{\sum_{x_i \in \mathcal{X}_i} \prod_{c \in \mathscr{C}(G)} \varphi_c(x_c)} \\ &= \frac{\left(\prod_{c \in \mathscr{C}(G) \text{ and } i \notin c} \varphi_c(x_c) \right) \cdot \left(\prod_{c \in \mathscr{C}(G) \text{ and } i \in c} \varphi_c(x_c) \right)}{\left(\prod_{c \in \mathscr{C}(G) \text{ and } i \notin c} \varphi_c(x_c) \right) \cdot \left(\sum_{x_i \in \mathcal{X}_i} \prod_{c \in \mathscr{C}(G) \text{ and } i \in c} \varphi_c(x_c) \right)} \\ &= \frac{\prod_{c \in \mathscr{C}(G) \text{ and } i \in c} \varphi_c(x_c)}{\sum_{x_i \in \mathcal{X}_i} \prod_{c \in \mathscr{C}(G) \text{ and } i \in c} \varphi_c(x_c)}. \end{aligned} \tag{5.2.5}$$

The factor cancellation in the third line of Eq. (5.2.5) is a generalization of the cancellation in Eq. (5.1.11) for the Ising model. It is easy to verify the following, which explains why we are interested in cliques

$$\mathcal{N}(i) = \left(\bigcup_{c \in \mathscr{C}(G) \text{ and } i \in c} c \right) - \{i\}. \tag{5.2.6}$$

The verification of Eq. (5.2.6) is left for homework. Eq. (5.2.6) and the last line of Eq. (5.2.5) imply that $p(x_i | x_{-i})$ depends only on the indices in $\{i\} \bigcup \mathcal{N}(i)$. We may represent this fact by the following formula

$$p(x_i | x_{-i}) = p(x_i | x_{\mathcal{N}(i)}). \tag{5.2.7}$$

The left-hand side of Eq. (5.2.7) involves all the coordinates of $(x_1, \ldots, x_d)^\intercal$, while the right-hand side involves only i and its neighbors. If $\max_{i \in V} \#\mathcal{N}(i)$ is small, the computation required for the conditional distribution $p(x_i | x_{-i})$ is substantially reduced, as illustrated by the improved Gibbs sampling method for the Ising model.

If we regard vertex i as the "future," its neighborhood $\mathcal{N}(i)$ as the "present," and the nonneighboring vertices as the "past," then Eq. (5.2.7) has the same conditional-independence structure as the Markov property. This observation motivates the following definition.

Definition 5.2.2. *Let $\boldsymbol{X} = (X_1, \ldots, X_d)^\intercal$ be a random vector and $G = (V, E)$ be a graph with $V = \{1, 2, \ldots, d\}$.*

Then, $\boldsymbol{X}$ is called a Markov random field (MRF) with respect to G if the Markov property in Eq. (5.2.7) is satisfied.

The following theorem shows that, under positivity, this local Markov property is equivalent to the GRF factorization.

Theorem 5.2.1. ***(Hammersley–Clifford, 1971)*** *Suppose that $p_{\boldsymbol{X}}(\boldsymbol{x}) > 0$ for every $\boldsymbol{x} \in \mathcal{X}$. Then $\boldsymbol{X}$ is a GRF w.r.t. G if and only if $\boldsymbol{X}$ is an MRF w.r.t. G.*

The theorem is named after John Hammersley and Peter Clifford, who proved the equivalence in an unpublished paper in 1971. Simpler proofs using the inclusion–exclusion principle were given independently by Grimmett (1973), Preston (1973) and Sherman (1973), with a further proof by Besag (1974).

Remark 5.2.1. *i) The proof of one direction of Theorem 5.2.1 has been provided in our discussion above. That an MRF is a GRF is a deep fact whose proof is mathematically involved and requires careful reasoning. The proof is omitted.*

ii) Strict positivity is essential for the MRF-to-GRF direction. The finite-temperature Ising PMF in Eq. (4.1.5) is strictly positive, so the theorem applies to it.

5.3 Homework for Chapter 5

Problem 5.3.1. *Consider the following 2D Ising model on the periodic lattice Λ_N^2:*

$$\pi_{N,\beta}^2(\boldsymbol{s}) = \frac{1}{Z_{N,\beta}} \exp\left\{\beta \sum_{\langle i,j\rangle} s_i s_j\right\}.$$

Its finite-volume average absolute magnetization is

$$m_N^2(\beta) := \mathbb{E}_{\pi_{N,\beta}^2}\left|\frac{1}{N^2}\sum_{i\in\Lambda_N^2} s_i\right|.$$

Set $N = 100$ and consider the twenty values

$$\mathcal{B} := \{0.2 + 0.02k : k = 1, 2, \ldots, 20\}.$$

Complete the following tasks using the single-site Gibbs update in Eq. (5.1.12).

i) For every $\beta \in \mathcal{B}$, simulate configurations

$$\left\{\boldsymbol{S}^{(n)} = (S_i^{(n)})_{i\in\Lambda_N^2}\right\}_{n=0}^{1000},$$

where n counts complete systematic sweeps. State the initial configuration and use periodic boundary conditions.

ii) Plot the final configurations $\boldsymbol{S}^{(1000)}$ for $\beta \in \{0.32, 0.40, 0.44, 0.60\}$ using the same color scale and orientation in all four panels.

iii) For every $\beta \in \mathcal{B}$, compute the raw estimate

$$\widehat{m}^2_{N,1000,0}(\beta) := \frac{1}{1000} \sum_{n=1}^{1000} \left| \frac{1}{N^2} \sum_{i \in \Lambda_N^2} S_i^{(n)} \right|.$$

Choose and justify a burn-in value $B < 1000$, and also compute the burn-in-adjusted estimate $\widehat{m}^2_{N,1000,B}(\beta)$ in Eq. (5.1.14). Explain why the two estimates may differ, especially near the critical region.

iv) Report the twenty values of both estimators and plot each estimator against β.

v) Use trace plots or another convergence diagnostic to assess the runs. Explain why these finite-lattice, finite-run results illustrate the transition but do not by themselves provide an exact estimate of the thermodynamic critical value β_C.

Include the code used for every task.

Problem 5.3.2. *Let $G = (V, E)$ be a graph with $V = \{1, 2, \ldots, d\}$. Prove the following equation (see Eq. (5.2.6))*

$$\mathcal{N}(i) = \left(\bigcup_{c \in \mathscr{C}(G) \text{ and } i \in c} c \right) - \{i\}.$$

Chapter 6

Dimension Reduction

High-dimensional data often contain useful low-dimensional structure. This chapter begins with the manifold hypothesis and distinguishes manifold learning from the broader task of dimension reduction. It then develops principal component analysis (PCA) as a linear dimension-reduction method: after reviewing the necessary covariance and eigenvalue tools, we derive PCA from fitting-error and variance viewpoints, discuss dimension selection, and conclude with practical computation.

6.1 The Manifold Hypothesis

Dimension reduction is the broad task of representing high-dimensional data in a lower-dimensional form while retaining information relevant to the analysis. Manifold learning is one family of dimension-reduction methods. It assumes a low-dimensional geometric structure and is motivated by the "manifold hypothesis": *high-dimensional data tend to lie in the vicinity of a low-dimensional manifold.* (Fefferman et al., 2016, 2025). The dimension of the ambient space is called the **extrinsic dimension**, whereas the dimension of the underlying manifold is called the **intrinsic dimension**.

6.2 Notation

The following notation will be used throughout the chapter.

- d and D are two positive integers satisfying $d < D$. Hereafter, d denotes an intrinsic dimension, and D denotes an extrinsic dimension.

The precise definition of a manifold is quite technical and beyond the scope of APMA 1690. For this course, it is enough to think of a manifold roughly as a surface or curve.

The first author, Charles Fefferman, became a full professor at the University of Chicago at age 22, making him the youngest full professor appointed in the United States. He entered the University of Maryland at age 14, graduated with degrees in mathematics and physics at 17, and earned his PhD in mathematics three years later from Princeton University.

- For any matrix $\boldsymbol{A}$, its transpose is denoted by $\boldsymbol{A}^\intercal$.
- For any $\boldsymbol{x} = (x_1, \ldots, x_D)^\intercal \in \mathbb{R}^D$, its norm is defined as $\|\boldsymbol{x}\| := \sqrt{\sum_{k=1}^D x_k^2}$. Furthermore, if the column vector $\boldsymbol{x}$ is viewed as a D-by-1 matrix, we have

$$\|\boldsymbol{x}\|^2 = \boldsymbol{x}^\intercal \boldsymbol{x}. \tag{6.2.1}$$

- All vectors in this chapter are viewed as column vectors, following a common convention in the literature.
- $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ are independent and identically distributed (iid) $\mathbb{R}^D$-valued random variables (i.e., D-dimensional random vectors).

6.3 A Glimpse of Manifold Learning

The manifold hypothesis from Section 6.1 can be expressed mathematically as follows:

- Latent d-dimensional random vectors $\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_n$ are generated in an iid way. They are not observed.
- There exists a function $g : \mathbb{R}^d \to \mathbb{R}^D$. The image of g, i.e.,

$$\mathcal{M} := \left\{ g(z) \,:\, z \in \mathbb{R}^d \right\}, \tag{6.3.1}$$

  is called a manifold. Neither g nor $\mathcal{M}$ is observed.
- The data points $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ that we observe in the high-dimensional space $\mathbb{R}^D$ are generated by the following mechanism

$$\boldsymbol{X}_i = g(\boldsymbol{Z}_i) + \boldsymbol{\varepsilon}_i, \qquad \text{for all } i = 1, 2, \ldots, n, \tag{6.3.2}$$

  where $\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_n$ are iid D-dimensional random vectors playing the role of random noise. We assume that the noise sequence $(\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_n)$ is independent of the latent sequence $(\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_n)$.

The ultimate goal of manifold learning is to learn the low-dimensional manifold $\mathcal{M}$ defined in Eq. (6.3.1). The observed high-dimensional data points $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ can then be reduced to their projections $\pi_{\mathcal{M}}(\boldsymbol{X}_1), \ldots, \pi_{\mathcal{M}}(\boldsymbol{X}_n)$ on $\mathcal{M}$. Figure 6.1 illustrates this latent-structure viewpoint. Before specializing to PCA, the next section places manifold learning within the broader landscape of dimension reduction.

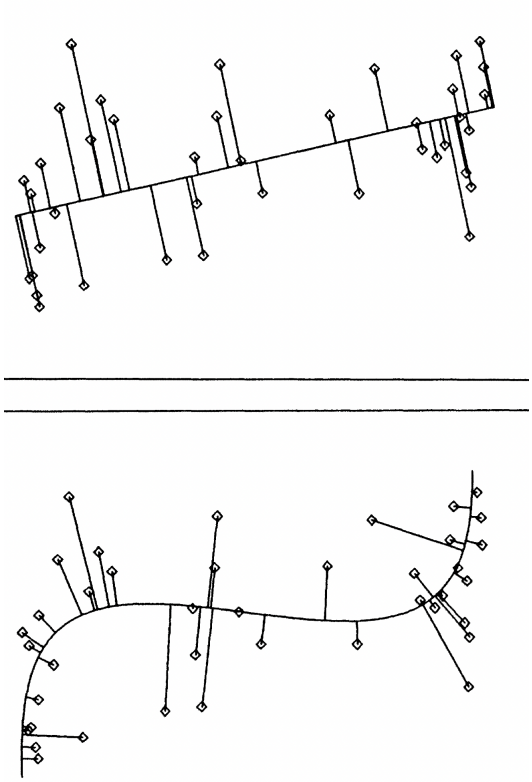


Figure 6.1: (2-dimensional) data points $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ and their projections $\pi_{\mathcal{M}}(\boldsymbol{X}_1), \ldots, \pi_{\mathcal{M}}(\boldsymbol{X}_n)$ on (1-dimensional) manifolds. The upper panel illustrates projection onto a linear manifold, and the lower panel illustrates projection onto a nonlinear manifold. This figure comes from Hastie and Stuetzle (1989).

6.4 Dimension Reduction and Manifold Learning

As the preceding model illustrates, manifold learning is one way to obtain a lower-dimensional representation, but it is not the only one. Dimension reduction also includes methods that do not posit or learn a manifold. Two broad classes are:

- Linear dimension reduction, e.g., principal component analysis (PCA, Pearson, 1901). Although a linear subspace is a manifold, PCA is conventionally treated as a linear dimension-reduction method and is not generally labeled manifold learning.
- Nonlinear dimension reduction, including many manifold-learning methods such as principal curves (Hastie and Stuetzle, 1989), Isomap (Tenenbaum et al., 2000), locally linear embedding (Roweis and Saul, 2000), Laplacian eigenmaps (Belkin and Niyogi, 2001), diffusion maps (Coifman et al., 2005; Coifman and Lafon, 2006), and principal manifolds (Smola et al., 2001; Meng and Eloyan, 2021).

This chapter focuses on the most widely used linear dimension-reduction approach—PCA—while using the manifold hypothesis to motivate why low-dimensional representations can be useful.

6.5 Mathematical Preparations

To analyze PCA, we first collect the covariance and linear-algebra identities used in its derivation. The material in this section is drawn from Sections 1.4 and 1.5 of Seber and Lee (2012).

For any D-dimensional random vector $\boldsymbol{X} = (X_1, \ldots, X_D)^\intercal$, we define its **covariance matrix** $\mathbb{V}(\boldsymbol{X})$ as follows (Seber and Lee, 2012, Definition 1.3)

$$\mathbb{V}(\boldsymbol{X}) := \begin{pmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_D) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) & \cdots & \mathrm{Cov}(X_2, X_D) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_D, X_1) & \mathrm{Cov}(X_D, X_2) & \cdots & \mathrm{Cov}(X_D, X_D) \end{pmatrix} = \Big(\mathrm{Cov}(X_i, X_j)\Big)_{1 \le i,j \le D}, \tag{6.5.1}$$

which is a D-by-D symmetric matrix. It also has the following properties.

Claim 6.5.1. *The covariance matrix $\mathbb{V}(\boldsymbol{X})$ defined in Eq. (6.5.1) is positive semi-definite.*

Proof. The proof is assigned in Section 6.7. □

One may apply the following claim to prove Claim 6.5.1.

Claim 6.5.2 (Theorem 1.3 of Seber and Lee (2012))**.** *Let $\boldsymbol{A}$ be any m-by-D deterministic matrix. Then,*

$$\mathbb{V}(\boldsymbol{A}\boldsymbol{X}) = \boldsymbol{A}\mathbb{V}(\boldsymbol{X})\boldsymbol{A}^\intercal.$$

Claim 6.5.3 (Theorem 1.5 of Seber and Lee (2012))**.** *Let $\boldsymbol{A}$ be any D-by-D symmetric matrix. Suppose $\boldsymbol{\Sigma} := \mathbb{V}(\boldsymbol{X})$ and $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_D)^\intercal = \mathbb{E}\boldsymbol{X}$. Then, we have*

$$\mathbb{E}(\boldsymbol{X}^\intercal \boldsymbol{A}\boldsymbol{X}) = \mathrm{tr}(\boldsymbol{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\intercal \boldsymbol{A}\boldsymbol{\mu}, \tag{6.5.2}$$

where $\mathrm{tr}(\cdot)$ denotes the trace of a square matrix.

6.6 Principal Component Analysis

With the necessary mathematical preparation in place, we now develop PCA. Without loss of generality, we assume that the data points $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ have been centered. That is, throughout this section we work under the following assumption.

Assumption 1. *Data points $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ satisfy*

$$\mathbb{E}\boldsymbol{X}_i = (0, 0, \ldots, 0)^\intercal =: \boldsymbol{0}, \quad \text{for all } i = 1, 2, \ldots, n. \tag{6.6.1}$$

Assumption 1 is widely adopted in the PCA literature. It can be satisfied by centering the data, namely by replacing $\boldsymbol{X}_i$ with $\boldsymbol{X}_i - \frac{1}{n}\sum_{j=1}^n \boldsymbol{X}_j$.

The PCA framework assumes that $\mathcal{M}$, defined in Eq. (6.3.1), is a d-dimensional linear subspace of $\mathbb{R}^D$. Hereafter, we use V to denote such a subspace.

6.6.1 PCA from the Viewpoint of Fitting Errors

Let V be a d-dimensional linear subspace of $\mathbb{R}^D$. For any $\boldsymbol{x} \in \mathbb{R}^D$, its projection onto V is denoted by $\boldsymbol{P}_V \boldsymbol{x}$.

Let $\lambda_1 \geq \cdots \geq \lambda_D$ be the eigenvalues of $\mathbb{V}(\boldsymbol{X})$. For the chosen dimension $d < D$, we assume the eigengap $\lambda_d > \lambda_{d+1}$; this condition ensures that the optimal d-dimensional linear subspace below is unique.

PCA models the underlying manifold (see Eq. (6.3.1)) as the linear subspace V^* that minimizes the fitting error $\mathbb{E}\left(\|\boldsymbol{X} - \boldsymbol{P}_V \boldsymbol{X}\|^2\right)$:

$$V^* = \operatorname*{argmin}_V \left\{\mathbb{E}\left(\|\boldsymbol{X} - \boldsymbol{P}_V \boldsymbol{X}\|^2\right) : V \text{ is a } d\text{-dimensional linear subspace of } \mathbb{R}^D\right\}. \tag{6.6.2}$$

Hereafter, all minimizations and maximizations are taken over all d-dimensional linear subspaces V of $\mathbb{R}^D$.

6.6.2 PCA from the Viewpoint of Variance

The fitting-error formulation has an equivalent variance interpretation. By the Pythagorean theorem, we have $\|\boldsymbol{X} - \boldsymbol{P}_V \boldsymbol{X}\|^2 = \|\boldsymbol{X}\|^2 - \|\boldsymbol{P}_V \boldsymbol{X}\|^2$. Therefore,

$$\min_V \left\{\mathbb{E}\left(\|\boldsymbol{X} - \boldsymbol{P}_V \boldsymbol{X}\|^2\right)\right\} = \mathbb{E}\left(\|\boldsymbol{X}\|^2\right) - \max_V \left\{\mathbb{E}\left(\|\boldsymbol{P}_V \boldsymbol{X}\|^2\right)\right\}.$$

Therefore, we have the following representations

$$\boxed{V^* = \operatorname*{argmin}_V \left\{\mathbb{E}\left(\|\boldsymbol{X} - \boldsymbol{P}_V \boldsymbol{X}\|^2\right)\right\} = \operatorname*{argmax}_V \left\{\mathbb{E}\left(\|\boldsymbol{P}_V \boldsymbol{X}\|^2\right)\right\}.} \tag{6.6.3}$$

Eq. (6.6.3) provides the following two interpretations of the optimal linear subspace V^*:

- V^* is the linear subspace that minimizes the average fitting error $\mathbb{E}\left(\|\boldsymbol{X} - \boldsymbol{P}_V \boldsymbol{X}\|^2\right)$.
- V^* is the linear subspace that maximizes the variance of the projection $\boldsymbol{P}_{V^*} \boldsymbol{X}$.

Both interpretations indicate that the optimal linear subspace V^* passes through the "middle" of the centered data points; see the upper panel of Figure 6.1 for an illustration.

Using a linear subspace, rather than an affine subspace, corresponds to the centering condition in Assumption 1.

6.6.3 Formulas for V^* and $\boldsymbol{P}_{V^*}$

The dual characterization becomes explicit through the eigendecomposition of the covariance matrix. Since $\mathbb{V}(\boldsymbol{X})$ is positive semi-definite (see Claim 6.5.1), it has the following eigenstructure:

- eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_d > \lambda_{d+1} \geq \ldots \geq \lambda_D \geq 0$;
- corresponding eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_D$ satisfying $\mathbb{V}(\boldsymbol{X})\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ for all $i = 1, 2, \ldots, D$ (the eigenvectors are viewed as column vectors);
- the eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_D$ are orthonormal, i.e.,

$$\boldsymbol{v}_i^\intercal \boldsymbol{v}_j = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases} \tag{6.6.4}$$

Then, the optimal linear subspace V^* defined in Eq. (6.6.3) is

$$\boxed{V^* = \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_d\} = \left\{ \sum_{k=1}^{d} \alpha_k \boldsymbol{v}_k \,:\, \alpha_1, \ldots, \alpha_d \in \mathbb{R} \right\},}$$

where span denotes the linear span. Define a D-by-d matrix $\boldsymbol{W}$ by

$$\boldsymbol{W} := (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d),$$

i.e., the column vector $\boldsymbol{v}_i$ is the i-th column of $\boldsymbol{W}$. Then, the projection $\boldsymbol{P}_{V^*}$ can be represented by the following matrix:

$$\boxed{\boldsymbol{P}_{V^*} = \boldsymbol{W}\boldsymbol{W}^\intercal,} \tag{6.6.5}$$

i.e., $\boldsymbol{P}_{V^*}\boldsymbol{X} = \boldsymbol{W}\boldsymbol{W}^\intercal\boldsymbol{X}$.

6.6.4 Choosing the Intrinsic Dimension d

Once the principal directions have been identified, the next question is how many of them to retain. The following claim leads to a standard explained-variance criterion.

Claim 6.6.1. *Let $\boldsymbol{X} = (X_1, \ldots, X_D)^\intercal$ be a D-dimensional random vector satisfying $\mathbb{E}X_i = 0$ for all $i = 1, 2, \ldots, D$. The projection matrix $\boldsymbol{P}_{V^*}$ is defined by Eq. (6.6.5). Then, we have*

i) $\mathbb{E}\left(\|\boldsymbol{X}\|^2\right) = \sum_{k=1}^{D} \lambda_k$.

ii) $\mathbb{E}\left(\|\boldsymbol{P}_{V^*}\boldsymbol{X}\|^2\right) = \sum_{k=1}^{d} \lambda_k$.

Proof. The proof is assigned in Section 6.7. □

The following ratio is interpreted as the proportion of variance explained by PCA:

$$\boxed{r_d := \frac{\mathbb{E}\left(\|\boldsymbol{P}_{V^*}\boldsymbol{X}\|^2\right)}{\mathbb{E}\left(\|\boldsymbol{X}\|^2\right)} = \frac{\sum_{k=1}^{d}\lambda_k}{\sum_{k=1}^{D}\lambda_k}.}$$

This ratio r_d provides a criterion for choosing the intrinsic dimension d in applications. A widely adopted rule is to choose the smallest dimension whose explained-variance ratio is at least 95%:

$$d := \min\left\{k \in \{1,\dots,D\} : r_k \geq 0.95\right\}.$$

6.6.5 Further Interpretation of r_d

To interpret the explained-variance ratio under a concrete signal-plus-noise model, suppose that $\boldsymbol{X}$ is generated by the following mechanism, a special case of Eq. (6.3.2):

$$\boldsymbol{X} = \boldsymbol{L}\boldsymbol{Z} + \boldsymbol{\varepsilon},$$

- $\boldsymbol{Z}$ is an unobserved latent d-dimensional random vector;
- $\boldsymbol{L}$ is an unobserved deterministic D-by-d matrix;
- $\boldsymbol{\varepsilon} = (\varepsilon_1,\dots,\varepsilon_D)^\intercal$ is a D-dimensional random vector representing noise; we assume $\varepsilon_1,\dots,\varepsilon_D \overset{iid}{\sim} N(0,\sigma^2)$, which implies

$$\mathbb{V}(\boldsymbol{\varepsilon}) = \begin{pmatrix} \sigma^2 & & \\ & \ddots & \\ & & \sigma^2 \end{pmatrix}, \tag{6.6.6}$$

  which is a D-by-D diagonal matrix;
- The latent random variable $\boldsymbol{Z}$ and noise $\boldsymbol{\varepsilon}$ are independent;
- The D-dimensional random vector $\boldsymbol{X}$ represents the data we observe.

Claim 6.5.2, together with the independence between $\boldsymbol{Z}$ and $\boldsymbol{\varepsilon}$, implies

$$\mathbb{V}(\boldsymbol{X}) = \boldsymbol{L}\mathbb{V}(\boldsymbol{Z})\boldsymbol{L}^\intercal + \mathbb{V}(\boldsymbol{\varepsilon}). \tag{6.6.7}$$

The percentage 95% can be replaced with another threshold.

We have the following rank bound:

$$\operatorname{rank}(\boldsymbol{L}\mathbb{V}(\boldsymbol{Z})\boldsymbol{L}^\intercal) \le \operatorname{rank}(\mathbb{V}(\boldsymbol{Z})) \le d,$$

where the last inequality follows from the fact that $\mathbb{V}(\boldsymbol{Z})$ is a d-by-d matrix. Define

$$r := \operatorname{rank}(\boldsymbol{L}\mathbb{V}(\boldsymbol{Z})\boldsymbol{L}^\intercal) \le d.$$

The matrix $\boldsymbol{L}\mathbb{V}(\boldsymbol{Z})\boldsymbol{L}^\intercal$ therefore has exactly r positive eigenvalues, which we denote by

$$\tilde{\lambda}_1 \ge \cdots \ge \tilde{\lambda}_r > 0.$$

We have the following eigen decomposition:

$$\boldsymbol{L}\mathbb{V}(\boldsymbol{Z})\boldsymbol{L}^\intercal = \boldsymbol{U} \begin{pmatrix} \tilde{\lambda}_1 & & & & & \\ & \ddots & & & & \\ & & \tilde{\lambda}_r & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix} \boldsymbol{U}^\intercal, \tag{6.6.8}$$

where $\boldsymbol{U}$ is an orthogonal matrix (i.e., $\boldsymbol{U}\boldsymbol{U}^\intercal = \boldsymbol{U}^\intercal\boldsymbol{U} = \boldsymbol{I}_D =$ the D-by-D identity matrix). Combining Eq. (6.6.6), (6.6.7), and (6.6.8), we have

$$\mathbb{V}(\boldsymbol{X}) = \boldsymbol{U} \begin{pmatrix} \tilde{\lambda}_1 + \sigma^2 & & & & & \\ & \ddots & & & & \\ & & \tilde{\lambda}_r + \sigma^2 & & & \\ & & & \sigma^2 & & \\ & & & & \ddots & \\ & & & & & \sigma^2 \end{pmatrix} \boldsymbol{U}^\intercal. \tag{6.6.9}$$

Therefore, the eigenvalues $\{\lambda_k\}_{k=1}^D$ of $\mathbb{V}(\boldsymbol{X})$ are the following

$$\begin{aligned}
&\lambda_1 = \tilde{\lambda}_1 + \sigma^2, \\
&\vdots \\
&\lambda_r = \tilde{\lambda}_r + \sigma^2, \\
&\lambda_{r+1} = \sigma^2, \\
&\vdots \\
&\lambda_D = \sigma^2.
\end{aligned} \tag{6.6.10}$$

If the noise is very small (i.e., $\sigma^2 \approx 0$), then

$$\lambda_1 \geq \ldots \geq \lambda_r > \lambda_{r+1} = \ldots = \lambda_D = \sigma^2 \approx 0.$$

Furthermore, we have

$$r_d = \frac{\sum_{k=1}^d \lambda_k}{\sum_{k=1}^D \lambda_k} = \frac{\sum_{k=1}^r \tilde{\lambda}_k + d \cdot \sigma^2}{\sum_{k=1}^r \tilde{\lambda}_k + D \cdot \sigma^2}, \tag{6.6.11}$$

where we used $r \leq d$. The relevant comparison is between the *total* noise variance and the total signal variance, rather than between σ^2 and zero. Indeed, Eq. (6.6.11) gives the exact identity

$$1 - r_d = \frac{(D-d)\sigma^2}{\sum_{k=1}^r \tilde{\lambda}_k + D\sigma^2}.$$

Consequently, a sufficient condition for $r_d \approx 1$ is

$$D\sigma^2 \ll \sum_{k=1}^r \tilde{\lambda}_k.$$

Under this dimension-aware condition, the total signal dominates the noise terms in both the numerator and denominator of Eq. (6.6.11). Hence,

$$r_D = 1, \qquad r_{D-1} \approx \ldots \approx r_{d+1} \approx r_d \approx 1.$$

6.6.6 Computation in Applications

The preceding analysis is stated at the population level. After a centering convention has been chosen, the population covariance matrix $\mathbb{V}(\boldsymbol{X})$ determines the principal directions. In applications this matrix is generally unknown, so sample PCA centers the observed data and estimates the directions from the sample covariance matrix. Given

observed data points $\{\boldsymbol{x}_i = (x_{i,1}, x_{i,2}, \ldots, x_{i,D})^\intercal\}_{i=1}^n \subseteq \mathbb{R}^D$, the sample covariance matrix $\widehat{\mathbb{V}(\boldsymbol{X})}$ is

$$\widehat{\mathbb{V}(\boldsymbol{X})} := \begin{pmatrix} \widehat{\mathrm{Cov}}(X_1, X_1) & \widehat{\mathrm{Cov}}(X_1, X_2) & \cdots & \widehat{\mathrm{Cov}}(X_1, X_D) \\ \widehat{\mathrm{Cov}}(X_2, X_1) & \widehat{\mathrm{Cov}}(X_2, X_2) & \cdots & \widehat{\mathrm{Cov}}(X_2, X_D) \\ \vdots & \vdots & \ddots & \vdots \\ \widehat{\mathrm{Cov}}(X_D, X_1) & \widehat{\mathrm{Cov}}(X_D, X_2) & \cdots & \widehat{\mathrm{Cov}}(X_D, X_D) \end{pmatrix}, \tag{6.6.12}$$

$$\text{where } \widehat{\mathrm{Cov}}(X_i, X_j) := \frac{1}{n-1} \sum_{w=1}^{n} \left[\left(x_{w,i} - \frac{1}{n} \sum_{k=1}^{n} x_{k,i} \right) \left(x_{w,j} - \frac{1}{n} \sum_{l=1}^{n} x_{l,j} \right) \right].$$

6.6.7 A Numerical Example

This subsection illustrates the preceding theory with a simple numerical example.

- $Z \sim N(0, 1)$;
- $\boldsymbol{L} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$;
- $\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{pmatrix}$, where $\varepsilon_1, \varepsilon_2, \varepsilon_3 \overset{iid}{\sim} N(0,\, 0.04)$;
- $\boldsymbol{X} \overset{\text{def}}{=} \boldsymbol{L}Z + \boldsymbol{\varepsilon} = \begin{pmatrix} Z + \varepsilon_1 \\ Z + \varepsilon_2 \\ Z + \varepsilon_3 \end{pmatrix}$.

Then, we have

- $\mathbb{V}(\boldsymbol{L}Z) = \boldsymbol{L}\boldsymbol{L}^\intercal = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, whose eigenvalues are $\tilde{\lambda}_1 = 3, \tilde{\lambda}_2 = \tilde{\lambda}_3 = 0$ (you have learned how to compute eigenvalues in your linear algebra class);
- Eq. (6.6.10) implies that the eigenvalues of $\mathbb{V}(\boldsymbol{X})$ are

$$\lambda_1 = 3 + 0.04 = 3.04,$$

$$\lambda_2 = \lambda_3 = 0.04.$$

The results of a numerical experiment conducted using R are presented as follows, and they are compatible with our theoretical discussion.

```
> n=10000000
> sigma=0.2
>
> Z=rnorm(n)
> L=matrix(1, nrow = 3, ncol = 1)
> e=cbind(rnorm(n, sd=sigma), rnorm(n, sd=sigma), rnorm(n, sd=sigma))
> X=t(L%*%Z)+e
>
> covariance_matrix=var(X)
> eigenstructure=eigen(covariance_matrix)
> eigenstructure
eigen() decomposition
$values
[1] 3.04140759 0.04002077 0.03997760

$vectors
           [,1]       [,2]       [,3]
[1,] 0.5773425  0.1238912  0.8070481
[2,] 0.5772954 -0.7609286 -0.2961717
[3,] 0.5774129  0.6368977 -0.5108382
```

6.7 Homework for Chapter 6

Problem 6.7.1 (Positive semi-definiteness of covariance matrices)**.** *Prove Claim 6.5.1. In particular, show that for every deterministic vector* $\boldsymbol{a} \in \mathbb{R}^n$*,*

$$\boldsymbol{a}^\intercal \mathbb{V}(\boldsymbol{X})\boldsymbol{a} = \mathrm{Var}(\boldsymbol{a}^\intercal \boldsymbol{X}) \geq 0.$$

Problem 6.7.2 (Variance explained by an optimal PCA subspace)**.** *Prove both identities in Claim 6.6.1. You may use Eq. (6.2.1), Claim 6.5.3, the cyclic property of the trace, Eq. (6.6.4), and Eq. (6.6.5).*

Problem 6.7.3 (A numerical PCA experiment)**.** *Use a fixed random seed and complete the following experiment.*

i) Set $n = 100$ *and generate independent observations* $Z_1, \ldots, Z_n \sim \mathcal{N}(0, 1)$*.*

ii) For $i = 1, \ldots, n$*, independently generate*

$$\boldsymbol{\varepsilon}_i = \begin{pmatrix} \varepsilon_{i,1} \\ \varepsilon_{i,2} \end{pmatrix}, \qquad \varepsilon_{i,j} \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.04),$$

independently of $Z_1, \ldots, Z_n$, *and define*

$$\boldsymbol{X}_i = \begin{pmatrix} Z_i + \varepsilon_{i,1} \\ Z_i + \varepsilon_{i,2} \end{pmatrix}.$$

iii) Compute and report the sample covariance matrix $\widehat{\mathbb{V}(\boldsymbol{X})}$ *defined in Eq.* (6.6.12).

iv) Compute and report the eigenvalues $\widehat{\lambda}_1 \geq \widehat{\lambda}_2$ *and corresponding orthonormal eigenvectors* $\widehat{\boldsymbol{v}}_1$ *and* $\widehat{\boldsymbol{v}}_2$ *of* $\widehat{\mathbb{V}(\boldsymbol{X})}$.

v) Form the projection matrix

$$\widehat{\boldsymbol{P}} := \widehat{\boldsymbol{v}}_1 \widehat{\boldsymbol{v}}_1^{\intercal},$$

and report its entries.

vi) In a single figure, display the observations $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$, *the estimated principal line*

$$\widehat{V}^* := \{\alpha \widehat{\boldsymbol{v}}_1 : \alpha \in \mathbb{R}\},$$

the projected points $\widehat{\boldsymbol{P}}\boldsymbol{X}_1, \ldots, \widehat{\boldsymbol{P}}\boldsymbol{X}_n$, *and the line segment joining each observation to its projection. Use equal scaling on the two axes and a clear legend.*

vii) Submit the code used for the calculations and visualization.

Chapter 7

Appendix

7.A Abbreviations

- APMA: applied mathematics; the Division of Applied Mathematics at Brown University
- CDF: cumulative distribution function
- CLT: central limit theorem
- GRF: Gibbs random field
- HRNG: hardware random number generator
- iid: independent and identically distributed
- LLN: law of large numbers
- MCG: multiplicative congruential generator
- MCMC: Markov chain Monte Carlo
- MLE: maximum likelihood estimation
- MRF: Markov random field
- PCA: principal component analysis
- PDF: probability density function
- PMF: probability mass function
- PRNG: pseudorandom number generator

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